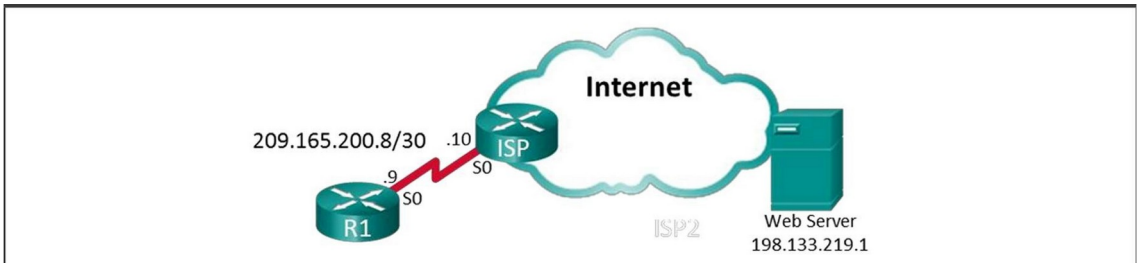
INDEX

Sr. No.	Practical	Date	Sign
1.	Implement IP SLA (IP Service Level Agreement)	01/08/26	
2.	Implement IPv4 ACLs a) Standard ACL b) Extended ACL	08/08/26	
3.	a) Implement SPAN Technologies (Switch Port Analyzer) b) Implement SNMP and Syslog c) Implement Flexible NetFlow	22/08/26	
4.	a) Implement a GRE Tunnel b) Implement VTP c) Implement NAT	05/09/26	
5.	Implement Inter-VLAN Routing	12/09/26	
6.	Observe STP Topology Changes and Implement RSTP a) Implement Advanced STP Modifications and Mechanisms b) Implement MST	03/10/26	
7.	a) Implement Ether Channel b) Tune and Optimize Ether Channel Operations	10/10/26	
8.	OSPF Implementation a) Implement Single-Area OSPFv2 b) Implement Multi-Area OSPFv2	17/10/26	
9.	a) Implement BGP Communities b) Implement MP-BGP c) Implement eBGP for IPv4 d) Implement BGP Path Manipulation	17/10/26	
10.	a) Implement IPsec Site-to-Site VPNs b) Implement GRE over IPsec Site-to-Site VPNs c) Implement VRF Lite	24/10/26	

PRACTICAL-01

Topology



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	S0/0/0	209.165.200.9	255.255.255.252	N/A
ISP	S0/0/0	209.165.200.10	255.255.255.252	N/A
	Lo0	198.133.219.1	255.255.255.255	N/A

Aim:

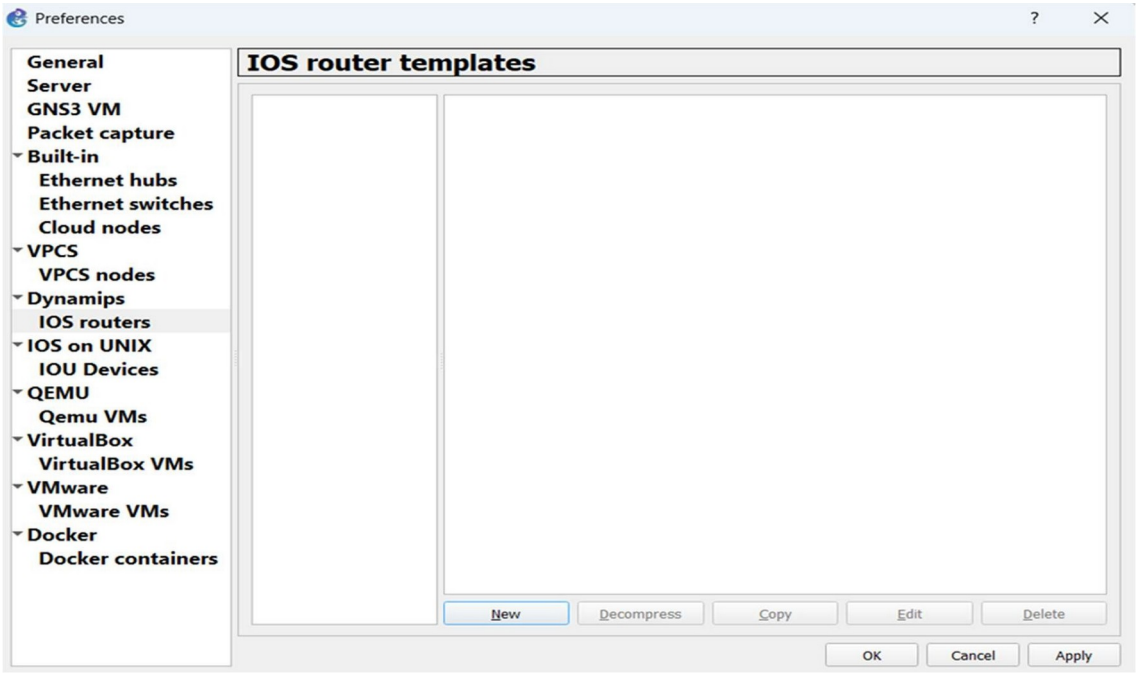
Implement IP SLA (IP Service Level Agreement)

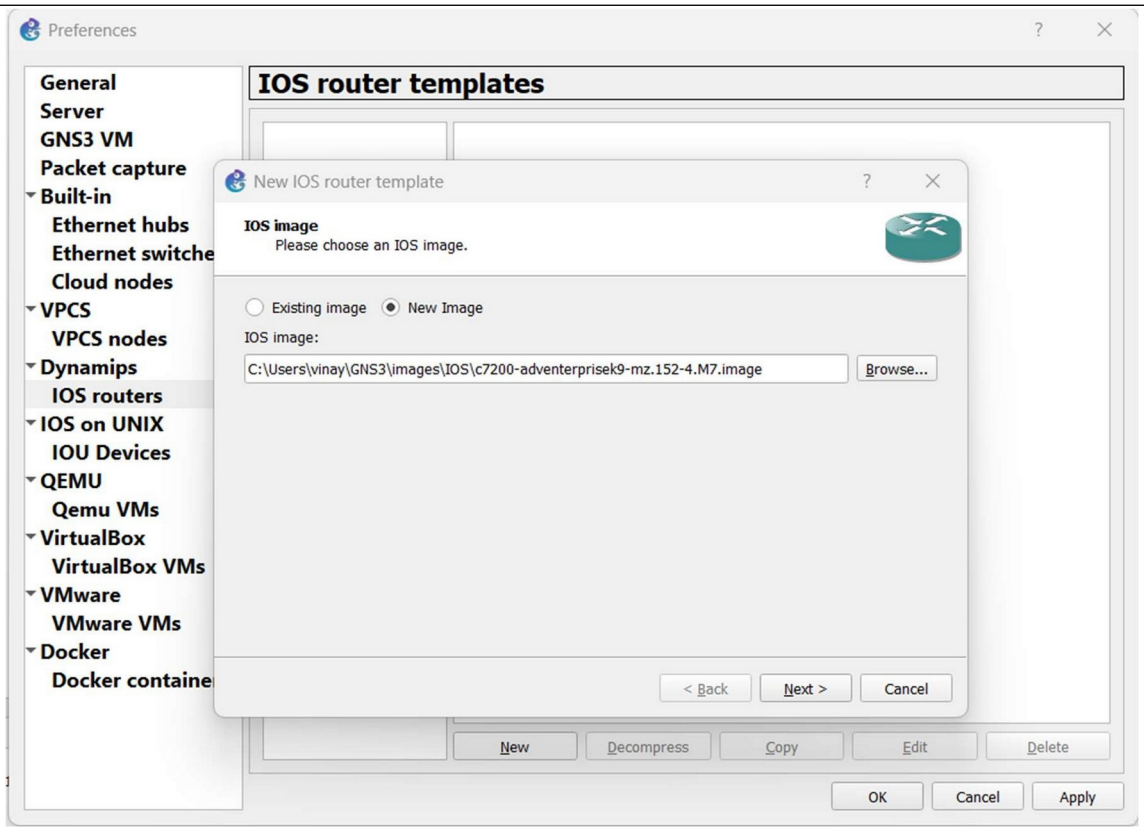
Theory:

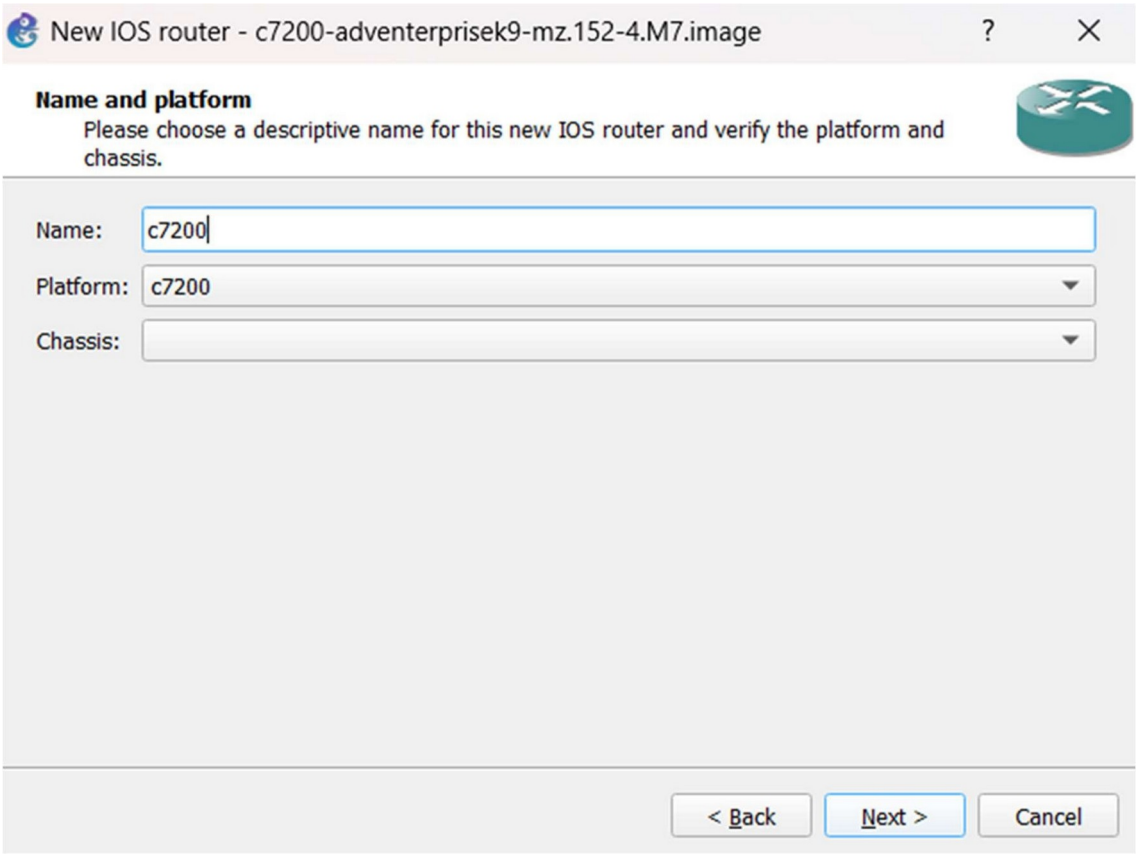
IP Service Level Agreement

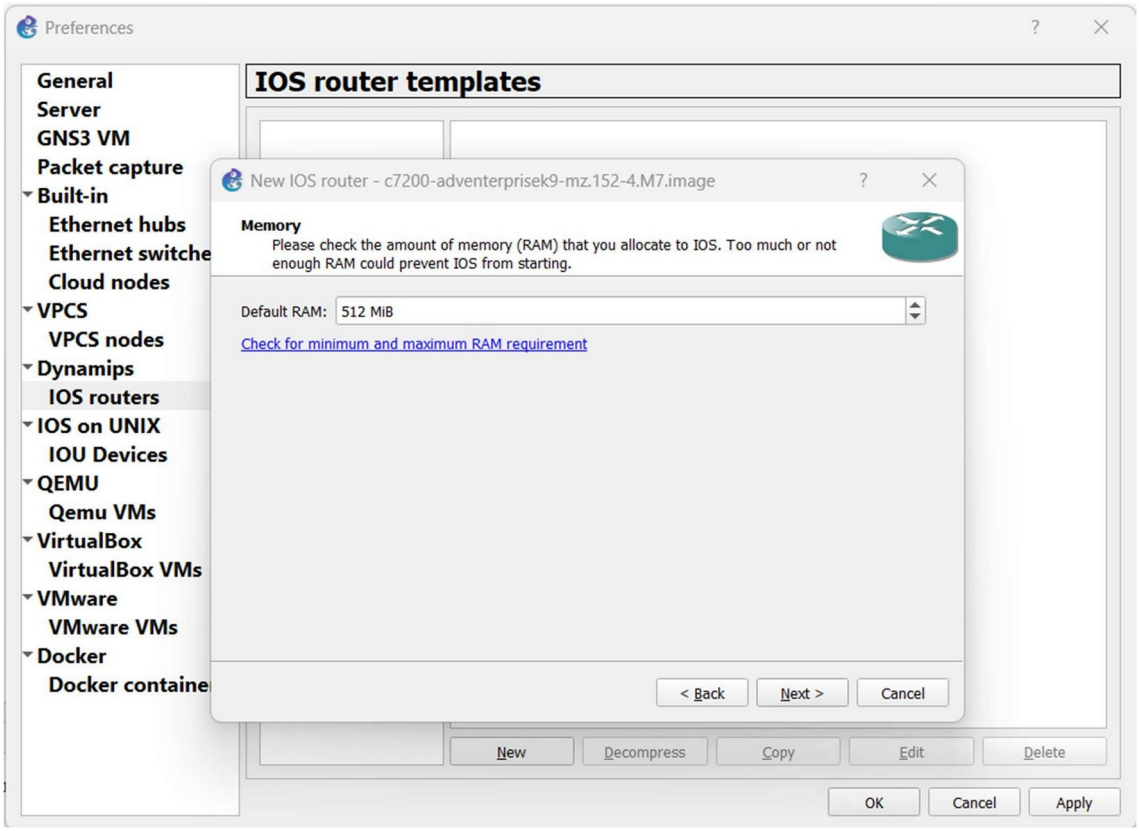
IP Service Level Agreement (IP SLA) is like a promise for your internet connection's performance. It sets expectations on speed and reliability, making it easier to manage and fix issues when needed.

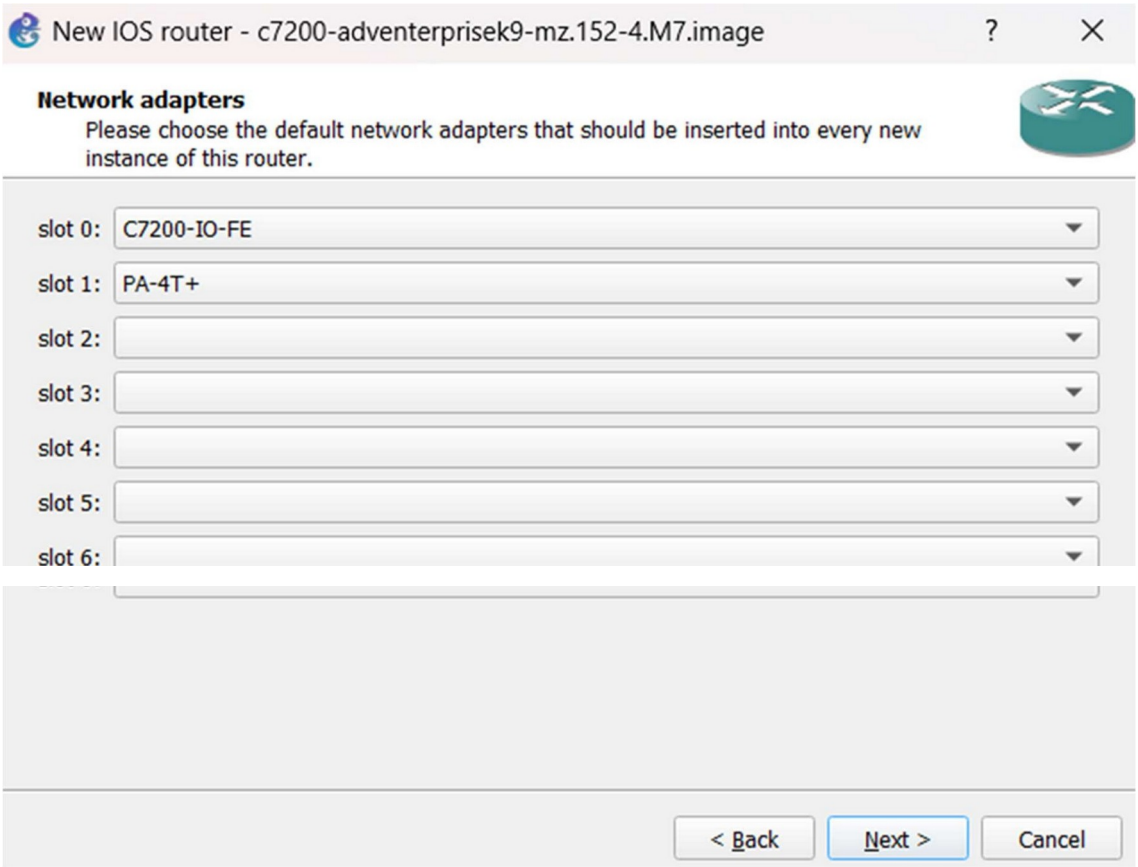
Step 1:

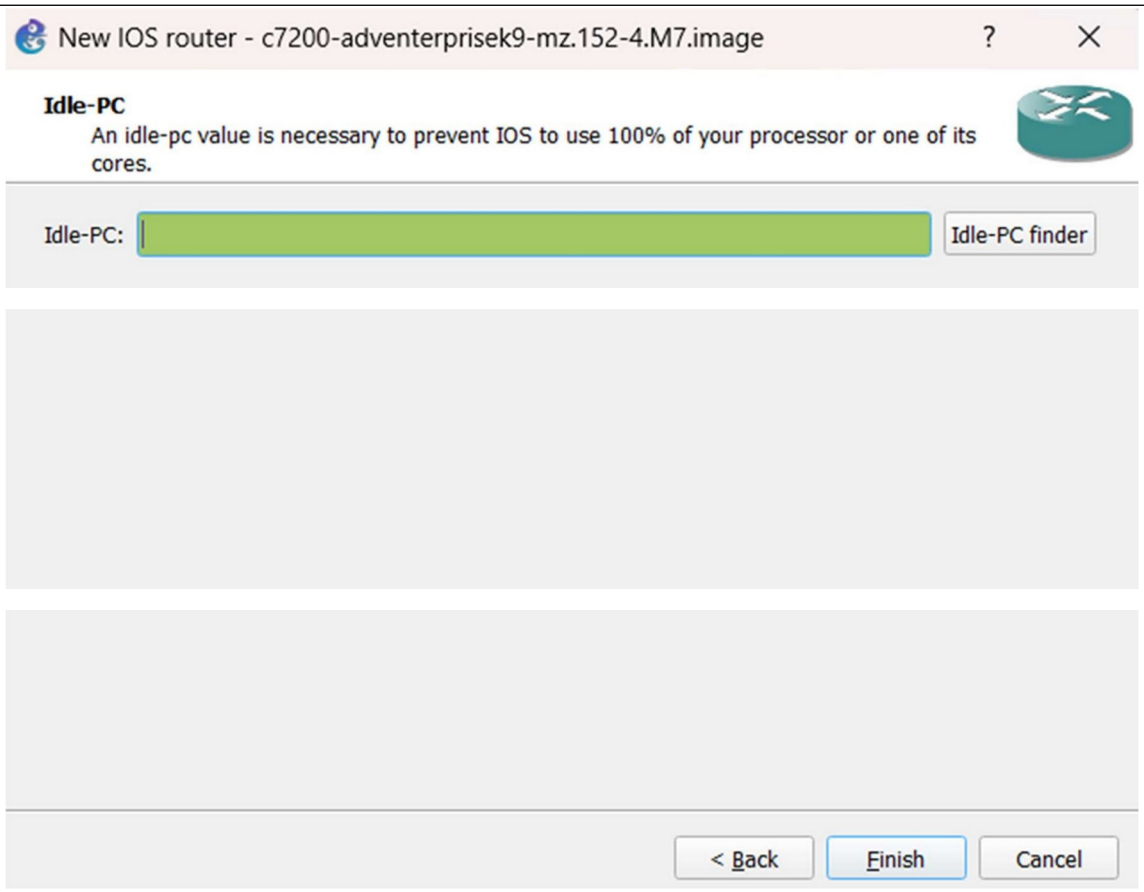


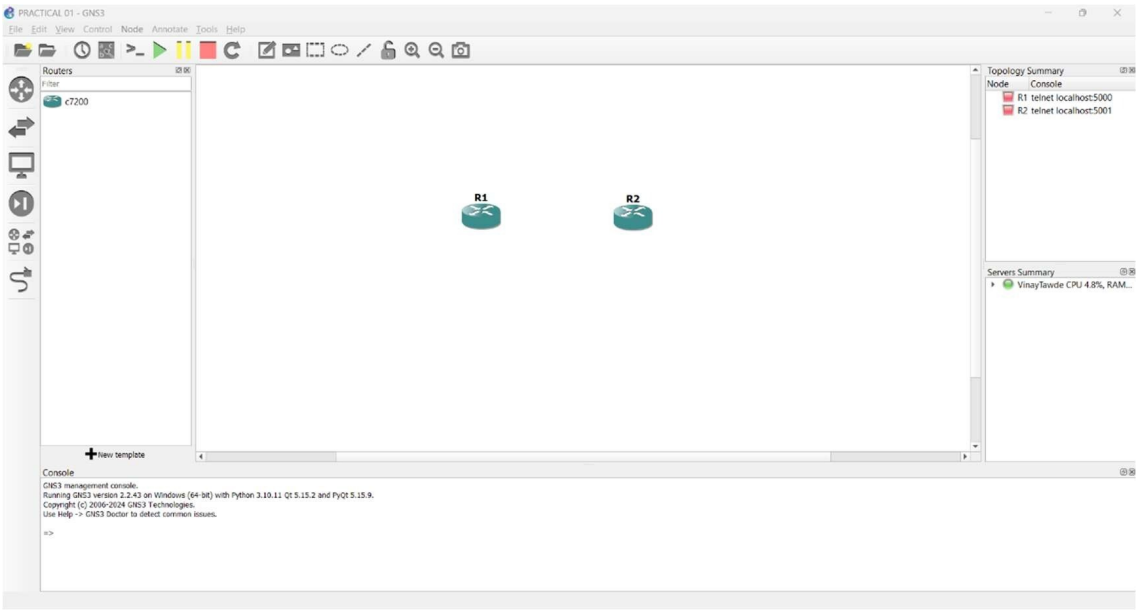

Preferences
General
Server
GNS3 VM
Packet capture
Built-in
Ethernet hubs
Ethernet switche
Cloud nodes
VPCS
VPCS nodes
Dynamips
IOS routers
IOS on UNIX
IOU Devices
QEMU
Qemu VMs
VirtualBox
VirtualBox VMs
VMware
VMware VMs
Docker
Docker containe
IOS router templates
New IOS router template
IOS image
Please choose an IOS image.
Existing image
New Image
IOS image:
C:\Users\vinay\GNS3\images\IOS\c7200-adventerprisek9-mz.152-4.M7.image
Browse...
< Back
Next >
Cancel
New
Decompress
Copy
Edit
Delete
OK
Cancel
Apply


New IOS router - c7200-adventerprisek9-mz.152-4.M7.image
Name and platform
Please choose a descriptive name for this new IOS router and verify the platform and chassis.
Name:
c7200
Platform:
c7200
Chassis:
< Back
Next >
Cancel

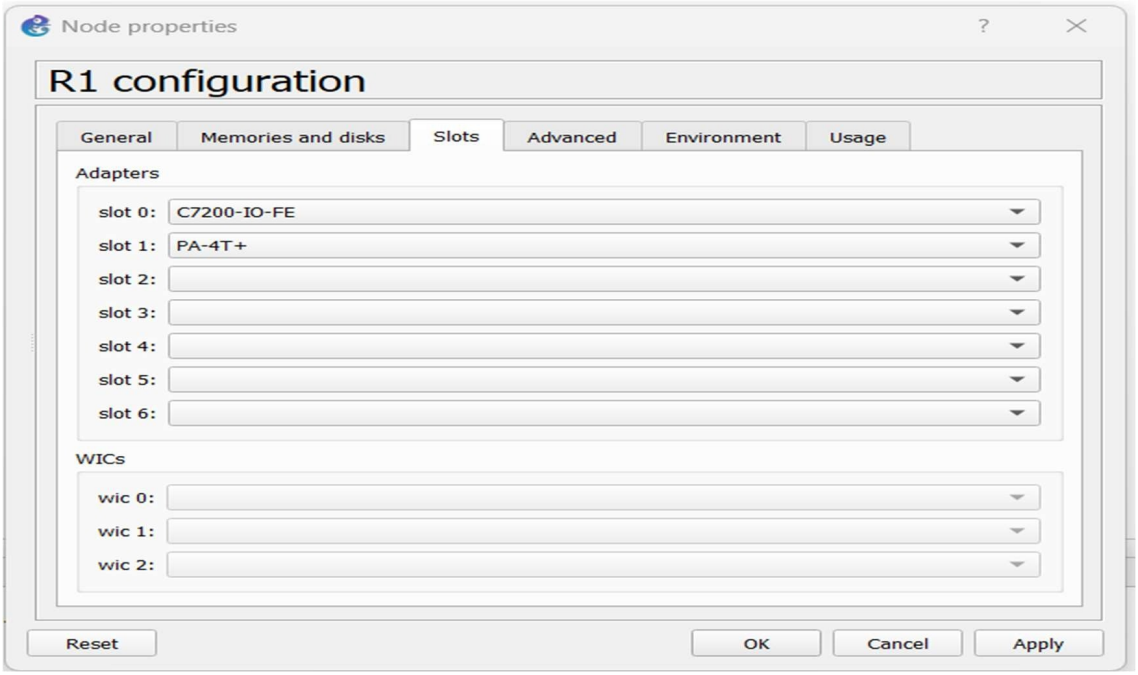

Preferences
General
Server
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Packet capture
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Ethernet hubs
Ethernet switche
Cloud nodes
VPCS
VPCS nodes
Dynamips
IOS routers
IOS on UNIX
IOU Devices
QEMU
Qemu VMs
VirtualBox
VirtualBox VMs
VMware
VMware VMs
Docker
Docker containe
IOS router templates
New IOS router - c7200-adventerprisek9-mz.152-4.M7.image
Memory
Please check the amount of memory (RAM) that you allocate to IOS. Too much or not enough RAM could prevent IOS from starting.
Default RAM: 512 MiB
Check for minimum and maximum RAM requirement
< Back
Next >
Cancel
New
Decompress
Copy
Edit
Delete
OK
Cancel
Apply


New IOS router - c7200-adventerprisek9-mz.152-4.M7.image
Network adapters
Please choose the default network adapters that should be inserted into every new instance of this router.
slot 0: C7200-IO-FE
slot 1: PA-4T+
slot 2:
slot 3:
slot 4:
slot 5:
slot 6:
< Back
Next >
Cancel

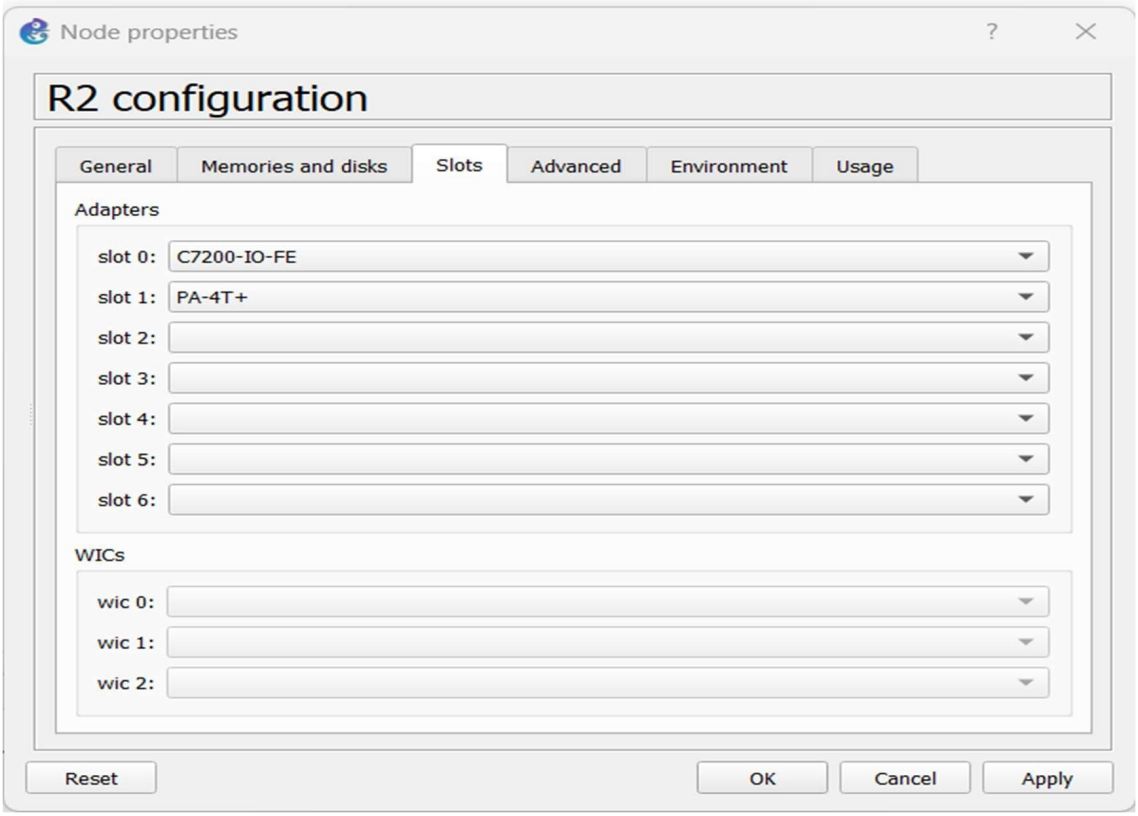

New IOS router - c7200-adventerprisek9-mz.152-4.M7.image
Idle-PC
An idle-pc value is necessary to prevent IOS to use 100% of your processor or one of its cores.
Idle-PC:
Idle-PC finder
< Back
Finish
Cancel


PRACTICAL 01 - GNS3
File Edit View Control Node Annotate Tools Help
Routers
c7200
New template
R1
R2
Topology Summary
Node Console
R1 telnet localhost:5000
R2 telnet localhost:5001
Servers Summary
VinayTawde CPU 4.8%, RAM...
Console
GNS3 management console.
Running GNS3 version 2.2.43 on Windows (64-bit) with Python 3.10.11 Qt 5.15.2 and PyQt 5.15.9.
Copyright (c) 2006-2024 GNS3 Technologies.
Use Help -> GNS3 Doctor to detect common issues.
=>

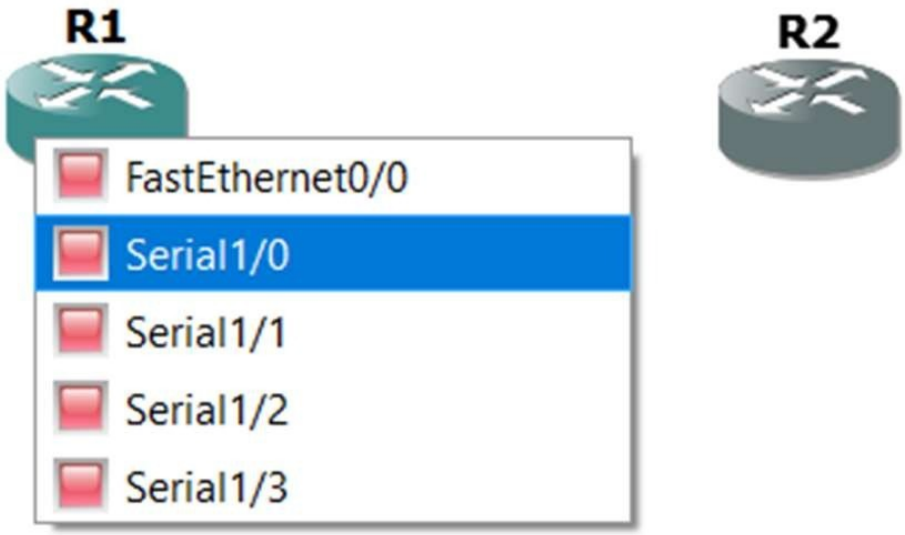
Step 2:

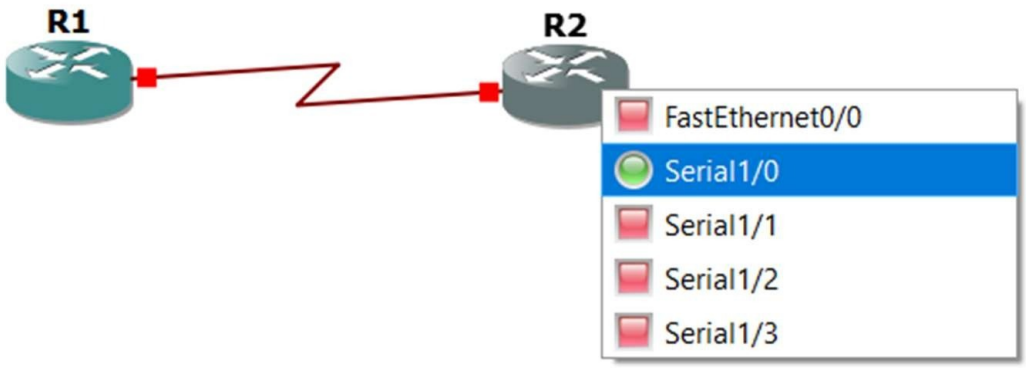

Node properties
R1 configuration
General
Memories and disks
Slots
Advanced
Environment
Usage
Adapters
slot 0: C7200-IO-FE
slot 1: PA-4T+
slot 2:
slot 3:
slot 4:
slot 5:
slot 6:
WICs
wic 0:
wic 1:
wic 2:
Reset
OK
Cancel
Apply

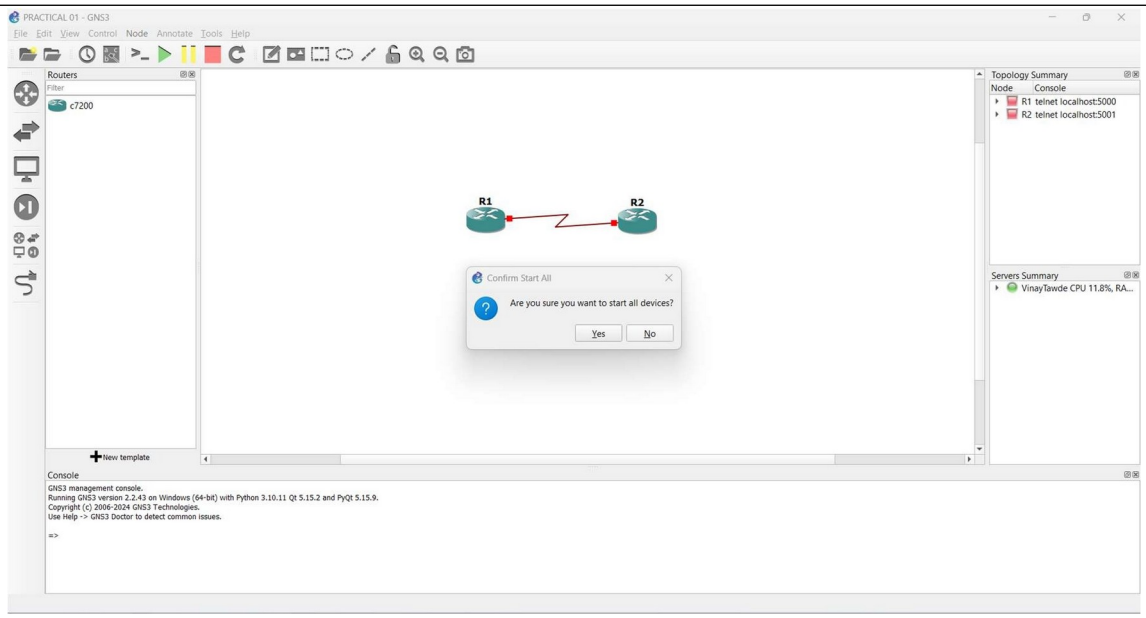
Step 3:


Node properties
R2 configuration
General
Memories and disks
Slots
Advanced
Environment
Usage
Adapters
slot 0: C7200-IO-FE
slot 1: PA-4T+
slot 2:
slot 3:
slot 4:
slot 5:
slot 6:
WICs
wic 0:
wic 1:
wic 2:
Reset
OK
Cancel
Apply

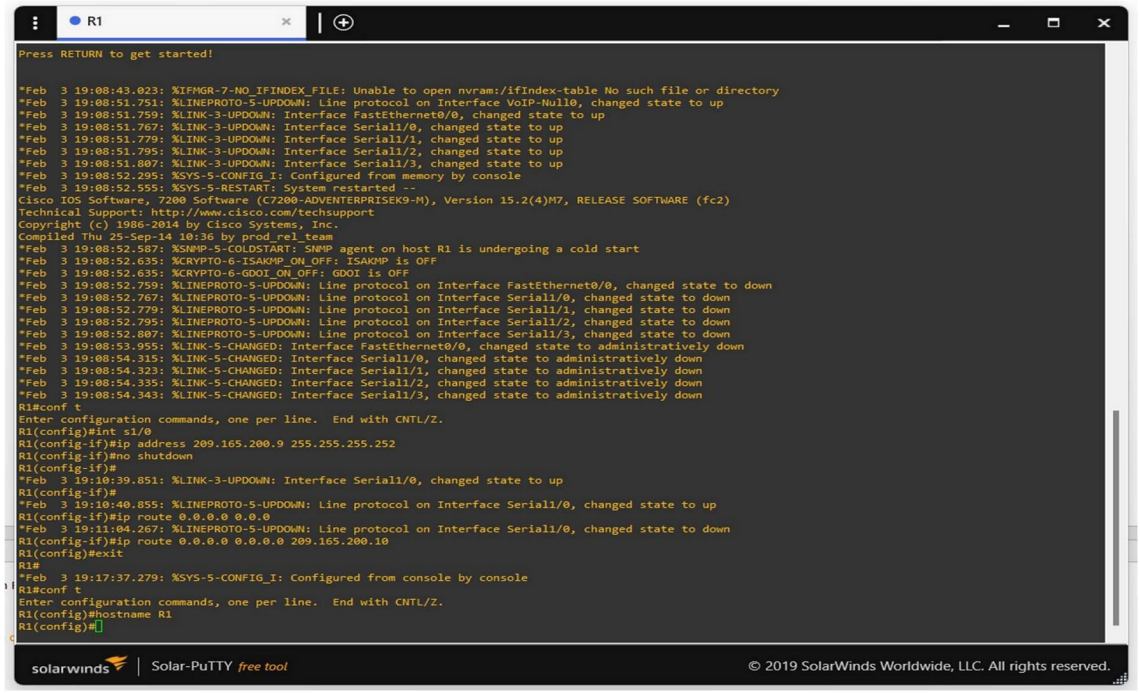
Step 4:


R1
R2
FastEthernet0/0
Serial1/0
Serial1/1
Serial1/2
Serial1/3

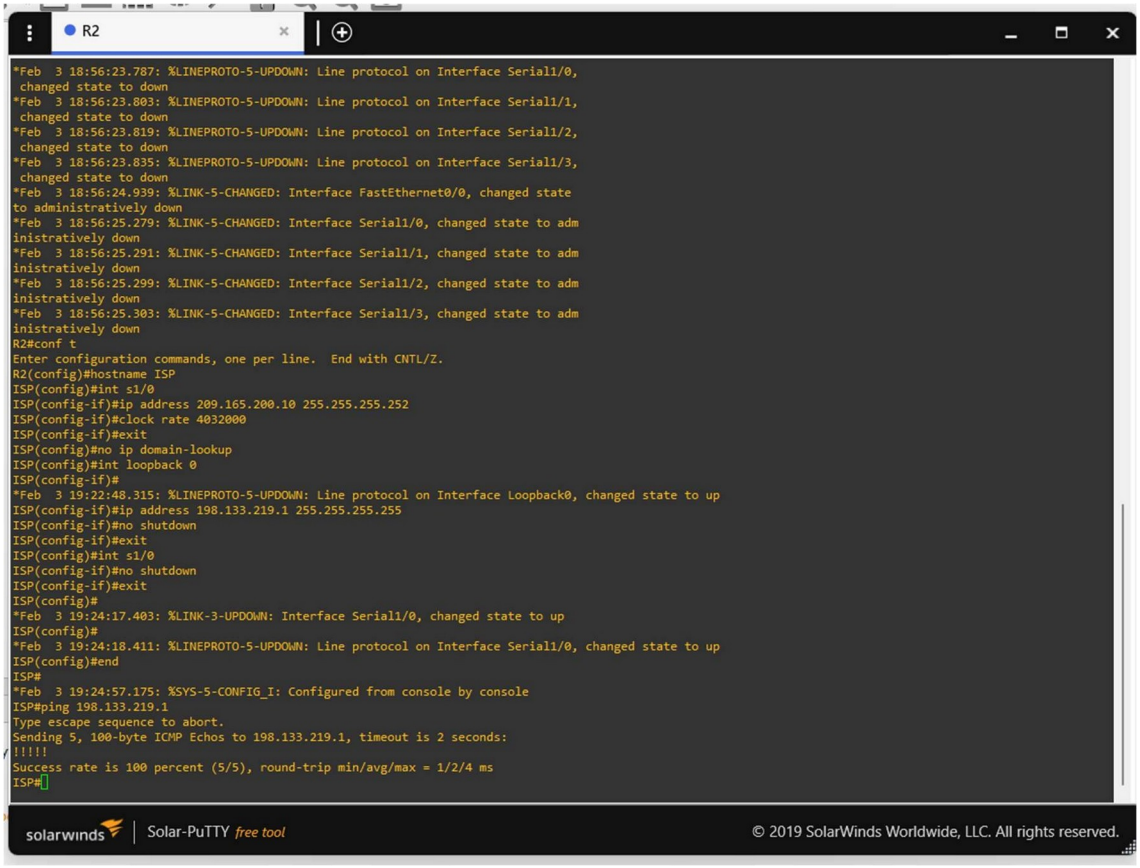

R1
R2
FastEthernet0/0
Serial1/0
Serial1/1
Serial1/2
Serial1/3


PRACTICAL 01 - GNS3
File Edit View Control Node Annotate Tools Help
Routers
Filter
c7200
New template
R1
R2
Confirm Start All
Are you sure you want to start all devices?
Yes
No
Topology Summary
Node Console
R1 telnet localhost:5000
R2 telnet localhost:5001
Servers Summary
VinayTawde CPU 11.8%, RA...
Console
GNS3 management console.
Running GNS3 version 2.2.43 on Windows (64-bit) with Python 3.10.11 Qt 5.15.2 and PyQt 5.15.9.
Copyright (c) 2006-2024 GNS3 Technologies.
Use Help -> GNS3 Doctor to detect common issues.
=>

Step 5:


R1
Press RETURN to get started!
*Feb 3 19:08:43.023: %IFMGR-7-NO_IFINDEX_FILE: Unable to open nvram:/ifIndex-table No such file or directory
*Feb 3 19:08:51.751: %LINEPROTO-5-UPDOWN: Line protocol on Interface VoIP-Null0, changed state to up
*Feb 3 19:08:51.759: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Feb 3 19:08:51.767: %LINK-3-UPDOWN: Interface Serial1/0, changed state to up
*Feb 3 19:08:51.779: %LINK-3-UPDOWN: Interface Serial1/1, changed state to up
*Feb 3 19:08:51.795: %LINK-3-UPDOWN: Interface Serial1/2, changed state to up
*Feb 3 19:08:51.807: %LINK-3-UPDOWN: Interface Serial1/3, changed state to up
*Feb 3 19:08:52.295: %SYS-5-CONFIG_I: Configured from memory by console
*Feb 3 19:08:52.555: %SYS-5-RESTART: System restarted --
Cisco IOS Software, 7200 Software (C7200-ADVENTERPRISEK9-M), Version 15.2(4)M7, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2014 by Cisco Systems, Inc.
Compiled Thu 25-Sep-14 10:36 by prod_rel_team
*Feb 3 19:08:52.587: %SNMP-5-COLDSTART: SNMP agent on host R1 is undergoing a cold start
*Feb 3 19:08:52.635: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is OFF
*Feb 3 19:08:52.635: %CRYPTO-6-GDOI_ON_OFF: GDOI is OFF
*Feb 3 19:08:52.759: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to down
*Feb 3 19:08:52.767: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/0, changed state to down
*Feb 3 19:08:52.779: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/1, changed state to down
*Feb 3 19:08:52.795: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/2, changed state to down
*Feb 3 19:08:52.807: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/3, changed state to down
*Feb 3 19:08:53.955: %LINK-5-CHANGED: Interface FastEthernet0/0, changed state to administratively down
*Feb 3 19:08:54.315: %LINK-5-CHANGED: Interface Serial1/0, changed state to administratively down
*Feb 3 19:08:54.323: %LINK-5-CHANGED: Interface Serial1/1, changed state to administratively down
*Feb 3 19:08:54.335: %LINK-5-CHANGED: Interface Serial1/2, changed state to administratively down
*Feb 3 19:08:54.343: %LINK-5-CHANGED: Interface Serial1/3, changed state to administratively down
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int s1/0
R1(config-if)#ip address 209.165.200.9 255.255.255.252
R1(config-if)#no shutdown
R1(config-if)#
*Feb 3 19:10:39.851: %LINK-3-UPDOWN: Interface Serial1/0, changed state to up
R1(config-if)#
*Feb 3 19:10:40.855: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/0, changed state to up
R1(config-if)#ip route 0.0.0.0 0.0.0.0
*Feb 3 19:11:04.267: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/0, changed state to down
R1(config-if)#ip route 0.0.0.0 0.0.0.0 209.165.200.10
R1(config)#exit
R1#
*Feb 3 19:17:37.279: %SYS-5-CONFIG_I: Configured from console by console
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#hostname R1
R1(config)#
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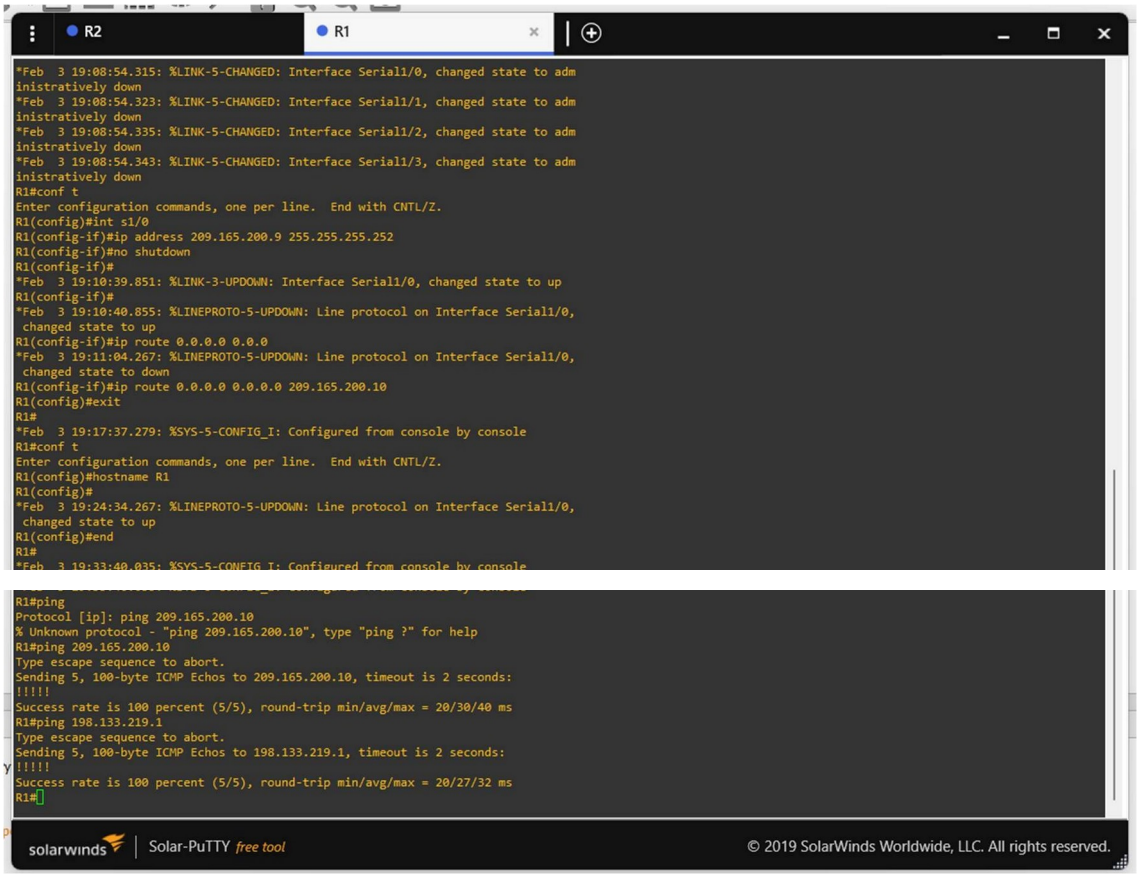
Step 6:


R2
*Feb 3 18:56:23.787: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/0, changed state to down
*Feb 3 18:56:23.803: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/1, changed state to down
*Feb 3 18:56:23.819: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/2, changed state to down
*Feb 3 18:56:23.835: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/3, changed state to down
*Feb 3 18:56:24.939: %LINK-5-CHANGED: Interface FastEthernet0/0, changed state to administratively down
*Feb 3 18:56:25.279: %LINK-5-CHANGED: Interface Serial1/0, changed state to administratively down
*Feb 3 18:56:25.291: %LINK-5-CHANGED: Interface Serial1/1, changed state to administratively down
*Feb 3 18:56:25.299: %LINK-5-CHANGED: Interface Serial1/2, changed state to administratively down
*Feb 3 18:56:25.303: %LINK-5-CHANGED: Interface Serial1/3, changed state to administratively down
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#hostname ISP
ISP(config)#int s1/0
ISP(config-if)#ip address 209.165.200.10 255.255.255.252
ISP(config-if)#clock rate 4032000
ISP(config-if)#exit
ISP(config)#no ip domain-lookup
ISP(config)#int loopback 0
ISP(config-if)#
*Feb 3 19:22:48.315: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
ISP(config-if)#ip address 198.133.219.1 255.255.255.255
ISP(config-if)#no shutdown
ISP(config-if)#exit
ISP(config)#int s1/0
ISP(config-if)#no shutdown
ISP(config-if)#exit
ISP(config)#
*Feb 3 19:24:17.403: %LINK-3-UPDOWN: Interface Serial1/0, changed state to up
ISP(config)#
*Feb 3 19:24:18.411: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/0, changed state to up
ISP(config)#end
ISP#
*Feb 3 19:24:57.175: %SYS-5-CONFIG_I: Configured from console by console
ISP#ping 198.133.219.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 198.133.219.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/4 ms
ISP#
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Solar-PuTTY free tool
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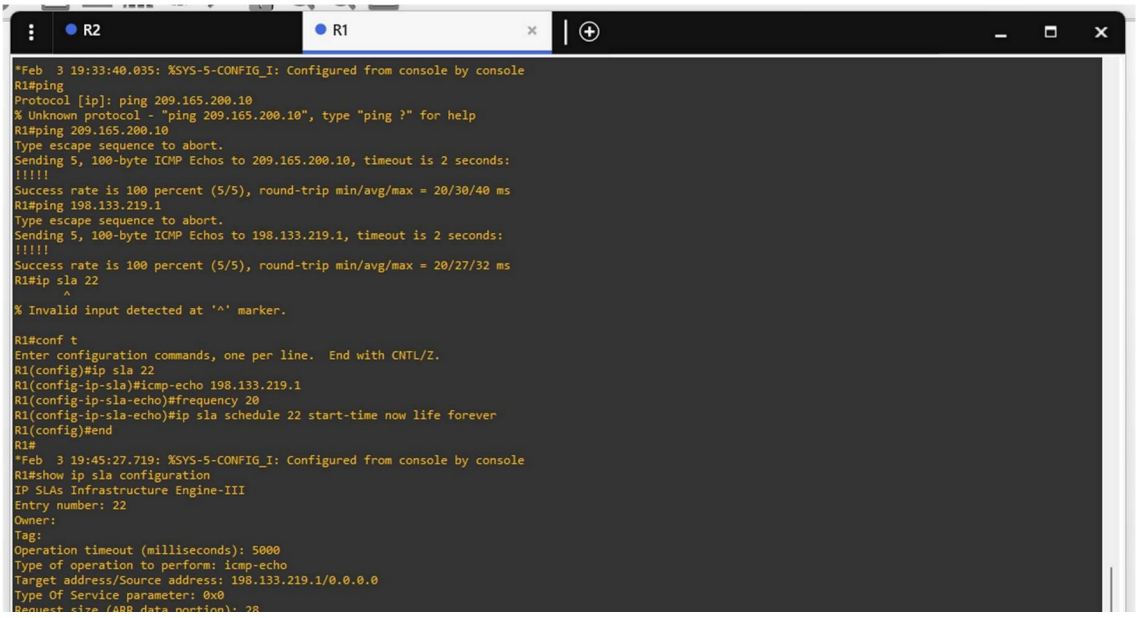
Step 7:


R2
R1
changed state to down
*Feb 3 19:08:52.807: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/3, changed state to down
*Feb 3 19:08:53.955: %LINK-5-CHANGED: Interface FastEthernet0/0, changed state to administratively down
*Feb 3 19:08:54.315: %LINK-5-CHANGED: Interface Serial1/0, changed state to administratively down
*Feb 3 19:08:54.323: %LINK-5-CHANGED: Interface Serial1/1, changed state to administratively down
*Feb 3 19:08:54.335: %LINK-5-CHANGED: Interface Serial1/2, changed state to administratively down
*Feb 3 19:08:54.343: %LINK-5-CHANGED: Interface Serial1/3, changed state to administratively down
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int s1/0
R1(config-if)#ip address 209.165.200.9 255.255.255.252
R1(config-if)#no shutdown
R1(config-if)#
*Feb 3 19:10:39.851: %LINK-3-UPDOWN: Interface Serial1/0, changed state to up
R1(config-if)#
*Feb 3 19:10:40.855: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/0, changed state to up
R1(config-if)#ip route 0.0.0.0 0.0.0
*Feb 3 19:11:04.267: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/0, changed state to down
R1(config-if)#ip route 0.0.0.0 0.0.0.0 209.165.200.10
R1(config)#exit
R1#
*Feb 3 19:17:37.279: %SYS-5-CONFIG_I: Configured from console by console
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#hostname R1
R1(config)#
*Feb 3 19:24:34.267: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/0, changed state to up
R1(config)#end
R1#
*Feb 3 19:33:40.035: %SYS-5-CONFIG_I: Configured from console by console
R1#ping
Protocol [ip]: ping 209.165.200.10
% Unknown protocol - "ping 209.165.200.10", type "ping ?" for help
R1#ping 209.165.200.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 209.165.200.10, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 20/30/40 ms
R1#
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Step 8:


R2
R1
*Feb 3 19:08:54.315: %LINK-5-CHANGED: Interface Serial1/0, changed state to administratively down
*Feb 3 19:08:54.323: %LINK-5-CHANGED: Interface Serial1/1, changed state to administratively down
*Feb 3 19:08:54.335: %LINK-5-CHANGED: Interface Serial1/2, changed state to administratively down
*Feb 3 19:08:54.343: %LINK-5-CHANGED: Interface Serial1/3, changed state to administratively down
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int s1/0
R1(config-if)#ip address 209.165.200.9 255.255.255.252
R1(config-if)#no shutdown
R1(config-if)#
*Feb 3 19:10:39.851: %LINK-3-UPDOWN: Interface Serial1/0, changed state to up
R1(config-if)#
*Feb 3 19:10:40.855: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/0, changed state to up
R1(config-if)#ip route 0.0.0.0 0.0.0.0
*Feb 3 19:11:04.267: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/0, changed state to down
R1(config-if)#ip route 0.0.0.0 0.0.0.0 209.165.200.10
R1(config)#exit
R1#
*Feb 3 19:17:37.279: %SYS-5-CONFIG_I: Configured from console by console
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#hostname R1
R1(config)#
*Feb 3 19:24:34.267: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/0, changed state to up
R1(config)#end
R1#
R1#ping
Protocol [ip]: ping 209.165.200.10
% Unknown protocol - "ping 209.165.200.10", type "ping ?" for help
R1#ping 209.165.200.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 209.165.200.10, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 20/30/40 ms
R1#ping 198.133.219.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 198.133.219.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 20/27/32 ms
R1#
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Step 9:


R2
R1
*Feb 3 19:33:40.035: %SYS-5-CONFIG_I: Configured from console by console
R1#ping
Protocol [ip]: ping 209.165.200.10
% Unknown protocol - "ping 209.165.200.10", type "ping ?" for help
R1#ping 209.165.200.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 209.165.200.10, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 20/30/40 ms
R1#ping 198.133.219.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 198.133.219.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 20/27/32 ms
R1#ip sla 22
^
% Invalid input detected at '^' marker.
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ip sla 22
R1(config-ip-sla)#icmp-echo 198.133.219.1
R1(config-ip-sla-echo)#frequency 20
R1(config-ip-sla-echo)#ip sla schedule 22 start-time now life forever
R1(config)#end
R1#
*Feb 3 19:45:27.719: %SYS-5-CONFIG_I: Configured from console by console
R1#show ip sla configuration
IP SLAs Infrastructure Engine-III
Entry number: 22
Owner:
Tag:
Operation timeout (milliseconds): 5000
Type of operation to perform: icmp-echo
Target address/Source address: 198.133.219.1/0.0.0.0
Type Of Service parameter: 0x0

```
Verify data: No
Vrf Name:
Schedule:
   Operation frequency (seconds): 20  (not considered if randomly scheduled)
   Next Scheduled Start Time: Start Time already passed
   Group Scheduled : FALSE
   Randomly Scheduled : FALSE
   Life (seconds): Forever
   Entry Ageout (seconds): never
   Recurring (Starting Everyday): FALSE
   Status of entry (SNMP RowStatus): Active
Threshold (milliseconds): 5000
Distribution Statistics:
   Number of statistic hours kept: 2
 --More--
```

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Step 10:

```
ISP#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
ISP(config)#int loopback 0
ISP(config-if)#shutdown
ISP(config-if)#
*Feb  3 20:03:08.887: %LINK-5-CHANGED: Interface Loopback0, changed state to administratively down
*Feb  3 20:03:09.887: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to down
ISP(config-if)#no shutdown
ISP(config-if)#
*Feb  3 20:03:47.543: %LINK-3-UPDOWN: Interface Loopback0, changed state to up
*Feb  3 20:03:48.543: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
ISP(config-if)#
```

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Step 11:

R2 R1

```
R1(config-ip-sla-echo)#frequency 20
R1(config-ip-sla-echo)#ip sla schedule 22 start-time now life forever
R1(config)#end
R1#
*Feb  3 19:45:27.719: %SYS-5-CONFIG_I: Configured from console by console
R1#show ip sla configuration
IP SLAs Infrastructure Engine-III
Entry number: 22
Owner:
Tag:
Operation timeout (milliseconds): 5000
Type of operation to perform: icmp-echo
Target address/Source address: 198.133.219.1/0.0.0.0
Type Of Service parameter: 0x0
Request size (ARR data portion): 28
Verify data: No
Vrf Name:
Schedule:
   Operation frequency (seconds): 20  (not considered if randomly scheduled)
   Next Scheduled Start Time: Start Time already passed
   Group Scheduled : FALSE
   Randomly Scheduled : FALSE
   Life (seconds): Forever
   Entry Ageout (seconds): never
   Recurring (Starting Everyday): FALSE
   Status of entry (SNMP RowStatus): Active
Threshold (milliseconds): 5000
Distribution Statistics:
   Number of statistic hours kept: 2
   Number of statistic distribution buckets kept: 1
   Statistic distribution interval (milliseconds): 20
Enhanced History:
History Statistics:
   Number of history Lives kept: 0
   Number of history Buckets kept: 15
   History Filter Type: None
```

```
R1#show ip sla statistics
IPSLAs Latest Operation Statistics

IPSLA operation id: 22
        Latest RTT: 40 milliseconds
Latest operation start time: 20:08:37 UTC Sat Feb 3 2024
Latest operation return code: OK
Number of successes: 72
Number of failures: 2
Operation time to live: Forever

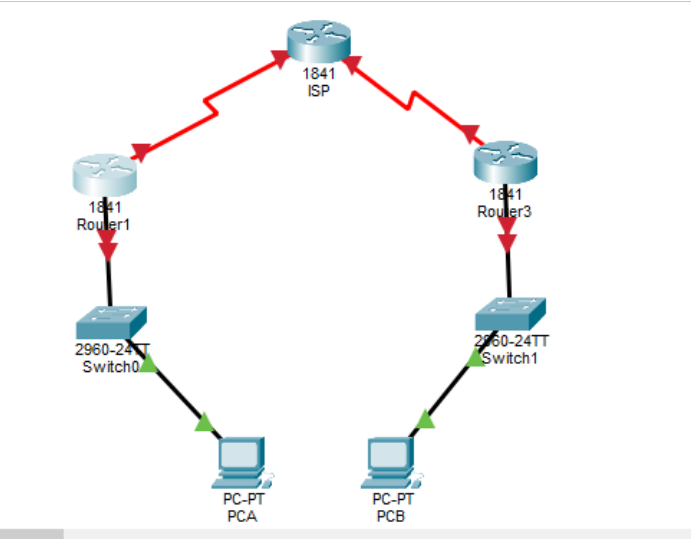
R1#
```

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Conclusion: The above practical Implementing the IP Service Level Agreement has been Implemented Successfully.

PRACTICAL-02

Topology



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
Router1	Fa0/0	192.168.10.1	255.255.255.0	N/A
	Lo0	192.168.20.1	255.255.255.0	N/A
	Se0/1/0	10.1.1.1	255.0.0.0	N/A
ISP	Se0/1/0	10.1.1.2	255.0.0.0	N/A
	Se0/1/1	20.2.2.2	255.0.0.0	N/A
	Lo0	209.165.200.225	255.255.255.224	N/A
Router3	Fa0/0	192.168.30.1	255.255.255.0	N/A
	Lo0	192.168.40.1	255.255.255.0	N/A
	Se0/1/0	20.2.2.1	255.0.0.0	N/A
Switch1	VLAN 1	192.168.10.11	255.255.255.0	192.168.10.1
Switch2	VLAN 1	192.168.30.11	255.255.255.0	192.168.30.1
PCA	NIC	192.168.10.3	255.255.255.0	192.168.10.1
PCB	NIC	192.168.30.3	255.255.255.0	192.168.30.1

Aim

Configuring and Verifying Standard IPv4 ACLs.

a) Set up the Topology and Initialize Devices.

b) Configure Devices and Verify Connectivity.

Theory

Internet Protocol version 4 (IPv4)

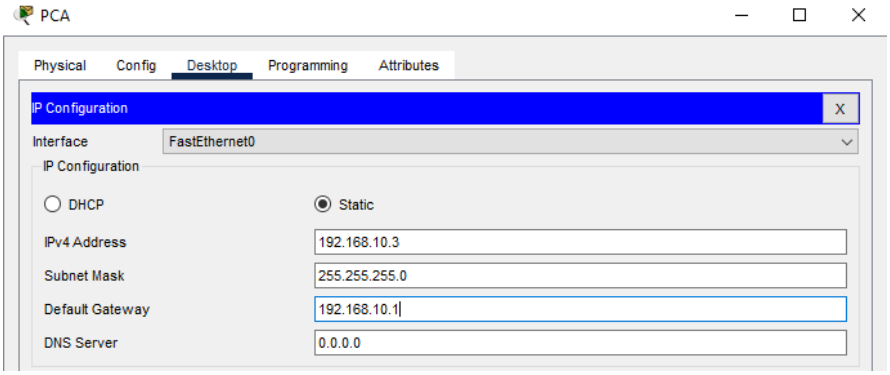
IPv4 is the fourth version of the Internet Protocol and the one that still carries most Internet traffic. It identifies each device with a 32-bit address and is used to route packets between networks.

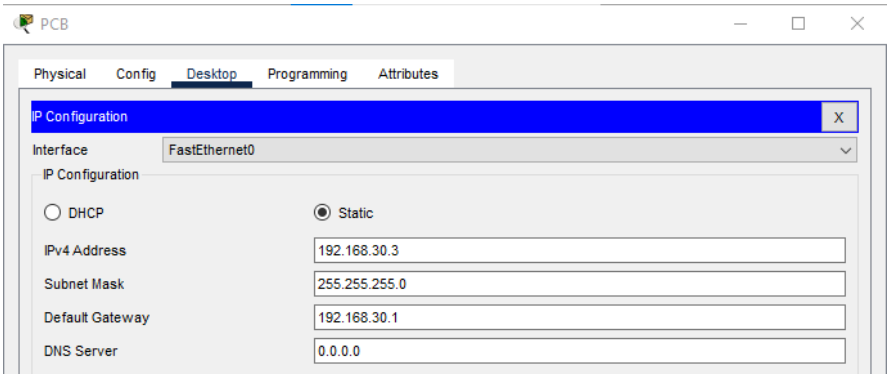
Access Control List (ACL)

An Access Control List is an ordered list of permit and deny statements that a router uses to filter traffic according to the security policy. A standard IPv4 ACL checks only the source address of a packet. The statements are checked from top to bottom, and an implicit deny at the end blocks everything that was not permitted.

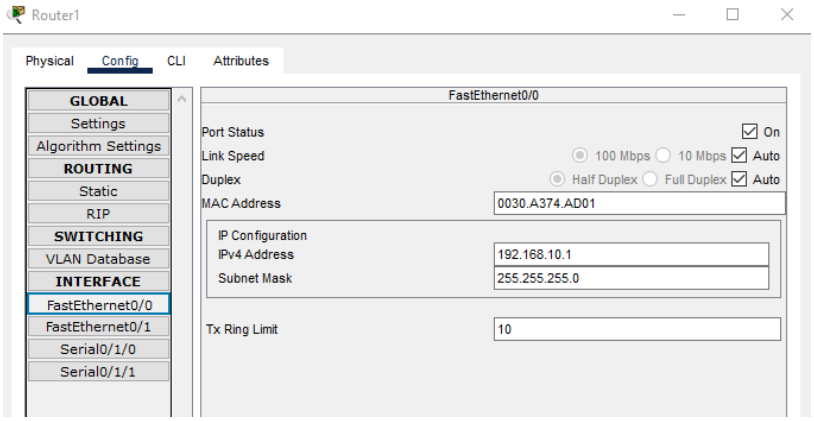
a) Set up the Topology and Initialize Devices

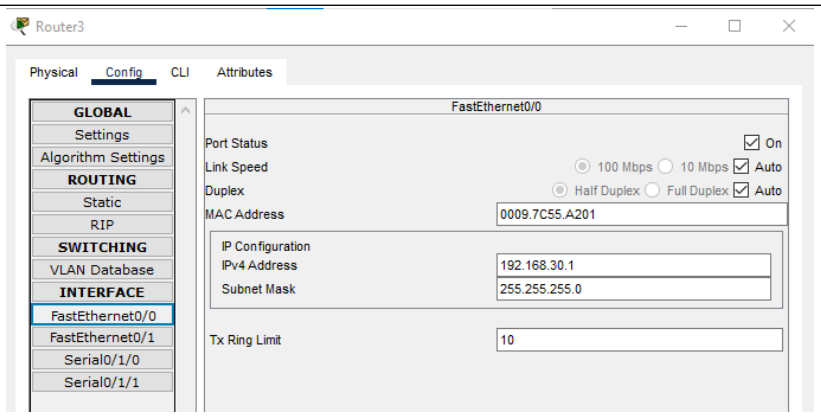
Step 1: Configure the IP addresses of PCA and PCB.

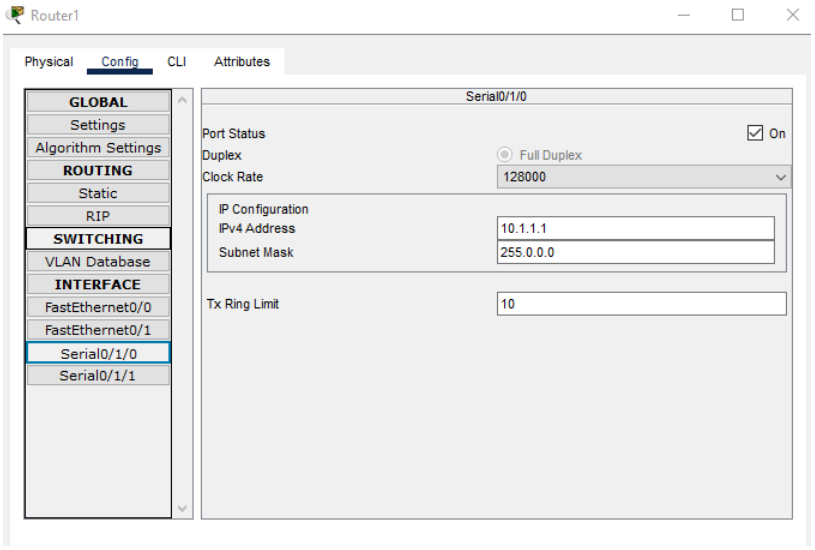


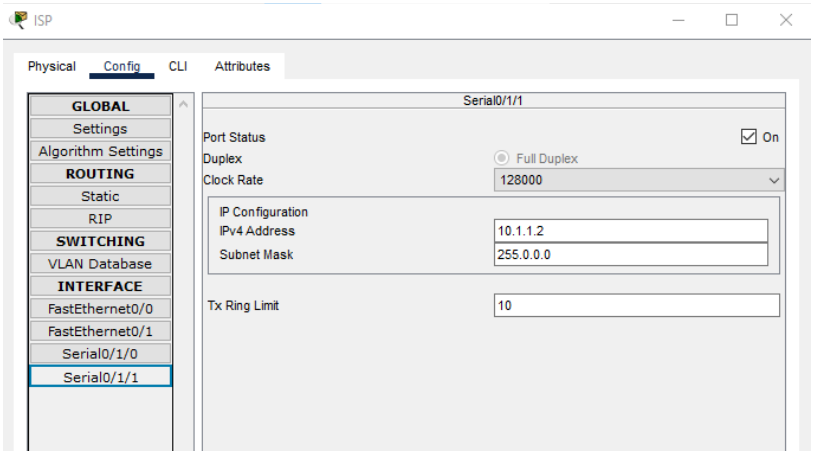


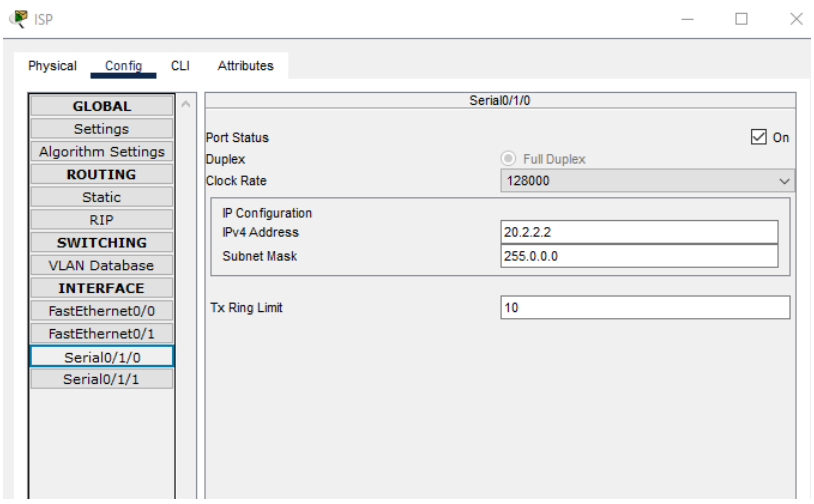
Step 2: Configure the FastEthernet and Serial interfaces of Router1, ISP and Router3.

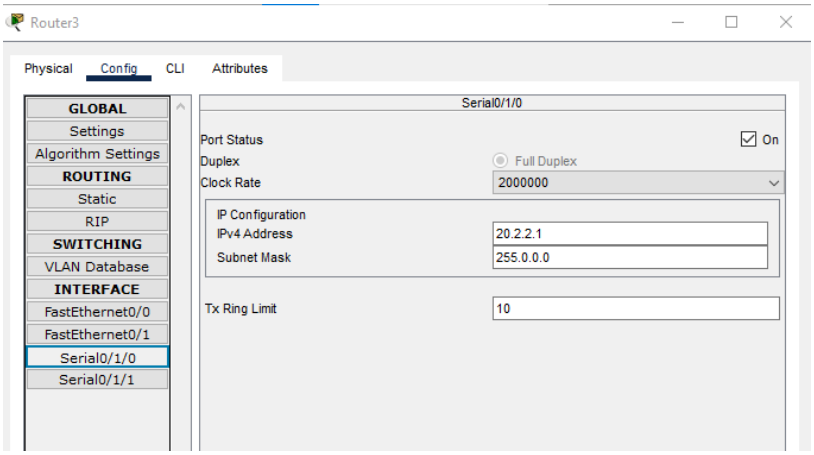



Router3
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
Serial0/1/0
Serial0/1/1
FastEthernet0/0
Port Status
On
Link Speed
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0009.7C55.A201
IP Configuration
IPv4 Address
192.168.30.1
Subnet Mask
255.255.255.0
Tx Ring Limit
10

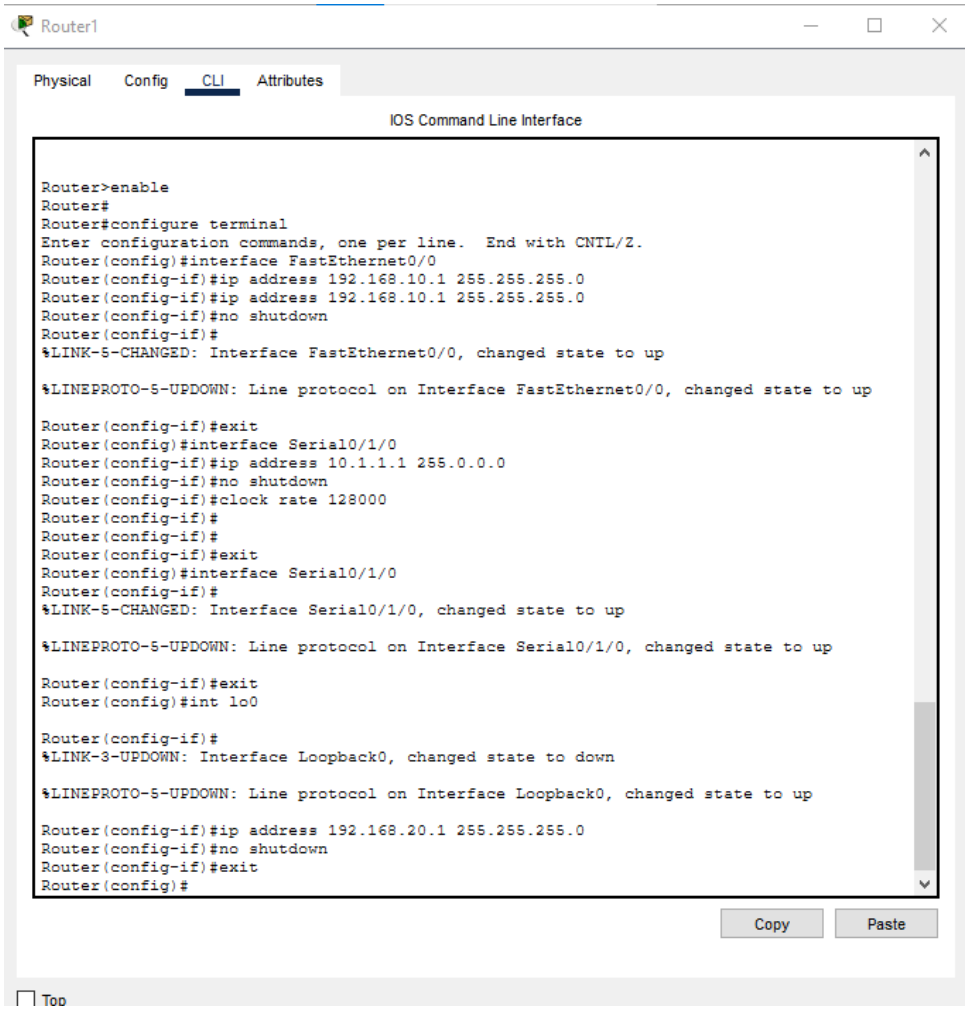

Router1
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
Serial0/1/0
Serial0/1/1
Serial0/1/0
Port Status
On
Duplex
Full Duplex
Clock Rate
128000
IP Configuration
IPv4 Address
10.1.1.1
Subnet Mask
255.0.0.0
Tx Ring Limit
10

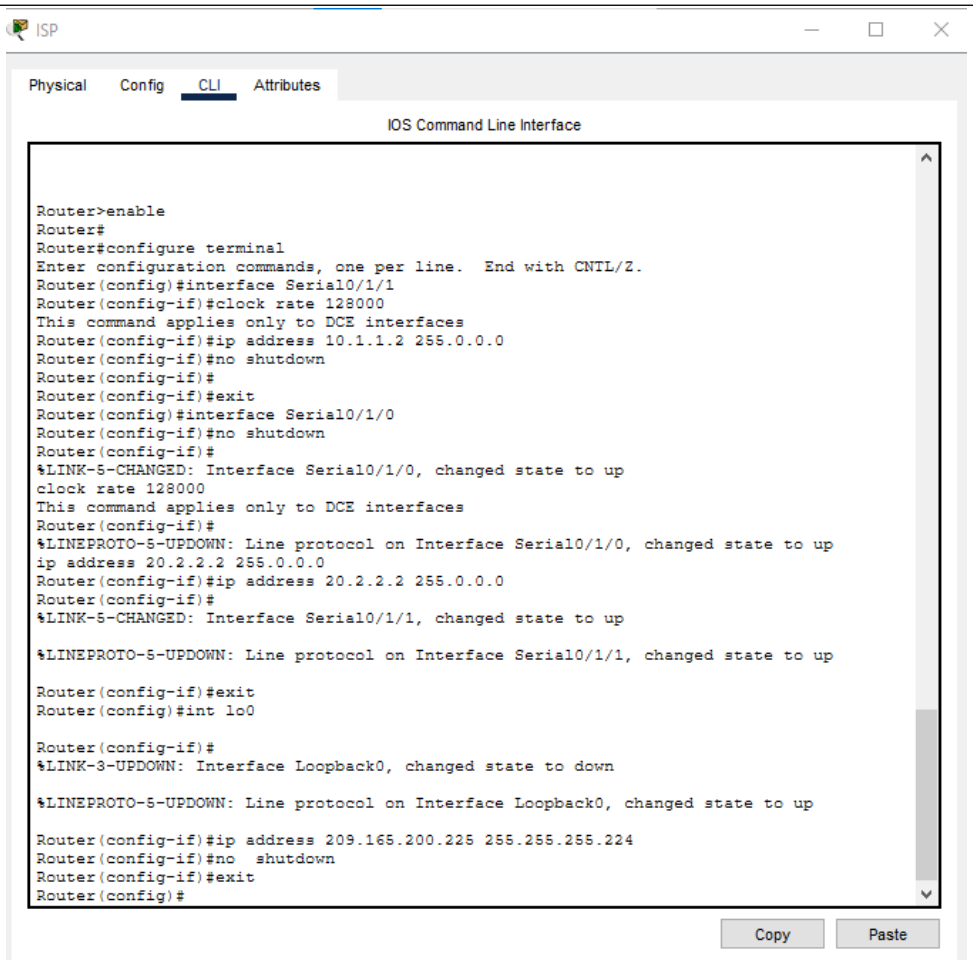

ISP
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
Serial0/1/0
Serial0/1/1
Serial0/1/1
Port Status
On
Duplex
Full Duplex
Clock Rate
128000
IP Configuration
IPv4 Address
10.1.1.2
Subnet Mask
255.0.0.0
Tx Ring Limit
10

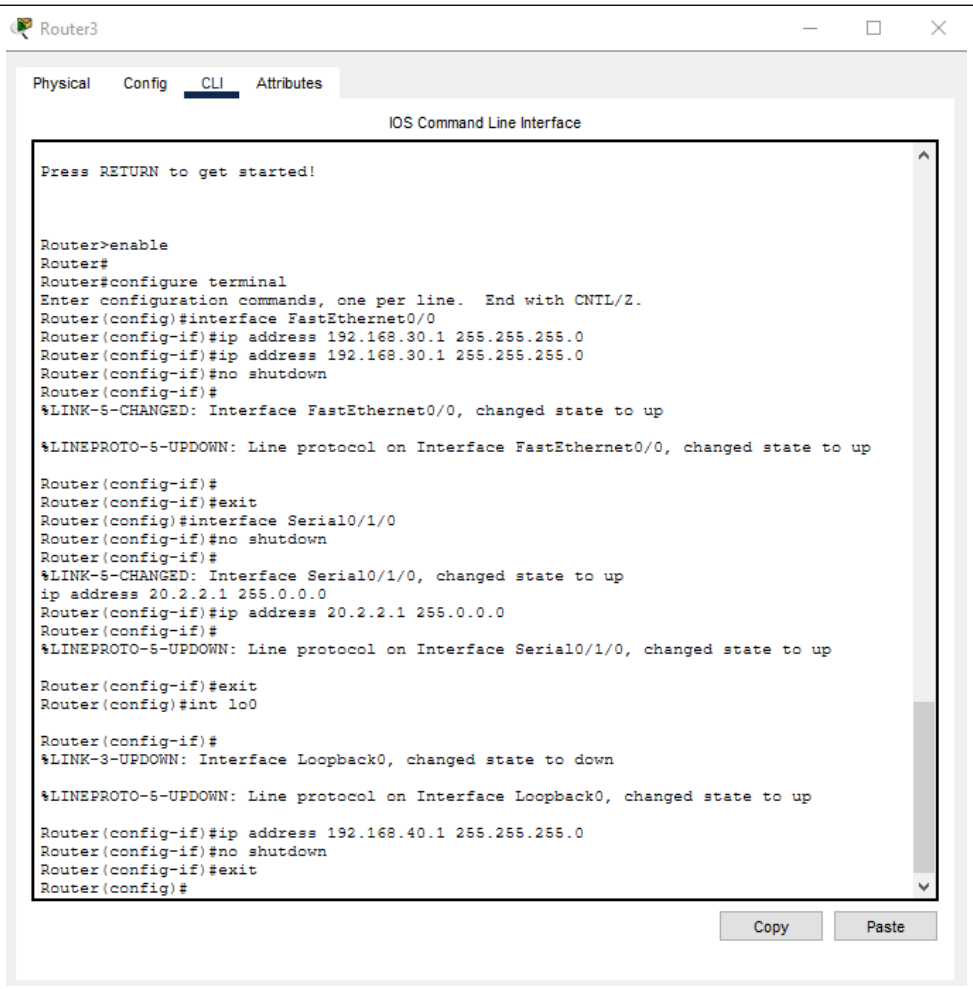

ISP
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
Serial0/1/0
Serial0/1/1
Serial0/1/0
Port Status
On
Duplex
Full Duplex
Clock Rate
128000
IP Configuration
IPv4 Address
20.2.2.2
Subnet Mask
255.0.0.0
Tx Ring Limit
10


Router3
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
Serial0/1/0
Serial0/1/1
Serial0/1/0
Port Status
On
Duplex
Full Duplex
Clock Rate
2000000
IP Configuration
IPv4 Address
20.2.2.1
Subnet Mask
255.0.0.0
Tx Ring Limit
10

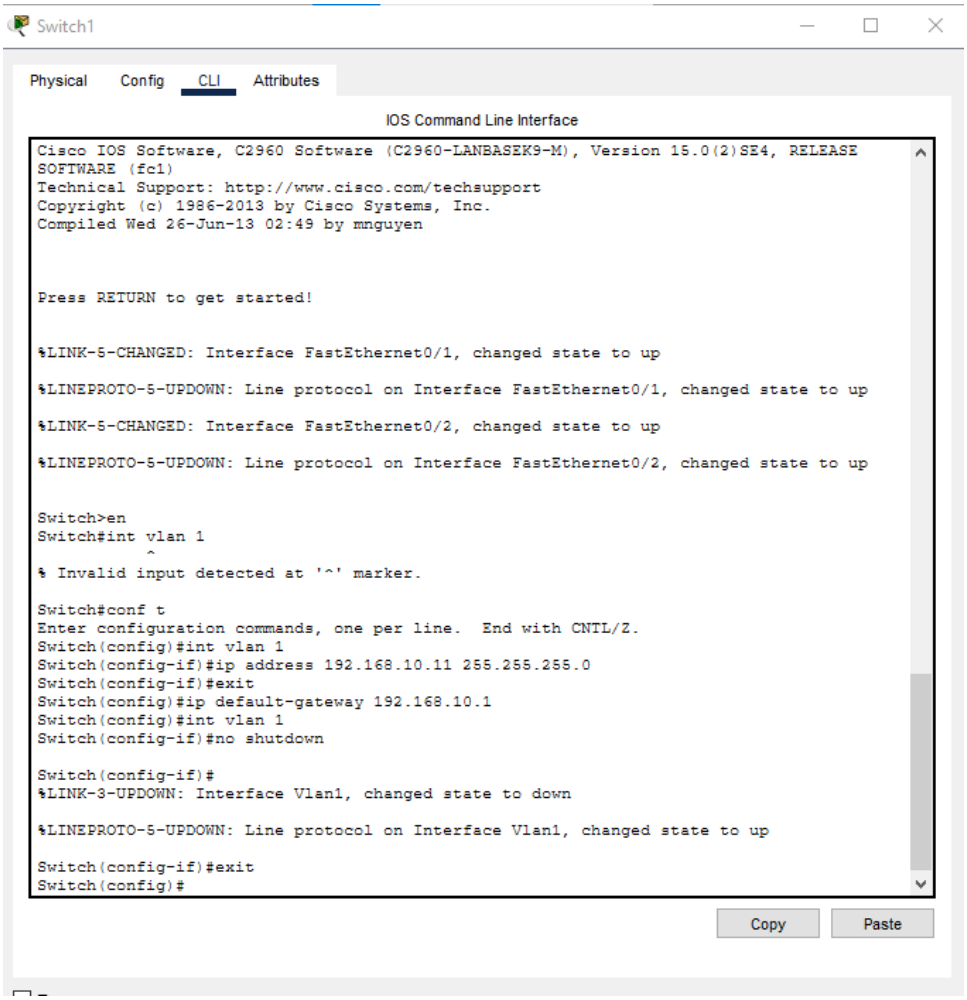
Step 3: Configure the interfaces and Loopback0 on Router1, ISP and Router3 from the CLI.

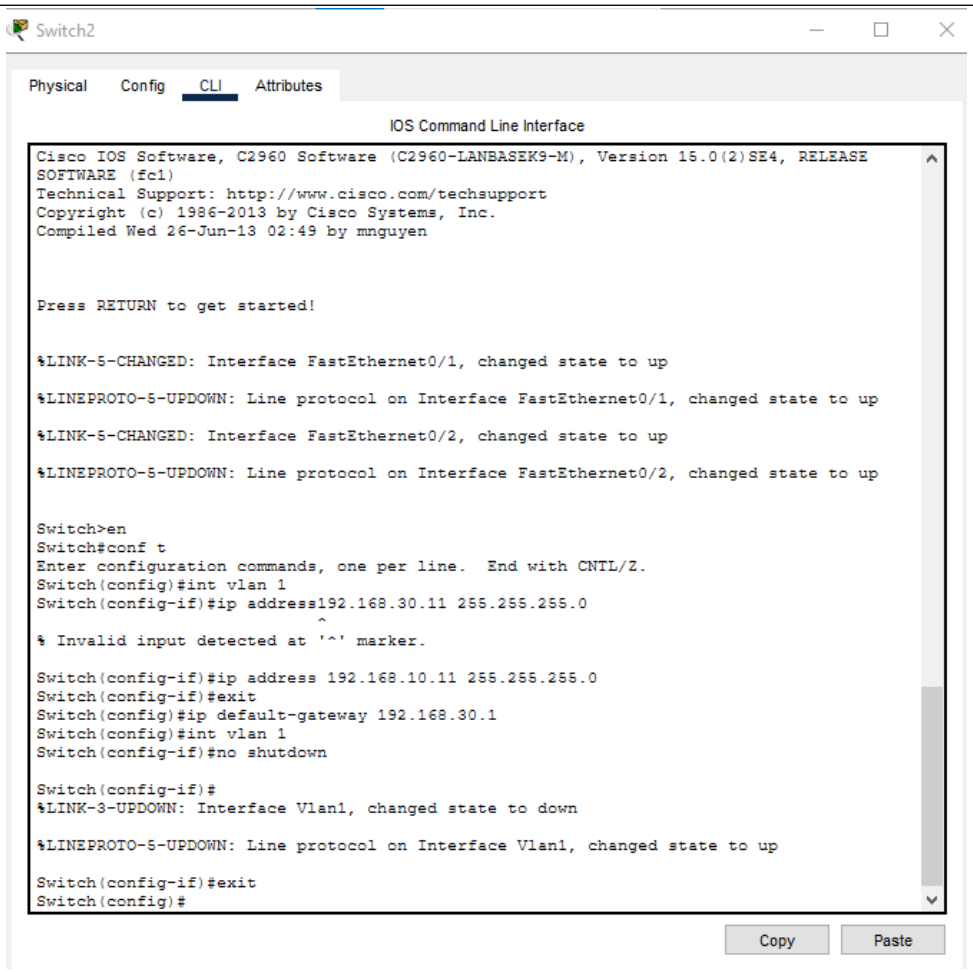



ISP
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface Serial0/1/1
Router(config-if)#clock rate 128000
This command applies only to DCE interfaces
Router(config-if)#ip address 10.1.1.2 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/1/0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/1/0, changed state to up
clock rate 128000
This command applies only to DCE interfaces
Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up
ip address 20.2.2.2 255.0.0.0
Router(config-if)#ip address 20.2.2.2 255.0.0.0
Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/1/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to up
Router(config-if)#exit
Router(config)#int lo0
Router(config-if)#
%LINK-3-UPDOWN: Interface Loopback0, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
Router(config-if)#ip address 209.165.200.225 255.255.255.224
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Copy
Paste

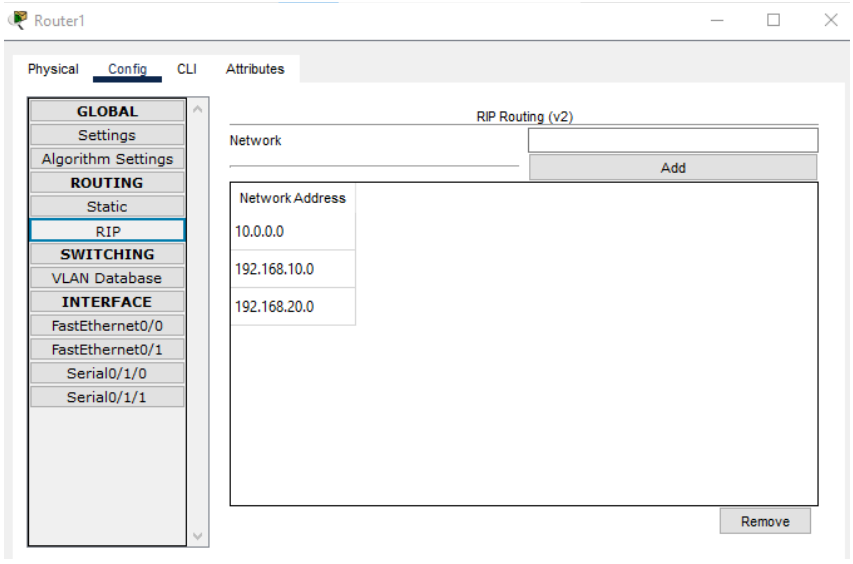

Router3
Physical
Config
CLI
Attributes
IOS Command Line Interface
Press RETURN to get started!
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 192.168.30.1 255.255.255.0
Router(config-if)#ip address 192.168.30.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/1/0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/1/0, changed state to up
ip address 20.2.2.1 255.0.0.0
Router(config-if)#ip address 20.2.2.1 255.0.0.0
Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up
Router(config-if)#exit
Router(config)#int lo0
Router(config-if)#
%LINK-3-UPDOWN: Interface Loopback0, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
Router(config-if)#ip address 192.168.40.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Copy
Paste

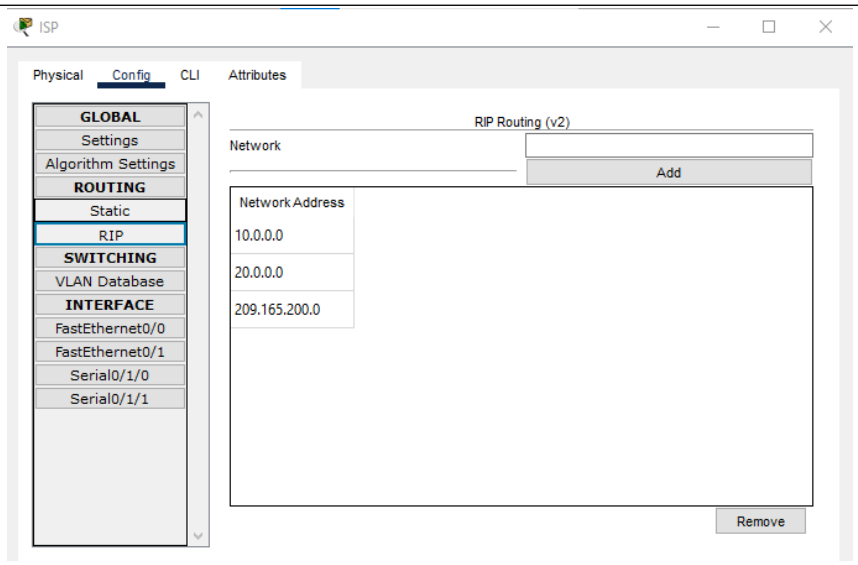
Step 4: Configure the VLAN 1 interface and default gateway on the switches.

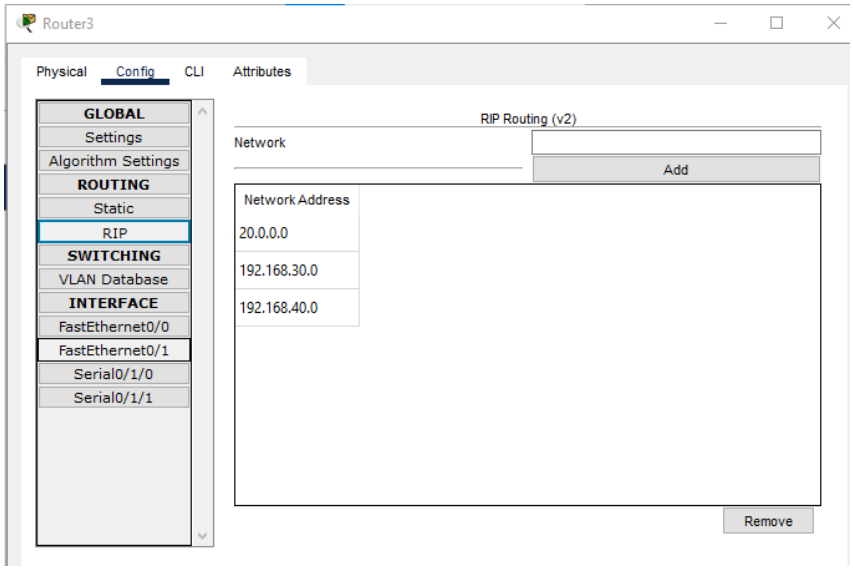




Step 5: Configure RIP version 2 on Router1, ISP and Router3.







Step 6: Verify connectivity by pinging 209.165.200.225 from the PCs.

```
C:\>ping 209.165.200.225

Pinging 209.165.200.225 with 32 bytes of data:

Reply from 209.165.200.225: bytes=32 time=1ms TTL=254
Reply from 209.165.200.225: bytes=32 time=1ms TTL=254
Reply from 209.165.200.225: bytes=32 time=1ms TTL=254
Reply from 209.165.200.225: bytes=32 time=1ms TTL=254

Ping statistics for 209.165.200.225:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 1ms, Average = 1ms
```

```
C:\>ping 209.165.200.225

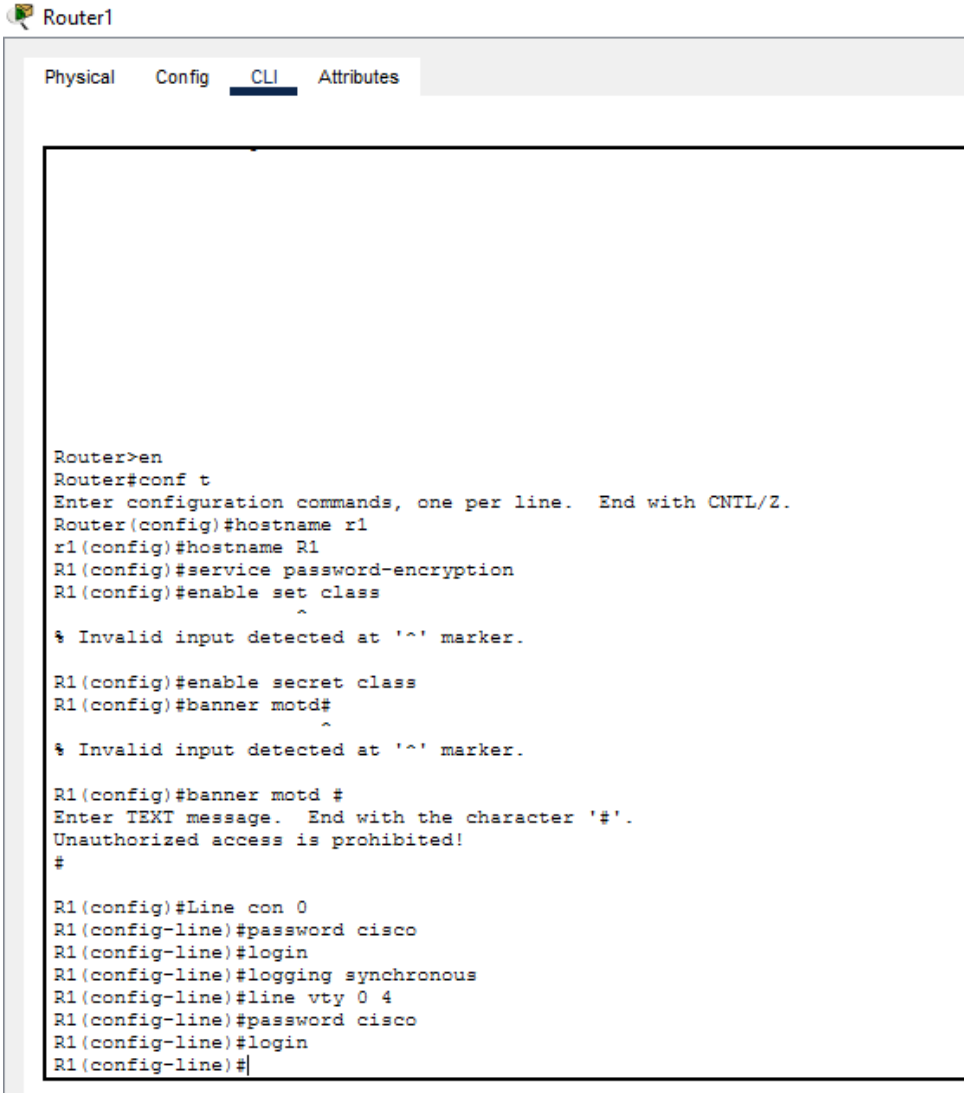
Pinging 209.165.200.225 with 32 bytes of data:

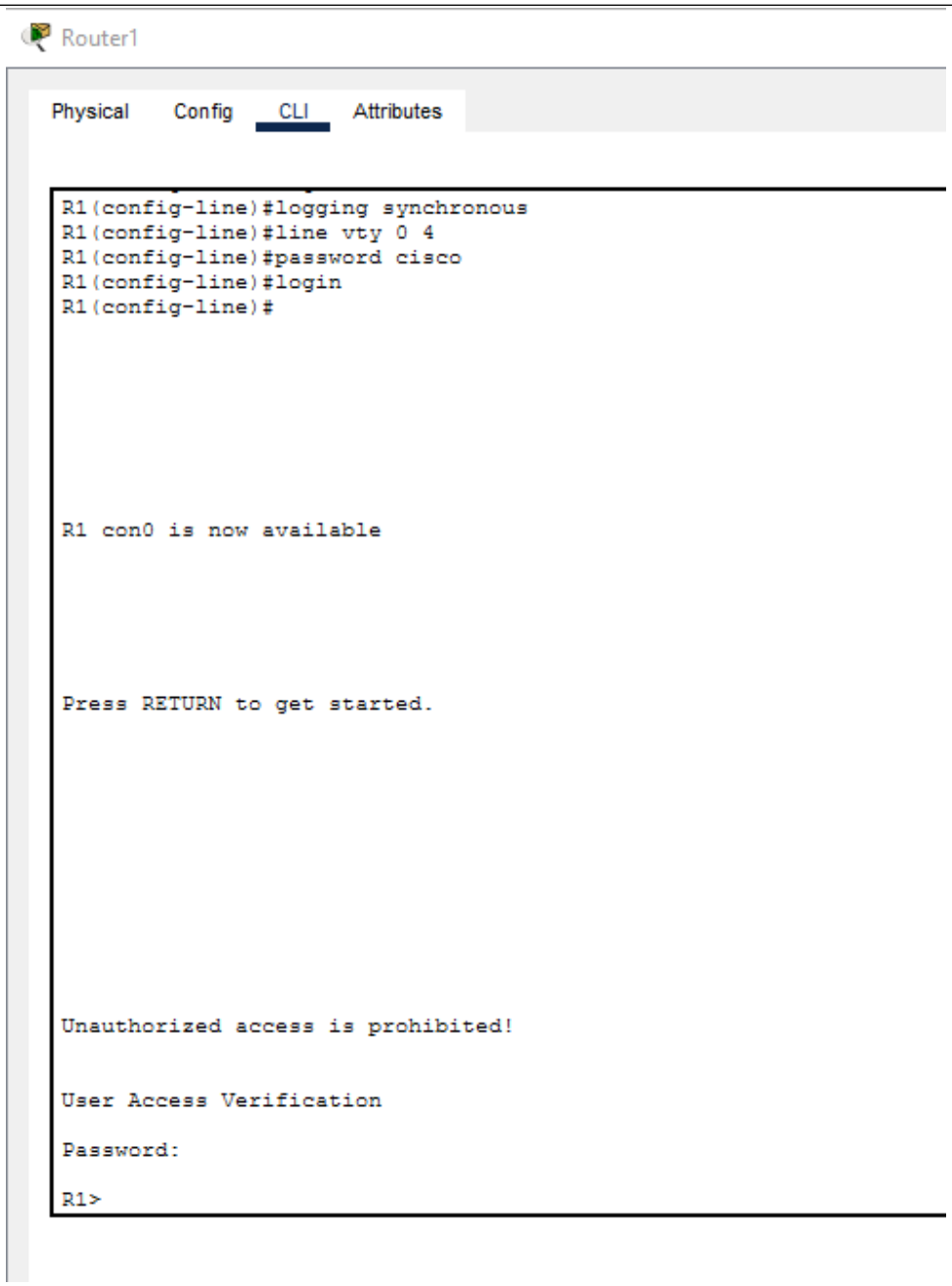
Reply from 209.165.200.225: bytes=32 time=2ms TTL=254
Reply from 209.165.200.225: bytes=32 time=1ms TTL=254
Reply from 209.165.200.225: bytes=32 time=1ms TTL=254
Reply from 209.165.200.225: bytes=32 time=1ms TTL=254

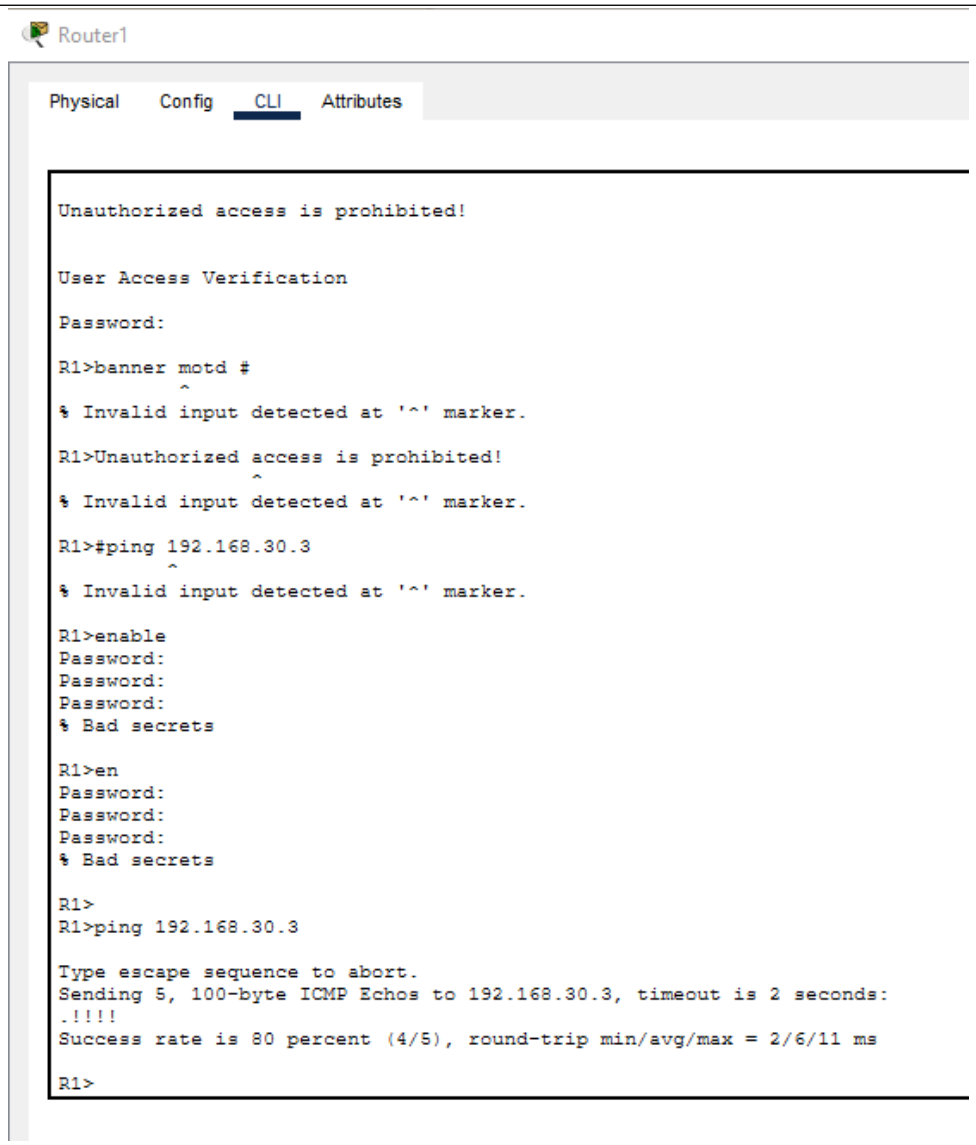
Ping statistics for 209.165.200.225:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2ms, Average = 1ms

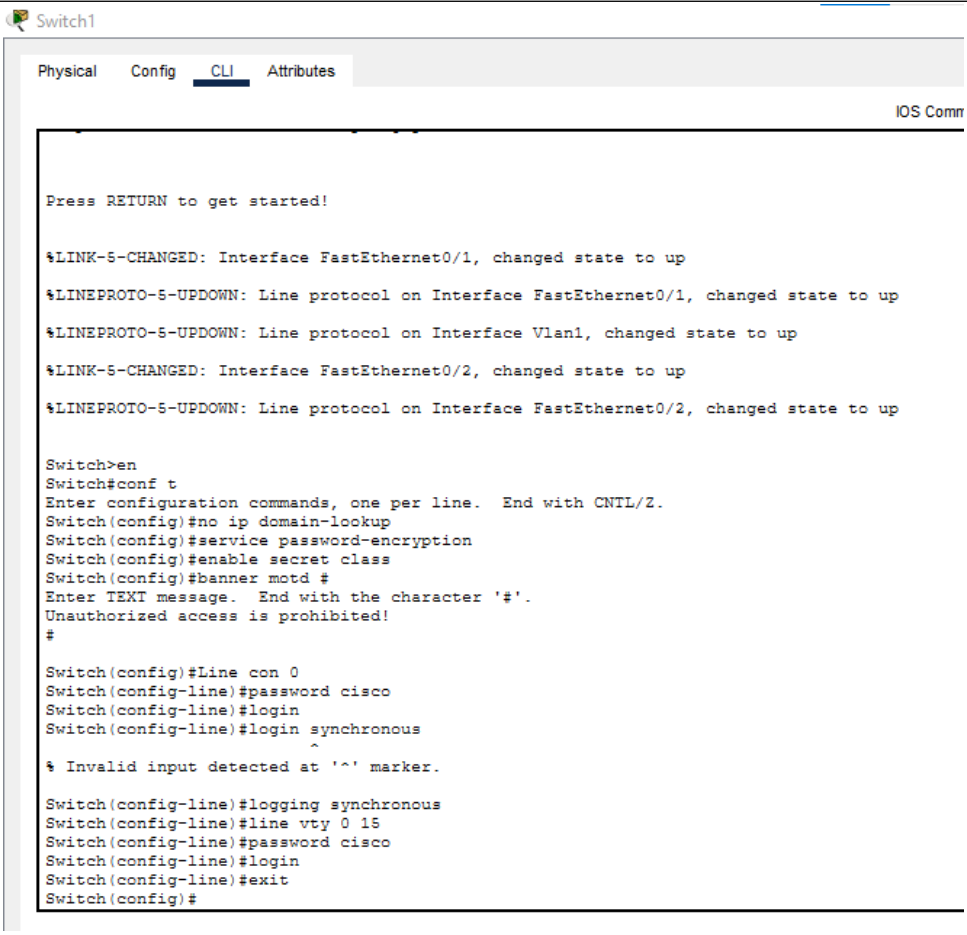
C:\>
```

Step 7: Configure the banner and passwords on the routers and switches.




Router1
Physical
Config
CLI
Attributes
R1(config-line)#logging synchronous
R1(config-line)#line vty 0 4
R1(config-line)#password cisco
R1(config-line)#login
R1(config-line)#
R1 con0 is now available
Press RETURN to get started.
Unauthorized access is prohibited!
User Access Verification
Password:
R1>

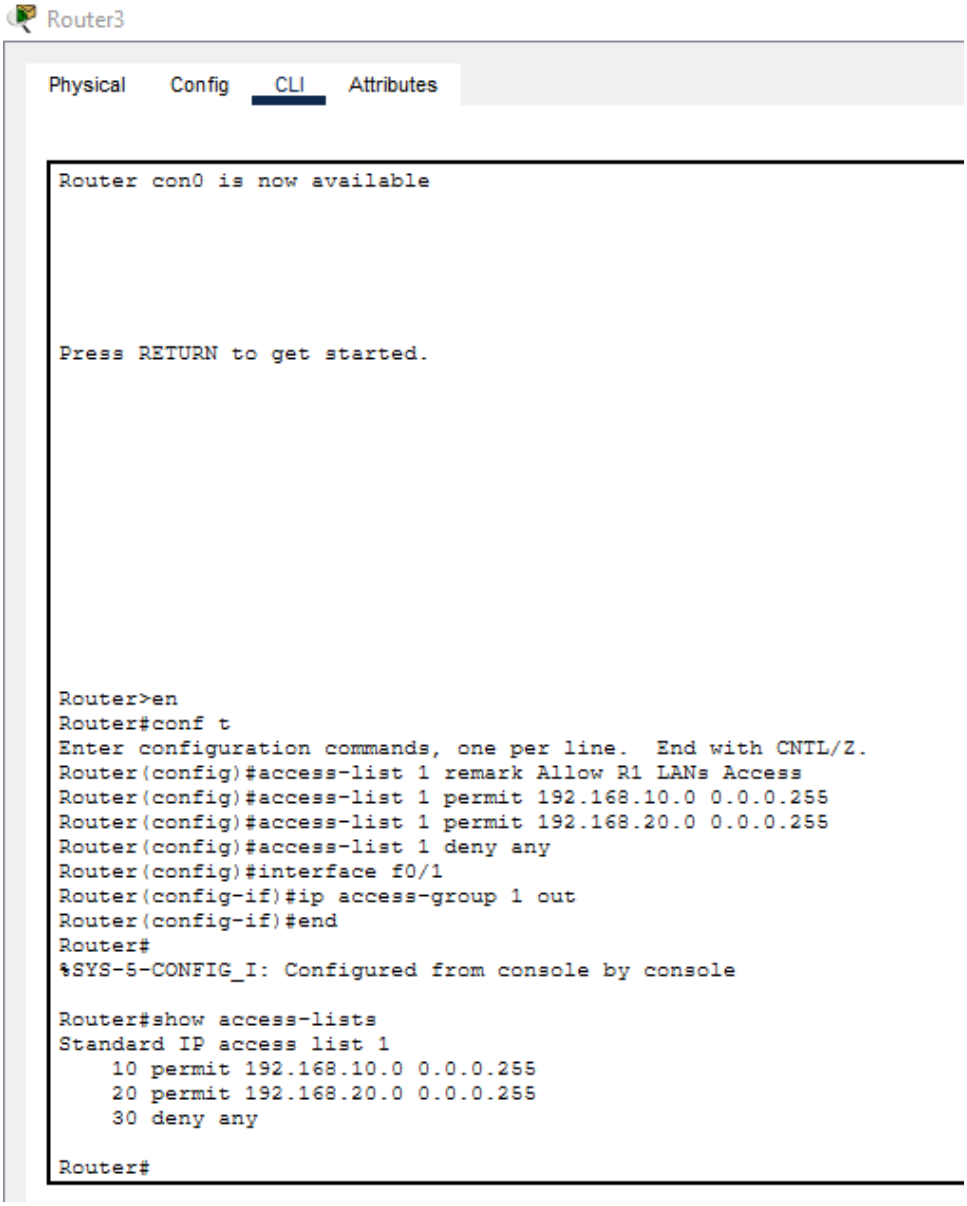

Router1
Physical
Config
CLI
Attributes
Unauthorized access is prohibited!
User Access Verification
Password:
R1>banner motd #
^
% Invalid input detected at '^' marker.
R1>Unauthorized access is prohibited!
^
% Invalid input detected at '^' marker.
R1>#ping 192.168.30.3
^
% Invalid input detected at '^' marker.
R1>enable
Password:
Password:
Password:
% Bad secrets
R1>en
Password:
Password:
Password:
% Bad secrets
R1>
R1>ping 192.168.30.3
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.30.3, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 2/6/11 ms
R1>

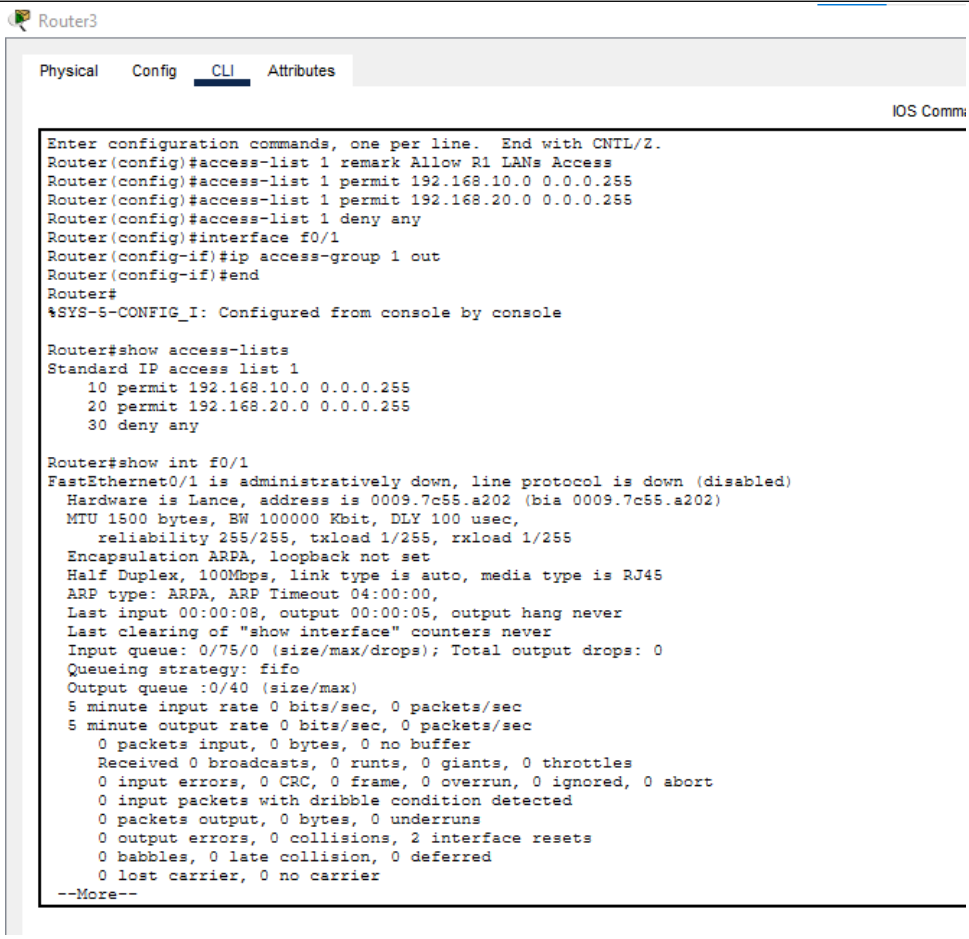

Switch1
Physical
Config
CLI
Attributes
IOS Comm
Press RETURN to get started!
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#no ip domain-lookup
Switch(config)#service password-encryption
Switch(config)#enable secret class
Switch(config)#banner motd #
Enter TEXT message. End with the character '#'.
Unauthorized access is prohibited!
#
Switch(config)#Line con 0
Switch(config-line)#password cisco
Switch(config-line)#login
Switch(config-line)#login synchronous
^
% Invalid input detected at '^' marker.
Switch(config-line)#logging synchronous
Switch(config-line)#line vty 0 15
Switch(config-line)#password cisco
Switch(config-line)#login
Switch(config-line)#exit
Switch(config)#

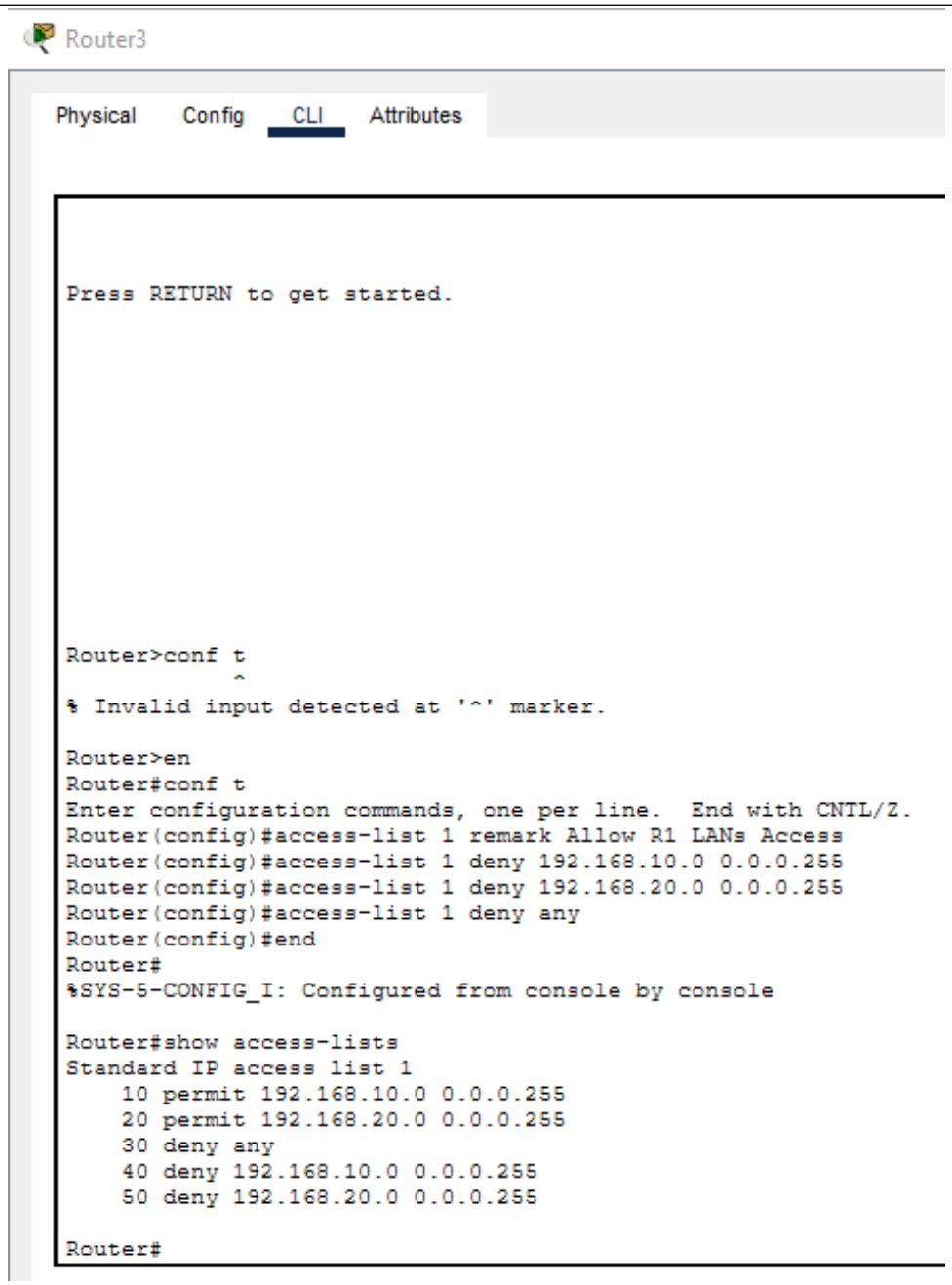

Switch2
Physical
Config
CLI
Attributes
IOS
Switch con0 is now available
Press RETURN to get started.
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#no ip domain-lookup
Switch(config)#service password-encryption
Switch(config)#enable secret class
Switch(config)# banner motd #
Enter TEXT message. End with the character '#'.
Unauthorized access is prohibited!
#
Switch(config)#Line con 0
Switch(config-line)#password cisco
Switch(config-line)#login
Switch(config-line)#logging synchronous
Switch(config-line)#line vty 0 15
Switch(config-line)#password cisco
Switch(config-line)#login
Switch(config-line)#exit
Switch(config)#

b) Configure Devices and Verify Connectivity

Step 1: Configure standard ACL 1 on Router3 and verify it with show access-lists.




Router3
Physical
Config
CLI
Attributes
IOS Comma
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#access-list 1 remark Allow R1 LANs Access
Router(config)#access-list 1 permit 192.168.10.0 0.0.0.255
Router(config)#access-list 1 permit 192.168.20.0 0.0.0.255
Router(config)#access-list 1 deny any
Router(config)#interface f0/1
Router(config-if)#ip access-group 1 out
Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#show access-lists
Standard IP access list 1
10 permit 192.168.10.0 0.0.0.255
20 permit 192.168.20.0 0.0.0.255
30 deny any
Router#show int f0/1
FastEthernet0/1 is administratively down, line protocol is down (disabled)
Hardware is Lance, address is 0009.7c55.a202 (bia 0009.7c55.a202)
MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Half Duplex, 100Mbps, link type is auto, media type is RJ45
ARP type: ARPA, ARP Timeout 04:00:00,
Last input 00:00:08, output 00:00:05, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0 (size/max/drops); Total output drops: 0
Queueing strategy: fifo
Output queue :0/40 (size/max)
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
0 packets input, 0 bytes, 0 no buffer
Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
0 input packets with dribble condition detected
0 packets output, 0 bytes, 0 underruns
0 output errors, 0 collisions, 2 interface resets
0 babbles, 0 late collision, 0 deferred
0 lost carrier, 0 no carrier
--More--


Router3
Physical
Config
CLI
Attributes
Press RETURN to get started.
Router>conf t
^
% Invalid input detected at '^' marker.
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#access-list 1 remark Allow R1 LANs Access
Router(config)#access-list 1 deny 192.168.10.0 0.0.0.255
Router(config)#access-list 1 deny 192.168.20.0 0.0.0.255
Router(config)#access-list 1 deny any
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#show access-lists
Standard IP access list 1
10 permit 192.168.10.0 0.0.0.255
20 permit 192.168.20.0 0.0.0.255
30 deny any
40 deny 192.168.10.0 0.0.0.255
50 deny 192.168.20.0 0.0.0.255
Router#

Step 2: Verify the target IP address operations from Router1.




Router1
Physical
Config
CLI
Attributes
IOS
R1>en
Password:
Password:
Password:
% Bad secrets
R1>piing
Translating "piing"...domain server (255.255.255.255)
% Unknown command or computer name, or unable to find computer address
R1>
R1>enable
Password:
R1#ping
Protocol [ip]:
Target IP address: 192.168.30.3
Repeat count [5]:
Datagram size [100]:
Timeout in seconds [2]:
Extended commands [n]: y
Source address or interface: 192.168.20.1
Type of service [0]:
Set DF bit in IP header? [no]:
Validate reply data? [no]:
Data pattern [0xABCD]:
Loose, Strict, Record, Timestamp, Verbose[none]:
Sweep range of sizes [n]:
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.30.3, timeout is 2 seconds:
Packet sent with a source address of 192.168.20.1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/9/22 ms
R1#ping 209.165.200.225
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 209.165.200.225, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/8/28 ms
R1#

PRACTICAL-03

Aim: Perform

1) Implement SPAN Technologies (Switch Port Analyzer)
2) Implement SNMP and Syslog
3) Implement Flexible NetFlow Theory:

Theory:

1) **SPAN technologies:**

SPAN (Switched Port Analyzer) is a dedicated port on a switch that takes a mirrored copy of network traffic from within the switch to be sent to a destination. The destination is typically a monitoring device or other tools used for troubleshooting or traffic analysis. SPAN ports present a readily available mechanism through which to access detailed packet information.

2) **SNMP and Syslog:**

SNMP is an application layer protocol that uses UDP port number 161/162.SNMP is used to monitor the network, detect network faults, and sometimes even to configure remote devices.

Syslog is a standard protocol for message logging that computer systems use to send event logs to a Syslog server for storage. On network devices, Syslog can be used to log events such as changes in interface status, system restarts, etc. A lot of different types of events can be logged. Logs are essential when troubleshooting issues, examining the cause of incidents, etc.

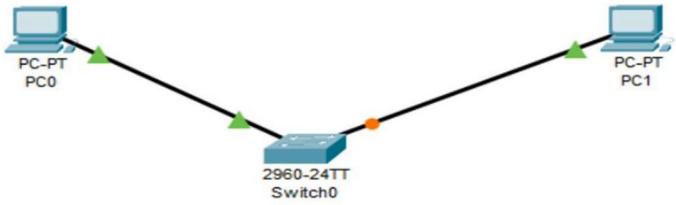
3) **Flexible NetFlow:**

Flexible NetFlow helps Cisco customers determine how to optimize resource usage, plan network capacity, and identify the optimal application layer for Quality of Service (QoS). It plays a vital role in network security by detecting Denial of Service (DoS) attacks and network-propagated worms.

Implementation:

1. **Implement Span Technologies**

Implement SPAN Technologies (Switch Port Analyzer)



Address Table

Device	IP Address
PC0	10.0.0.2
PC1	10.0.0.3

Configure PC addresses:

Click each PC → Desktop → IP Configuration.

Device	**IP address**	**Subnet mask**
PC0	10.0.0.2	255.0.0.0
PC1	10.0.0.3	255.0.0.0

Configure SPAN on the switch:

Open Switch → CLI:

enable

configure terminal

hostname S1

monitor session 1 source interface fastEthernet 0/1 both

monitor session 1 destination interface fastEthernet 0/2

end

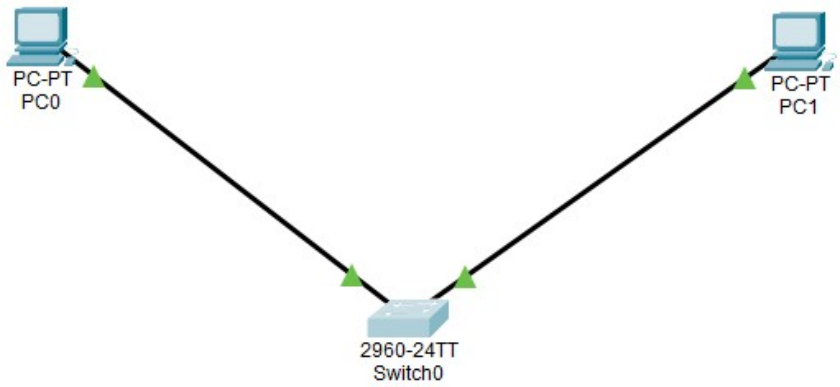
Verify SPAN:

show monitor session 1

Generate traffic:

ping 10.0.0.3

OUTPUT SCREENSHOT:



Switch0

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch>conf t
          ^
% Invalid input detected at '^' marker.

Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#monitor session 1 source into f0/1
                                  ^
% Invalid input detected at '^' marker.

Switch(config)#monitor session 1 source interface fastEthernet 0/1 both
Switch(config)#monitor session 1 destination interface fastEthernet 0/2
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#end
Translating "end"...domain server (255.255.255.255)
% Unknown command or computer name, or unable to find computer address

Switch#show monitor session 1
Session 1
---------
Type                   : Local Session
Description            : -
Source Ports           :
    Both               : Fa0/1
Destination Ports      : Fa0/2
    Encapsulation      : Native
          Ingress      : Disabled


Switch#show monitor detail
Session 1
---------
Type                   : Local Session
Description            : -
Source Ports           :
    RX Only            : None
```

Switch0

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#end
Translating "end"...domain server (255.255.255.255)
% Unknown command or computer name, or unable to find computer address

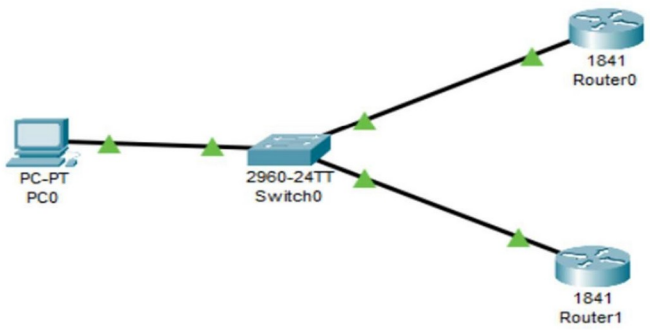
Switch#show monitor session 1
Session 1
---------
Type                   : Local Session
Description            : -
Source Ports           :
    Both               : Fa0/1
Destination Ports      : Fa0/2
    Encapsulation      : Native
          Ingress      : Disabled


Switch#show monitor detail
Session 1
---------
Type                   : Local Session
Description            : -
Source Ports           :
    RX Only            : None
    TX Only            : None
    Both               : Fa0/1
Source VLANs           :
    RX Only            : None
    TX Only            : None
    Both               : None
Source RSPAN VLAN      : None
Destination Ports      : Fa0/2
    Encapsulation      : Native
          Ingress      : Disabled
Filter VLANs           : None
Dest RSPAN VLAN        : None


Switch#
```

2. Implement SNMP

Implement SNMP



Device	Interface	IP Address
PC0	PC0	172.168.1.2 /172.168.1.1
R1	F0/0	172.168.1.1
R2	F0/0	172.168.2.1

Configure the PC:

Open PC → Desktop → IP Configuration:

Setting	Value
IP address	172.168.1.2
Subnet mask	255.255.255.0
Default gateway	172.168.1.1

Configure R1:

enable

configure terminal

hostname R1

interface fastEthernet 0/0

ip address 172.168.1.1 255.255.255.0

no shutdown

exit

snmp-server community read ro

snmp-server community write rw

end

Configure R2:

enable

configure terminal

hostname R2

interface fastEthernet 0/0

ip address 172.168.1.3 255.255.255.0

no shutdown

exit

snmp-server community read ro

snmp-server community write rw

end

Check connectivity:

ping 172.168.1.1

ping 172.168.1.3

Open the MIB Browser

Click PC → Desktop → MIB Browser → Advanced.

Enter:

Setting	Value
Address	172.168.1.1
Port	161
Read Community	read
Write Community	write
SNMP Version	v1

Click **OK**.

Retrieve the router hostname:

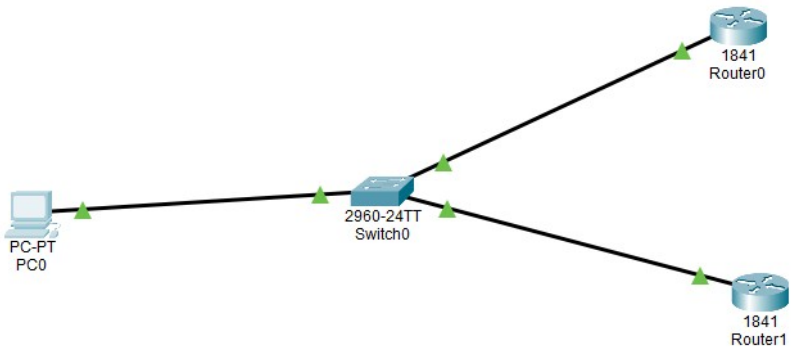
In the MIB Browser:

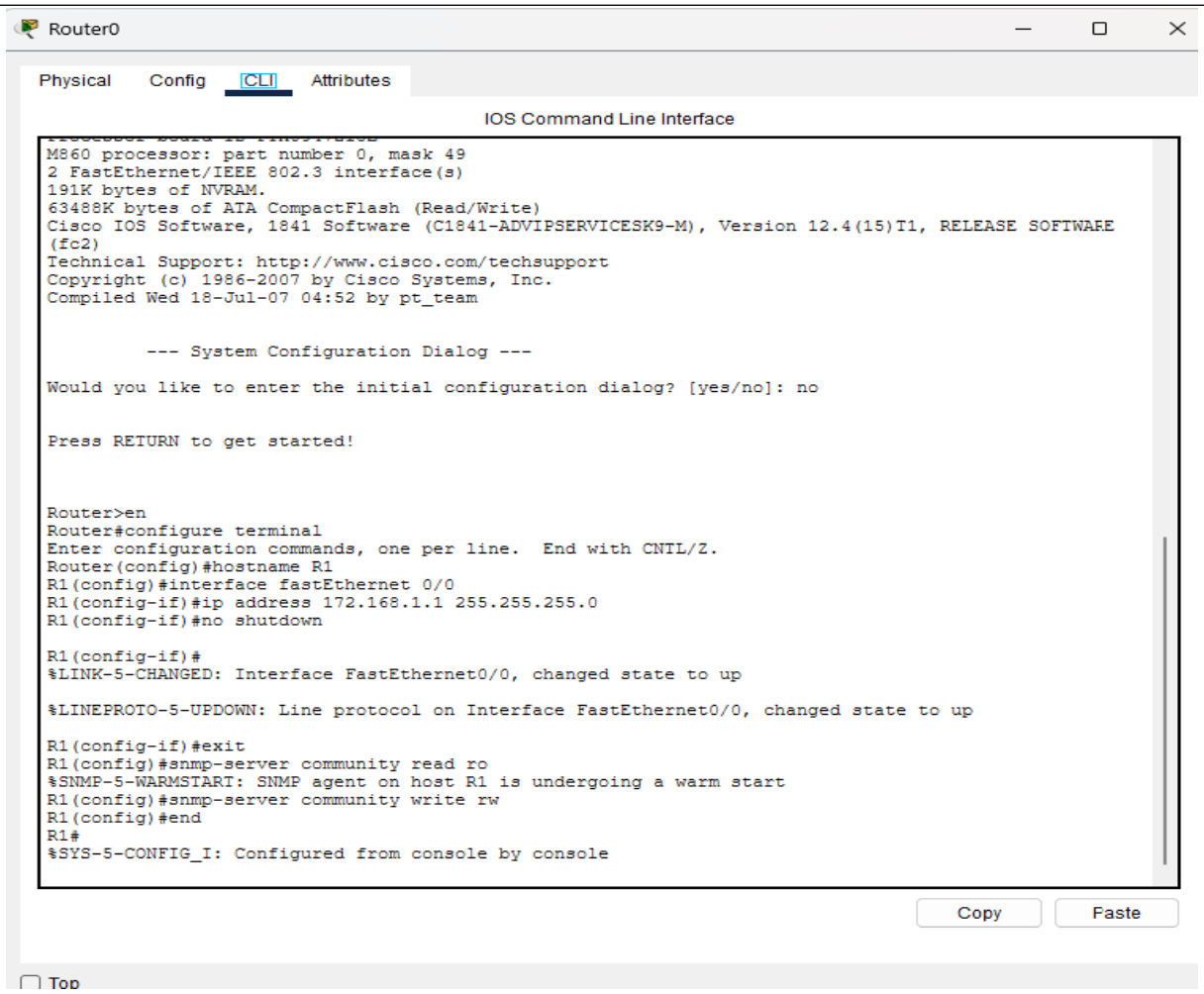
1. **Enter the OID:**

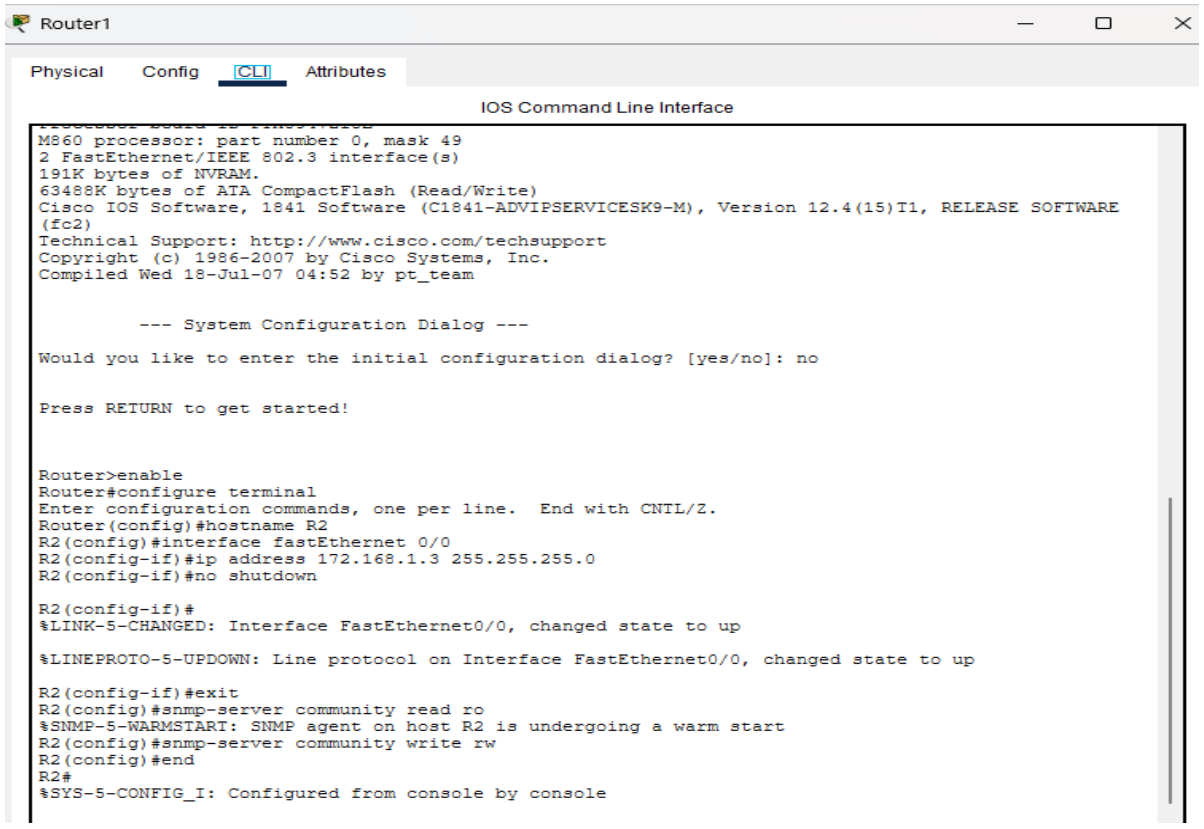
1.3.6.1.2.1.1.5.0

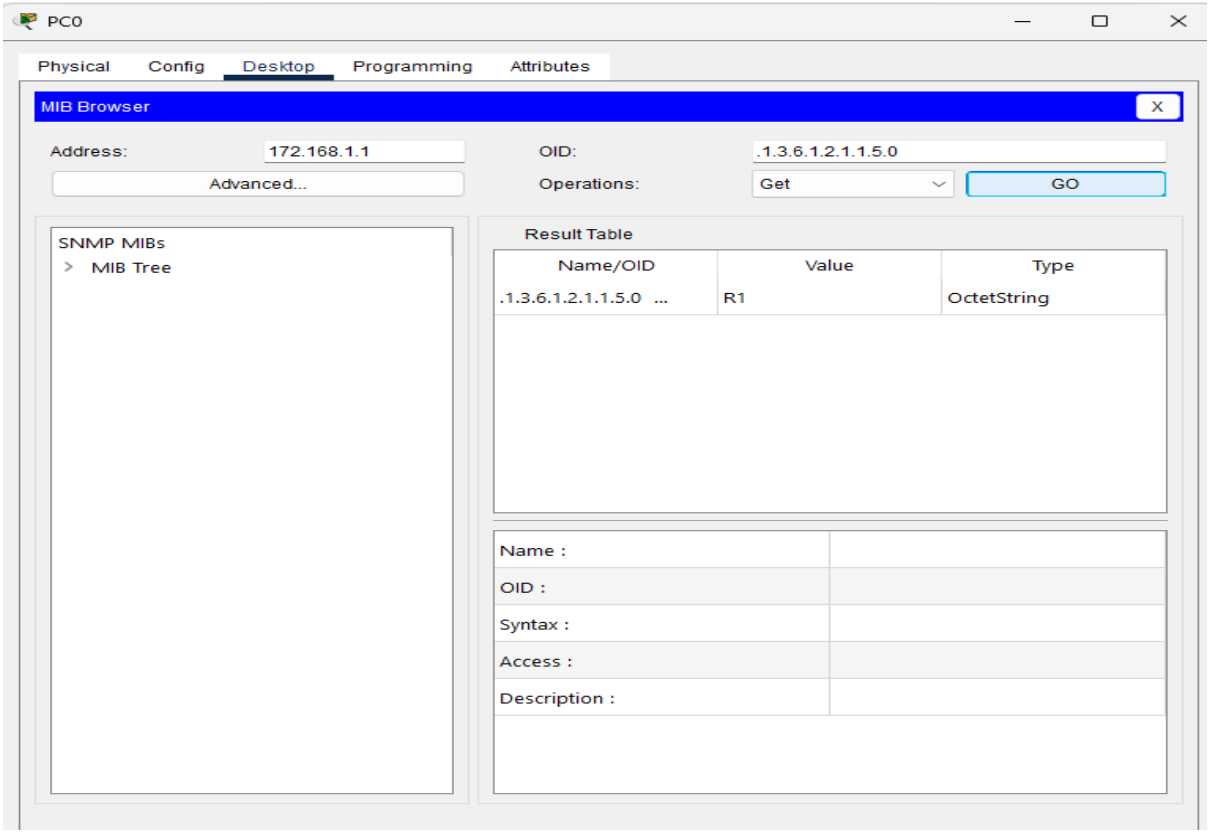
2. Select **Get**.
3. Click **GO**.

OUTPUT SCREENSHOTS:




Router0
Physical
Config
CLI
Attributes
IOS Command Line Interface
M860 processor: part number 0, mask 49
2 FastEthernet/IEEE 802.3 interface(s)
191K bytes of NVRAM.
63488K bytes of ATA CompactFlash (Read/Write)
Cisco IOS Software, 1841 Software (C1841-ADVIPSERVICESK9-M), Version 12.4(15)T1, RELEASE SOFTWARE
(fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2007 by Cisco Systems, Inc.
Compiled Wed 18-Jul-07 04:52 by pt_team
--- System Configuration Dialog ---
Would you like to enter the initial configuration dialog? [yes/no]: no
Press RETURN to get started!
Router>en
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#interface fastEthernet 0/0
R1(config-if)#ip address 172.168.1.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R1(config-if)#exit
R1(config)#snmp-server community read ro
%SNMP-5-WARMSTART: SNMP agent on host R1 is undergoing a warm start
R1(config)#snmp-server community write rw
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
Copy
Paste
Top

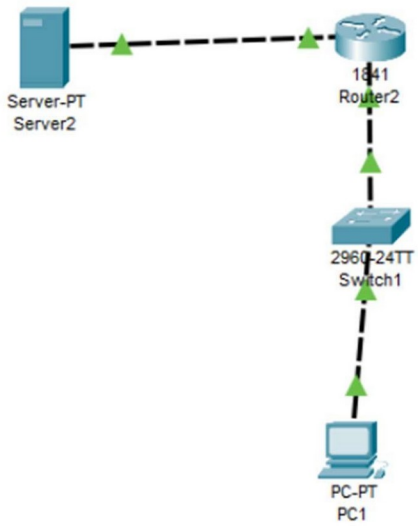

Router1
Physical
Config
CLI
Attributes
IOS Command Line Interface
M860 processor: part number 0, mask 49
2 FastEthernet/IEEE 802.3 interface(s)
191K bytes of NVRAM.
63488K bytes of ATA CompactFlash (Read/Write)
Cisco IOS Software, 1841 Software (C1841-ADVIPSERVICESK9-M), Version 12.4(15)T1, RELEASE SOFTWARE
(fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2007 by Cisco Systems, Inc.
Compiled Wed 18-Jul-07 04:52 by pt_team
--- System Configuration Dialog ---
Would you like to enter the initial configuration dialog? [yes/no]: no
Press RETURN to get started!
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R2
R2(config)#interface fastEthernet 0/0
R2(config-if)#ip address 172.168.1.3 255.255.255.0
R2(config-if)#no shutdown
R2(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R2(config-if)#exit
R2(config)#snmp-server community read ro
%SNMP-5-WARMSTART: SNMP agent on host R2 is undergoing a warm start
R2(config)#snmp-server community write rw
R2(config)#end
R2#
%SYS-5-CONFIG_I: Configured from console by console



3. Implement Syslog

Syslog sends router event messages to a server.

Implement Syslog



Device	Interface	IP Address
PC1		30.0.0.2/30.0.0.1
R2	F0/0 server	10.0.0.1
	F0/1 PC1	30.0.0.1
server		10.0.0.2/10.0.0.1

Configure the router interfaces:

enable

configure terminal

hostname R-SYSLOG

interface fastEthernet 0/0

ip address 10.0.0.1 255.0.0.0

no shutdown

exit

interface fastEthernet 0/1

ip address 30.0.0.1 255.0.0.0

no shutdown

exit

end

Test connectivity:

ping 30.0.0.1

ping 10.0.0.2

Enable the Syslog server:

Click:

Server → Services → SYSLOG → On

Configure logging on the router:

enable

configure terminal

service timestamps log datetime msec

logging on

logging 10.0.0.2

logging trap informational

end

Generate interface events:

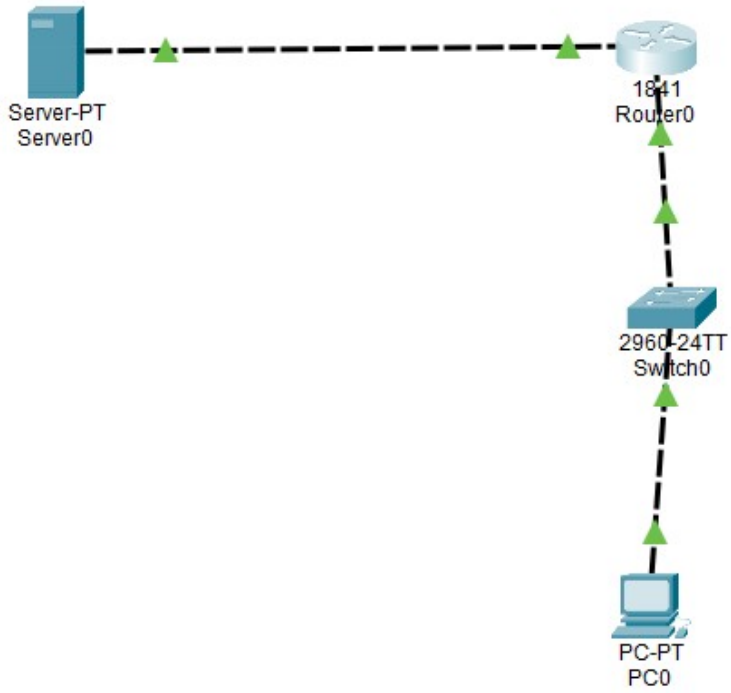
Shut down and re-enable the interface connected to the switch:

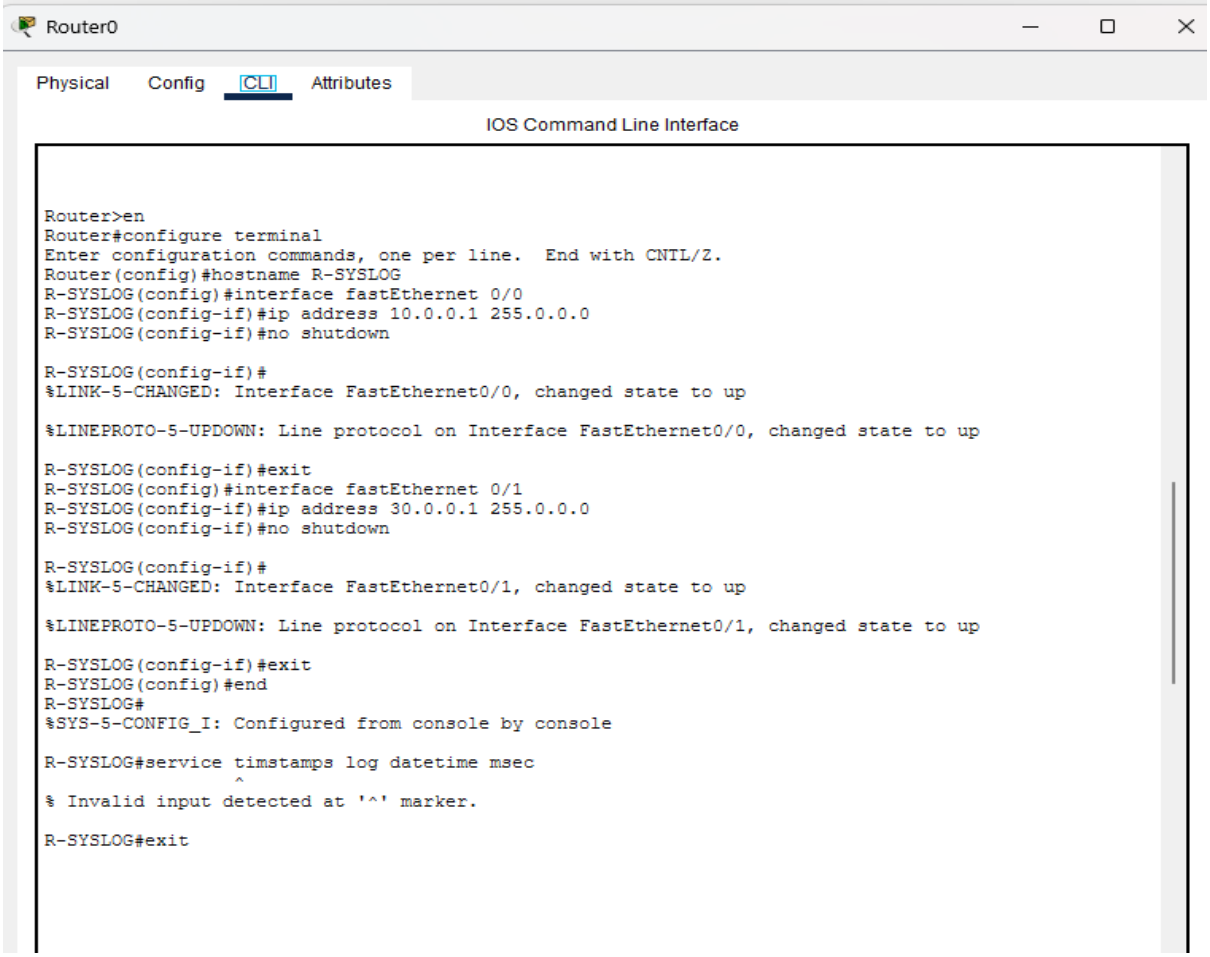
configure terminal

interface fastEthernet 0/1

shutdown

OUTPUT SCREENSHOTS:





Conclusion: We have successfully implemented,

- Implement SPAN Technologies (Switch Port Analyzer)
- Implement SNMP and Syslog
- Implement Flexible NetFlow

PRACTICAL-04

Aim

a) Implement a GRE Tunnel

b) Implement VTP

c) Implement NAT

Theory

Generic Routing Encapsulation Tunnel

Generic Routing Encapsulation (GRE) is a tunnelling protocol that encapsulates network-layer packets inside IP packets, creating a virtual point-to-point link between two routers across an intermediate network. In this practical a GRE tunnel (Tunnel 1) is built between Router1 and Router2 through Router0, using their physical interfaces as the tunnel source and destination. RIP is used so that the physical networks are reachable from each other.

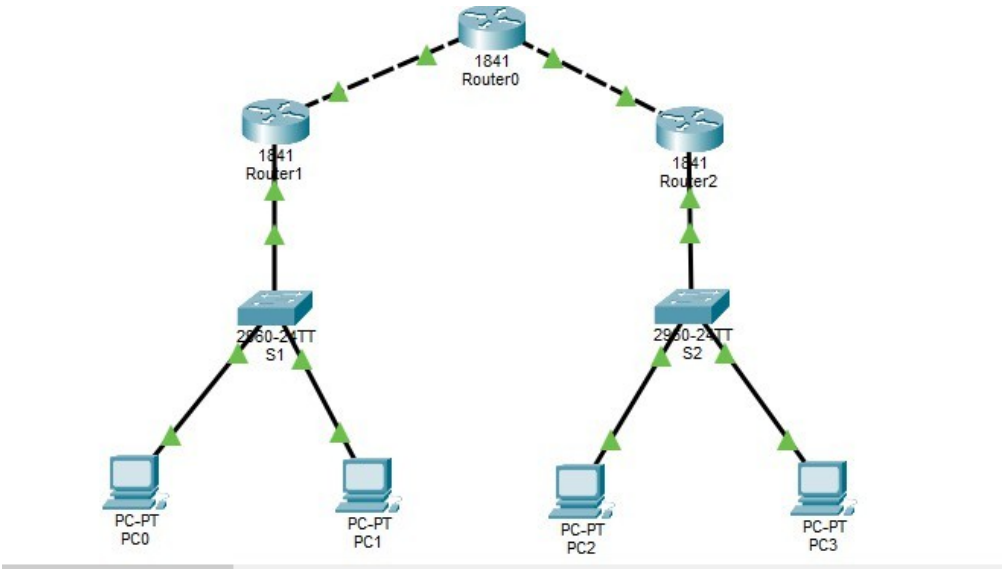
VLAN Trunking Protocol

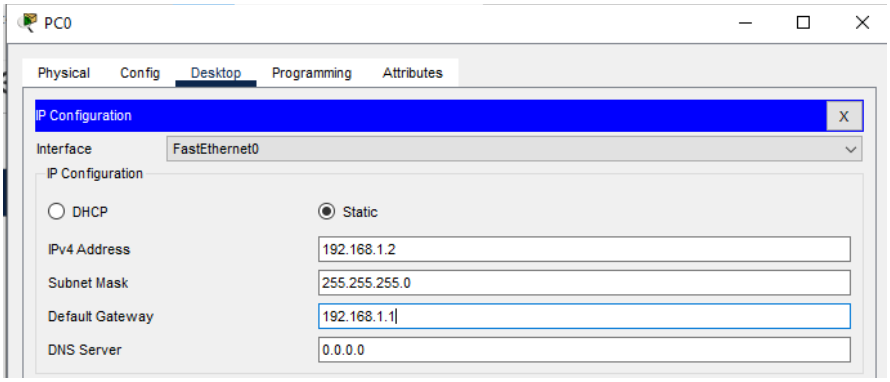
VLAN Trunking Protocol (VTP) is a Cisco protocol that distributes VLAN information from one switch to others in the same VTP domain over trunk links. A switch in server mode can create, modify and delete VLANs; a client receives and applies the VLAN database but cannot change it; a transparent switch keeps its own VLAN database and only forwards VTP advertisements. Here S1 is the server, S2 and S4 are clients, S3 is transparent, and the domain is VTPServer.

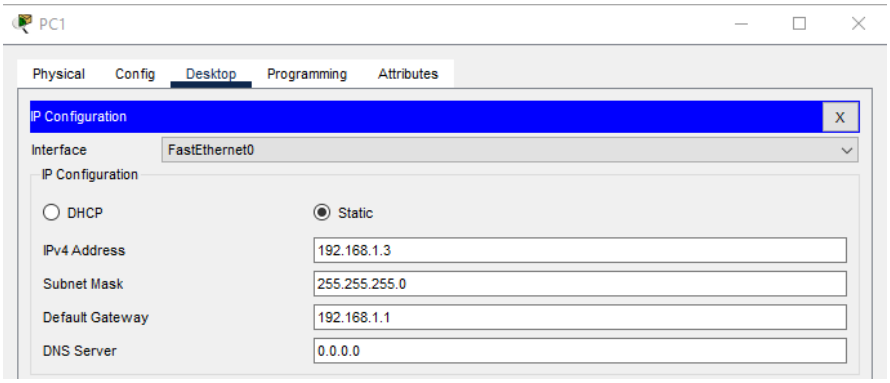
Network Address Translation

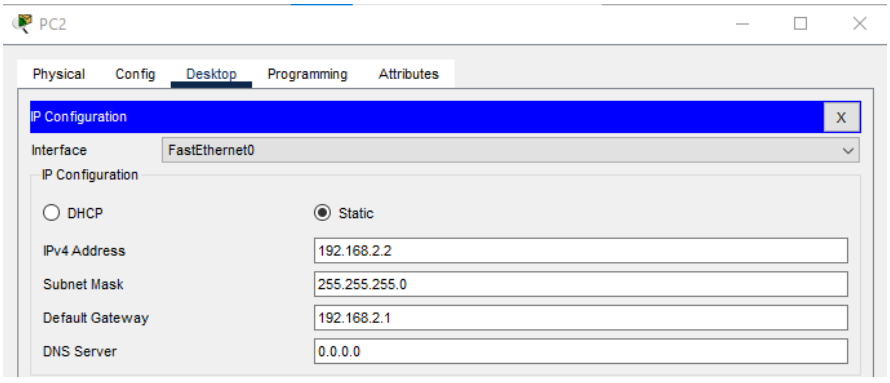
Network Address Translation (NAT) is a technique in which a router translates the IP addresses of packets passing between an inside (private) network and an outside network. It lets hosts with private addresses communicate with external networks, conserves public IPv4 addresses and hides the internal addressing from the outside. In this practical static NAT is configured on Router1, with FastEthernet0/0 as the inside interface and Serial0/1/0 as the outside interface, and RIP is used so that the PCs can reach Server0.

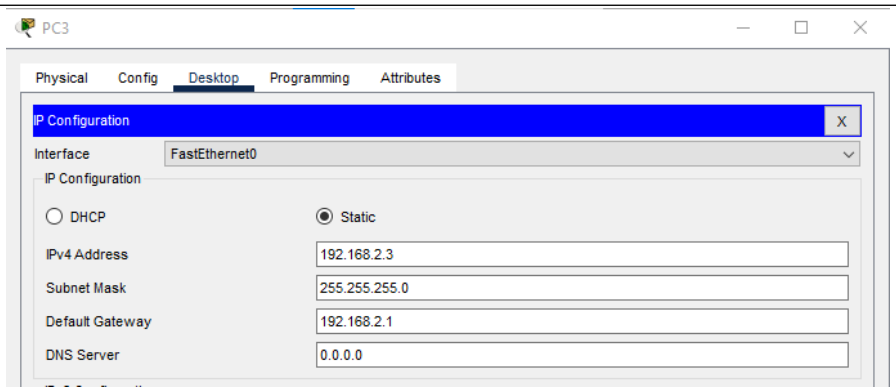
a) Implement a GRE Tunnel

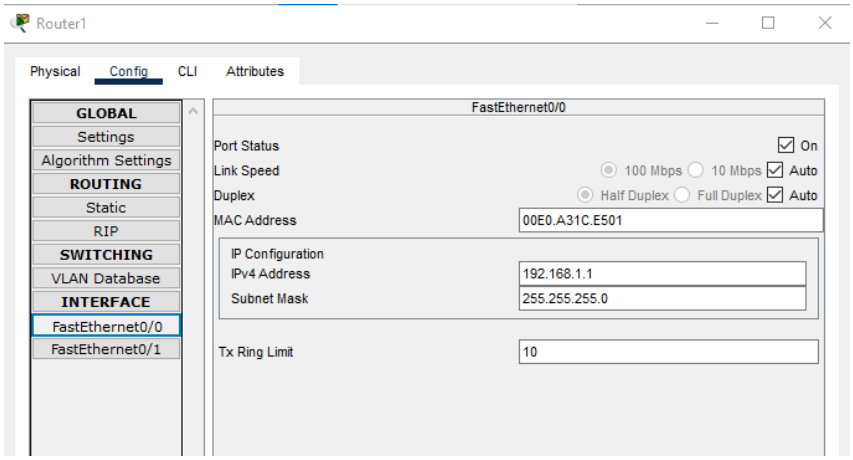


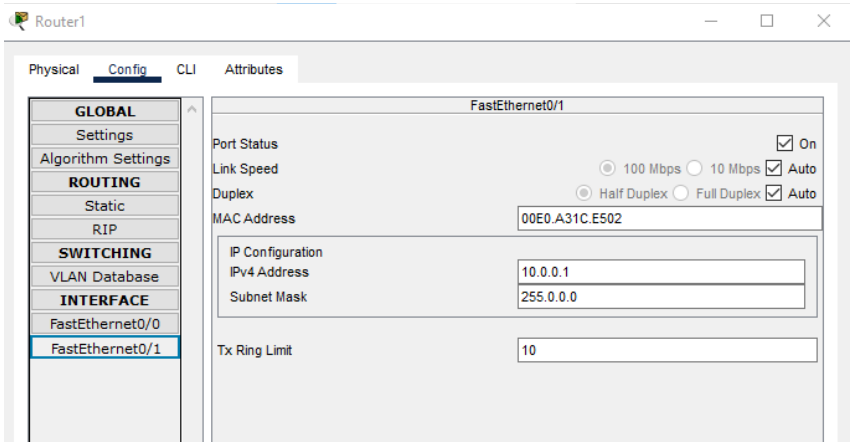


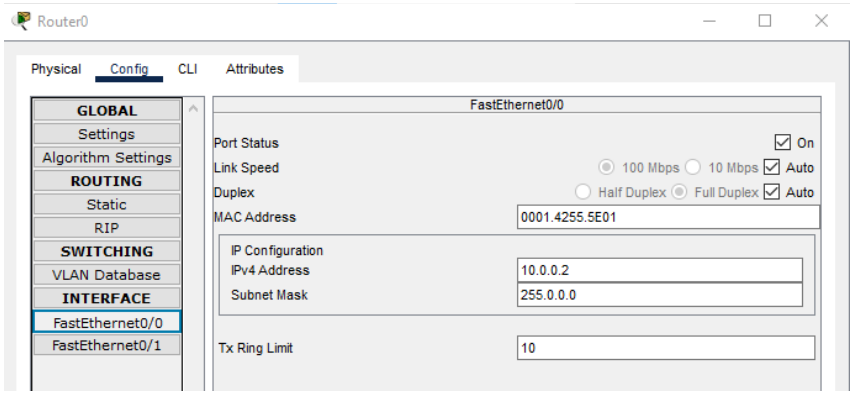


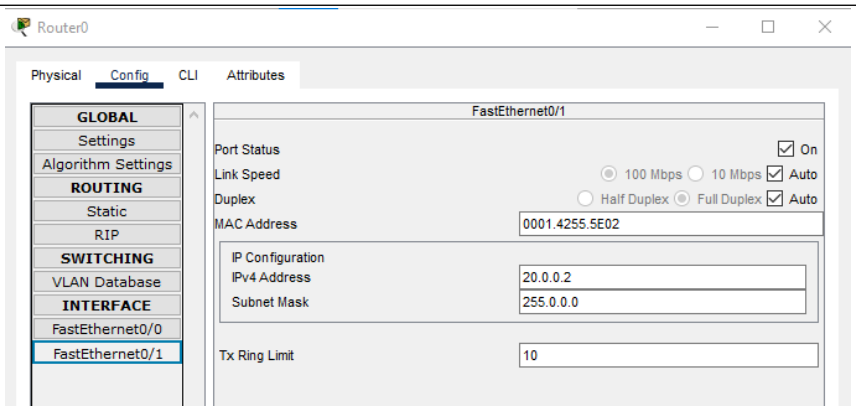


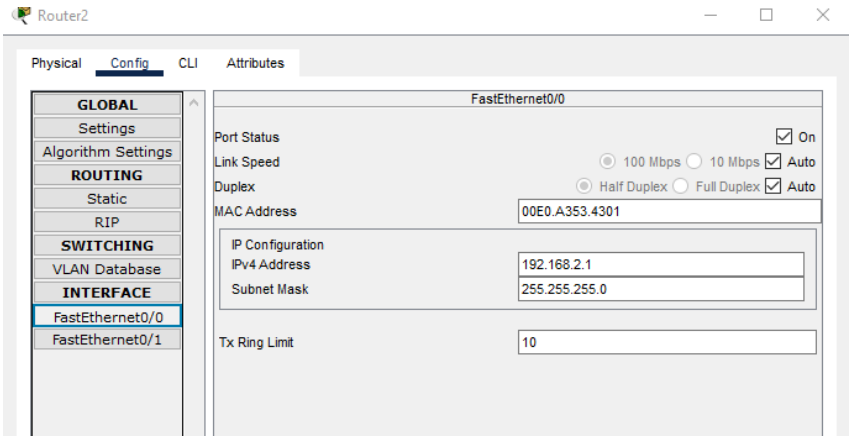

PC3
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.2.3
Subnet Mask
255.255.255.0
Default Gateway
192.168.2.1
DNS Server
0.0.0.0

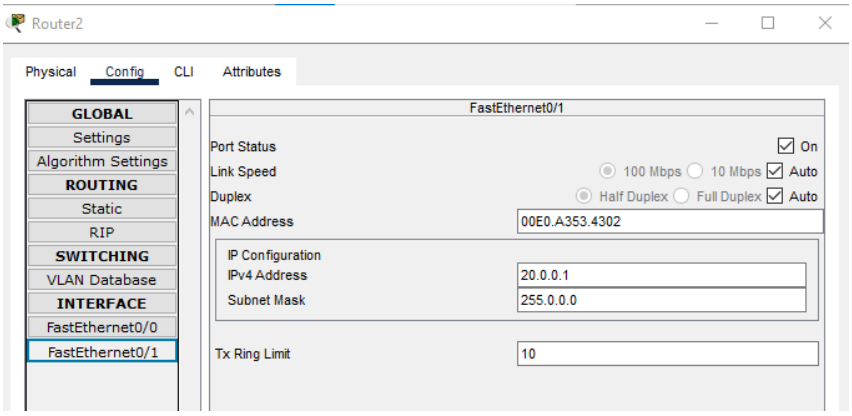

Router1
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
FastEthernet0/0
Port Status
On
Link Speed
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
00E0.A31C.E501
IP Configuration
IPv4 Address
192.168.1.1
Subnet Mask
255.255.255.0
Tx Ring Limit
10

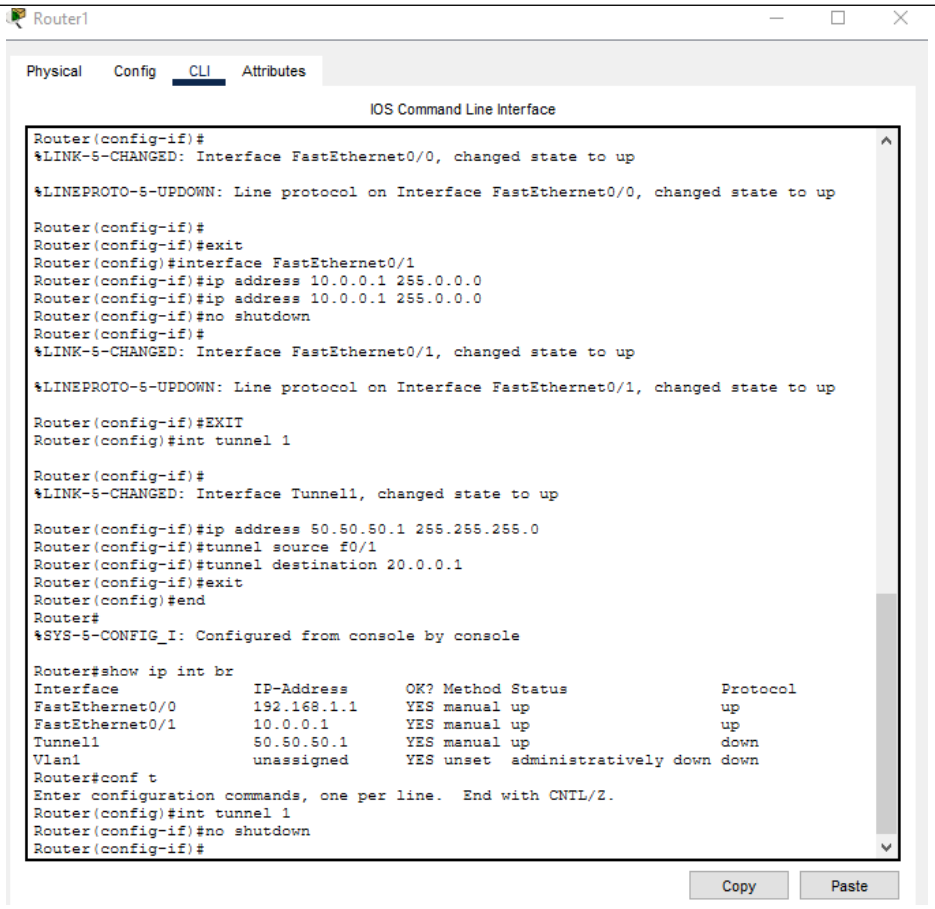

Router1
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
FastEthernet0/1
Port Status
On
Link Speed
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
00E0.A31C.E502
IP Configuration
IPv4 Address
10.0.0.1
Subnet Mask
255.0.0.0
Tx Ring Limit
10

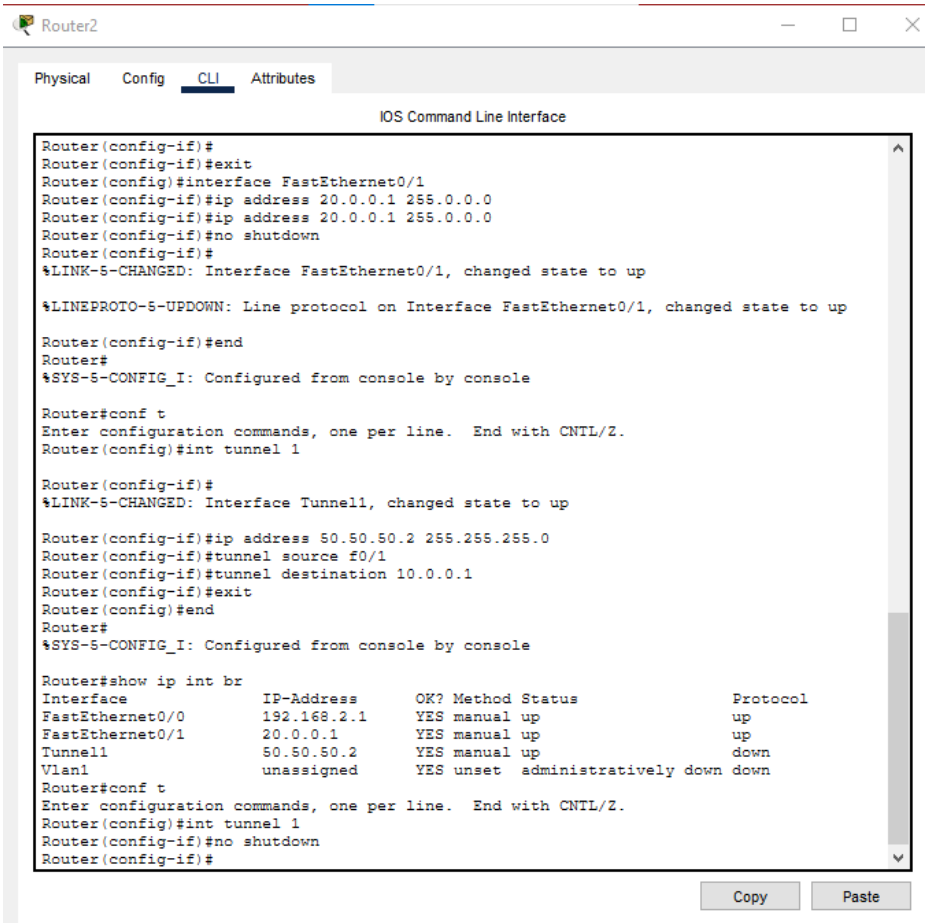

Router0
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
FastEthernet0/0
Port Status
On
Link Speed
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0001.4255.5E01
IP Configuration
IPv4 Address
10.0.0.2
Subnet Mask
255.0.0.0
Tx Ring Limit
10

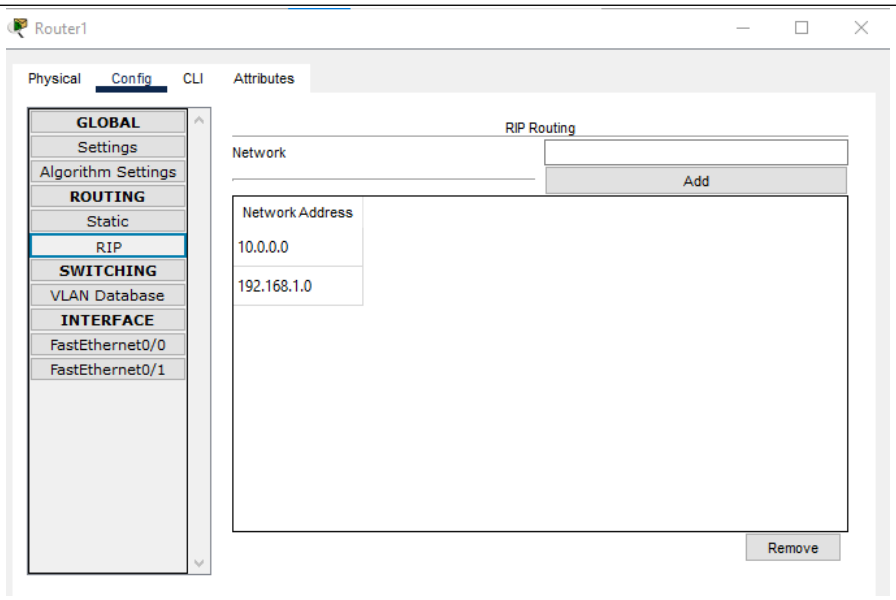

Router0
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
FastEthernet0/1
Port Status
On
Link Speed
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0001.4255.5E02
IP Configuration
IPv4 Address
20.0.0.2
Subnet Mask
255.0.0.0
Tx Ring Limit
10

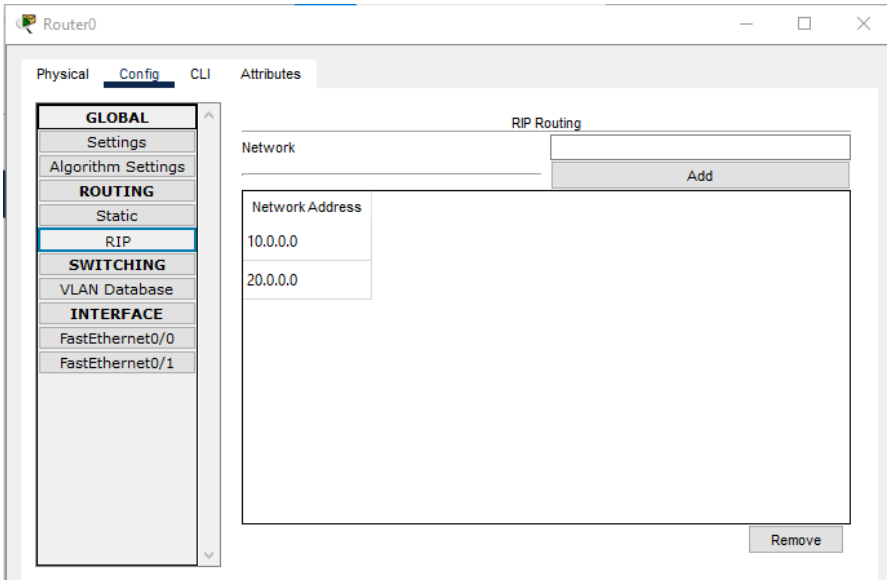

Router2
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
FastEthernet0/0
Port Status
On
Link Speed
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
00E0.A353.4301
IP Configuration
IPv4 Address
192.168.2.1
Subnet Mask
255.255.255.0
Tx Ring Limit
10

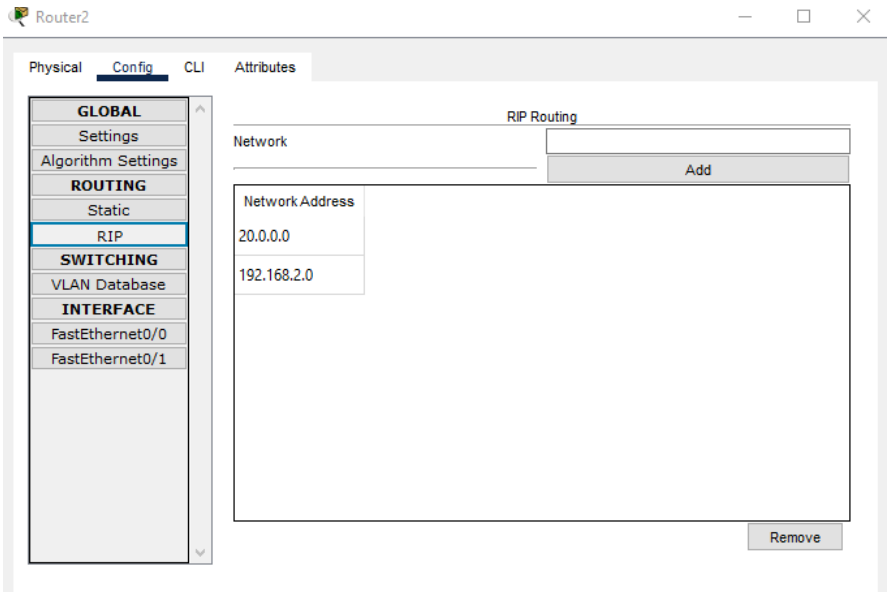

Router2
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
FastEthernet0/1
Port Status
On
Link Speed
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
00E0.A353.4302
IP Configuration
IPv4 Address
20.0.0.1
Subnet Mask
255.0.0.0
Tx Ring Limit
10

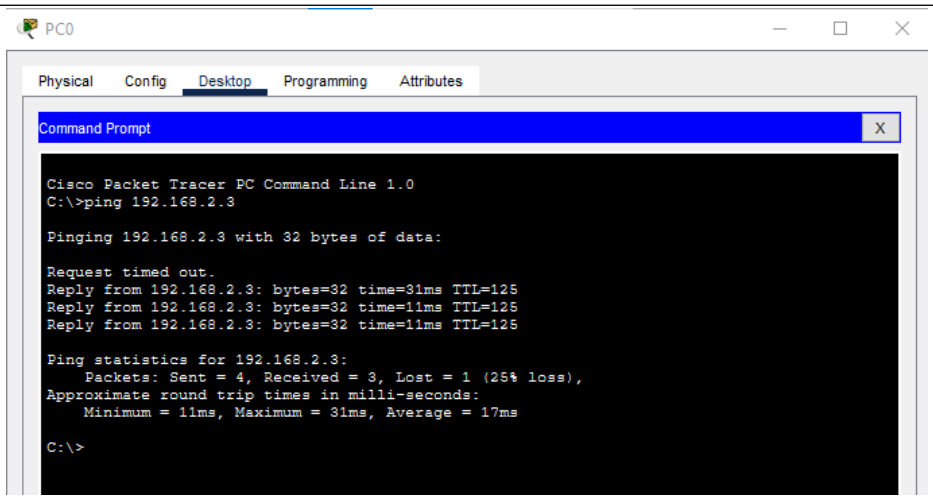

Router1
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#ip address 10.0.0.1 255.0.0.0
Router(config-if)#ip address 10.0.0.1 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
Router(config-if)#EXIT
Router(config)#int tunnel 1
Router(config-if)#
%LINK-5-CHANGED: Interface Tunnel1, changed state to up
Router(config-if)#ip address 50.50.50.1 255.255.255.0
Router(config-if)#tunnel source f0/1
Router(config-if)#tunnel destination 20.0.0.1
Router(config-if)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#show ip int br
Interface FastEthernet0/0 192.168.1.1 YES manual up up
FastEthernet0/1 10.0.0.1 YES manual up up
Tunnel1 50.50.50.1 YES manual up down
Vlan1 unassigned YES unset administratively down down
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int tunnel 1
Router(config-if)#no shutdown
Router(config-if)#
Copy
Paste


Router2
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#ip address 20.0.0.1 255.0.0.0
Router(config-if)#ip address 20.0.0.1 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int tunnel 1
Router(config-if)#
%LINK-5-CHANGED: Interface Tunnel1, changed state to up
Router(config-if)#ip address 50.50.50.2 255.255.255.0
Router(config-if)#tunnel source f0/1
Router(config-if)#tunnel destination 10.0.0.1
Router(config-if)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#show ip int br
Interface FastEthernet0/0 192.168.2.1 YES manual up up
FastEthernet0/1 20.0.0.1 YES manual up up
Tunnel1 50.50.50.2 YES manual up down
Vlan1 unassigned YES unset administratively down down
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int tunnel 1
Router(config-if)#no shutdown
Router(config-if)#
Copy
Paste

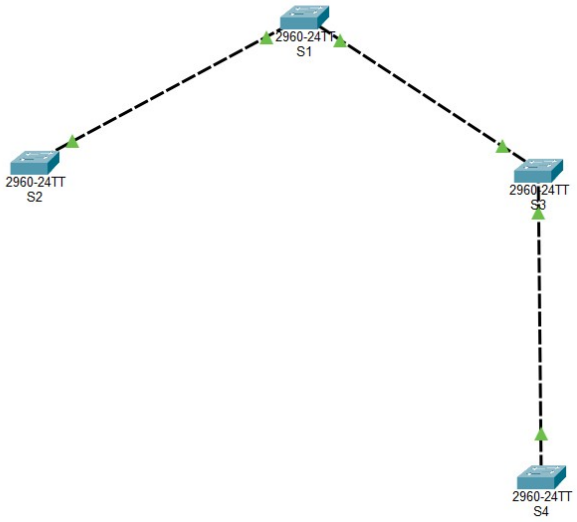

Router1
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
RIP Routing
Network
Add
Network Address
10.0.0.0
192.168.1.0
Remove

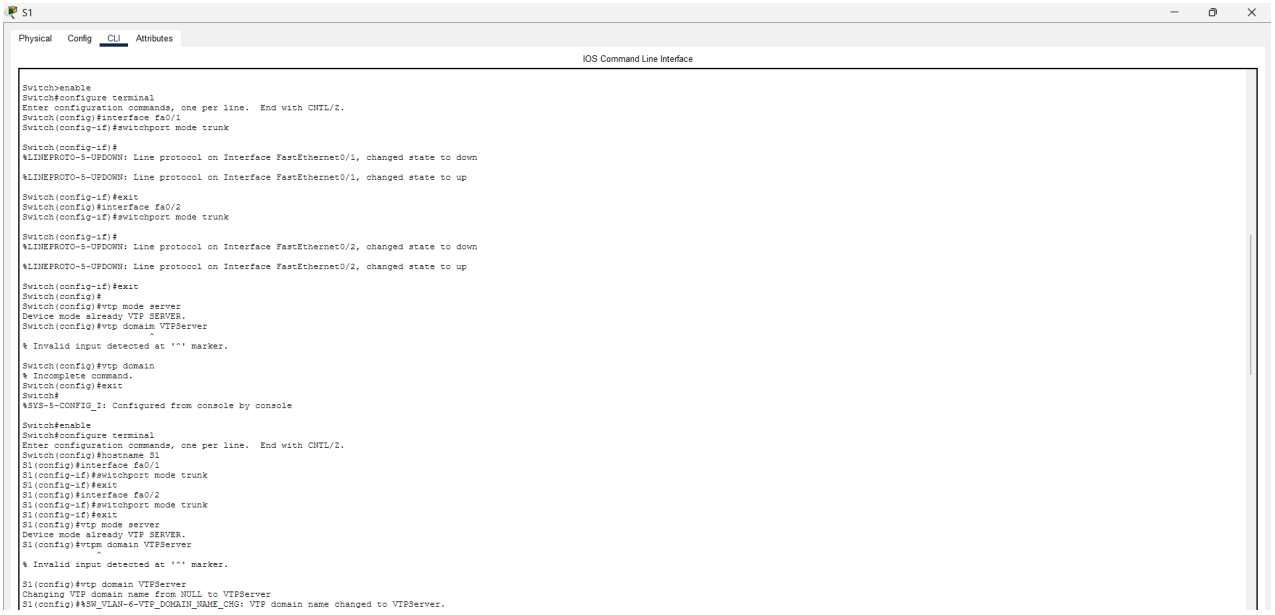

Router0
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
RIP Routing
Network
Add
Network Address
10.0.0.0
20.0.0.0
Remove


Router2
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
RIP Routing
Network
Add
Network Address
20.0.0.0
192.168.2.0
Remove



b) Implement VTP





S1

Physical Config CLI Attributes

IOS Command Line Interface

```
S1(config)#end
S1#
%SYS-5-CONFIG_I: Configured from console by console

S1#show vtp status
VTP Version capable             : 1 to 2
VTP version running             : 1
VTP Domain Name                 : VTPServer
VTP Pruning Mode                : Disabled
VTP Traps Generation            : Disabled
Device ID                       : 0000.0C92.E400
Configuration last modified by 0.0.0.0 at 0-0-00 00:00:00
Local updater ID is 0.0.0.0 (no valid interface found)

Feature VLAN :
--------------
VTP Operating Mode              : Server
Maximum VLANs supported locally : 255
Number of existing VLANs        : 5
Configuration Revision          : 0
MD5 digest                      : 0xB8 0x02 0x0F 0x8B 0xD8 0xFA 0x0B 0x13
                                  0xD1 0x7E 0x67 0x30 0xA2 0xC6 0x78 0x9F

S1#configure terminal
Enter configuration commands, one per line.  End with CNTL/2.
S1(config)#vlan 10
S1(config-vlan)#name University
S1(config-vlan)#end
S1#
%SYS-5-CONFIG_I: Configured from console by console

S1#show vtp status
VTP Version capable             : 1 to 2
VTP version running             : 1
VTP Domain Name                 : VTPServer
VTP Pruning Mode                : Disabled
VTP Traps Generation            : Disabled
Device ID                       : 0000.0C92.E400
Configuration last modified by 0.0.0.0 at 3-1-93 00:12:56
Local updater ID is 0.0.0.0 (no valid interface found)

Feature VLAN :
--------------
VTP Operating Mode              : Server
Maximum VLANs supported locally : 255
Number of existing VLANs        : 6
Configuration Revision          : 2
MD5 digest                      : 0x4B 0x36 0x83 0x7A 0x90 0xDD 0x95 0x9A
                                  0x2D 0xE2 0x94 0x30 0x4B 0x66 0xF4 0xF9

S1#show vlan

VLAN Name                              Status    Ports
---- --------------------------------- --------- -------------------------------
1    default                           active    Fa0/3, Fa0/4, Fa0/5, Fa0/6
                                                 0x2D 0xE2 0x94 0x30 0x4B 0x66 0xF4 0xF9

S1#show vlan

VLAN Name                              Status    Ports
---- --------------------------------- --------- -------------------------------
1    default                           active    Fa0/3, Fa0/4, Fa0/5, Fa0/6
                                                 Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                 Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                 Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                 Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                 Fa0/23, Fa0/24, Gig0/1, Gig0/2
10   University                        active
1002 fddi-default                      active
1003 token-ring-default                active
1004 fddinet-default                   active
1005 trnet-default                     active

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------
1    enet  100001     1500  -      -      -        -    -        0      0
10   enet  100010     1500  -      -      -        -    -        0      0
1002 fddi  101002     1500  -      -      -        -    -        0      0
1003 tr    101003     1500  -      -      -        -    -        0      0
1004 fdnet 101004     1500  -      -      -        ieee -        0      0
1005 trnet 101005     1500  -      -      -        ibm  -        0      0

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------

Remote SPAN VLANs
------------------------------------------------------------------------------

Primary Secondary Type              Ports
------- --------- ----------------- ------------------------------------------
S1#
S1#
S1#
S1#
S1#
```

S2

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/2.
Switch(config)#hostname S2
S2(config)#interface fa0/1
S2(config-if)#switchport mode trunk
S2(config-if)#exit
S2(config)#vtp mode client
Setting device to VTP CLIENT mode.
S2(config)#end
S2#
%SYS-5-CONFIG_I: Configured from console by console

S2#show vtp status
VTP Version capable             : 1 to 2
VTP version running             : 1
VTP Domain Name                 : VTPServer
VTP Pruning Mode                : Disabled
VTP Traps Generation            : Disabled
Device ID                       : 0050.0F87.DC00
Configuration last modified by 0.0.0.0 at 3-1-93 00:12:56

Feature VLAN :
--------------
VTP Operating Mode              : Client
Maximum VLANs supported locally : 255
Number of existing VLANs        : 6
Configuration Revision          : 2
MD5 digest                      : 0x4B 0x36 0x83 0x7A 0x90 0xDD 0x95 0x9A
                                  0x2D 0xE2 0x94 0x30 0x4B 0x66 0xF4 0xF9

S2#
S2#
S2#




S2 con0 is now available




Press RETURN to get started.
```

S3

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname S3
S3(config)#interface fa0/1
S3(config-if)#switchport mode trunk
S3(config-if)#exit
S3(config)#interface fa0/2
S3(config-if)#switchport mode trunk

S3(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

S3(config-if)#exit
S3(config)#vtp doamin VTPServer
                ^
% Invalid input detected at '^' marker.

S3(config)#vtp domain VTPServer
Domain name already set to VTPServer.
S3(config)#vtp mode transparent
Setting device to VTP TRANSPARENT mode.
S3(config)#end
S3#
%SYS-5-CONFIG_I: Configured from console by console

S3#show vtp status
VTP Version capable             : 1 to 2
VTP version running             : 1
VTP Domain Name                 : VTPServer
VTP Pruning Mode                : Disabled
VTP Traps Generation            : Disabled
Device ID                       : 0002.177A.3E00
Configuration last modified by 0.0.0.0 at 3-1-93 00:12:56

Feature VLAN :
--------------
VTP Operating Mode              : Transparent
Maximum VLANs supported locally : 255
Number of existing VLANs        : 6
Configuration Revision          : 0
MD5 digest                      : 0xDA 0xDB 0xFF 0xC5 0x11 0x6B 0xF5 0xA3
                                  0x4D 0xAF 0x32 0x4A 0xAB 0x5B 0x91 0xEF

S3#show vlan

VLAN Name                              Status    Ports
---- --------------------------------- --------- -------------------------------
1    default                           active    Fa0/3, Fa0/4, Fa0/5, Fa0/6
                                                 Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                 Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                 Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                 Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                 Fa0/23, Fa0/24, Gig0/1, Gig0/2
10   University                        active
1002 fddi-default                      active
1003 token-ring-default                active
1004 fddinet-default                   active
1005 trnet-default                     active

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------
1    enet  100001     1500  -      -      -        -    -        0      0
10   enet  100010     1500  -      -      -        -    -        0      0
1002 fddi  101002     1500  -      -      -        -    -        0      0
1003 tr    101003     1500  -      -      -        -    -        0      0
1004 fdnet 101004     1500  -      -      -        ieee -        0      0
1005 trnet 101005     1500  -      -      -        ibm  -        0      0

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------

Remote SPAN VLANs
------------------------------------------------------------------------------

Primary Secondary Type              Ports
------- --------- ----------------- ------------------------------------------

S3#
S3#
S3#
```

S4

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname S4
S4(config)#interface fa0/1
S4(config-if)#switchport mode trunk
S4(config-if)#exit
S4(config)#vtp domain VTPServer
Domain name already set to VTPServer.
S4(config)#vtp mode client
Setting device to VTP CLIENT mode.
S4(config)#end
S4#
%SYS-5-CONFIG_I: Configured from console by console

S4#show vtp status
VTP Version capable             : 1 to 2
VTP version running             : 1
VTP Domain Name                 : VTPServer
VTP Pruning Mode                : Disabled
VTP Traps Generation            : Disabled
Device ID                       : 0060.5C74.0D00
Configuration last modified by 0.0.0.0 at 3-1-93 00:12:56

Feature VLAN :
--------------
VTP Operating Mode              : Client
Maximum VLANs supported locally : 255
Number of existing VLANs        : 6
Configuration Revision          : 2
MD5 digest                      : 0x4B 0x36 0x83 0x7A 0x90 0xDD 0x95 0x9A
                                  0x2D 0xF2 0x94 0x30 0x4B 0x66 0xF4 0xF9

S4#show vlan

VLAN Name                              Status    Ports
---- --------------------------------- --------- -------------------------------
1    default                           active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                                 Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                 Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                 Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                 Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                 Fa0/22, Fa0/23, Fa0/24, Gig0/1
                                                 Gig0/2
10   University                        active
1002 fddi-default                      active
1003 token-ring-default                active
1004 fddinet-default                   active
1005 trnet-default                     active

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------
1    enet  100001     1500  -      -      -        -    -        0      0
10   enet  100010     1500  -      -      -        -    -        0      0
1002 fddi  101002     1500  -      -      -        -    -        0      0
1003 tr    101003     1500  -      -      -        -    -        0      0
1004 fdnet 101004     1500  -      -      -        ieee -        0      0
1005 trnet 101005     1500  -      -      -        ibm  -        0      0

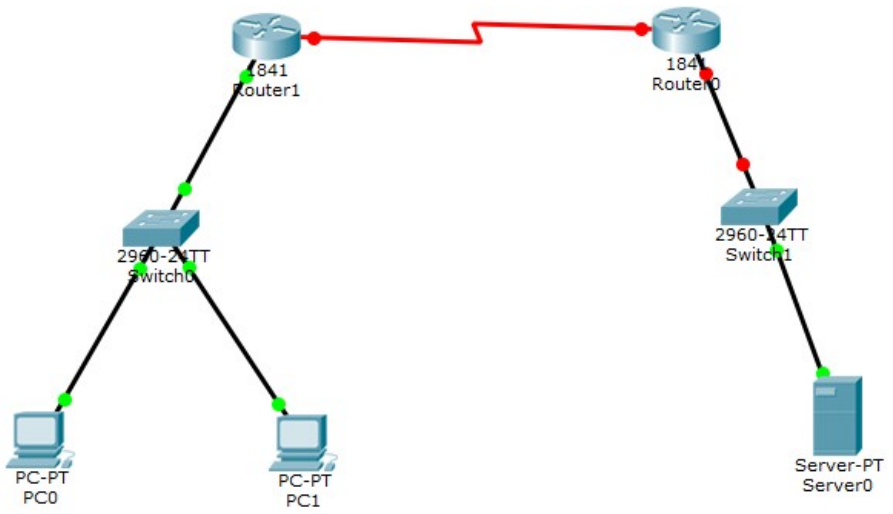
VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------

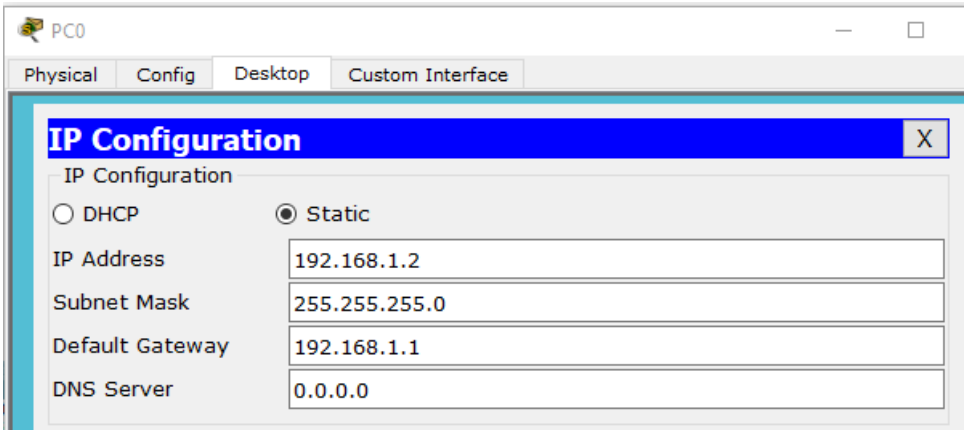
Remote SPAN VLANs
------------------------------------------------------------------------------

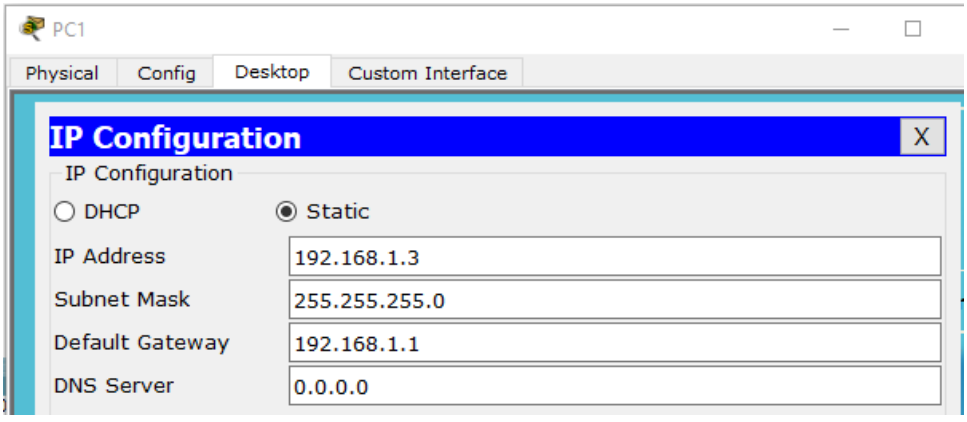
Primary Secondary Type              Ports
------- --------- ----------------- ------------------------------------------

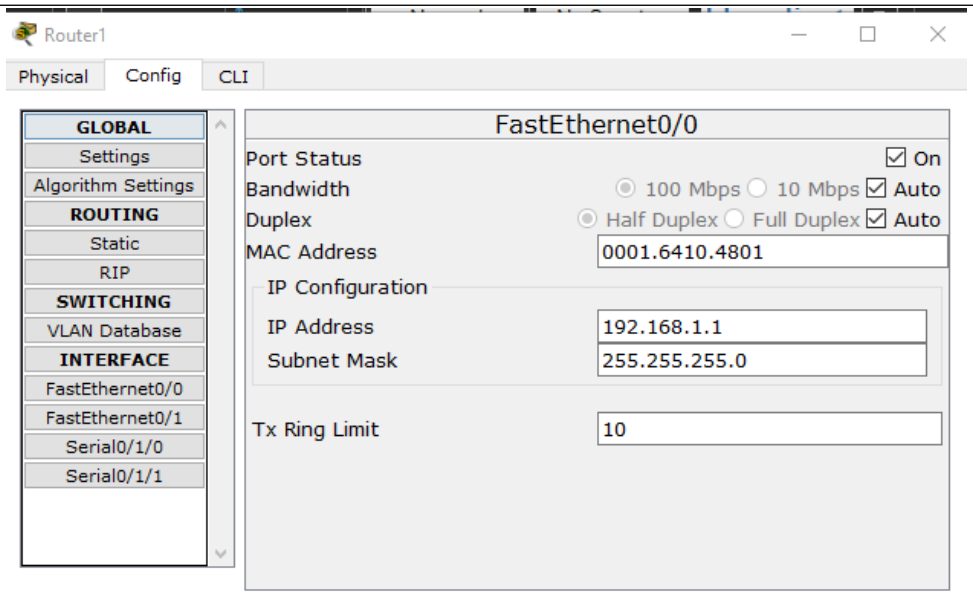
S4#
S4#
S4#
S4#
```

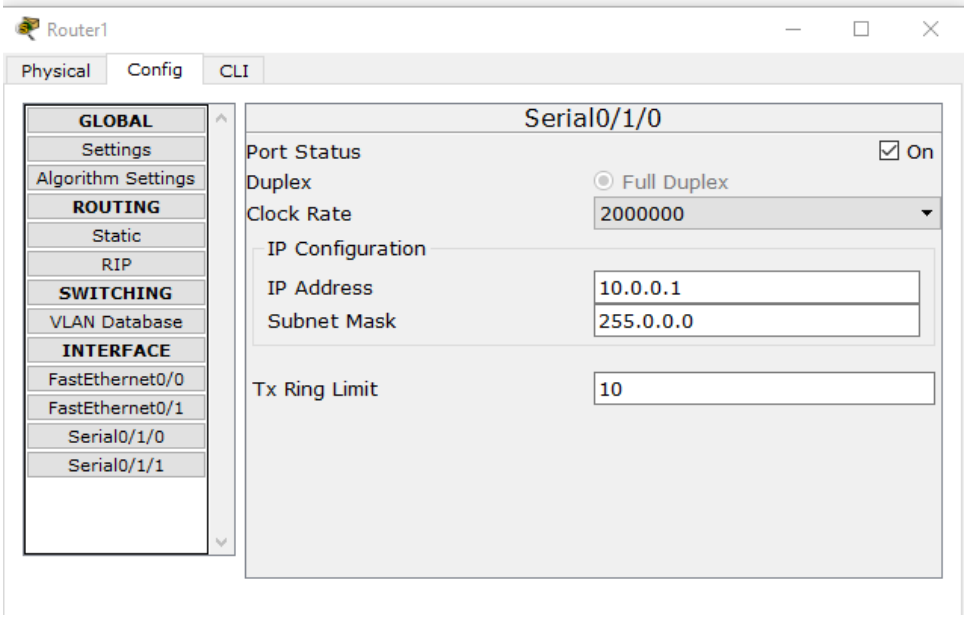
c) Implement NAT

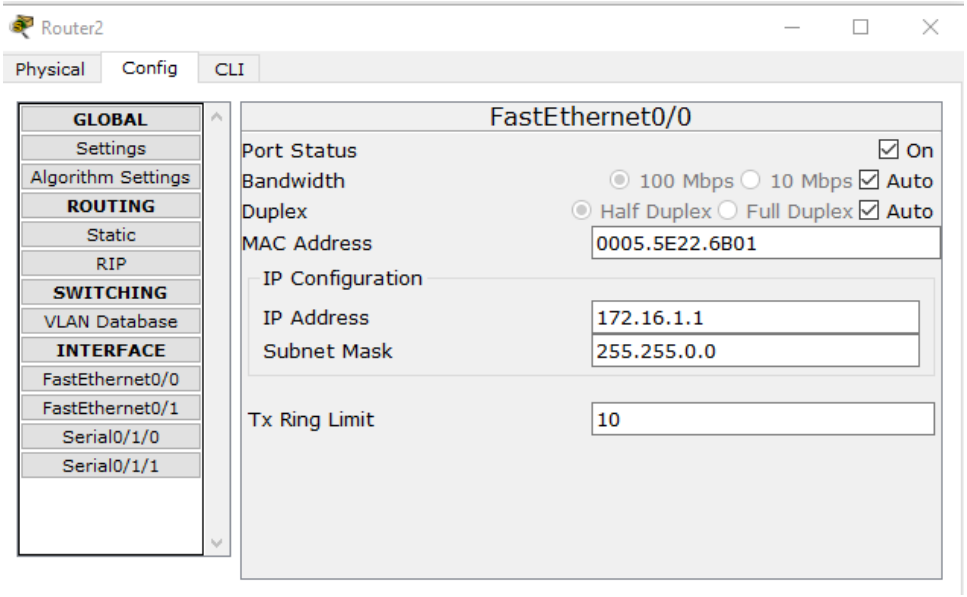


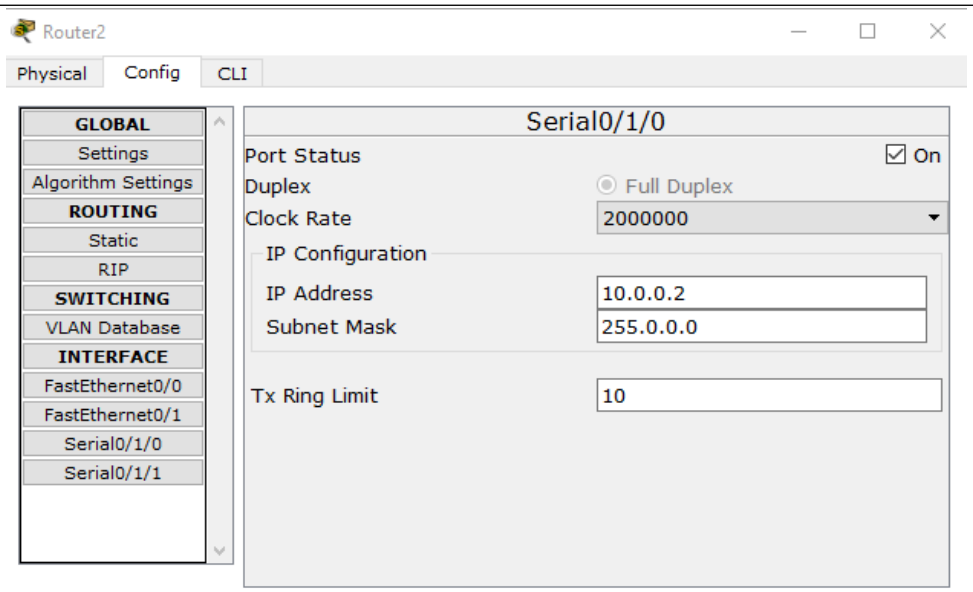


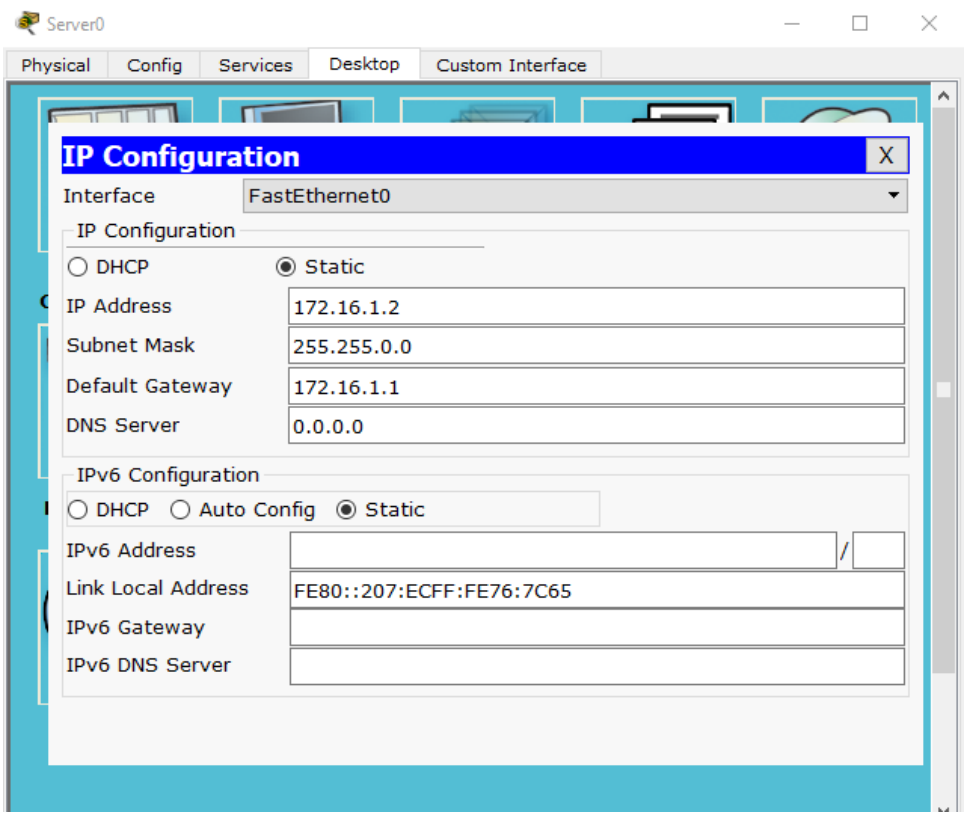


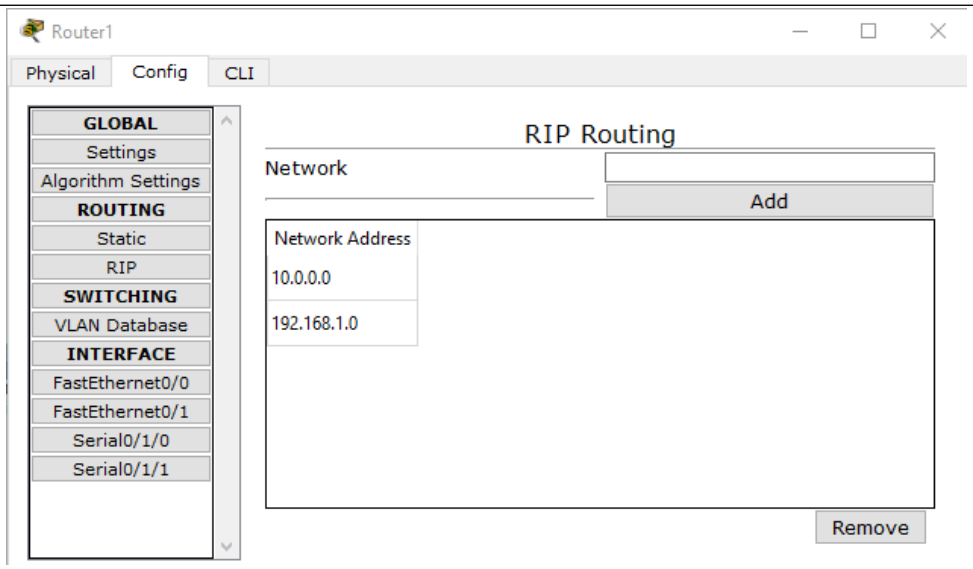

Router1
Physical
Config
CLI
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
Serial0/1/0
Serial0/1/1
FastEthernet0/0
Port Status
On
Bandwidth
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0001.6410.4801
IP Configuration
IP Address
192.168.1.1
Subnet Mask
255.255.255.0
Tx Ring Limit
10

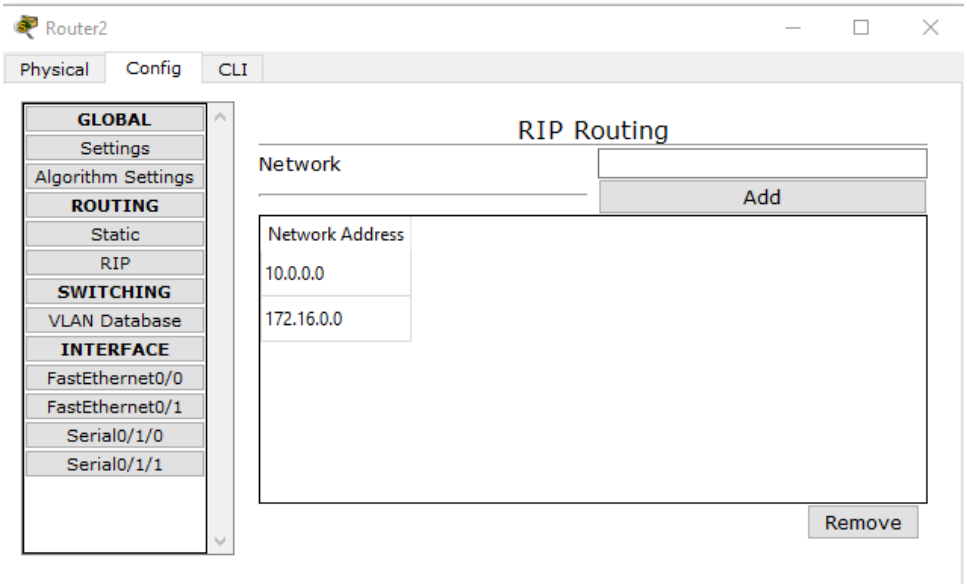

Router1
Physical
Config
CLI
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
Serial0/1/0
Serial0/1/1
Serial0/1/0
Port Status
On
Duplex
Full Duplex
Clock Rate
2000000
IP Configuration
IP Address
10.0.0.1
Subnet Mask
255.0.0.0
Tx Ring Limit
10

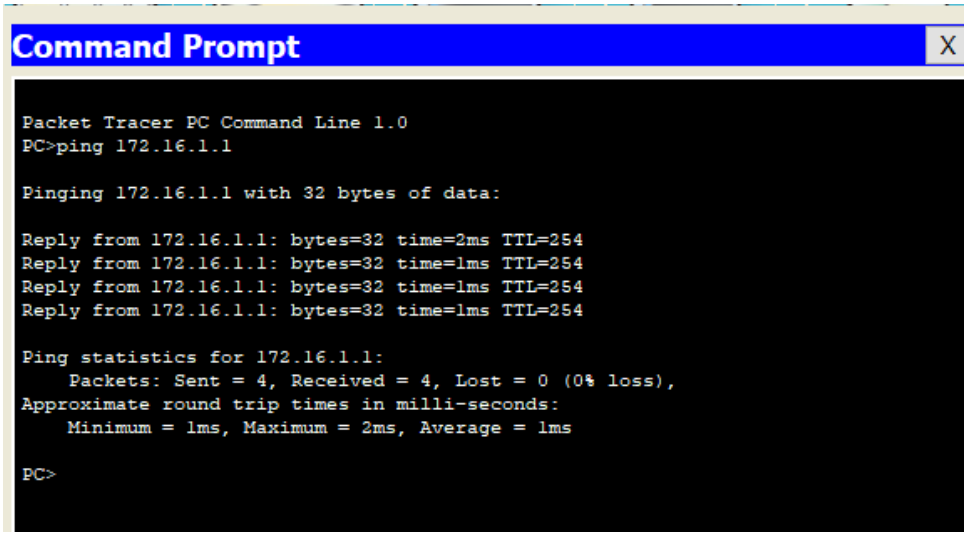

Router2
Physical
Config
CLI
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
Serial0/1/0
Serial0/1/1
FastEthernet0/0
Port Status
On
Bandwidth
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0005.5E22.6B01
IP Configuration
IP Address
172.16.1.1
Subnet Mask
255.255.0.0
Tx Ring Limit
10

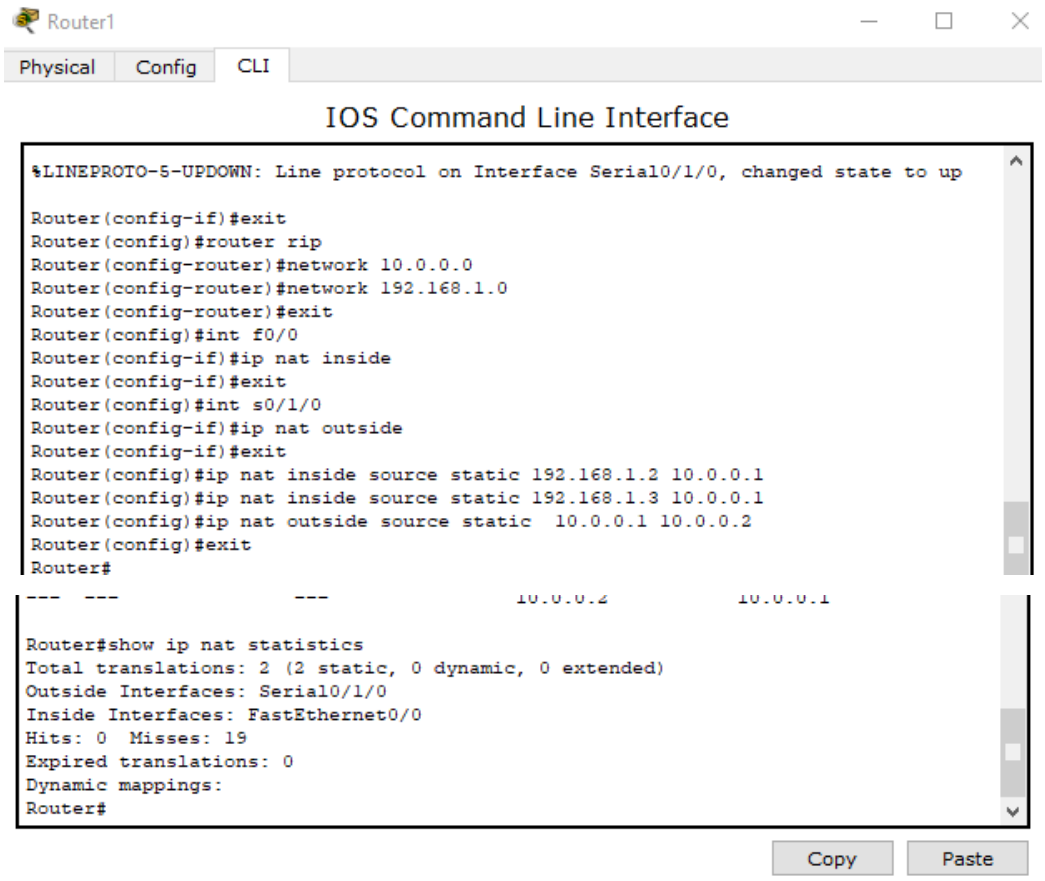

Router2
Physical
Config
CLI
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
Serial0/1/0
Serial0/1/1
Serial0/1/0
Port Status
On
Duplex
Full Duplex
Clock Rate
2000000
IP Configuration
IP Address
10.0.0.2
Subnet Mask
255.0.0.0
Tx Ring Limit
10

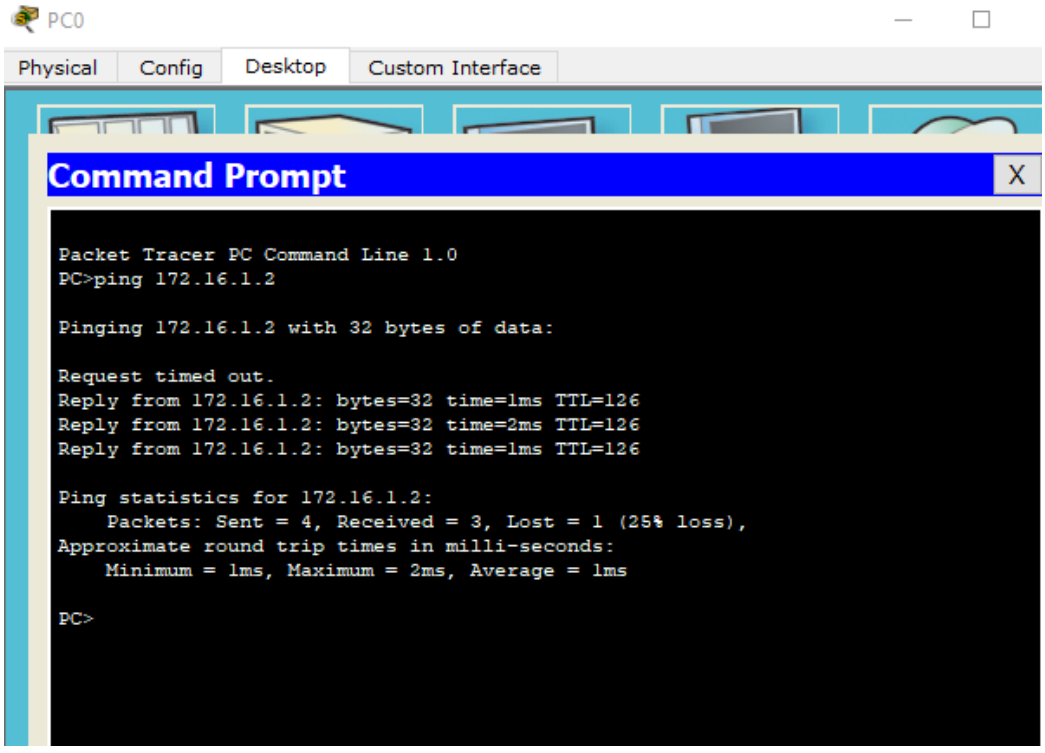

Server0
Physical
Config
Services
Desktop
Custom Interface
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IP Address
172.16.1.2
Subnet Mask
255.255.0.0
Default Gateway
172.16.1.1
DNS Server
0.0.0.0
IPv6 Configuration
DHCP
Auto Config
Static
IPv6 Address
Link Local Address
FE80::207:ECFF:FE76:7C65
IPv6 Gateway
IPv6 DNS Server


Router1
Physical
Config
CLI
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
Serial0/1/0
Serial0/1/1
RIP Routing
Network
Add
Network Address
10.0.0.0
192.168.1.0
Remove

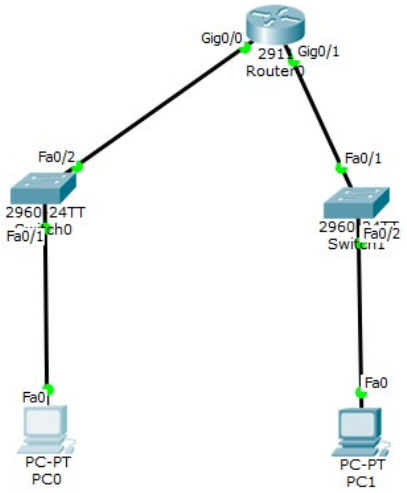

Router2
Physical
Config
CLI
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
Serial0/1/0
Serial0/1/1
RIP Routing
Network
Add
Network Address
10.0.0.0
172.16.0.0
Remove


Command Prompt
X
Packet Tracer PC Command Line 1.0
PC>ping 172.16.1.1
Pinging 172.16.1.1 with 32 bytes of data:
Reply from 172.16.1.1: bytes=32 time=2ms TTL=254
Reply from 172.16.1.1: bytes=32 time=1ms TTL=254
Reply from 172.16.1.1: bytes=32 time=1ms TTL=254
Reply from 172.16.1.1: bytes=32 time=1ms TTL=254
Ping statistics for 172.16.1.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 2ms, Average = 1ms
PC>


Router1
Physical
Config
CLI
IOS Command Line Interface
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up
Router(config-if)#exit
Router(config)#router rip
Router(config-router)#network 10.0.0.0
Router(config-router)#network 192.168.1.0
Router(config-router)#exit
Router(config)#int f0/0
Router(config-if)#ip nat inside
Router(config-if)#exit
Router(config)#int s0/1/0
Router(config-if)#ip nat outside
Router(config-if)#exit
Router(config)#ip nat inside source static 192.168.1.2 10.0.0.1
Router(config)#ip nat inside source static 192.168.1.3 10.0.0.1
Router(config)#ip nat outside source static 10.0.0.1 10.0.0.2
Router(config)#exit
Router#
Router#show ip nat statistics
Total translations: 2 (2 static, 0 dynamic, 0 extended)
Outside Interfaces: Serial0/1/0
Inside Interfaces: FastEthernet0/0
Hits: 0 Misses: 19
Expired translations: 0
Dynamic mappings:
Router#
Copy
Paste


PC0
Physical
Config
Desktop
Custom Interface
Command Prompt
X
Packet Tracer PC Command Line 1.0
PC>ping 172.16.1.2
Pinging 172.16.1.2 with 32 bytes of data:
Request timed out.
Reply from 172.16.1.2: bytes=32 time=1ms TTL=126
Reply from 172.16.1.2: bytes=32 time=2ms TTL=126
Reply from 172.16.1.2: bytes=32 time=1ms TTL=126
Ping statistics for 172.16.1.2:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 2ms, Average = 1ms
PC>

PRACTICAL-05



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/0/1.10	192.168.10.1	255.255.255.0	N/A
	G0/0/1.20	192.168.20.1	255.255.255.0	
	G0/0/1.30	192.168.30.1	255.255.255.0	
	G0/0/1.1000	N/A	N/A	
S1	VLAN 10	192.168.10.11	255.255.255.0	192.168.10.1
S2	VLAN 10	192.168.10.12	255.255.255.0	192.168.10.1
PC-A	NIC	192.168.20.3	255.255.255.0	192.168.20.1
PC-B	NIC	192.168.30.3	255.255.255.0	192.168.30.1

VLAN Table

VLAN	Name	Interface Assigned
10	Management	S1: VLAN 10 S2: VLAN 10
20	Sales	S1: F0/3
30	Operations	S2: F0/2
999	Parking_Lot	S1: F0/4-24, G0/1-2 S2: F0/3-24, G0/1-2
1000	Native	N/A

Aim:

Implement Inter-VLAN Routing

Theory:

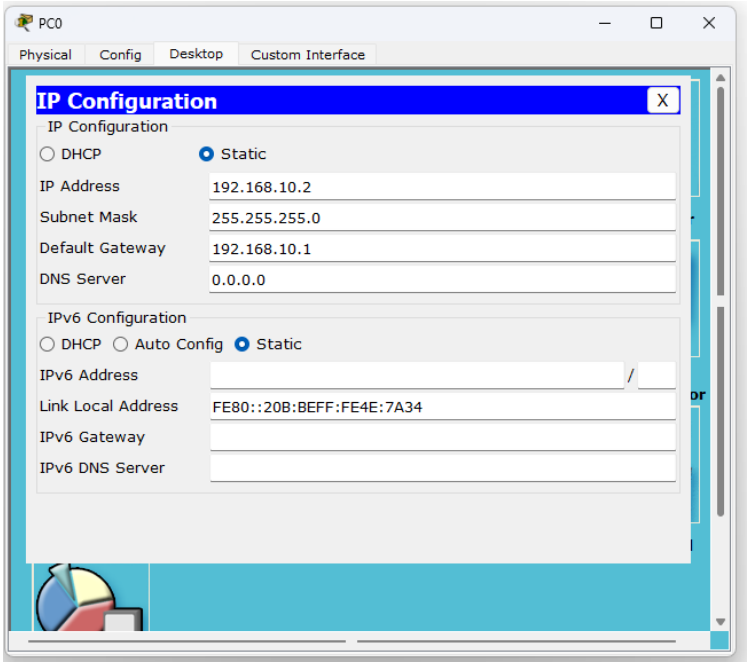
Modern switches use virtual local-area networks (VLANs) to improve network performance by separating large Layer 2 broadcast domains into smaller ones. VLANs can also be used as a security measure by separating sensitive data traffic from the rest of the network. In general, VLANs make it easier to design a network to support the goals of an organization. Communication between VLANs requires a device operating at Layer 3 of the OSI model. Adding an inter-VLAN router allows the organization to segregate and separate broadcast domains while simultaneously allowing them to communicate with each other.

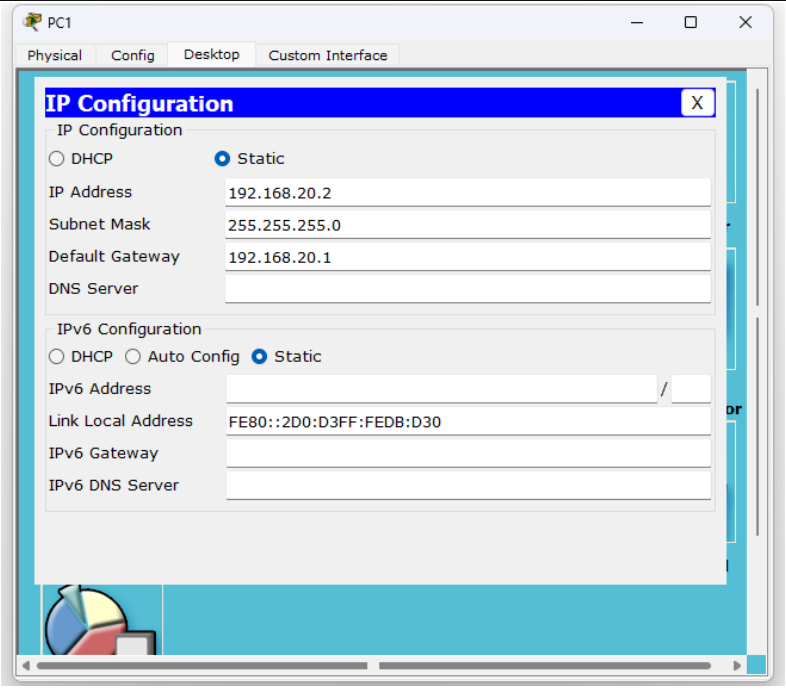
VLAN trunks are used to span VLANs across multiple devices. Trunks allow the traffic from multiple VLANs to travel over a single link, while keeping the VLAN identification and segmentation intact. A particular kind of inter-VLAN routing, called “Router-on-a-Stick”, uses a trunk from the router to the switch to enable all VLANs to pass to the router.

Procedure:

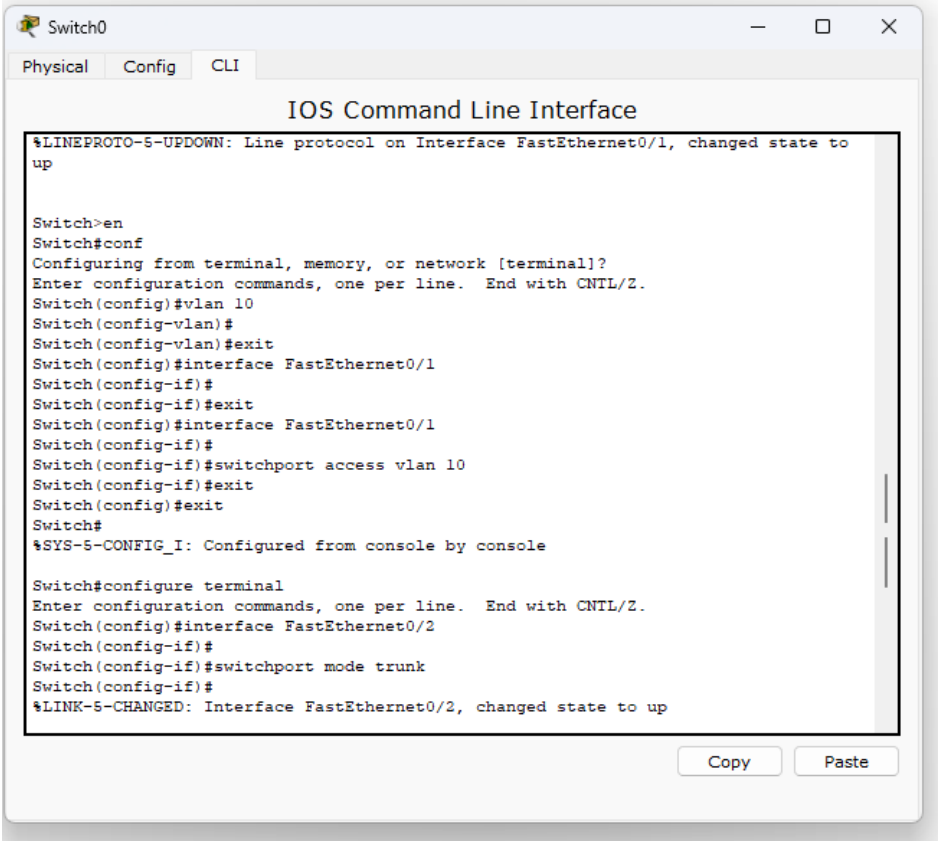
Step 1: Cable the network as shown in the topology.

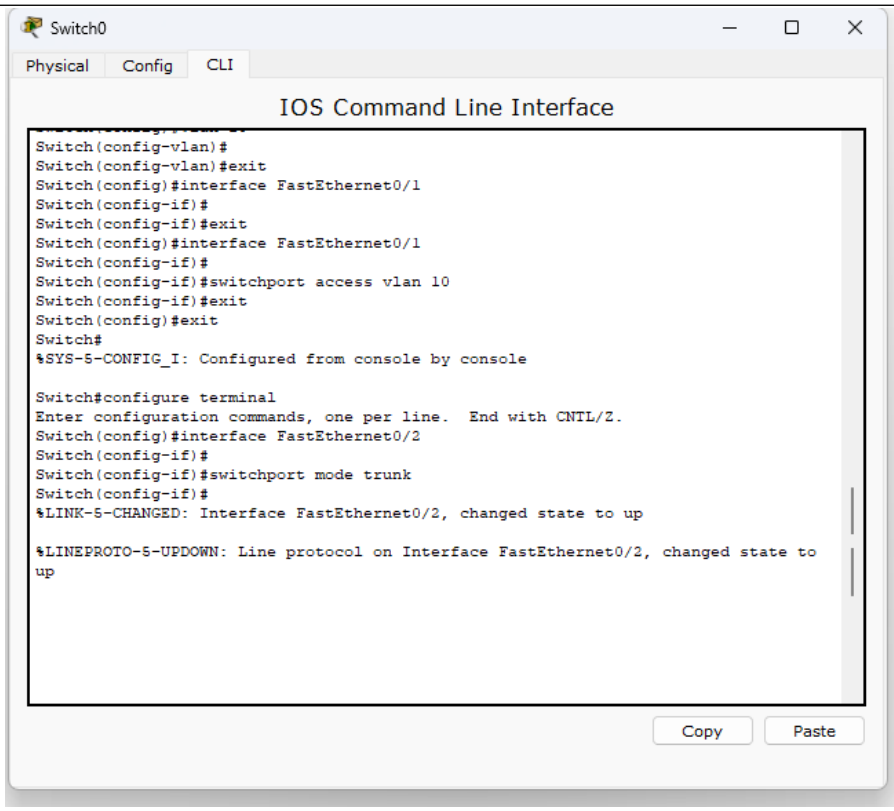
Step 2: Configure PC hosts as follows:

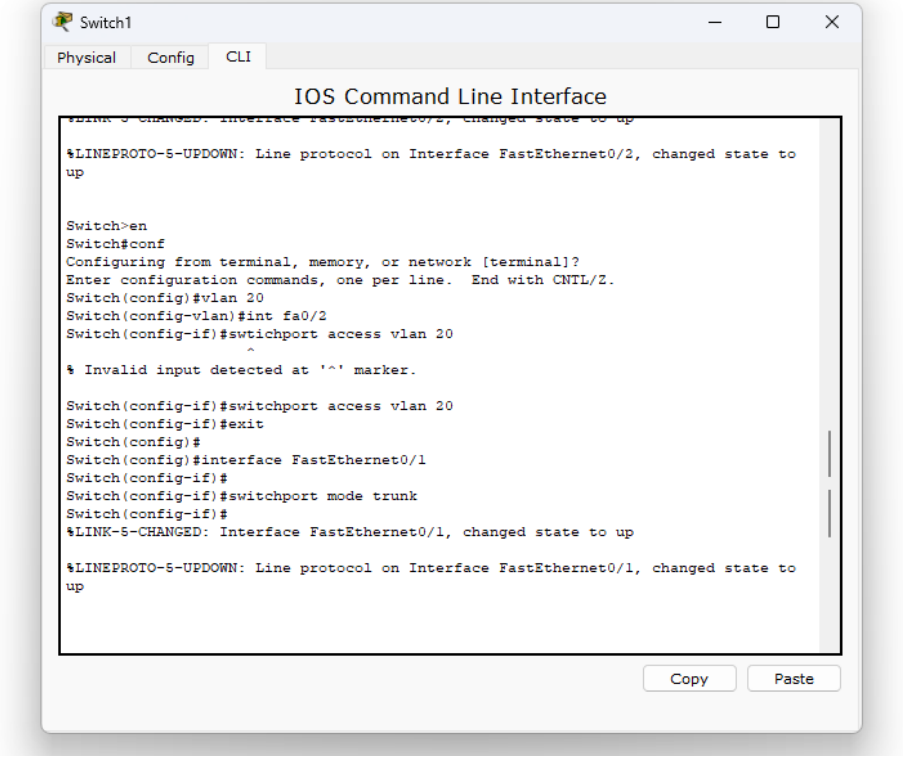




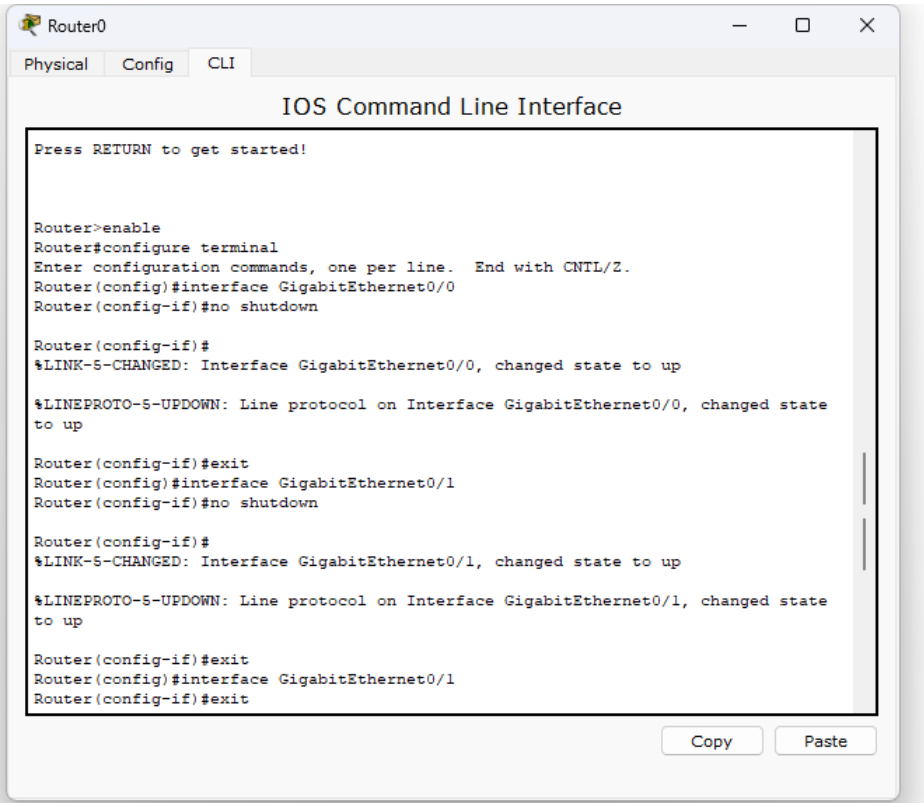
Step 3: Create VLANs on both switches

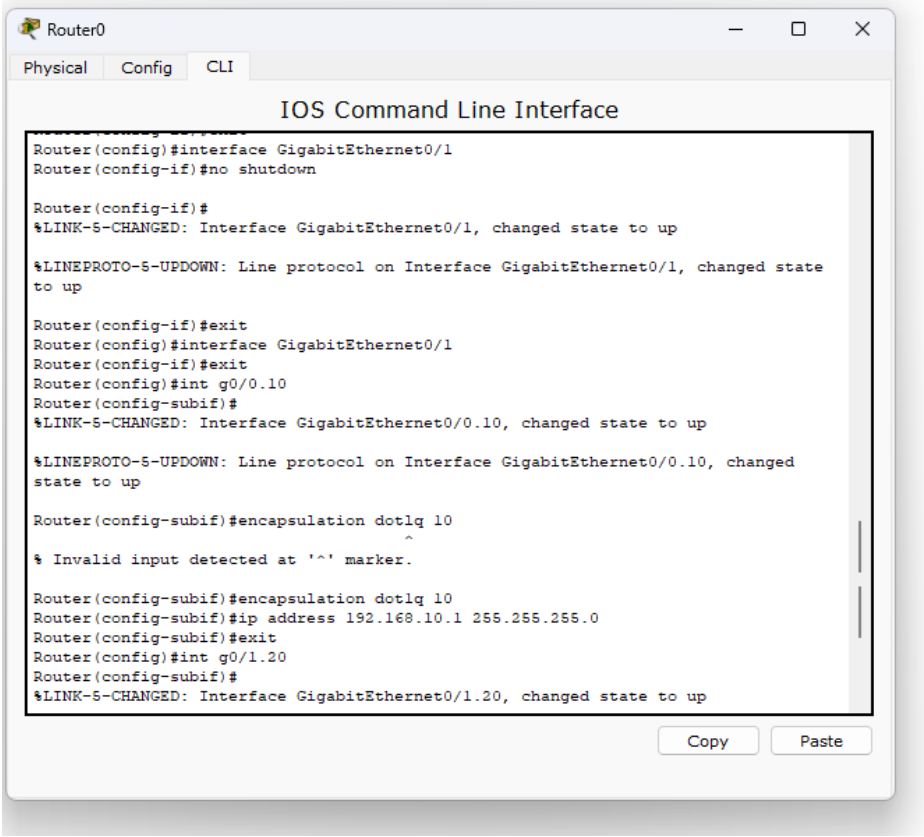



Switch0
Physical
Config
CLI
IOS Command Line Interface
Switch(config-vlan)#
Switch(config-vlan)#exit
Switch(config)#interface FastEthernet0/1
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#interface FastEthernet0/1
Switch(config-if)#
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface FastEthernet0/2
Switch(config-if)#
Switch(config-if)#switchport mode trunk
Switch(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to
up
Copy
Paste


Switch1
Physical
Config
CLI
IOS Command Line Interface
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to
up
Switch>en
Switch#conf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 20
Switch(config-vlan)#int fa0/2
Switch(config-if)#swtichport access vlan 20
^
% Invalid input detected at '^' marker.
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface FastEthernet0/1
Switch(config-if)#
Switch(config-if)#switchport mode trunk
Switch(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to
up
Copy
Paste

Step 4: Configure the Router.



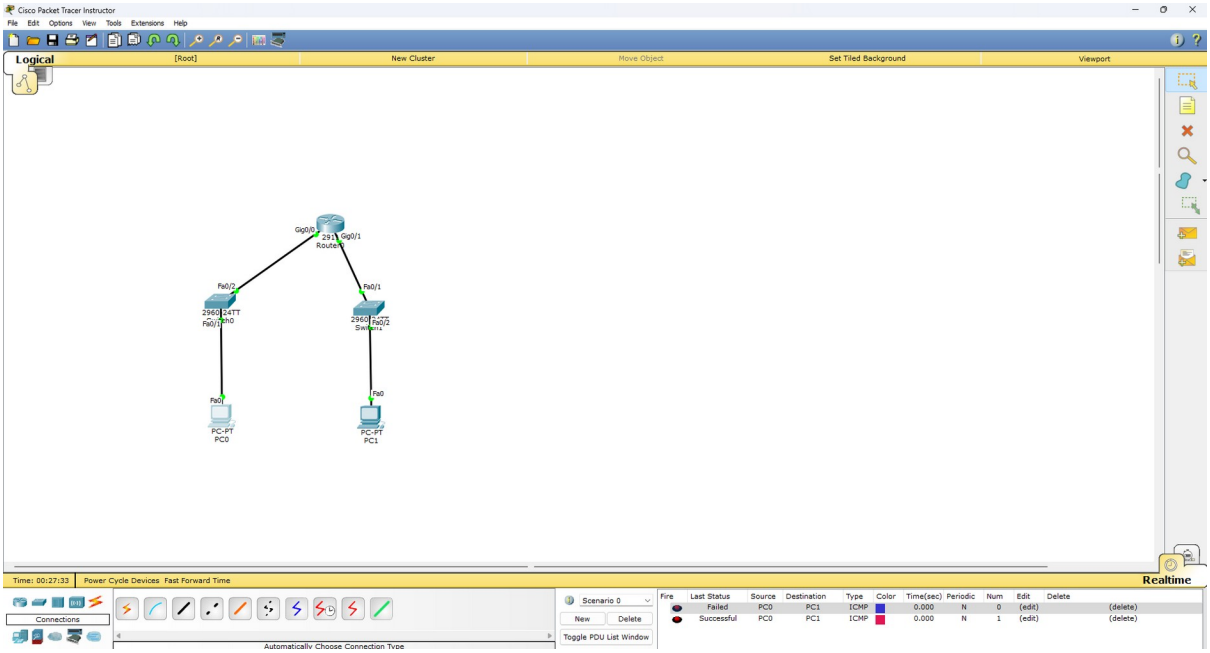


Step 8: Next, we make 3 pings to ensure the routing is working perfectly:

- Ping from PC-A to its default gateway.

- Ping from PC-A to PC-B
- Ping from PC-A to S2

All were executed successfully.



Conclusion: The above practical Implementing Inter-VLAN Routing has been completed Successfully.

PRACTICAL-06

(a): Implement Advanced STP Modifications and Mechanisms

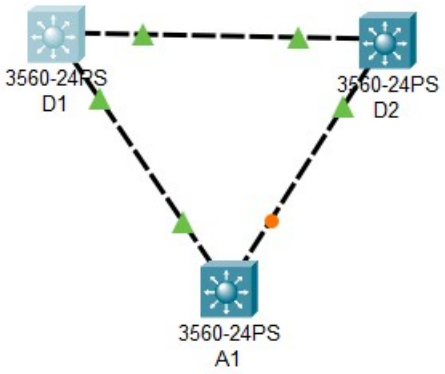
(b) Implement MST

Aim: To observe Spanning Tree Protocol (STP) and implement Rapid Spanning Tree Protocol (RSTP) using Cisco Packet Tracer.

Theory: STP prevents loops in switched networks by blocking redundant links. RSTP is an improved version of STP that provides faster network recovery when a link fails.

(a): Implement Advanced STP Modifications and Mechanisms

1. Topology



2. D1 Root Bridge Configuration

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#spanning-tree vlan 1 root primary
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
show spanning-tree vlan 1
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID     Priority    24577
              Address     00D0.BA91.B371
              This bridge is the root
              Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID   Priority    24577  (priority 24576 sys-id-ext 1)
              Address     00D0.BA91.B371
              Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Gi0/2            Altn BLK 19        128.26   P2p
Gi0/1            Desg LRN 4         128.25   P2p
```

3. D2 Secondary Root Bridge Configuration

```
Switch#show spanning-tree vlan 1
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID     Priority    24577
              Address     00D0.BA91.B371
              Cost        4
              Port        25(GigabitEthernet0/1)
              Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID   Priority    28673  (priority 28672 sys-id-ext 1)
              Address     0060.4711.E3C5
              Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Gi0/2            Desg FWD 19        128.26   P2p
Gi0/1            Root FWD 4         128.25   P2p
```

4. Enable Rapid PVST+ on D1

```
Switch#enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#spanning-tree mode rapid-pvst
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
show spanning-tree summary
Switch is in rapid-pvst mode
Root bridge for: default
Extended system ID           is enabled
Portfast Default             is disabled
PortFast BPDU Guard Default  is disabled
Portfast BPDU Filter Default is disabled
Loopguard Default            is disabled
EtherChannel misconfig guard is disabled
UplinkFast                   is disabled
BackboneFast                 is disabled
Configured Pathcost method used is short

Name                   Blocking Listening Learning Forwarding STP Active
---------------------- -------- --------- -------- ---------- ----------
VLAN0001                    0         0        0          2          2

---------------------- -------- --------- -------- ---------- ----------
1 vlans                     0         0        0          2          2
```

5. Enable Rapid PVST+ on D2

```
Switch#enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#spanning-tree mode rapid-pvst
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
show spanning-tree summary
Switch is in rapid-pvst mode
Root bridge for:
Extended system ID           is enabled
Portfast Default             is disabled
PortFast BPDU Guard Default  is disabled
Portfast BPDU Filter Default is disabled
Loopguard Default            is disabled
EtherChannel misconfig guard is disabled
UplinkFast                   is disabled
BackboneFast                 is disabled
Configured Pathcost method used is short

Name                   Blocking Listening Learning Forwarding STP Active
---------------------- -------- --------- -------- ---------- ----------
VLAN0001                      2         0        0          0          2

---------------------- -------- --------- -------- ---------- ----------
1 vlans                       2         0        0          0          2
```

6. Enable Rapid PVST+ on A1

```
Switch#enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#spanning-tree mode rapid-pvst
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
show spanning-tree summary
Switch is in rapid-pvst mode
Root bridge for:
Extended system ID           is enabled
Portfast Default             is disabled
PortFast BPDU Guard Default  is disabled
Portfast BPDU Filter Default is disabled
Loopguard Default            is disabled
EtherChannel misconfig guard is disabled
UplinkFast                   is disabled
BackboneFast                 is disabled
Configured Pathcost method used is short

Name                   Blocking Listening Learning Forwarding STP Active
---------------------- -------- --------- -------- ---------- ----------
VLAN0001                      1         0        0          1          2

---------------------- -------- --------- -------- ---------- ----------
1 vlans                       1         0        0          1          2
```

7. RSTP Verification on D1, D2 and A1

D1

```
Switch#show spanning-tree vlan 1
VLAN0001
  Spanning tree enabled protocol rstp
  Root ID     Priority    24577
              Address     00D0.BA91.B371
              This bridge is the root
              Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID   Priority    24577  (priority 24576 sys-id-ext 1)
              Address     00D0.BA91.B371
              Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Gi0/2            Desg FWD 19        128.26   P2p
Gi0/1            Desg FWD 4         128.25   P2p
```

D2

```
Switch#show spanning-tree vlan 1
VLAN0001
  Spanning tree enabled protocol rstp
  Root ID     Priority    24577
              Address     00D0.BA91.B371
              Cost        4
              Port        25(GigabitEthernet0/1)
              Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID   Priority    28673  (priority 28672 sys-id-ext 1)
              Address     0060.4711.E3C5
              Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------
Gi0/2            Desg FWD 19        128.26   P2p
Gi0/1            Root FWD 4         128.25   P2p
```

A1

```
Switch#show spanning-tree vlan 1
VLAN0001
  Spanning tree enabled protocol rstp
  Root ID     Priority    24577
              Address     00D0.BA91.B371
              Cost        19
              Port        1(FastEthernet0/1)
              Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID   Priority    32769  (priority 32768 sys-id-ext 1)
              Address     0001.642C.E14D
              Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- ----------------------------
Fa0/1            Root FWD 19        128.1    P2p
Fa0/2            Altn BLK 19        128.2    P2p
```

(b) Implement MST

In Cisco Packet Tracer, the available switch models did not support the MST command. Therefore, PVST was configured and STP operation was observed using the available commands.

Result: STP and RSTP were configured and observed in Cisco Packet Tracer. Root bridge selection and blocking of redundant links were verified. MST configuration was not supported by the switch models used.

Conclusion: STP helps prevent switching loops by blocking redundant paths. RSTP provides faster convergence, while MST could not be implemented due to Packet Tracer switch model limitations.

PRACTICAL-07

Packet Tracer – Configure EtherChannel

1. Aim

a) Implement EtherChannel.
b) Tune and optimize EtherChannel operations.

2. Objectives

1. Configure basic switch settings.
2. Configure an EtherChannel using Cisco Port Aggregation Protocol (PAgP).
3. Configure an 802.3ad Link Aggregation Control Protocol (LACP) EtherChannel.
4. Configure and verify a redundant EtherChannel link.

3. Theory

EtherChannel, also known as link aggregation, combines multiple physical Ethernet links into one logical link. It increases available bandwidth and provides redundancy. In this practical, three switches (S1, S2 and S3) are connected using three EtherChannels. PAgP is used for Port-channel 1, while LACP is used for Port-channel 2 and Port-channel 3. Spanning Tree Protocol (STP) prevents switching loops by selecting forwarding and blocking paths.

4. Port-channel Table

Channel Group	Physical Connections	Protocol	Switches
1	Fa0/21 and Fa0/22	PAgP (desirable)	S1 ↔ S3
2	Gi0/1 and Gi0/2	LACP (active)	S1 ↔ S2
3	Fa0/23 and Fa0/24	LACP (S2 passive, S3 active)	S2 ↔ S3

Note: The interface numbers above are the ports used in this Packet Tracer activity. Both ends of each EtherChannel must use compatible protocol modes and matching trunk settings.

5. Network Topology

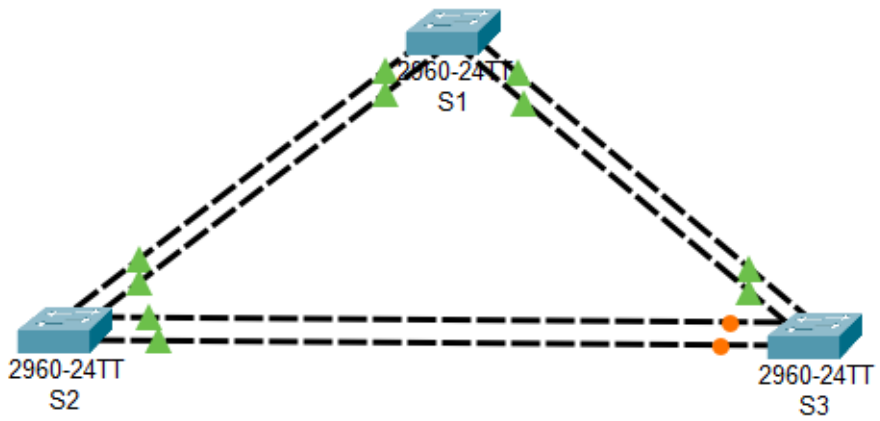


Figure 1: Final Packet Tracer topology showing S1, S2 and S3 connected through three EtherChannels.

6. Procedure and CLI Configuration

Open each switch, select the CLI tab, press Enter if prompted, and enter the commands one line at a time. Configure both ends of every EtherChannel.

6.1 Basic hostname configuration

```
S1:
enable
configure terminal
hostname S1
end

S2:
enable
configure terminal
hostname S2
end

S3:
enable
configure terminal
hostname S3
end
```

6.2 S1 – Port-channel 1 (PAgP) and Port-channel 2 (LACP)

```
enable
configure terminal
interface range fastEthernet 0/21 - 22
shutdown
switchport mode trunk
channel-group 1 mode desirable
no shutdown
exit
interface port-channel 1
switchport mode trunk
exit
interface range gigabitEthernet 0/1 - 2
shutdown
switchport mode trunk
channel-group 2 mode active
no shutdown
exit
interface port-channel 2
switchport mode trunk
end
```

6.3 S3 – Port-channel 1 (PAgP) and Port-channel 3 (LACP)

```
enable
configure terminal
interface range fastEthernet 0/21 - 22
shutdown
switchport mode trunk
channel-group 1 mode desirable
no shutdown
exit
```

```
interface port-channel 1
switchport mode trunk
exit
interface range fastEthernet 0/23 - 24
shutdown
switchport mode trunk
channel-group 3 mode active
no shutdown
exit
interface port-channel 3
switchport mode trunk
end
```

6.4 S2 – Port-channel 2 (LACP) and Port-channel 3 (LACP)

```
enable
configure terminal
interface range gigabitEthernet 0/1 - 2
shutdown
switchport mode trunk
channel-group 2 mode active
no shutdown
exit
interface port-channel 2
switchport mode trunk
exit
interface range fastEthernet 0/23 - 24
shutdown
switchport mode trunk
channel-group 3 mode passive
no shutdown
exit
interface port-channel 3
switchport mode trunk
end
```

6.5 Set S1 as the STP root bridge

```
enable
configure terminal
spanning-tree vlan 1 root primary
end
```

6.6 Verification commands (run on each switch)

```
show etherchannel summary
show interfaces trunk
show spanning-tree
show spanning-tree active
```

6.7 Save the configuration (run on each switch)

```
enable
copy running-config startup-config
(Press Enter to accept the default destination filename.)
```

7. Expected Verification Results

The output of `show etherchannel summary` should show the expected groups on each switch. The status `(SU)` indicates a Layer 2 port-channel in use, and `(P)` beside a physical interface indicates that the interface has joined the bundle.

Switch	Port-channel	Protocol	Expected member ports
S1	Po1	PAgP	Fa0/21(P), Fa0/22(P)
S1	Po2	LACP	Gi0/1(P), Gi0/2(P)
S2	Po2	LACP	Gi0/1(P), Gi0/2(P)
S2	Po3	LACP	Fa0/23(P), Fa0/24(P)
S3	Po1	PAgP	Fa0/21(P), Fa0/22(P)
S3	Po3	LACP	Fa0/23(P), Fa0/24(P)

Spanning Tree may show one redundant logical link in the alternate/blocking state. This is expected in a redundant triangular topology and prevents a Layer 2 loop.

8. Output Screenshots

The following screenshots were supplied from the Packet Tracer practical and are included as evidence of the topology, EtherChannel summary, trunk status and spanning-tree verification.

Network topology

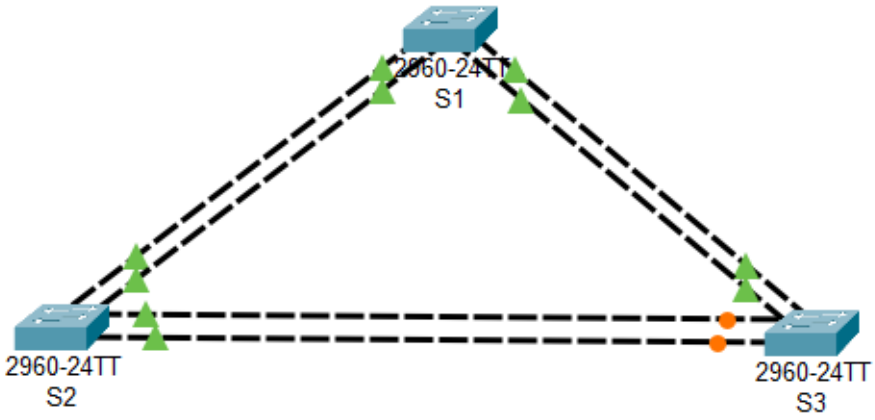


Figure 2: Final topology output from Cisco Packet Tracer. The orange indicators show the redundant path blocked by Spanning Tree.

S1 – EtherChannel summary and trunk verification

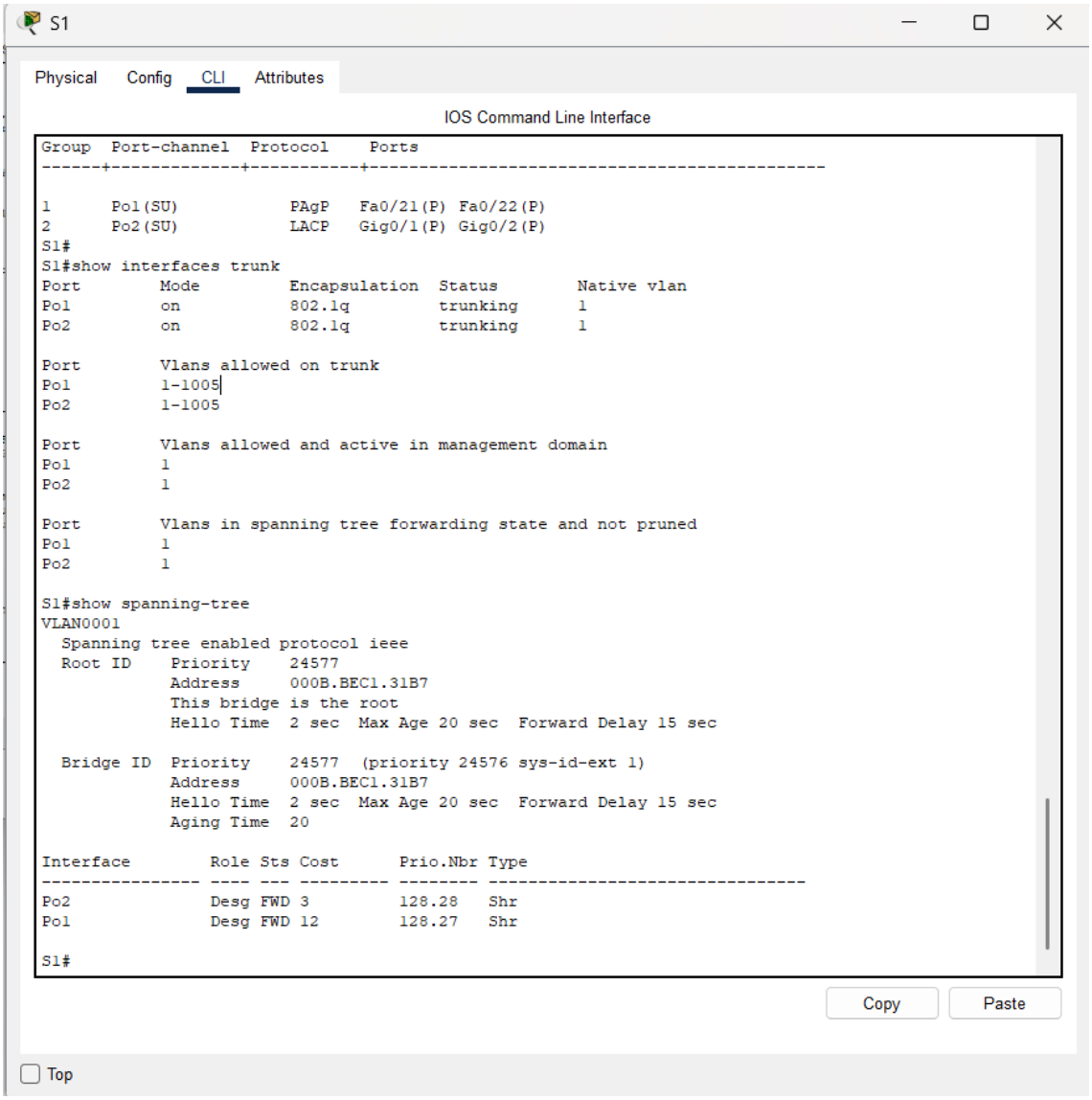


Figure 3: S1 Port-channel 1 and Port-channel 2 verification.

S1 – LACP Port-channel 2 configuration

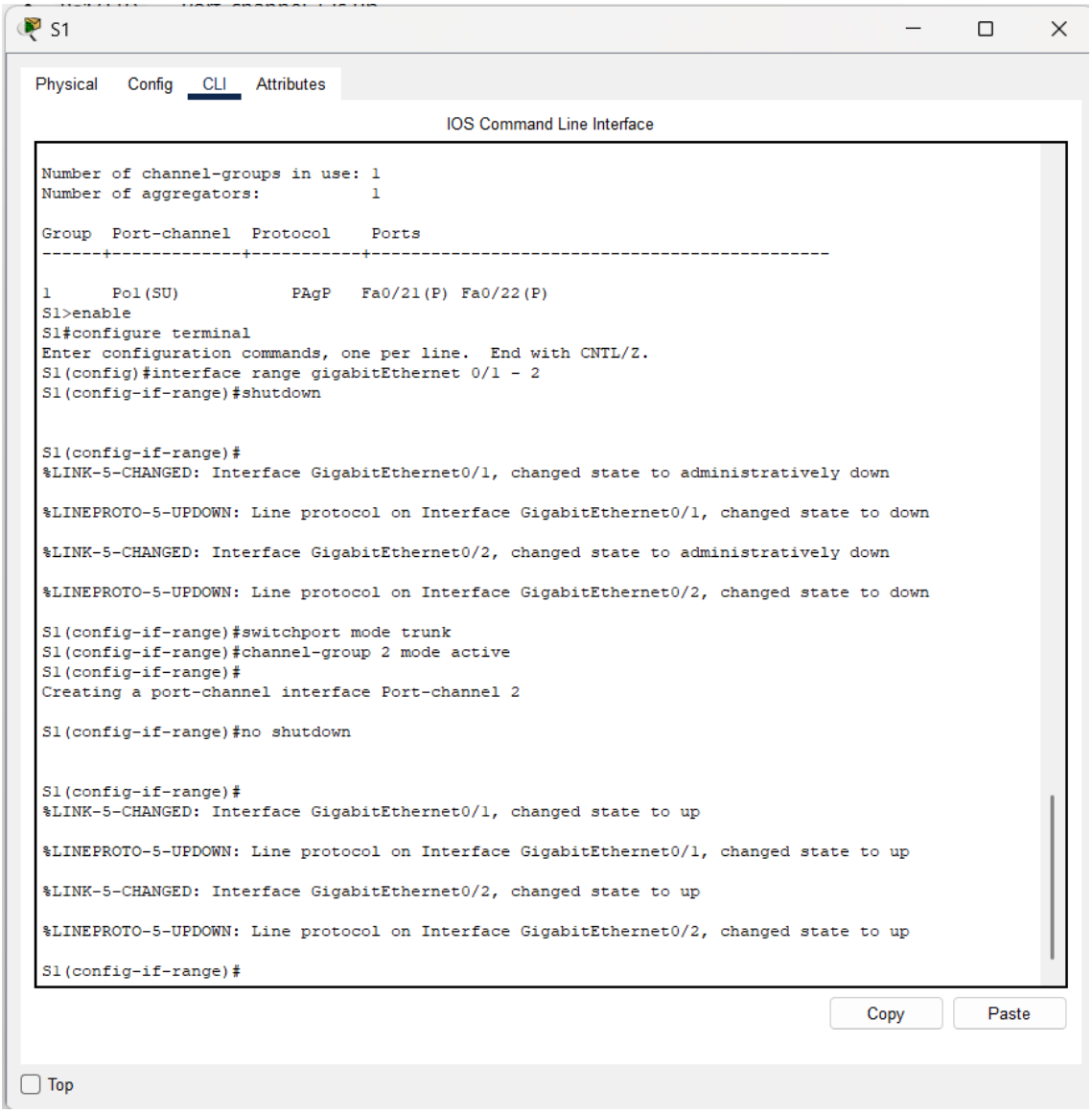


Figure 4: S1 Port-channel 2 configuration and status.

S1 – Spanning-tree root

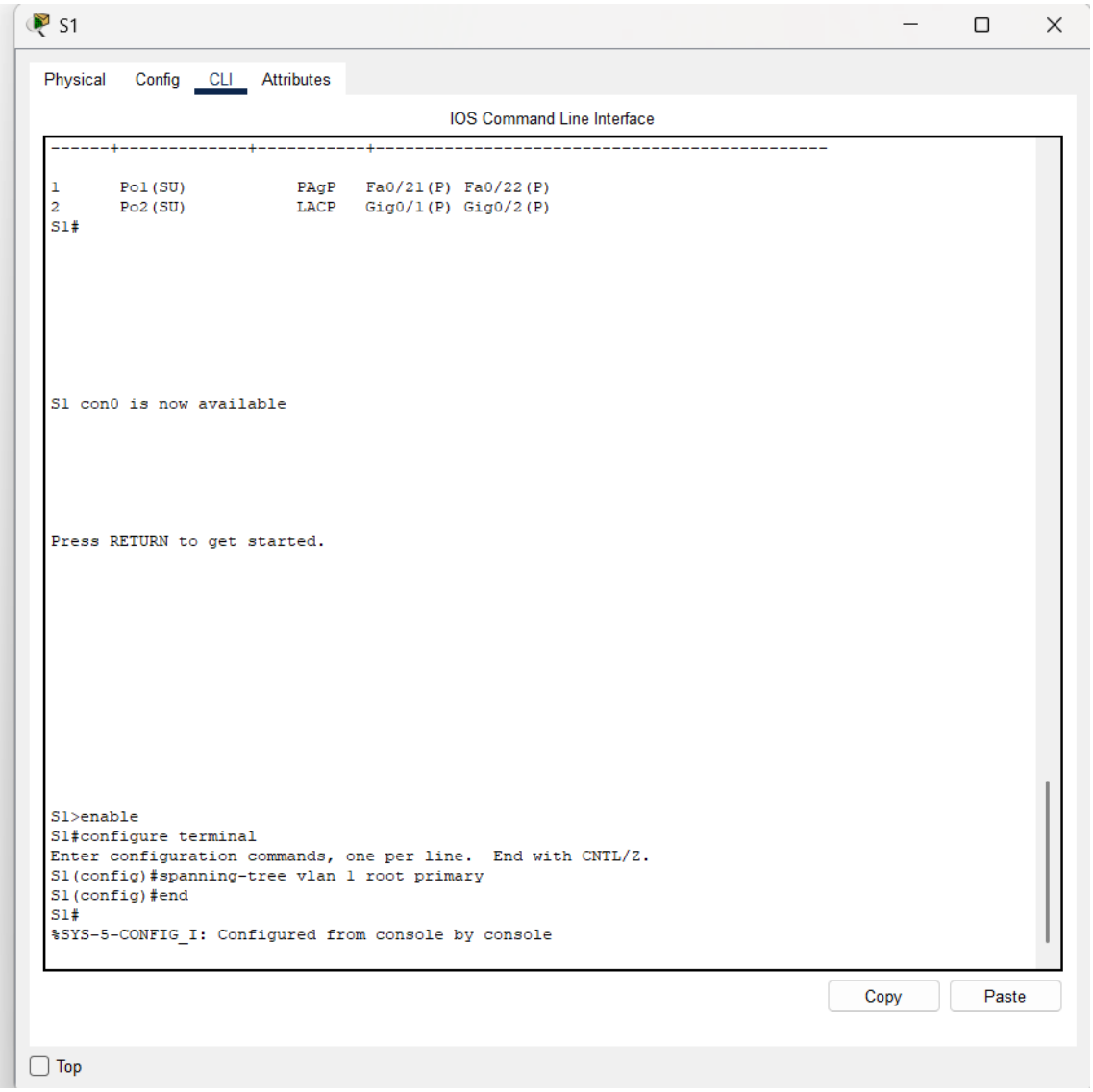


Figure 5: S1 configured as the root bridge.

S2 – EtherChannel summary and trunk verification

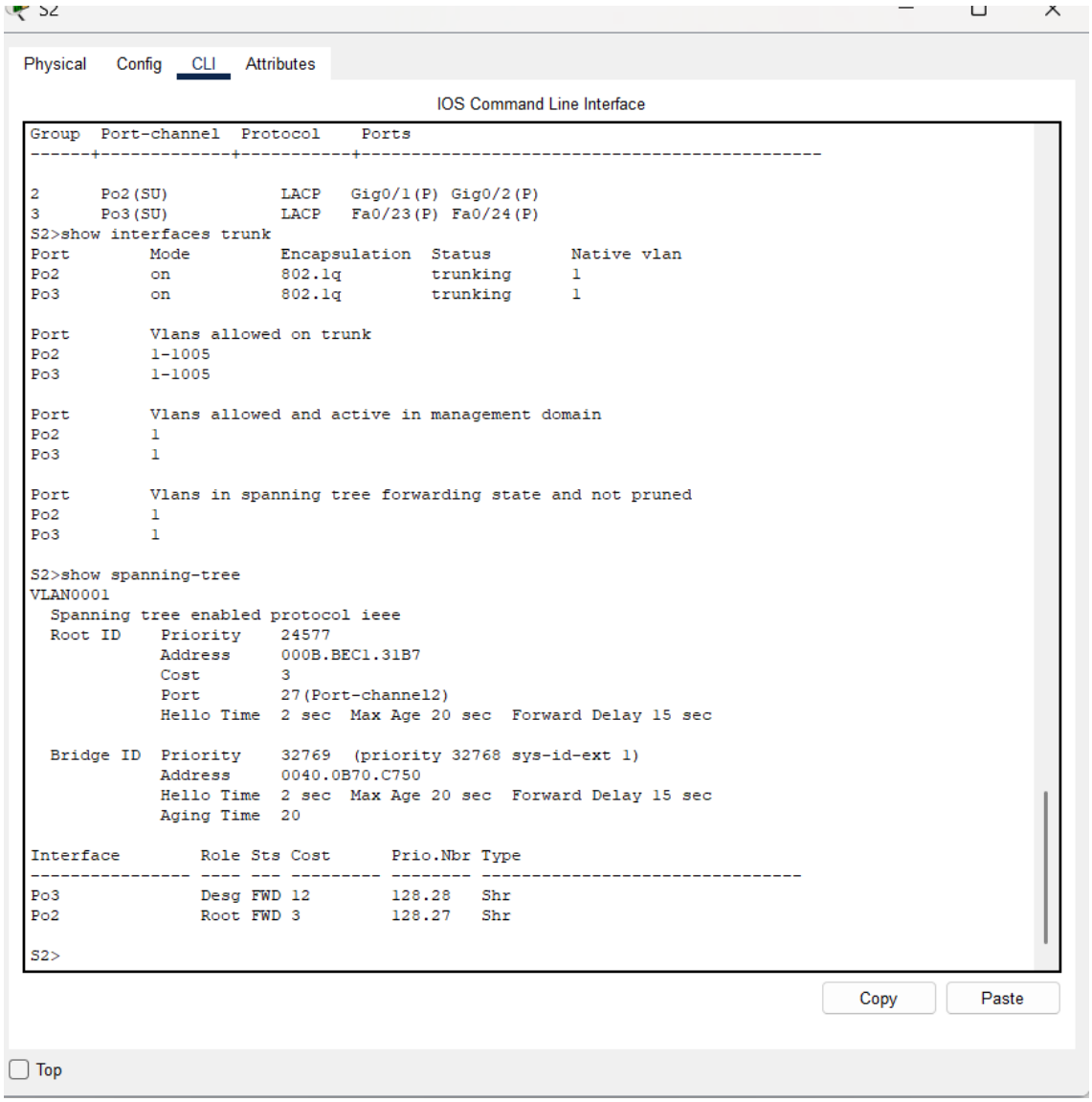


Figure 6: S2 Port-channel 2 and Port-channel 3 verification.

S2 – Port-channel 2 configuration

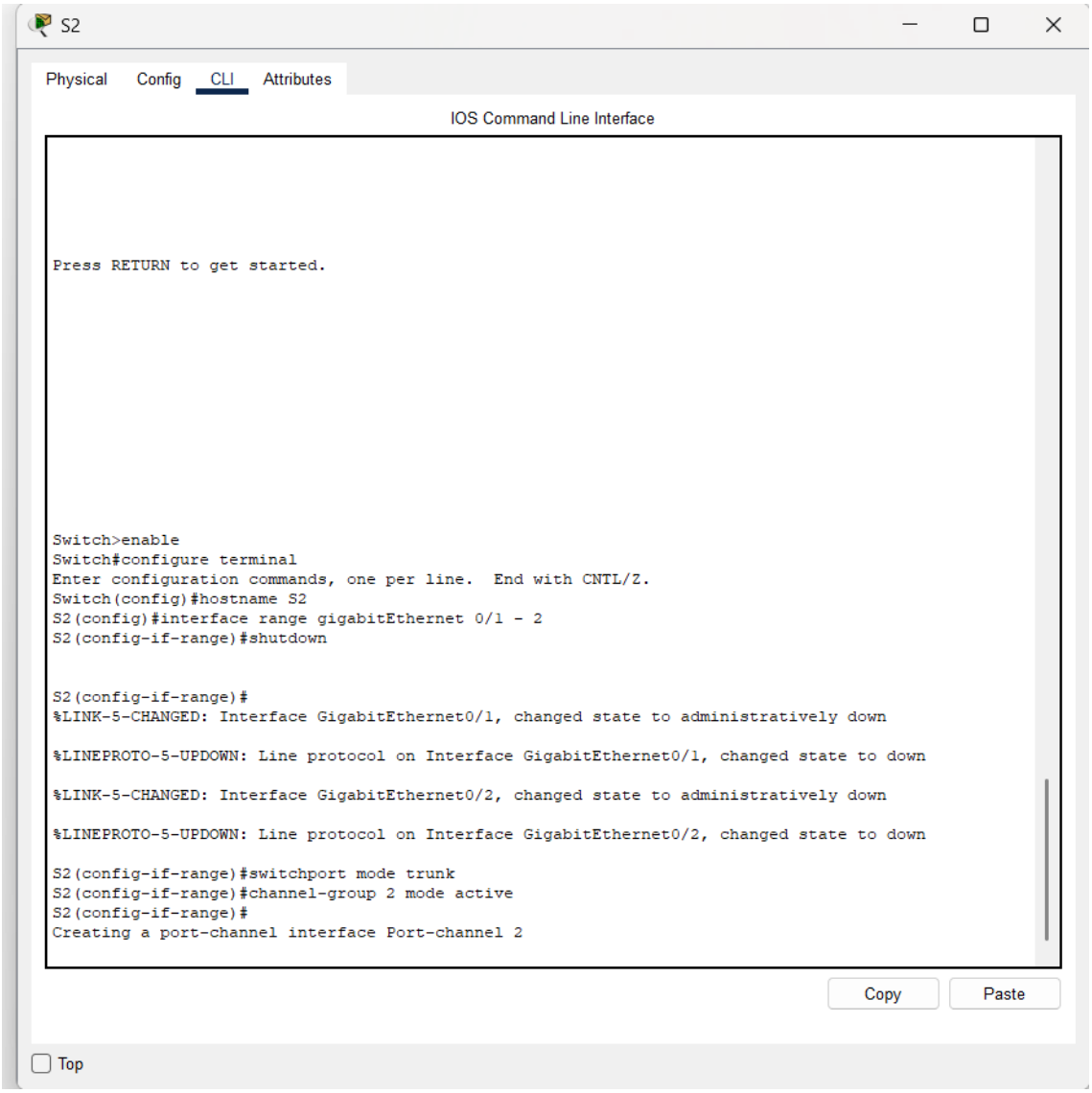


Figure 7: S2 LACP Port-channel 2.

S2 – Port-channel 3 configuration

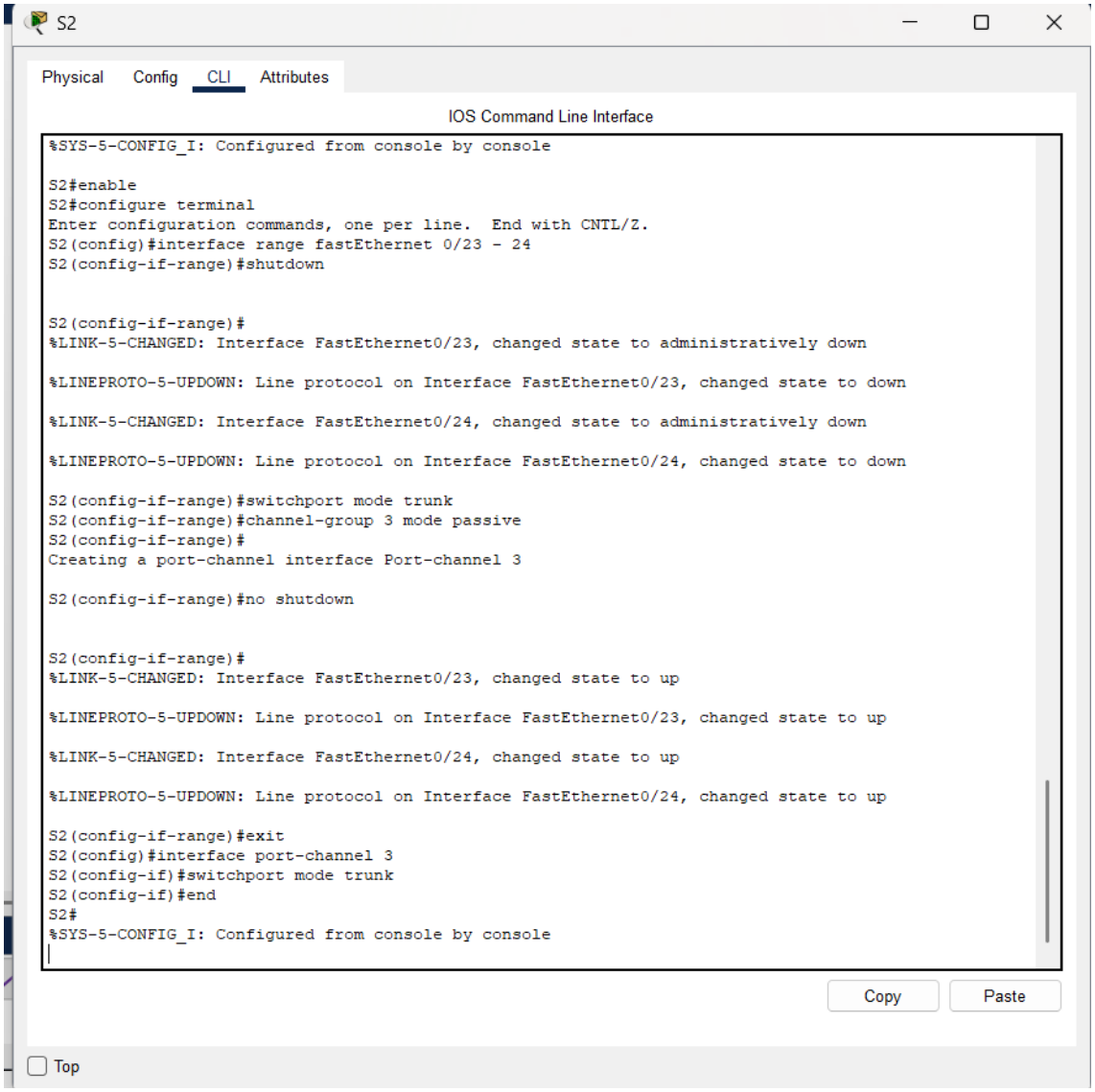


Figure 8: S2 LACP Port-channel 3.

S3 – EtherChannel summary and trunk verification

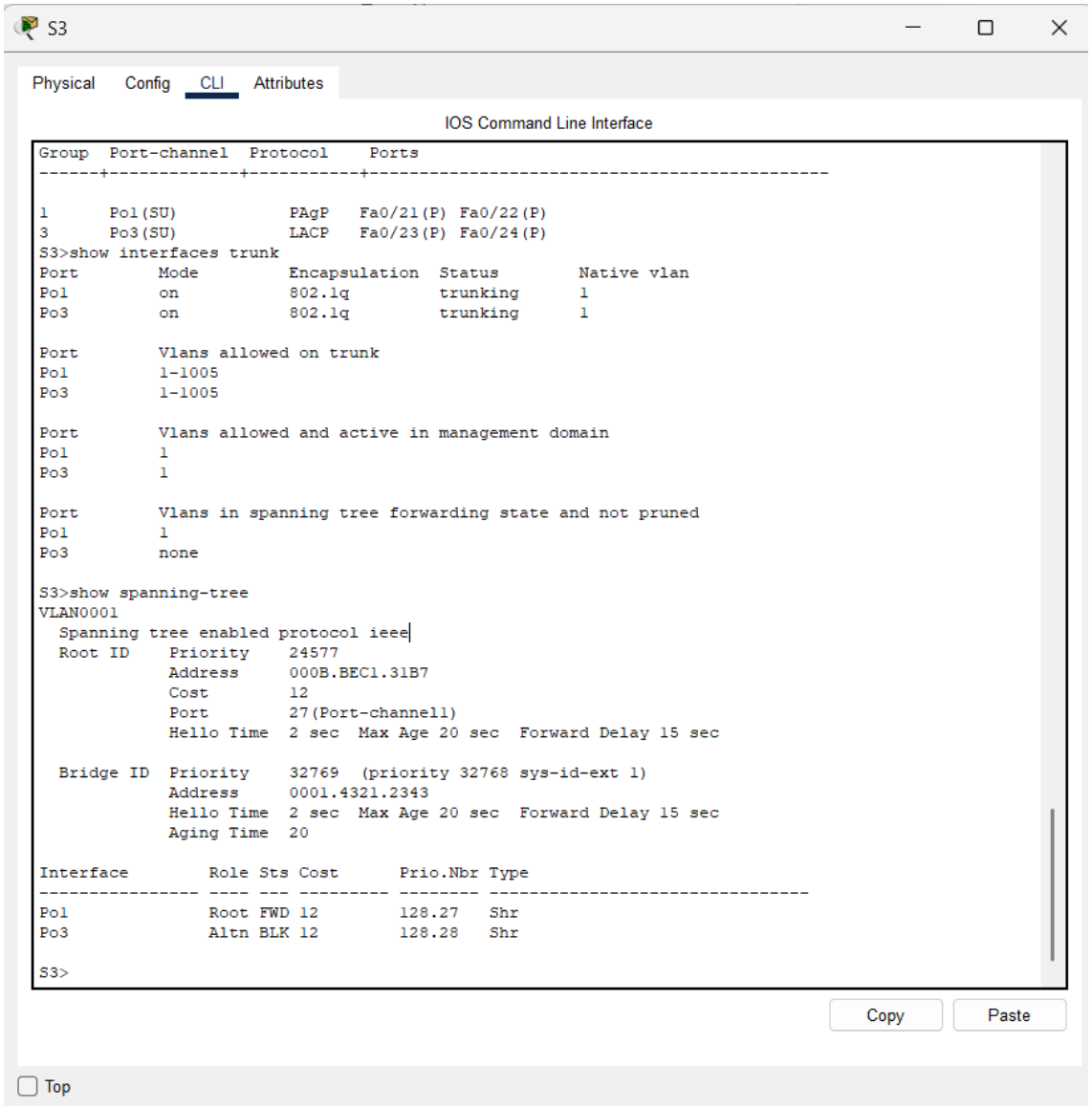


Figure 9: S3 Port-channel 1 and Port-channel 3 verification.

S3 – Port-channel 3 configuration

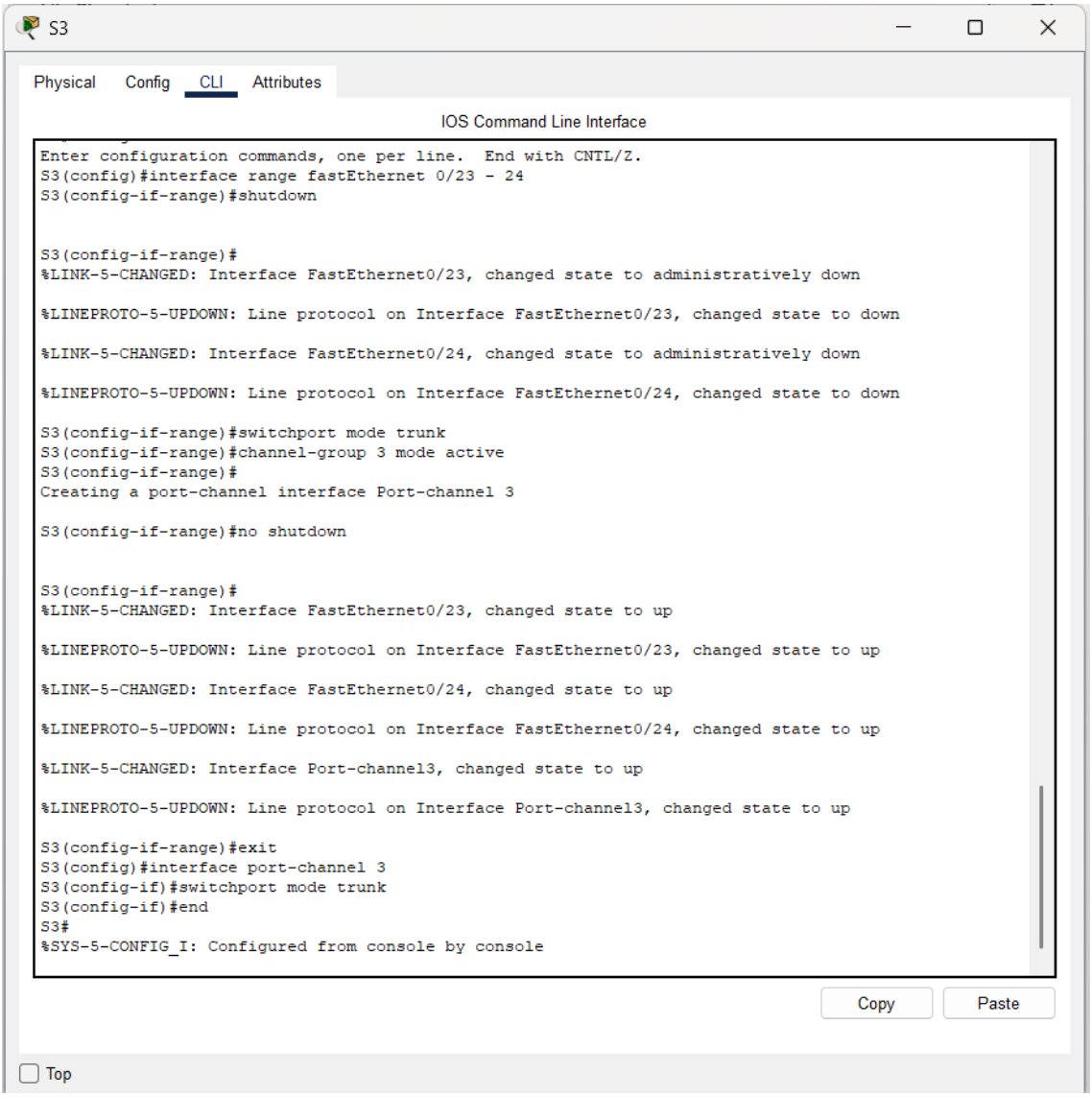


Figure 10: S3 LACP Port-channel 3.

S3 – Final spanning-tree state

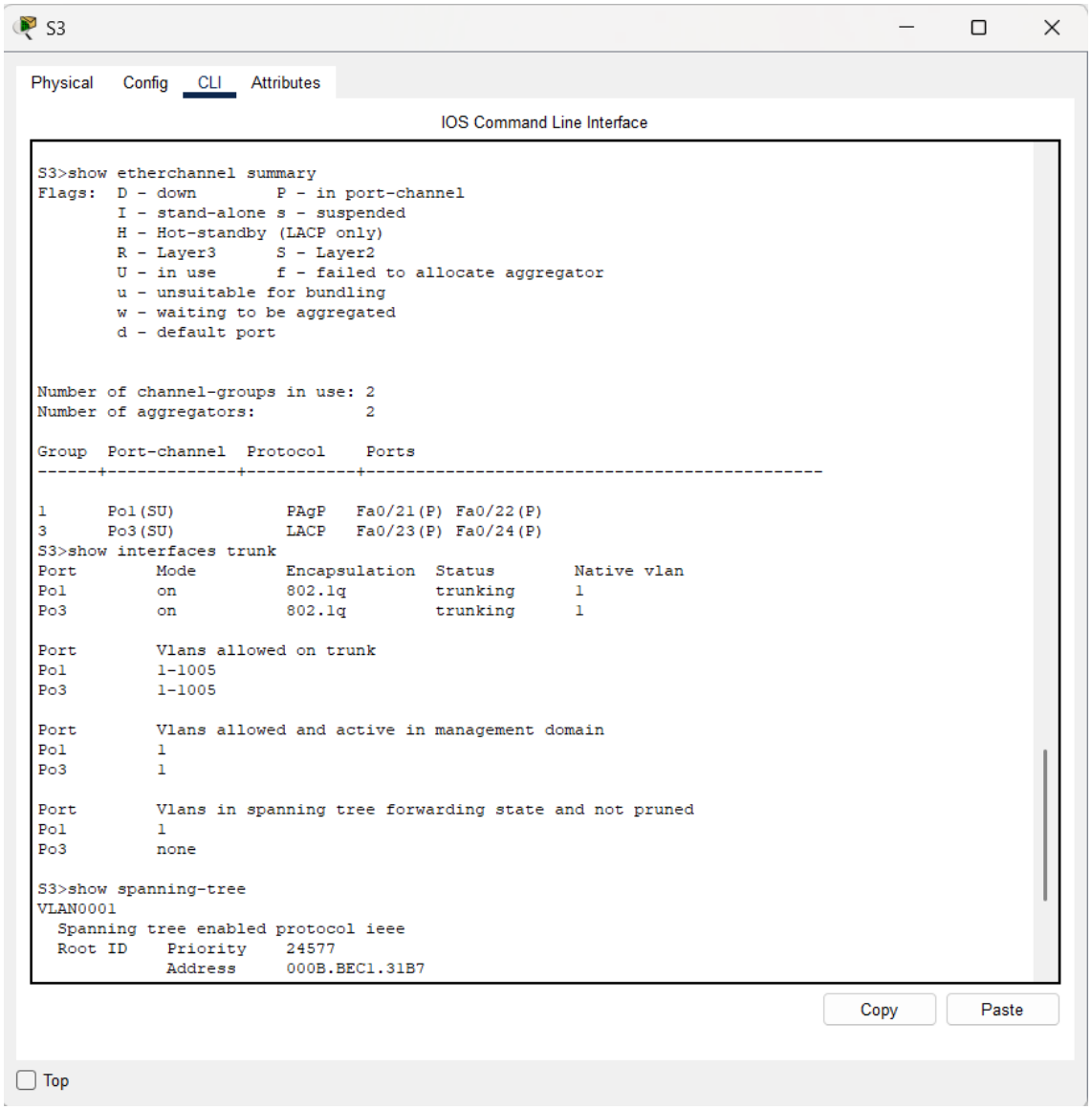


Figure 11: S3 spanning-tree output showing forwarding and alternate path states.

9. Result

EtherChannel was configured successfully among switches S1, S2 and S3. Port-channel 1 was configured using PAgP, while Port-channel 2 and Port-channel 3 were configured using LACP. The EtherChannel summary, trunk interfaces and spanning-tree state were verified.

10. Conclusion

The practical demonstrated link aggregation using PAgP and LACP. Multiple physical links were combined into logical port-channels to provide redundancy and increased aggregate bandwidth. Spanning Tree Protocol maintained a loop-free topology by selecting forwarding and blocking paths.

PRACTICAL-08

Aim:

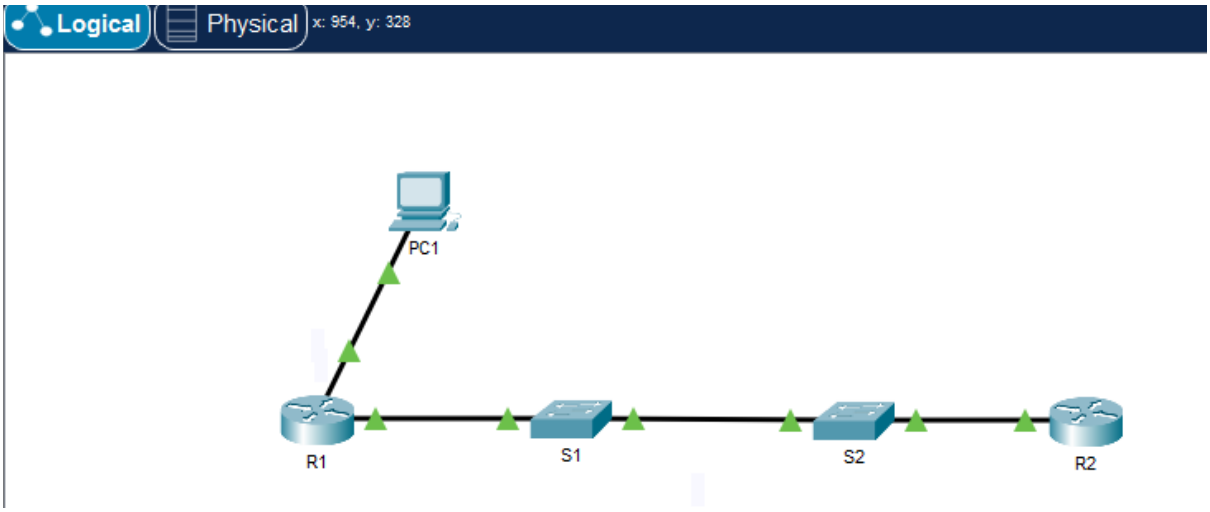
A) Configure Single-Area OSPFv2

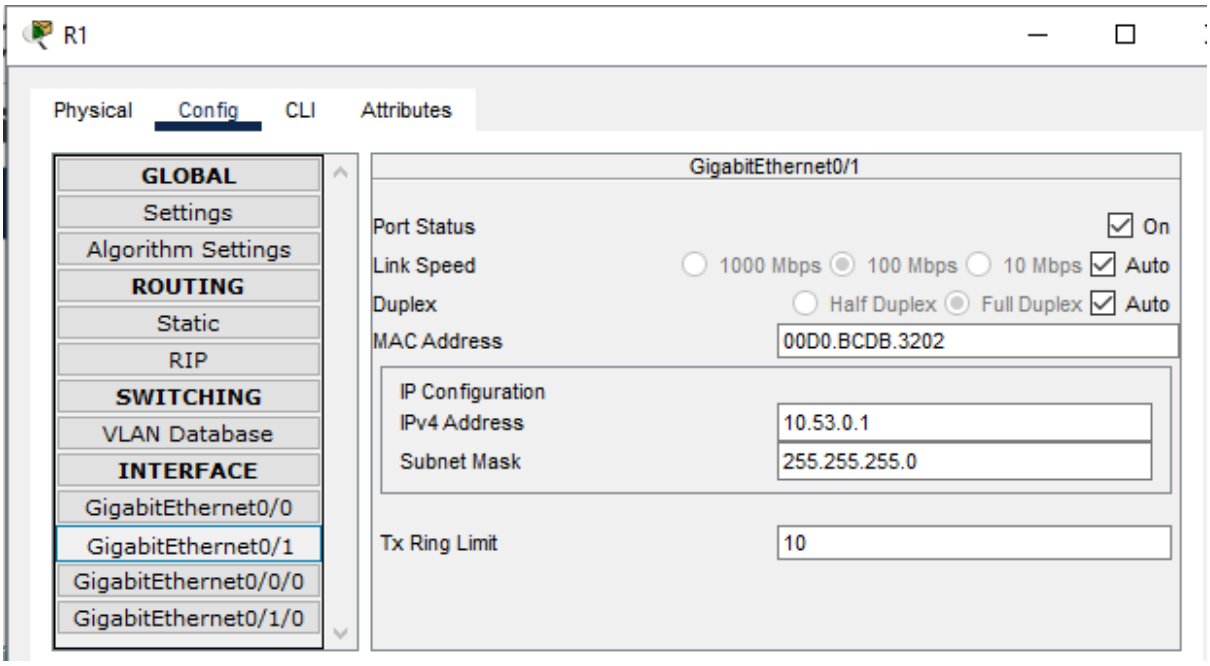
Theory:

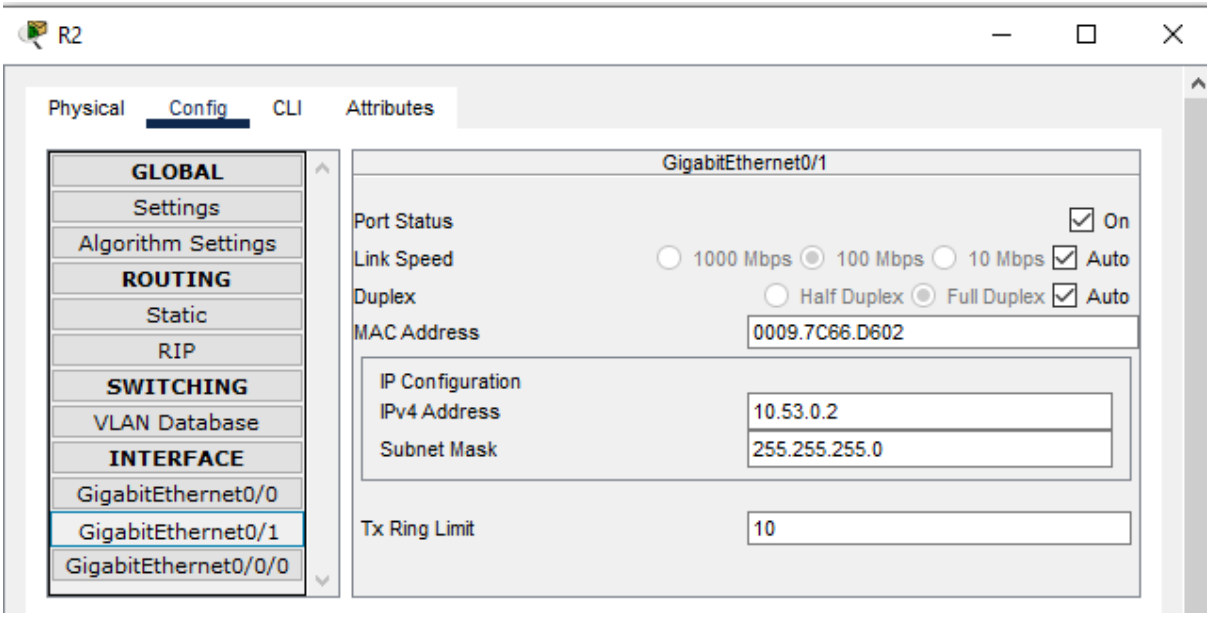
The OSPF (Open Shortest Path First) protocol is one of a family of IP Routing protocols for the Internet, used to distribute IP routing information throughout a single Autonomous System (AS) in an IP network.

A single area OSPF design is a network in which all router in same area

Topology -







R1 –

```
R1(config)#interface g0/1

R1(config-if)#ip address 10.53.0.1 255.255.255.0
R1(config-if)#no shut

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

R1(config-if)#exit
```

```
R1(config)#interface loopback 1

R1(config-if)#
%LINK-3-UPDOWN: Interface Loopback1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback1, changed state to up

R1(config-if)#ip address 172.16.1.1 255.255.255.0
R1(config-if)#exit
```

```
R1(config)#router ospf 56
R1(config-router)#do sh ip ospf
 Routing Process "ospf 56" with ID 1.1.1.1
 Supports only single TOS(TOS0) routes
 Supports opaque LSA
 SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
 Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
 Number of external LSA 0. Checksum Sum 0x000000
 Number of opaque AS LSA 0. Checksum Sum 0x000000
 Number of DCbitless external and opaque AS LSA 0
 Number of DoNotAge external and opaque AS LSA 0
 Number of areas in this router is 1. 1 normal 0 stub 0 nssa
 External flood list length 0
    Area BACKBONE(0)
        Number of interfaces in this area is 1
        Area has no authentication
        SPF algorithm executed 4 times
        Area ranges are
        Number of LSA 3. Checksum Sum 0x013a80
        Number of opaque link LSA 0. Checksum Sum 0x000000
        Number of DCbitless LSA 0
        Number of indication LSA 0
        Number of DoNotAge LSA 0
        Flood list length 0
 --More-- |
```

```
R1(config-router)#do show ip route connected
 C   10.53.0.0/24  is directly connected, GigabitEthernet0/1
 C   172.16.1.0/24  is directly connected, Loopback1

R1(config-router)#network 10.53.0.0 0.0.0.255 area 0
R1(config-router)#
00:52:06: %OSPF-5-ADJCHG: Process 56, Nbr 2.2.2.2 on GigabitEthernet0/1 from LOADING to
FULL, Loading Done

R1(config-router)#end
R1#
```

```
R1#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
2.2.2.2           1   FULL/BDR        00:00:38    10.53.0.2       GigabitEthernet0/1
R1#show ip route ospf
     192.168.1.0/32 is subnetted, 1 subnets
O       192.168.1.1 [110/2] via 10.53.0.2, 00:04:35, GigabitEthernet0/1
```

R2 —

```
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#interface g0/1
R2(config-if)#ip address 10.53.0.2 255.255.255.0
R2(config-if)#no shut

R2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

R2(config-if)#exit
```

```
R2(config)#interface loopback 1

R2(config-if)#
%LINK-3-UPDOWN: Interface Loopback1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback1, changed state to up

R2(config-if)#ip address 192.168.1.1 255.255.255.0
R2(config-if)#exit
```

```
R2(config)#router ospf 56
R2(config-router)#do sh ip ospf
 Routing Process "ospf 56" with ID 2.2.2.2
 Supports only single TOS(TOS0) routes
 Supports opaque LSA
 SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
 Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
 Number of external LSA 0. Checksum Sum 0x000000
 Number of opaque AS LSA 0. Checksum Sum 0x000000
 Number of DCbitless external and opaque AS LSA 0
 Number of DoNotAge external and opaque AS LSA 0
 Number of areas in this router is 1. 1 normal 0 stub 0 nssa
 External flood list length 0
    Area BACKBONE(0)
        Number of interfaces in this area is 2
        Area has no authentication
        SPF algorithm executed 5 times
        Area ranges are
        Number of LSA 3. Checksum Sum 0x0103e6
        Number of opaque link LSA 0. Checksum Sum 0x000000
        Number of DCbitless LSA 0
        Number of indication LSA 0
        Number of DoNotAge LSA 0
        Flood list length 0
 --More--
```

```
R2(config)#router ospf 56
R2(config-router)#router-id 2.2.2.2
R2(config-router)#do show ip route connected
 C    10.53.0.0/24  is directly connected, GigabitEthernet0/1
 C    192.168.1.0/24  is directly connected, Loopback1

R2(config-router)#netwrok 10.53.0.0 0.0.0.255 area 0
                  ^
% Invalid input detected at '^' marker.

R2(config-router)#network 10.53.0.0 0.0.0.255 area 0
R2(config-router)#
00:52:00: %OSPF-5-ADJCHG: Process 56, Nbr 1.1.1.1 on GigabitEthernet0/1 from LOADING to
FULL, Loading Done

R2(config-router)#network 192.168.1.0 0.0.0.255 area 0
R2(config-router)#end
```

```
R2#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           1   FULL/DR         00:00:37    10.53.0.1       GigabitEthernet0/1
```

```
C:\>
C:\>
C:\>ping 10.53.0.1

Pinging 10.53.0.1 with 32 bytes of data:

Reply from 10.53.0.1: bytes=32 time=12ms TTL=128
Reply from 10.53.0.1: bytes=32 time=19ms TTL=128
Reply from 10.53.0.1: bytes=32 time=5ms TTL=128
Reply from 10.53.0.1: bytes=32 time<1ms TTL=128

Ping statistics for 10.53.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 19ms, Average = 9ms

C:\>
```

Conclusion:

The Above Practical Configure Single-Area OSPFv2 is configured Successfully.

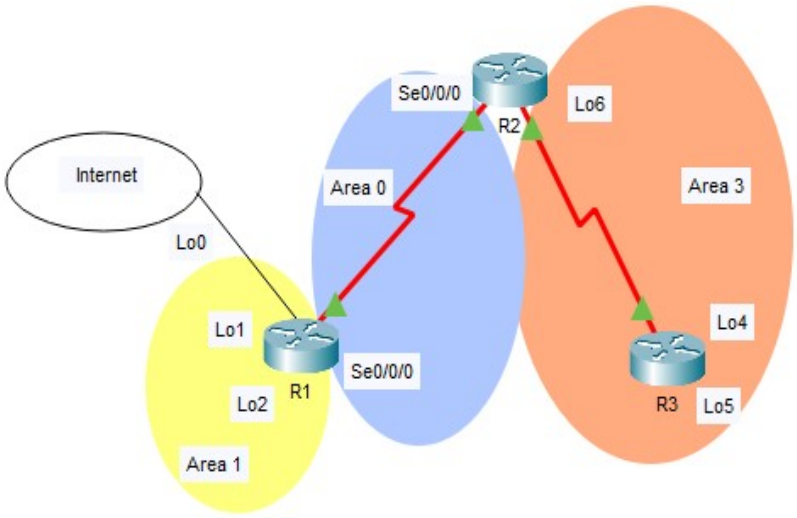
Aim:

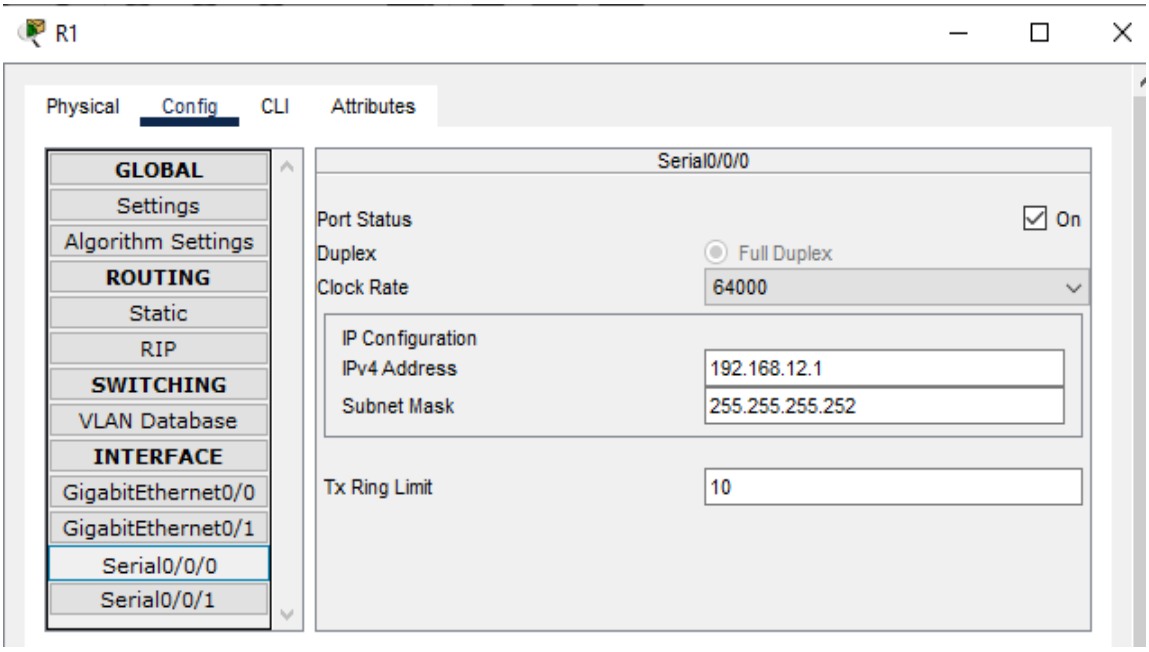
B) Configure multi-Area OSPFv2

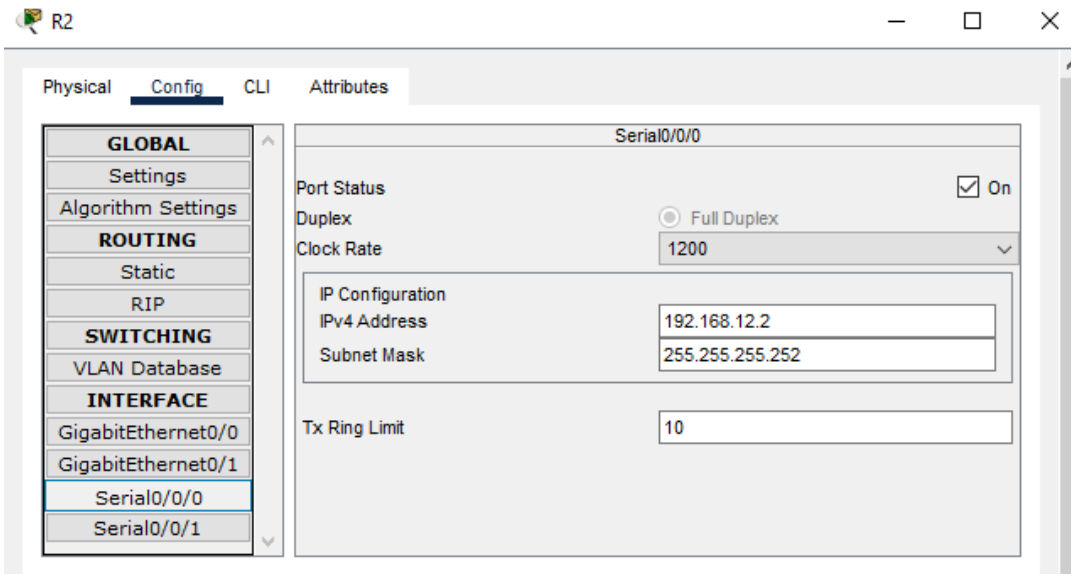
Theory:

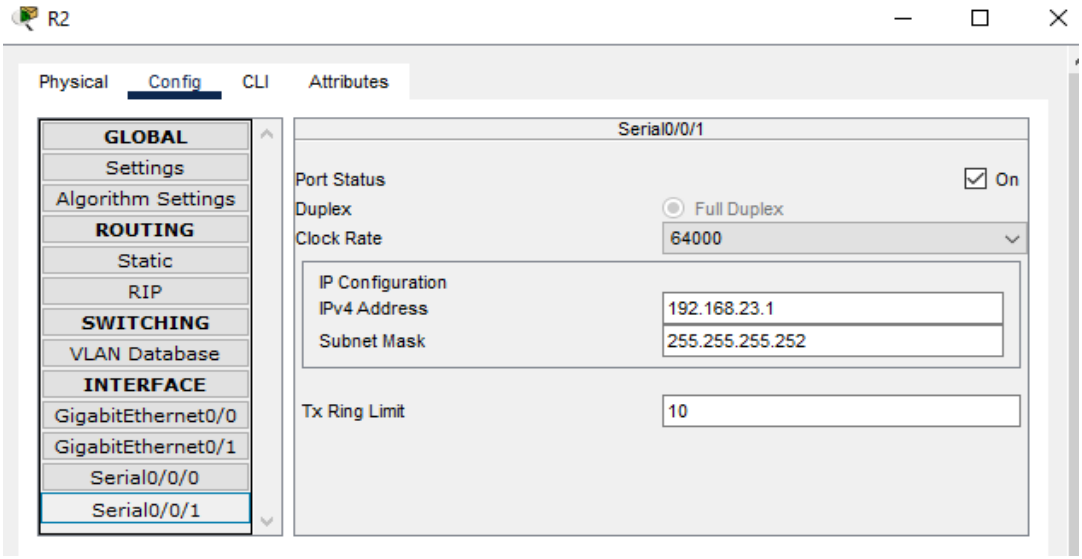
A single physical Link (Router Interface) can belong to multiple OSPF area.

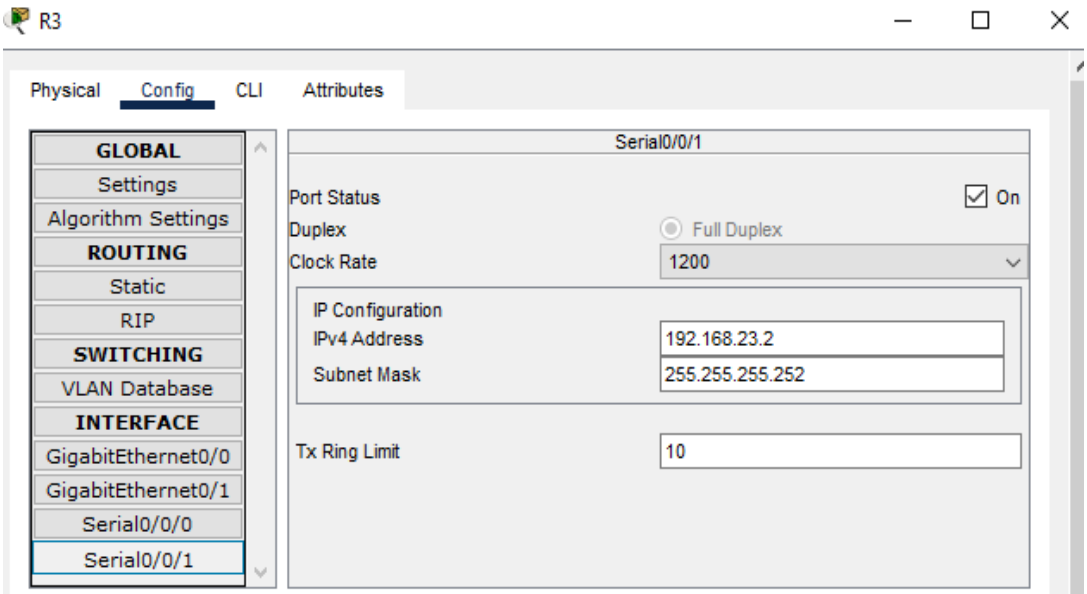
Topology :






R2
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
Serial0/0/0
Serial0/0/1
Serial0/0/0
Port Status
On
Duplex
Full Duplex
Clock Rate
1200
IP Configuration
IPv4 Address
192.168.12.2
Subnet Mask
255.255.255.252
Tx Ring Limit
10


R2
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
Serial0/0/0
Serial0/0/1
Serial0/0/1
Port Status
On
Duplex
Full Duplex
Clock Rate
64000
IP Configuration
IPv4 Address
192.168.23.1
Subnet Mask
255.255.255.252
Tx Ring Limit
10


R3
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
Serial0/0/0
Serial0/0/1
Serial0/0/1
Port Status
On
Duplex
Full Duplex
Clock Rate
1200
IP Configuration
IPv4 Address
192.168.23.2
Subnet Mask
255.255.255.252
Tx Ring Limit
10

R1 –

```
R1#enable
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface loopback 0
R1(config-if)#ip address 209.165.200.225 255.255.255.252
R1(config-if)#exit
R1(config)#interface loopback 1

R1(config-if)#
%LINK-3-UPDOWN: Interface Loopback1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback1, changed state to up

R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#exit
```

```
R1(config)#interface loopback 2

R1(config-if)#
%LINK-3-UPDOWN: Interface Loopback2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback2, changed state to up

R1(config-if)#ip address 192.168.2.1 255.255.255.0
R1(config-if)#exit
R1(config)#interface s0/0/0
R1(config-if)#ip address 192.168.12.1 255.255.255.252
R1(config-if)#clock rate 64000
R1(config-if)#no shutdown
R1(config-if)#exit
```

```
R1>show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
2.2.2.2           0   FULL/  -        00:00:35    192.168.12.2    Serial0/0/0
R1>
R1>show ip route ospf
     192.168.4.0/32 is subnetted, 1 subnets
O IA    192.168.4.1 [110/129] via 192.168.12.2, 00:07:21, Serial0/0/0
     192.168.5.0/32 is subnetted, 1 subnets
O IA    192.168.5.1 [110/129] via 192.168.12.2, 00:07:21, Serial0/0/0
     192.168.6.0/32 is subnetted, 1 subnets
O IA    192.168.6.1 [110/65] via 192.168.12.2, 00:18:36, Serial0/0/0
     192.168.23.0/30 is subnetted, 1 subnets
O IA    192.168.23.0 [110/128] via 192.168.12.2, 00:14:49, Serial0/0/0
```

R2 –

```
R2>enable
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#hostname R2
R2(config)#
R2(config)#interface loopback 6
R2(config-if)#ip address 192.168.6.1 255.255.255.0
R2(config-if)#exit
R2(config)#interface s0/0/0
R2(config-if)#ip address 192.168.12.2 255.255.255.252
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#interface s0/0/1
R2(config-if)#ip address 192.168.23.1 255.255.255.252
R2(config-if)#clock rate 64000
R2(config-if)#no shutdown
R2(config-if)#exit
```

```
R2(config)#router ospf 1
R2(config-router)#router-id 2.2.2.2
R2(config-router)#network 192.168.6.0 0.0.0.255 area 3
R2(config-router)#network 192.168.12.0 0.0.0.3 area 0
R2(config-router)#network 192.168.23.0 0.0.0.3 area 0
R2(config-router)#network 192.168.23.0 0.0.0.3 area 1
R2(config-router)#
02:19:44: %OSPF-6-AREACHG: 192.168.23.0/0 changed from area 0 to area 1

R2(config-router)#network 192.168.23.0 0.0.0.3 area 3
R2(config-router)#
02:19:56: %OSPF-6-AREACHG: 192.168.23.0/0 changed from area 1 to area 3

R2(config-router)#passive-interface lo6
R2(config-router)#
02:27:37: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on Serial0/0/1 from LOADING to
FULL, Loading Done
```

```
R2>show ip ospf
 Routing Process "ospf 1" with ID 2.2.2.2
 Supports only single TOS(TOS0) routes
 Supports opaque LSA
 It is an area border router
 SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
 Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
 Number of external LSA 0. Checksum Sum 0x000000
 Number of opaque AS LSA 0. Checksum Sum 0x000000
 Number of DCbitless external and opaque AS LSA 0
 Number of DoNotAge external and opaque AS LSA 0
 Number of areas in this router is 2. 2 normal 0 stub 0 nssa
 External flood list length 0
    Area 3
        Number of interfaces in this area is 2
        Area has no authentication
        SPF algorithm executed 6 times
        Area ranges are
        Number of LSA 5. Checksum Sum 0x0203ba
        Number of opaque link LSA 0. Checksum Sum 0x000000
        Number of DCbitless LSA 0
        Number of indication LSA 0
        Number of DoNotAge LSA 0
 --More-- |
```

```
R2>show ip ospf ne

Neighbor ID     Pri   State           Dead Time   Address         Interface
3.3.3.3           0   FULL/  -        00:00:30    192.168.23.2    Serial0/0/1
1.1.1.1           0   FULL/  -        00:00:34    192.168.12.1    Serial0/0/0
```

R3 –

```
Router>enable
Router#conf term
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R3
R3(config)#
R3(config)#interface loopback 4

R3(config-if)#
%LINK-3-UPDOWN: Interface Loopback4, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback4, changed state to up

R3(config-if)#ip address 192.168.4.1 255.255.255.0
R3(config-if)#exit
R3(config)#interface loopback 5

R3(config-if)#
%LINK-3-UPDOWN: Interface Loopback5, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback5, changed state to up

R3(config-if)#ip address 192.168.5.1 255.255.255.0
```

```
R3(config)#router ospf 1
R3(config-router)#router-id 3.3.3.3
R3(config-router)#network 192.168.4.0 0.0.0.255 area 3
R3(config-router)#network 192.168.5.0 0.0.0.255 area 3
R3(config-router)#network 192.168.23.0 0.0.0.3 area 3
R3(config-router)#
02:27:12: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on Serial0/0/1 from LOADING to
FULL, Loading Done

R3(config-router)#passive-interface lo4
R3(config-router)#passive-interface lo5
```

```
R3>show ip ospf
 Routing Process "ospf 1" with ID 3.3.3.3
 Supports only single TOS(TOS0) routes
 Supports opaque LSA
 SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
 Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
 Number of external LSA 0. Checksum Sum 0x000000
 Number of opaque AS LSA 0. Checksum Sum 0x000000
 Number of DCbitless external and opaque AS LSA 0
 Number of DoNotAge external and opaque AS LSA 0
 Number of areas in this router is 1. 1 normal 0 stub 0 nssa
 External flood list length 0
    Area 3
        Number of interfaces in this area is 3
        Area has no authentication
        SPF algorithm executed 3 times
        Area ranges are
        Number of LSA 5. Checksum Sum 0x0203ba
        Number of opaque link LSA 0. Checksum Sum 0x000000
        Number of DCbitless LSA 0
        Number of indication LSA 0
        Number of DoNotAge LSA 0
        Flood list length 0
```

```
R3> show ip ospf ne

Neighbor ID     Pri   State           Dead Time   Address         Interface
2.2.2.2           0   FULL/  -        00:00:36    192.168.23.1    Serial0/0/1
R3>
```

```
R1#enable
R1#ping 192.168.12.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.12.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/9 ms

R1#ping 192.168.23.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.23.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/5/18 ms

R1#ping 192.168.4.1

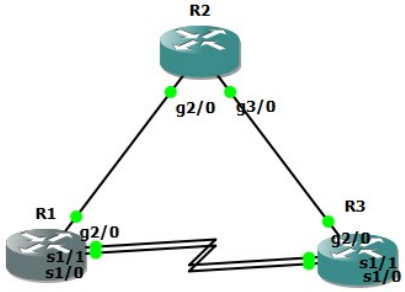
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.4.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/5/18 ms
```

Conclusion:

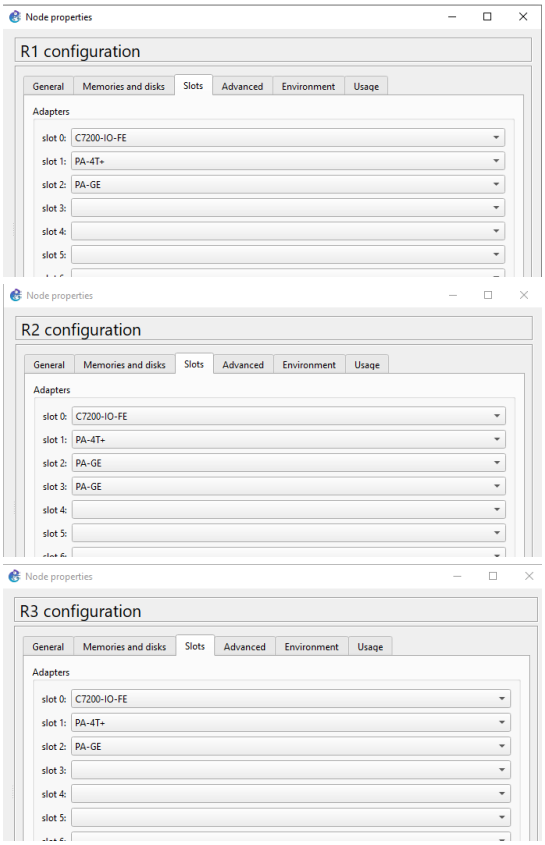
The Above Practical Configure Multi-area OSPFv2 is configured Successfully

PRACTICAL-09

Topology



Router slot configuration (GNS3)



Addressing Table

Device	Interface	IPv4 Address	IPv6 Address
R1	Gi2/0	10.1.2.1/24	2001:db8:acad:1012::1/64
	Se1/0	10.1.3.1/25	2001:db8:acad:1013::1/64
	Se1/1	10.1.3.129/25	2001:db8:acad:1014::1/64
	Loopback0	192.168.1.1/27	2001:db8:acad:1000::1/64
	Loopback1	192.168.1.65/26	2001:db8:acad:1001::1/64
R2	Gi2/0	10.1.2.2/24	2001:db8:acad:1012::2/64
	Gi3/0	10.2.3.2/24	2001:db8:acad:1023::2/64
	Loopback0	192.168.2.1/27	2001:db8:acad:2000::1/64
	Loopback1	192.168.2.65/26	2001:db8:acad:2001::1/64
R3	Gi2/0	10.2.3.3/24	2001:db8:acad:1023::3/64
	Se1/0	10.1.3.3/25	2001:db8:acad:1013::3/64
	Se1/1	10.1.3.130/25	2001:db8:acad:1014::3/64
	Loopback0	192.168.3.1/27	2001:db8:acad:3000::1/64

Device	Interface	IPv4 Address	IPv6 Address
	Loopback1	192.168.3.65/26	2001:db8:acad:3001::1/64

Aim:

a) Implement BGP Communities

b) Implement MP-BGP

c) Implement eBGP for IPv4

d) Implement BGP Path Manipulation

Theory:

BGP (Border Gateway Protocol) is the routing protocol used to exchange routes between different autonomous systems (AS). BGP between routers in different autonomous systems is called eBGP, and the best path is chosen using attributes such as AS path and local preference.

BGP Communities are tags attached to routes so that routers can treat many routes the same way. The well-known communities are Internet, No-Advertise and No-Export. MP-BGP lets one BGP session carry more than one address family, such as IPv4 and IPv6. BGP path manipulation uses ACLs, prefix-lists, AS-PATH filters, summarization and local preference to control which routes are sent, accepted and preferred.

a) Implement BGP Communities

Step 1: Configure interfaces on R1, R2 and R3

R1

```
 changed state to down
*Oct 27 20:54:24.075: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/3,
 changed state to down
*Oct 27 20:54:24.079: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEth
ernet2/0, changed state to down
R1#enable
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#hostname R1
R1(config)#no ip domain-lookup
R1(config)#ipv6 unicast-routing
R1(config)#interface g2/0
R1(config-if)# ip address 10.1.2.1 255.255.255.0
R1(config-if)# ipv6 address 2001:db8:acad:1012::1/64
R1(config-if)# no shutdown
R1(config-if)#interface s1/0
R1(config-if)# ip address 10.1.3.1 255.255.255.128
R1(config-if)# ipv6 address 2001:db8:acad:1013::1/64
R1(config-if)# no shutdown
R1(config-if)#interface s1/1
R1(config-if)# ip address 10.1.3.129 255.255.255.128
R1(config-if)# ipv6 address 2001:db8:acad:1014::1/64
R1(config-if)# no shutdown
R1(config-if)#interface loopback 0
R1(config-if)# ip address 192.168.1.1 255.255.255.224
R1(config-if)# ipv6 address 2001:db8:acad:1000::1/64
R1(config-if)#interface loopback 1
R1(config-if)# ip address 192.168.1.65 255.255.255.192
R1(config-if)# ipv6 address 2001:db8:acad:1001::1/64
R1(config-if)#end
*Oct 27 20:57:26.723: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
*Oct 27 20:57:27.299: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback1, changed state to up
R1(config-if)#end
*Oct 27 20:57:27.563: %LINK-3-UPDOWN: Interface GigabitEthernet2/0, changed state to up
*Oct 27 20:57:27.987: %LINK-3-UPDOWN: Interface Serial1/0, changed state to up
*Oct 27 20:57:28.499: %LINK-3-UPDOWN: Interface Serial1/1, changed state to up
*Oct 27 20:57:28.563: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet2/0, changed state to up
R1(config-if)#end
*Oct 27 20:57:28.999: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/0, changed state to up
*Oct 27 20:57:29.515: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/1, changed state to up
R1(config-if)#end
*Oct 27 20:57:50.551: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/0, changed state to down
*Oct 27 20:57:50.587: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/1, changed state to down
R1(config-if)#
```

R2

```
*Oct 27 20:54:23.467: %LINK-5-CHANGED: Interface GigabitEthernet3/0, changed sta
te to administratively down
*Oct 27 20:54:24.315: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/0,
 changed state to down
*Oct 27 20:54:24.379: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/1,
 changed state to down
*Oct 27 20:54:24.415: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/2,
 changed state to down
*Oct 27 20:54:24.423: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/3,
 changed state to down
*Oct 27 20:54:24.463: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEth
ernet2/0, changed state to down
*Oct 27 20:54:24.467: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEth
ernet3/0, changed state to down
R2#enable
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#hostname R2
R2(config)#no ip domain-lookup
R2(config)#ipv6 unicast-routing
R2(config)#interface g2/0
R2(config-if)# ip address 10.1.2.2 255.255.255.0
R2(config-if)# ipv6 address 2001:db8:acad:1012::2/64
R2(config-if)# no shutdown
R2(config-if)#interface g3/0
R2(config-if)# ip address 10.2.3.2 255.255.255.0
R2(config-if)# ipv6 address 2001:db8:acad:1023::2/64
R2(config-if)# no shutdown
R2(config-if)#interface loopback 0
R2(config-if)# ip address 192.168.2.1 255.255.255.224
R2(config-if)# ipv6 address 2001:db8:acad:2000::1/64
R2(config-if)#interface loopback 1
R2(config-if)# ip address 192.168.2.65 255.255.255.192
R2(config-if)# ipv6 address 2001:db8:acad:2001::1/64
R2(config-if)#end
*Oct 27 21:00:21.095: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
*Oct 27 21:00:21.671: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback1, changed state to up
R2(config-if)#end
*Oct 27 21:00:22.391: %LINK-3-UPDOWN: Interface GigabitEthernet2/0, changed state to up
*Oct 27 21:00:22.879: %LINK-3-UPDOWN: Interface GigabitEthernet3/0, changed state to up
*Oct 27 21:00:23.391: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet2/0, changed state to up
R2(config-if)#end
*Oct 27 21:00:23.879: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet3/0, changed state to up
R2(config-if)#
```

R3

```
*Oct 27 20:54:24.167: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/1,
 changed state to down
*Oct 27 20:54:24.203: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/2,
 changed state to down
*Oct 27 20:54:24.207: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/3,
 changed state to down
*Oct 27 20:54:24.211: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEth
ernet2/0, changed state to down
R3#enable
R3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#hostname R3
R3(config)#no ip domain-lookup
R3(config)#ipv6 unicast-routing
R3(config)#interface g2/0
R3(config-if)# ip address 10.2.3.3 255.255.255.0
R3(config-if)# ipv6 address 2001:db8:acad:1023::3/64
R3(config-if)# no shutdown
R3(config-if)#interface s1/0
R3(config-if)# ip address 10.1.3.3 255.255.255.128
R3(config-if)# ipv6 address 2001:db8:acad:1013::3/64
R3(config-if)# no shutdown
R3(config-if)#interface s1/1
R3(config-if)# ip address 10.1.3.130 255.255.255.128
R3(config-if)# ipv6 address 2001:db8:acad:1014::3/64
R3(config-if)# no shutdown
R3(config-if)#interface loopback 0
R3(config-if)# ip address 192.168.3.1 255.255.255.224
R3(config-if)# ipv6 address 2001:db8:acad:3000::1/64
R3(config-if)#interface loopback 1
R3(config-if)# ip address 192.168.3.65 255.255.255.192
R3(config-if)# ipv6 address 2001:db8:acad:3001::1/64
R3(config-if)#end
*Oct 27 21:01:44.047: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
*Oct 27 21:01:44.635: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback1, changed state to up
R3(config-if)#end
*Oct 27 21:01:44.891: %LINK-3-UPDOWN: Interface GigabitEthernet2/0, changed state to up
*Oct 27 21:01:45.311: %LINK-3-UPDOWN: Interface Serial1/0, changed state to up
*Oct 27 21:01:45.819: %LINK-3-UPDOWN: Interface Serial1/1, changed state to up
*Oct 27 21:01:45.891: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet2/0, changed state to up
R3(config-if)#end
*Oct 27 21:01:46.323: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/0, changed state to up
*Oct 27 21:01:46.835: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/1, changed state to up
R3(config-if)#
```

```
R1(config-if)#do show ip interface brief
Interface                  IP-Address      OK? Method Status                Protocol
FastEthernet0/0            unassigned      YES unset  administratively down down
Serial1/0                  10.1.3.1        YES manual up                    up
Serial1/1                  10.1.3.129      YES manual up                    up
Serial1/2                  unassigned      YES unset  administratively down down
Serial1/3                  unassigned      YES unset  administratively down down
GigabitEthernet2/0         10.1.2.1        YES manual up                    up
Loopback0                  192.168.1.1     YES manual up                    up
Loopback1                  192.168.1.65    YES manual up                    up
R1(config-if)#do show ipv6 interface brief
FastEthernet0/0            [administratively down/down]
    unassigned
Serial1/0                  [up/up]
    FE80::C801:21FF:FE38:0
    2001:DB8:ACAD:1013::1
Serial1/1                  [up/up]
    FE80::C801:21FF:FE38:0
    2001:DB8:ACAD:1014::1
Serial1/2                  [administratively down/down]
    unassigned
Serial1/3                  [administratively down/down]
    unassigned
GigabitEthernet2/0         [up/up]
    FE80::C801:21FF:FE38:38
    2001:DB8:ACAD:1012::1
Loopback0                  [up/up]
    FE80::C801:21FF:FE38:0
    2001:DB8:ACAD:1000::1
Loopback1                  [up/up]
    FE80::C801:21FF:FE38:0
    2001:DB8:ACAD:1001::1
R1(config-if)#
```

```
R2(config-if)#do show ip interface brief
Interface                  IP-Address      OK? Method Status                Protocol
FastEthernet0/0            unassigned      YES unset  administratively down down
Serial1/0                  unassigned      YES unset  administratively down down
Serial1/1                  unassigned      YES unset  administratively down down
Serial1/2                  unassigned      YES unset  administratively down down
Serial1/3                  unassigned      YES unset  administratively down down
GigabitEthernet2/0         10.1.2.2        YES manual up                    up
GigabitEthernet3/0         10.2.3.2        YES manual up                    up
Loopback0                  192.168.2.1     YES manual up                    up
Loopback1                  192.168.2.65    YES manual up                    up
R2(config-if)#do show ipv6 interface brief
FastEthernet0/0            [administratively down/down]
    unassigned
Serial1/0                  [administratively down/down]
    unassigned
Serial1/1                  [administratively down/down]
    unassigned
Serial1/2                  [administratively down/down]
    unassigned
Serial1/3                  [administratively down/down]
    unassigned
GigabitEthernet2/0         [up/up]
    FE80::C802:32FF:FE58:38
    2001:DB8:ACAD:1012::2
GigabitEthernet3/0         [up/up]
    FE80::C802:32FF:FE58:54
    2001:DB8:ACAD:1023::2
Loopback0                  [up/up]
    FE80::C802:32FF:FE58:0
    2001:DB8:ACAD:2000::1
Loopback1                  [up/up]
    FE80::C802:32FF:FE58:0
    2001:DB8:ACAD:2001::1
R2(config-if)#
```

```
R3(config-if)#do show ip interface brief
Interface                  IP-Address      OK? Method Status                Protocol
FastEthernet0/0            unassigned      YES unset  administratively down down
Serial1/0                  10.1.3.3        YES manual up                    up
Serial1/1                  10.1.3.130      YES manual up                    up
Serial1/2                  unassigned      YES unset  administratively down down
Serial1/3                  unassigned      YES unset  administratively down down
GigabitEthernet2/0         10.2.3.3        YES manual up                    up
Loopback0                  192.168.3.1     YES manual up                    up
Loopback1                  192.168.3.65    YES manual up                    up
R3(config-if)#do show ipv6 interface brief
FastEthernet0/0            [administratively down/down]
    unassigned
Serial1/0                  [up/up]
    FE80::C803:1AFF:FE34:0
    2001:DB8:ACAD:1013::3
Serial1/1                  [up/up]
    FE80::C803:1AFF:FE34:0
    2001:DB8:ACAD:1014::3
Serial1/2                  [administratively down/down]
    unassigned
Serial1/3                  [administratively down/down]
    unassigned
GigabitEthernet2/0         [up/up]
    FE80::C803:1AFF:FE34:38
    2001:DB8:ACAD:1023::3
Loopback0                  [up/up]
    FE80::C803:1AFF:FE34:0
    2001:DB8:ACAD:3000::1
Loopback1                  [up/up]
    FE80::C803:1AFF:FE34:0
    2001:DB8:ACAD:3001::1
R3(config-if)#
```

R1

```
GigabitEthernet2/0      10.1.2.1        YES manual up                    up
Loopback0               192.168.1.1     YES manual up                    up
Loopback1               192.168.1.65    YES manual up                    up
R1(config-if)#do show ipv6 interface brief
FastEthernet0/0         [administratively down/down]
    unassigned
Serial1/0               [up/up]
    FE80::C801:21FF:FE38:0
    2001:DB8:ACAD:1013::1
Serial1/1               [up/up]
    FE80::C801:21FF:FE38:0
    2001:DB8:ACAD:1014::1
Serial1/2               [administratively down/down]
    unassigned
Serial1/3               [administratively down/down]
    unassigned
GigabitEthernet2/0      [up/up]
    FE80::C801:21FF:FE38:38
    2001:DB8:ACAD:1012::1
Loopback0               [up/up]
    FE80::C801:21FF:FE38:0
    2001:DB8:ACAD:1000::1
Loopback1               [up/up]
    FE80::C801:21FF:FE38:0
    2001:DB8:ACAD:1001::1
R1(config-if)#
R1#
*Oct 27 21:43:17.707: %SYS-5-CONFIG_I: Configured from console by console
R1#ping 10.1.2.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.2.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 72/138/220 ms
R1#ping 10.1.3.3
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.3.3, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 40/72/128 ms
R1#ping 10.1.3.130
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.3.130, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 40/89/128 ms
R1#
```

Step 2: BGP base (R1 = AS 6500, R2 = AS 500, R3 = AS 300)

R1

```
Sending 5, 100-byte ICMP Echos to 10.1.3.3, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 40/72/128 ms
R1#ping 10.1.3.130
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.3.130, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 40/89/128 ms
R1#copy running-config startup-config
Destination filename [startup-config]?
Warning: Attempting to overwrite an NVRAM configuration previously written
by a different version of the system image.
Overwrite the previous NVRAM configuration?[confirm]
Building configuration...
[OK]
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#router bgp 6500
R1(config-router)# bgp router-id 1.1.1.1
R1(config-router)# no bgp default ipv4-unicast
R1(config-router)# neighbor 10.1.2.2 remote-as 500
R1(config-router)# neighbor 10.1.3.3 remote-as 300
R1(config-router)# neighbor 10.1.3.130 remote-as 300
R1(config-router)# neighbor 2001:db8:acad:1012::2 remote-as 500
R1(config-router)# neighbor 2001:db8:acad:1013::3 remote-as 300
R1(config-router)# neighbor 2001:db8:acad:1014::3 remote-as 300
R1(config-router)# address-family ipv4 unicast
R1(config-router-af)#  network 192.168.1.0 mask 255.255.255.224
R1(config-router-af)#  network 192.168.1.64 mask 255.255.255.192
R1(config-router-af)#  neighbor 10.1.2.2 activate
R1(config-router-af)#  neighbor 10.1.3.3 activate
R1(config-router-af)#  neighbor 10.1.3.130 activate
R1(config-router-af)# exit-address-family
R1(config-router)# address-family ipv6 unicast
R1(config-router-af)#  network 2001:db8:acad:1000::/64
R1(config-router-af)#  network 2001:db8:acad:1001::/64
R1(config-router-af)#  neighbor 2001:db8:acad:1012::2 activate
R1(config-router-af)#  neighbor 2001:db8:acad:1013::3 activate
R1(config-router-af)#  neighbor 2001:db8:acad:1014::3 activate
R1(config-router-af)# exit-address-family
R1(config-router)#end
R1#
*Oct 27 21:49:48.143: %SYS-5-CONFIG_I: Configured from console by console
R1#
```

R2

```
R2(config-if)#conf t
             ^
% Invalid input detected at '^' marker.

R2(config-if)#
R2#
*Oct 27 21:47:47.183: %SYS-5-CONFIG_I: Configured from console by console
R2#copy running--confi startup-config
%Error parsing filename (No such device)
R2#copy running-config startup-config
Destination filename [startup-config]?
Warning: Attempting to overwrite an NVRAM configuration previously written
by a different version of the system image.
Overwrite the previous NVRAM configuration?[confirm]
Building configuration...
[OK]
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#router bgp 500
R2(config-router)# bgp router-id 2.2.2.2
R2(config-router)# no bgp default ipv4-unicast
R2(config-router)# neighbor 10.1.2.1 remote-as 6500
R2(config-router)# neighbor 10.2.3.3 remote-as 300
R2(config-router)# neighbor 2001:db8:acad:1012::1 remote-as 6500
R2(config-router)# neighbor 2001:db8:acad:1023::3 remote-as 300
R2(config-router)# address-family ipv4 unicast
R2(config-router-af)#  network 192.168.2.0 mask 255.255.255.224
R2(config-router-af)#  network 192.168.2.64 mask 255.255.255.192
R2(config-router-af)#  neighbor 10.1.2.1 activate
R2(config-router-af)#  neighbor 10.2.3.3 activate
R2(config-router-af)# exit-address-family
R2(config-router)# address-family ipv6 unicast
R2(config-router-af)#  network 2001:db8:acad:2000::/64
R2(config-router-af)#  network 2001:db8:acad:2001::/64
R2(config-router-af)#  neighbor 2001:db8:acad:1012::1 activate
R2(config-router-af)#  neighbor 2001:db8:acad:1023::3 activate
R2(config-router-af)# exit-address-family
R2(config-router)#end
R2#
*Oct 27 21:51:14.215: %SYS-5-CONFIG_I: Configured from console by console
*Oct 27 21:51:14.303: %BGP-5-ADJCHANGE: neighbor 2001:DB8:ACAD:1012::1 Up
R2#
*Oct 27 21:51:16.251: %BGP-5-ADJCHANGE: neighbor 10.1.2.1 Up
R2#
```

R3

```
     FE80::C803:1AFF:FE34:0
     2001:DB8:ACAD:3001::1
R3(config-if)#
R3#configure terminal
*Oct 27 21:48:54.267: %SYS-5-CONFIG_I: Configured from console by console
R3#copy running-config startup-config
Destination filename [startup-config]?
Warning: Attempting to overwrite an NVRAM configuration previously written
by a different version of the system image.
Overwrite the previous NVRAM configuration?[confirm]
Building configuration...
[OK]
R3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#router bgp 300
R3(config-router)# bgp router-id 3.3.3.3
R3(config-router)# no bgp default ipv4-unicast
R3(config-router)# neighbor 10.1.3.1 remote-as 6500
R3(config-router)# neighbor 10.1.3.129 remote-as 6500
R3(config-router)# neighbor 10.2.3.2 remote-as 500
R3(config-router)# neighbor 2001:db8:acad:1013::1 remote-as 6500
R3(config-router)# neighbor 2001:db8:acad:1014::1 remote-as 6500
R3(config-router)# neighbor 2001:db8:acad:1023::2 remote-as 500
R3(config-router)# address-family ipv4 unicast
R3(config-router-af)#  network 192.168.3.0 mask 255.255.255.224
R3(config-router-af)#  network 192.168.3.64 mask 255.255.255.192
R3(config-router-af)#  neighbor 10.1.3.1 activate
R3(config-router-af)#  neighbor 10.1.3.129 activate
R3(config-router-af)#  neighbor 10.2.3.2 activate
R3(config-router-af)# exit-address-family
R3(config-router)# address-family ipv6 unicast
R3(config-router-af)#  network 2001:db8:acad:3000::/64
R3(config-router-af)#  network 2001:db8:acad:3001::/64
R3(config-router-af)#  neighbor 2001:db8:acad:1013::1 activate
R3(config-router-af)#  neighbor 2001:db8:acad:1014::1 activate
R3(config-router-af)#  neighbor 2001:db8:acad:1023::2 activate
R3(config-router-af)# exit-address-family
R3(config-router)#end
R3#
*Oct 27 21:52:20.603: %BGP-5-ADJCHANGE: neighbor 10.1.3.129 Up
*Oct 27 21:52:21.587: %BGP-5-ADJCHANGE: neighbor 10.1.3.1 Up
R3#
*Oct 27 21:52:21.711: %SYS-5-CONFIG_I: Configured from console by console
R3#
```

R1

```
*Oct 27 21:51:16.255: %BGP-5-ADJCHANGE: neighbor 10.1.2.2 Up
R1#
*Oct 27 21:52:20.723: %BGP-5-ADJCHANGE: neighbor 10.1.3.130 Up
*Oct 27 21:52:21.679: %BGP-5-ADJCHANGE: neighbor 10.1.3.3 Up
R1#
*Oct 27 21:52:24.003: %BGP-5-ADJCHANGE: neighbor 2001:DB8:ACAD:1013::3 Up
R1#
*Oct 27 21:52:30.983: %BGP-5-ADJCHANGE: neighbor 2001:DB8:ACAD:1014::3 Up
R1#show bgp ipv4 unicast summary
BGP router identifier 1.1.1.1, local AS number 6500
BGP table version is 7, main routing table version 7
6 network entries using 864 bytes of memory
14 path entries using 1120 bytes of memory
5/3 BGP path/bestpath attribute entries using 680 bytes of memory
4 BGP AS-PATH entries using 96 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 2760 total bytes of memory
BGP activity 12/0 prefixes, 28/0 paths, scan interval 60 secs

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
10.1.2.2        4          500       6       8        7    0    0 00:02:00        4
10.1.3.3        4          300       7       9        7    0    0 00:00:54        4
10.1.3.130      4          300       7       9        7    0    0 00:00:55        4
R1#show bgp ipv6 unicast summary
BGP router identifier 1.1.1.1, local AS number 6500
BGP table version is 7, main routing table version 7
6 network entries using 1008 bytes of memory
14 path entries using 1456 bytes of memory
5/3 BGP path/bestpath attribute entries using 680 bytes of memory
4 BGP AS-PATH entries using 96 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 3240 total bytes of memory
BGP activity 12/0 prefixes, 28/0 paths, scan interval 60 secs

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
2001:DB8:ACAD:1012::2
                4          500       6       6        5    0    0 00:02:05        4
2001:DB8:ACAD:1013::3
                4          300       7       6        5    0    0 00:00:55        4
2001:DB8:ACAD:1014::3
                4          300       7       8        5    0    0 00:00:48        4
R1#
```

Step 3: Send communities (all three routers)

R1

```
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 2760 total bytes of memory
BGP activity 12/0 prefixes, 28/0 paths, scan interval 60 secs

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
10.1.2.2        4          500       6       8        7    0    0 00:02:00        4
10.1.3.3        4          300       7       9        7    0    0 00:00:54        4
10.1.3.130      4          300       7       9        7    0    0 00:00:55        4
R1#show bgp ipv6 unicast summary
BGP router identifier 1.1.1.1, local AS number 6500
BGP table version is 7, main routing table version 7
6 network entries using 1008 bytes of memory
14 path entries using 1456 bytes of memory
5/3 BGP path/bestpath attribute entries using 680 bytes of memory
4 BGP AS-PATH entries using 96 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 3240 total bytes of memory
BGP activity 12/0 prefixes, 28/0 paths, scan interval 60 secs

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
2001:DB8:ACAD:1012::2
                4          500       6       6        5    0    0 00:02:05        4
2001:DB8:ACAD:1013::3
                4          300       7       6        5    0    0 00:00:55        4
2001:DB8:ACAD:1014::3
                4          300       7       8        5    0    0 00:00:48        4
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ip bgp-community new-format
R1(config)#router bgp 6500
R1(config-router)# address-family ipv4 unicast
R1(config-router-af)#  neighbor 10.1.2.2 send-community
R1(config-router-af)#  neighbor 10.1.3.3 send-community
R1(config-router-af)#  neighbor 10.1.3.130 send-community
R1(config-router-af)# address-family ipv6 unicast
R1(config-router-af)#  neighbor 2001:db8:acad:1012::2 send-community
R1(config-router-af)#  neighbor 2001:db8:acad:1013::3 send-community
R1(config-router-af)#  neighbor 2001:db8:acad:1014::3 send-community
R1(config-router-af)#end
R1#
*Oct 27 21:55:28.655: %SYS-5-CONFIG_I: Configured from console by console
R1#
```

R2

```
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#router bgp 500
R2(config-router)# bgp router-id 2.2.2.2
R2(config-router)# no bgp default ipv4-unicast
R2(config-router)# neighbor 10.1.2.1 remote-as 6500
R2(config-router)# neighbor 10.2.3.3 remote-as 300
R2(config-router)# neighbor 2001:db8:acad:1012::1 remote-as 6500
R2(config-router)# neighbor 2001:db8:acad:1023::3 remote-as 300
R2(config-router)# address-family ipv4 unicast
R2(config-router-af)#  network 192.168.2.0 mask 255.255.255.224
R2(config-router-af)#  network 192.168.2.64 mask 255.255.255.192
R2(config-router-af)#  neighbor 10.1.2.1 activate
R2(config-router-af)#  neighbor 10.2.3.3 activate
R2(config-router-af)# exit-address-family
R2(config-router)# address-family ipv6 unicast
R2(config-router-af)#  network 2001:db8:acad:2000::/64
R2(config-router-af)#  network 2001:db8:acad:2001::/64
R2(config-router-af)#  neighbor 2001:db8:acad:1012::1 activate
R2(config-router-af)#  neighbor 2001:db8:acad:1023::3 activate
R2(config-router-af)# exit-address-family
R2(config-router)#end
R2#
*Oct 27 21:51:14.215: %SYS-5-CONFIG_I: Configured from console by console
*Oct 27 21:51:14.303: %BGP-5-ADJCHANGE: neighbor 2001:DB8:ACAD:1012::1 Up
R2#
*Oct 27 21:51:16.251: %BGP-5-ADJCHANGE: neighbor 10.1.2.1 Up
R2#
*Oct 27 21:52:25.227: %BGP-5-ADJCHANGE: neighbor 2001:DB8:ACAD:1023::3 Up
R2#
*Oct 27 21:52:27.267: %BGP-5-ADJCHANGE: neighbor 10.2.3.3 Up
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#ip bgp-community new-format
R2(config)#router bgp 500
R2(config-router)# address-family ipv4 unicast
R2(config-router-af)#  neighbor 10.1.2.1 send-community
R2(config-router-af)#  neighbor 10.2.3.3 send-community
R2(config-router-af)# address-family ipv6 unicast
R2(config-router-af)#  neighbor 2001:db8:acad:1012::1 send-community
R2(config-router-af)#  neighbor 2001:db8:acad:1023::3 send-community
R2(config-router-af)#end
R2#
*Oct 27 21:56:36.075: %SYS-5-CONFIG_I: Configured from console by console
R2#
```

R3

```
R3(config-router)# neighbor 2001:db8:acad:1014::1 remote-as 6500
R3(config-router)# neighbor 2001:db8:acad:1023::2 remote-as 500
R3(config-router)# address-family ipv4 unicast
R3(config-router-af)#  network 192.168.3.0 mask 255.255.255.224
R3(config-router-af)#  network 192.168.3.64 mask 255.255.255.192
R3(config-router-af)#  neighbor 10.1.3.1 activate
R3(config-router-af)#  neighbor 10.1.3.129 activate
R3(config-router-af)#  neighbor 10.2.3.2 activate
R3(config-router-af)# exit-address-family
R3(config-router)# address-family ipv6 unicast
R3(config-router-af)#  network 2001:db8:acad:3000::/64
R3(config-router-af)#  network 2001:db8:acad:3001::/64
R3(config-router-af)#  neighbor 2001:db8:acad:1013::1 activate
R3(config-router-af)#  neighbor 2001:db8:acad:1014::1 activate
R3(config-router-af)#  neighbor 2001:db8:acad:1023::2 activate
R3(config-router-af)# exit-address-family
R3(config-router)#end
R3#
*Oct 27 21:52:20.603: %BGP-5-ADJCHANGE: neighbor 10.1.3.129 Up
*Oct 27 21:52:21.587: %BGP-5-ADJCHANGE: neighbor 10.1.3.1 Up
R3#
*Oct 27 21:52:21.711: %SYS-5-CONFIG_I: Configured from console by console
R3#
*Oct 27 21:52:23.591: %BGP-5-ADJCHANGE: neighbor 2001:DB8:ACAD:1013::1 Up
R3#
*Oct 27 21:52:25.179: %BGP-5-ADJCHANGE: neighbor 2001:DB8:ACAD:1023::2 Up
R3#
*Oct 27 21:52:27.071: %BGP-5-ADJCHANGE: neighbor 10.2.3.2 Up
R3#
*Oct 27 21:52:31.295: %BGP-5-ADJCHANGE: neighbor 2001:DB8:ACAD:1014::1 Up
R3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#ip bgp-community new-format
R3(config)#router bgp 300
R3(config-router)# address-family ipv4 unicast
R3(config-router-af)#  neighbor 10.1.3.1 send-community
R3(config-router-af)#  neighbor 10.1.3.129 send-community
R3(config-router-af)#  neighbor 10.2.3.2 send-community
R3(config-router-af)# address-family ipv6 unicast
R3(config-router-af)#  neighbor 2001:db8:acad:1013::1 send-community
R3(config-router-af)#  neighbor 2001:db8:acad:1014::1 send-community
R3(config-router-af)#  neighbor 2001:db8:acad:1023::2 send-community
R3(config-router-af)#end
R3#
```

Step 4: no-export community

R2

```
R2(config-router-af)#   network 2001:db8:acad:2000::/64
R2(config-router-af)#   network 2001:db8:acad:2001::/64
R2(config-router-af)#   neighbor 2001:db8:acad:1012::1 activate
R2(config-router-af)#   neighbor 2001:db8:acad:1023::3 activate
R2(config-router-af)# exit-address-family
R2(config-router)#end
R2#
*Oct 27 21:51:14.215: %SYS-5-CONFIG_I: Configured from console by console
*Oct 27 21:51:14.303: %BGP-5-ADJCHANGE: neighbor 2001:DB8:ACAD:1012::1 Up
R2#
*Oct 27 21:51:16.251: %BGP-5-ADJCHANGE: neighbor 10.1.2.1 Up
R2#
*Oct 27 21:52:25.227: %BGP-5-ADJCHANGE: neighbor 2001:DB8:ACAD:1023::3 Up
R2#
*Oct 27 21:52:27.267: %BGP-5-ADJCHANGE: neighbor 10.2.3.3 Up
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#ip bgp-community new-format
R2(config)#router bgp 500
R2(config-router)# address-family ipv4 unicast
R2(config-router-af)#   neighbor 10.1.2.1 send-community
R2(config-router-af)#   neighbor 10.2.3.3 send-community
R2(config-router-af)# address-family ipv6 unicast
R2(config-router-af)#   neighbor 2001:db8:acad:1012::1 send-community
R2(config-router-af)#   neighbor 2001:db8:acad:1023::3 send-community
R2(config-router-af)#end
R2#
*Oct 27 21:56:36.075: %SYS-5-CONFIG_I: Configured from console by console
R2#show bgp ipv4 unicast 192.168.3.0/27
BGP routing table entry for 192.168.3.0/27, version 6
Paths: (2 available, best #2, table default)
  Advertised to update-groups:
     2
  Refresh Epoch 1
  6500 300
    10.1.2.1 from 10.1.2.1 (1.1.1.1)
      Origin IGP, localpref 100, valid, external
      rx pathid: 0, tx pathid: 0
  Refresh Epoch 2
  300
    10.2.3.3 from 10.2.3.3 (3.3.3.3)
      Origin IGP, metric 0, localpref 100, valid, external, best
      rx pathid: 0, tx pathid: 0x0
R2#
```

R3

```
R3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#ip bgp-community new-format
R3(config)#router bgp 300
R3(config-router)# address-family ipv4 unicast
R3(config-router-af)#  neighbor 10.1.3.1 send-community
R3(config-router-af)#  neighbor 10.1.3.129 send-community
R3(config-router-af)#  neighbor 10.2.3.2 send-community
R3(config-router-af)# address-family ipv6 unicast
R3(config-router-af)#  neighbor 2001:db8:acad:1013::1 send-community
R3(config-router-af)#  neighbor 2001:db8:acad:1014::1 send-community
R3(config-router-af)#  neighbor 2001:db8:acad:1023::2 send-community
R3(config-router-af)#end
R3#
*Oct 27 21:58:02.535: %SYS-5-CONFIG_I: Configured from console by console
R3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#$ist LOCAL_NETWORK_COMMSET seq 5 permit 192.168.3.0/24 le 27
R3(config)#$L_6_NETWORK_COMMSET seq 5 permit 2001:db8:acad:3000::/64
R3(config)#$L_6_NETWORK_COMMSET seq 10 permit 2001:db8:acad:3001::/64
R3(config)#route-map COMMSET permit 10
R3(config-route-map)# match ip address prefix-list LOCAL_NETWORK_COMMSET
R3(config-route-map)# set community no-export additive
R3(config-route-map)#route-map COMMSET permit 20
R3(config-route-map)# set community internet additive
R3(config-route-map)#route-map COMMSET_6 permit 10
R3(config-route-map)# match ipv6 address prefix-list LOCAL_6_NETWORK_COMMSET
R3(config-route-map)# set community no-export additive
R3(config-route-map)#route-map COMMSET_6 permit 20
R3(config-route-map)# set community internet additive
R3(config-route-map)#exit
R3(config)#router bgp 300
R3(config-router)# address-family ipv4 unicast
R3(config-router-af)#  neighbor 10.1.3.1 route-map COMMSET out
R3(config-router-af)#  neighbor 10.1.3.129 route-map COMMSET out
R3(config-router-af)# address-family ipv6 unicast
R3(config-router-af)#  neighbor 2001:db8:acad:1013::1 route-map COMMSET_6 out
R3(config-router-af)#  neighbor 2001:db8:acad:1014::1 route-map COMMSET_6 out
R3(config-router-af)#end
R3#clear bgp ipv4 unicast 6500 out
R3#clear bgp ipv6 unicast 6500 out
*Oct 27 22:01:58.423: %SYS-5-CONFIG_I: Configured from console by console
R3#clear bgp ipv6 unicast 6500 out
R3#
```

R2

```
R2(config-router-af)#  neighbor 10.1.2.1 send-community
R2(config-router-af)#  neighbor 10.2.3.3 send-community
R2(config-router-af)# address-family ipv6 unicast
R2(config-router-af)#  neighbor 2001:db8:acad:1012::1 send-community
R2(config-router-af)#  neighbor 2001:db8:acad:1023::3 send-community
R2(config-router-af)#end
R2#
*Oct 27 21:56:36.075: %SYS-5-CONFIG_I: Configured from console by console
R2#show bgp ipv4 unicast 192.168.3.0/27
BGP routing table entry for 192.168.3.0/27, version 6
Paths: (2 available, best #2, table default)
  Advertised to update-groups:
     2
  Refresh Epoch 1
  6500 300
    10.1.2.1 from 10.1.2.1 (1.1.1.1)
      Origin IGP, localpref 100, valid, external
      rx pathid: 0, tx pathid: 0
  Refresh Epoch 2
  300
    10.2.3.3 from 10.2.3.3 (3.3.3.3)
      Origin IGP, metric 0, localpref 100, valid, external, best
      rx pathid: 0, tx pathid: 0x0
R2#show bgp ipv4 unicast 192.168.3.0/27
BGP routing table entry for 192.168.3.0/27, version 6
Paths: (1 available, best #1, table default)
  Advertised to update-groups:
     2
  Refresh Epoch 2
  300
    10.2.3.3 from 10.2.3.3 (3.3.3.3)
      Origin IGP, metric 0, localpref 100, valid, external, best
      rx pathid: 0, tx pathid: 0x0
R2#show bgp ipv6 unicast 2001:db8:acad:3000::/64
BGP routing table entry for 2001:DB8:ACAD:3000::/64, version 6
Paths: (1 available, best #1, table default)
  Advertised to update-groups:
     2
  Refresh Epoch 1
  300
    2001:DB8:ACAD:1023::3 (FE80::C803:1AFF:FE34:38) from 2001:DB8:ACAD:1023::3 (3.3.3.3)
      Origin IGP, metric 0, localpref 100, valid, external, best
      rx pathid: 0, tx pathid: 0x0
R2#
```

Step 5: Private communities from R1

R1

```
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 3240 total bytes of memory
BGP activity 12/0 prefixes, 28/0 paths, scan interval 60 secs

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
2001:DB8:ACAD:1012::2
                4          500       6       6        5    0    0 00:02:05        4
2001:DB8:ACAD:1013::3
                4          300       7       6        5    0    0 00:00:55        4
2001:DB8:ACAD:1014::3
                4          300       7       8        5    0    0 00:00:48        4
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ip bgp-community new-format
R1(config)#router bgp 6500
R1(config-router)# address-family ipv4 unicast
R1(config-router-af)#  neighbor 10.1.2.2 send-community
R1(config-router-af)#  neighbor 10.1.3.3 send-community
R1(config-router-af)#  neighbor 10.1.3.130 send-community
R1(config-router-af)# address-family ipv6 unicast
R1(config-router-af)#  neighbor 2001:db8:acad:1012::2 send-community
R1(config-router-af)#  neighbor 2001:db8:acad:1013::3 send-community
R1(config-router-af)#  neighbor 2001:db8:acad:1014::3 send-community
R1(config-router-af)#end
R1#
*Oct 27 21:55:28.655: %SYS-5-CONFIG_I: Configured from console by console
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#route-map ADDCOMM permit 10
R1(config-route-map)# set community 650:400 additive
R1(config-route-map)#route-map ADDCOMM_6 permit 10
R1(config-route-map)# set community 650:600 additive
R1(config-route-map)#exit
R1(config)#router bgp 6500
R1(config-router)# address-family ipv4 unicast
R1(config-router-af)#  neighbor 10.1.2.2 route-map ADDCOMM out
R1(config-router-af)# address-family ipv6 unicast
R1(config-router-af)#  neighbor 2001:db8:acad:1012::2 route-map ADDCOMM_6 out
R1(config-router-af)#end
R1#clear bgp ipv4 unicast 500 out
R1#clear bgp ipv6 unicast 500 out
*Oct 27 22:04:31.455: %SYS-5-CONFIG_I: Configured from console by console
R1#clear bgp ipv6 unicast 500 out
R1#
```

R2

```
R2(config-router-af)#  neighbor 2001:db8:acad:1023::3 send-community
R2(config-router-af)#end
R2#
*Oct 27 21:56:36.075: %SYS-5-CONFIG_I: Configured from console by console
R2#show bgp ipv4 unicast 192.168.3.0/27
BGP routing table entry for 192.168.3.0/27, version 6
Paths: (2 available, best #2, table default)
  Advertised to update-groups:
     2
  Refresh Epoch 1
  6500 300
    10.1.2.1 from 10.1.2.1 (1.1.1.1)
      Origin IGP, localpref 100, valid, external
      rx pathid: 0, tx pathid: 0
  Refresh Epoch 2
  300
    10.2.3.3 from 10.2.3.3 (3.3.3.3)
      Origin IGP, metric 0, localpref 100, valid, external, best
      rx pathid: 0, tx pathid: 0x0
R2#show bgp ipv4 unicast 192.168.3.0/27
BGP routing table entry for 192.168.3.0/27, version 6
Paths: (1 available, best #1, table default)
  Advertised to update-groups:
     2
  Refresh Epoch 2
  300
    10.2.3.3 from 10.2.3.3 (3.3.3.3)
      Origin IGP, metric 0, localpref 100, valid, external, best
      rx pathid: 0, tx pathid: 0x0
R2#show bgp ipv6 unicast 2001:db8:acad:3000::/64
BGP routing table entry for 2001:DB8:ACAD:3000::/64, version 6
Paths: (1 available, best #1, table default)
  Advertised to update-groups:
     2
  Refresh Epoch 1
  300
    2001:DB8:ACAD:1023::3 (FE80::C803:1AFF:FE34:38) from 2001:DB8:ACAD:1023::3 (3.3.3.3)
      Origin IGP, metric 0, localpref 100, valid, external, best
      rx pathid: 0, tx pathid: 0x0
R2#show bgp ipv4 unicast 192.168.1.0/27 | i Community
      Community: 650:400
R2#show bgp ipv6 unicast 2001:db8:acad:1000::/64 | i Community
      Community: 650:600
R2#
```

Step 6: Filter by community on R3

R3

```
R3(config-route-map)# set community no-export additive
R3(config-route-map)#route-map COMMSET permit 20
R3(config-route-map)# set community internet additive
R3(config-route-map)#route-map COMMSET_6 permit 10
R3(config-route-map)# match ipv6 address prefix-list LOCAL_6_NETWORK_COMMSET
R3(config-route-map)# set community no-export additive
R3(config-route-map)#route-map COMMSET_6 permit 20
R3(config-route-map)# set community internet additive
R3(config-route-map)#exit
R3(config)#router bgp 300
R3(config-router)# address-family ipv4 unicast
R3(config-router-af)#  neighbor 10.1.3.1 route-map COMMSET out
R3(config-router-af)#  neighbor 10.1.3.129 route-map COMMSET out
R3(config-router-af)# address-family ipv6 unicast
R3(config-router-af)#  neighbor 2001:db8:acad:1013::1 route-map COMMSET_6 out
R3(config-router-af)#  neighbor 2001:db8:acad:1014::1 route-map COMMSET_6 out
R3(config-router-af)#end
R3#clear bgp ipv4 unicast 6500 out
R3#clear bgp ipv6 unicast 6500 out
*Oct 27 22:01:58.423: %SYS-5-CONFIG_I: Configured from console by console
R3#clear bgp ipv6 unicast 6500 out
R3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#ip community-list 100 permit 650:400
R3(config)#ip community-list 101 permit 650:600
R3(config)#route-map COMMCHECK_4 deny 10
R3(config-route-map)# match community 100
R3(config-route-map)#route-map COMMCHECK_4 permit 20
R3(config-route-map)#route-map COMMCHECK_6 permit 10
R3(config-route-map)# match community 101
R3(config-route-map)# set local-preference 250
R3(config-route-map)#route-map COMMCHECK_6 permit 20
R3(config-route-map)#exit
R3(config)#router bgp 300
R3(config-router)# address-family ipv4 unicast
R3(config-router-af)#  neighbor 10.2.3.2 route-map COMMCHECK_4 in
R3(config-router-af)# address-family ipv6 unicast
R3(config-router-af)#$2001:db8:acad:1023::2 route-map COMMCHECK_6 in
R3(config-router-af)#end
R3#clear bgp ipv4 unicast 500 in
R3#clear bgp ipv6 unicast 500 in
*Oct 27 22:07:10.331: %SYS-5-CONFIG_I: Configured from console by console
R3#clear bgp ipv6 unicast 500 in
R3#
```

R3

```
R3#clear bgp ipv6 unicast 500 in
R3#show bgp ipv4 unicast
BGP table version is 7, local router ID is 3.3.3.3
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>  192.168.1.0/27   10.1.3.1                 0             0 6500 i
 *                    10.1.3.129               0             0 6500 i
 *>  192.168.1.64/26  10.1.3.1                 0             0 6500 i
 *                    10.1.3.129               0             0 6500 i
 *   192.168.2.0/27   10.1.3.129                             0 6500 500 i
 *>                   10.2.3.2                 0             0 500 i
 *                    10.1.3.1                                0 6500 500 i
 *   192.168.2.64/26  10.1.3.129                             0 6500 500 i
 *>                   10.2.3.2                 0             0 500 i
 *                    10.1.3.1                                0 6500 500 i
 *>  192.168.3.0/27   0.0.0.0                  0         32768 i
 *>  192.168.3.64/26  0.0.0.0                  0         32768 i
R3#show bgp ipv6 unicast
BGP table version is 9, local router ID is 3.3.3.3
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *   2001:DB8:ACAD:1000::/64
                       2001:DB8:ACAD:1014::1
                                                0             0 6500 i
 *>                    2001:DB8:ACAD:1023::2
                                              250             0 500 6500 i
 *                     2001:DB8:ACAD:1013::1
                                                0             0 6500 i
 *   2001:DB8:ACAD:1001::/64
                       2001:DB8:ACAD:1014::1
                                                0             0 6500 i
 *>                    2001:DB8:ACAD:1023::2
                                              250             0 500 6500 i
 *                     2001:DB8:ACAD:1013::1
```

R3

```
BGP table version is 7, local router ID is 3.3.3.3
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>  192.168.1.0/27   10.1.3.1                 0             0 6500 i
 *                    10.1.3.129               0             0 6500 i
 *>  192.168.1.64/26  10.1.3.1                 0             0 6500 i
 *                    10.1.3.129               0             0 6500 i
 *   192.168.2.0/27   10.1.3.129                             0 6500 500 i
 *>                   10.2.3.2                 0             0 500 i
 *                    10.1.3.1                                0 6500 500 i
 *   192.168.2.64/26  10.1.3.129                             0 6500 500 i
 *>                   10.2.3.2                 0             0 500 i
 *                    10.1.3.1                                0 6500 500 i
 *>  192.168.3.0/27   0.0.0.0                  0         32768 i
 *>  192.168.3.64/26  0.0.0.0                  0         32768 i
R3#show bgp ipv6 unicast
BGP table version is 9, local router ID is 3.3.3.3
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *   2001:DB8:ACAD:1000::/64
                       2001:DB8:ACAD:1014::1
                                                0             0 6500 i
 *>                    2001:DB8:ACAD:1023::2
                                              250             0 500 6500 i
 *                     2001:DB8:ACAD:1013::1
                                                0             0 6500 i
 *   2001:DB8:ACAD:1001::/64
                       2001:DB8:ACAD:1014::1
                                                0             0 6500 i
 *>                    2001:DB8:ACAD:1023::2
                                              250             0 500 6500 i
 *                     2001:DB8:ACAD:1013::1
                                                0             0 6500 i
 --More--
```

Conclusion: The above practical Implementing BGP Communities has been Implemented Successfully.

b) Implement MP-BGP

Step 1: Remove the old BGP configuration

R1

```
inistratively down
*Oct 28 08:26:07.691: %LINK-5-CHANGED: Interface Serial1/3, changed state to adm
inistratively down
*Oct 28 08:26:09.135: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/2,
 changed state to down
*Oct 28 08:26:09.143: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/3,
 changed state to down
*Oct 28 08:26:09.279: %SYS-5-CONFIG_I: Configured from memory by console
*Oct 28 08:26:14.275: %SYS-5-RESTART: System restarted --
Cisco IOS Software, 7200 Software (C7200-ADVENTERPRISEK9-M), Version 15.2(4)S6,
RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2014 by Cisco Systems, Inc.
Compiled Fri 08-Aug-14 04:05 by prod_rel_team
*Oct 28 08:26:21.611: %BGP-5-ADJCHANGE: neighbor 10.1.3.130 Up
*Oct 28 08:26:21.615: %BGP-3-NOTIFICATION: received from neighbor 2001:DB8:ACAD:
1012::2 active 2/8 (no supported AFI/SAFI) 3 bytes 000000
*Oct 28 08:26:21.615: %BGP-5-NBR_RESET: Neighbor 2001:DB8:ACAD:1012::2 active reset (BGP Notification received)
*Oct 28 08:26:21.619: %BGP-5-ADJCHANGE: neighbor 2001:DB8:ACAD:1012::2 active Down BGP Notification received
*Oct 28 08:26:21.623: %BGP_SESSION-5-ADJCHANGE: neighbor 2001:DB8:ACAD:1012::2 IPv6 Unicast topology base removed from session  BGP No
*Oct 28 08:26:21.631: %BGP-5-ADJCHANGE: neighbor 2001:DB8:ACAD:1014::3 Up
*Oct 28 08:26:22.099: %BGP-5-NBR_RESET: Neighbor 2001:DB8:ACAD:1012::2 passive reset (Peer closed the session)
*Oct 28 08:26:22.103: %BGP-5-ADJCHANGE: neighbor 2001:DB8:ACAD:1012::2 passive Down AFI/SAFI not supported
*Oct 28 08:26:23.275: %BGP-5-ADJCHANGE: neighbor 2001:DB8:ACAD:1013::3 Up
*Oct 28 08:26:27.419: %BGP-5-ADJCHANGE: neighbor 10.1.3.3 Up
*Oct 28 08:26:27.747: %BGP-5-ADJCHANGE: neighbor 10.1.2.2 Up
*Oct 28 08:26:29.579: %BGP-5-ADJCHANGE: neighbor 2001:DB8:ACAD:1012::2 Up
R1#enable
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#no router bgp 6500
R1(config)#end
*Oct 28 08:36:19.859: %BGP-5-ADJCHANGE: neighbor 10.1.2.2 Down Neighbor deleted
*Oct 28 08:36:19.863: %BGP_SESSION-5-ADJCHANGE: neighbor 10.1.2.2 IPv4 Unicast topology base removed from session  Neighbor deleted
*Oct 28 08:36:19.867: %BGP-5-ADJCHANGE: neighbor 10.1.3.3 Down Neighbor deleted
*Oct 28 08:36:19.871: %BGP_SESSION-5-ADJCHANGE: neighbor 10.1.3.3 IPv4 Unicast topology base removed from session  Neighbor deleted
*Oct 28 08:36:19.875: %BGP-5-ADJCHANGE: neighbor 10.1.3.130 Down Neighbor deleted
*Oct 28 08:36:19.875: %BGP_SESSION-5-ADJCHANGE: neighbor 10.1.3.130 IPv4 Unicast topology base removed from session  Neighbor deleted
*Oct 28 08:36:19.879: %BGP-5-ADJCHANGE: neighbor 2001:DB8:ACAD:1012::2 Down Unknown path error
*Oct 28 08:36:19.883: %BGP_SESSION-5-ADJCHANGE: neighbor 2001:DB8:ACAD:1012::2 IPv6 Unicast topology base removed from session  Unknow
*Oct 28 08:36:19.887: %BGP-5-ADJCHANGE: neighbor 2001:DB8:ACAD:1013::3 Down Unknown path error
*Oct 28 08:
R1(config)#end36:19.887: %BGP_SESSION-5-ADJCHANGE: neighbor 2001:DB8:ACAD:1013::3 IPv6 Unicast topology base removed from session  Unk
R1(config)#end
```

```
R2#enable
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#no router bgp 500
R2(config)#end
*Oct 28 08:37:41.203: %BGP-5-ADJCHANGE: neighbor 10.2.3.3 Down Neighbor deleted
*Oct 28 08:37:41.207: %BGP_SESSION-5-ADJCHANGE: neighbor 10.2.3.3 IPv4 Unicast topology base removed from session  Neighbor deleted
*Oct 28 08:37:41.211: %BGP-5-ADJCHANGE: neighbor 2001:DB8:ACAD:1023::3 Down Unknown path error
*Oct 28 08:37:41.211: %BGP_SESSION-5-ADJCHANGE: neighbor 2001:DB8:ACAD:1023::3 IPv6 Unicast topology base removed from session  Unknown path error
R2(config)#end
```

Activate Windo
Go to Settings to a

```
R3#enable
R3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#no router bgp 300
R3(config)#end
R3#
*Oct 28 08:40:03.623: %SYS-5-CONFIG_I: Configured from console by console
R3#
```

Step 2: Configure MP-BGP (R1 = AS 1000, R2 = AS 500, R3 = AS 300)

```
R1(config)#configure terminal
                ^
% Invalid input detected at '^' marker.

R1(config)#router bgp 1000
R1(config-router)# bgp router-id 1.1.1.1
R1(config-router)# no bgp default ipv4-unicast
R1(config-router)# neighbor 10.1.2.2 remote-as 500
R1(config-router)# neighbor 10.1.3.3 remote-as 300
R1(config-router)# neighbor 10.1.3.130 remote-as 300
R1(config-router)# neighbor 2001:db8:acad:1012::2 remote-as 500
R1(config-router)# neighbor 2001:db8:acad:1013::3 remote-as 300
R1(config-router)# neighbor 2001:db8:acad:1014::3 remote-as 300
R1(config-router)# address-family ipv4 unicast
R1(config-router-af)#  network 192.168.1.0 mask 255.255.255.224
R1(config-router-af)#  network 192.168.1.64 mask 255.255.255.192
R1(config-router-af)#  neighbor 10.1.2.2 activate
R1(config-router-af)#  neighbor 10.1.3.3 activate
R1(config-router-af)#  neighbor 10.1.3.130 activate
R1(config-router-af)# exit-address-family
R1(config-router)# address-family ipv6 unicast
R1(config-router-af)#  network 2001:db8:acad:1000::/64
R1(config-router-af)#  network 2001:db8:acad:1001::/64
R1(config-router-af)#  neighbor 2001:db8:acad:1012::2 activate
R1(config-router-af)#  neighbor 2001:db8:acad:1013::3 activate
R1(config-router-af)#  neighbor 2001:db8:acad:1014::3 activate
R1(config-router-af)# exit-address-family
R1(config-router)#end
R1#
*Oct 28 08:42:48.723: %SYS-5-CONFIG_I: Configured from console by console
R1#
```

```
R2(config)#configure terminal
                ^
% Invalid input detected at '^' marker.

R2(config)#router bgp 500
R2(config-router)# bgp router-id 2.2.2.2
R2(config-router)# no bgp default ipv4-unicast
R2(config-router)# neighbor 10.1.2.1 remote-as 1000
R2(config-router)# neighbor 10.2.3.3 remote-as 300
R2(config-router)# neighbor 2001:db8:acad:1012::1 remote-as 1000
R2(config-router)# neighbor 2001:db8:acad:1023::3 remote-as 300
R2(config-router)# address-family ipv4 unicast
R2(config-router-af)#  network 192.168.2.0 mask 255.255.255.224
R2(config-router-af)#  network 192.168.2.64 mask 255.255.255.192
R2(config-router-af)#  neighbor 10.1.2.1 activate
R2(config-router-af)#  neighbor 10.2.3.3 activate
R2(config-router-af)# exit-address-family
R2(config-router)# address-family ipv6 unicast
R2(config-router-af)#  network 2001:db8:acad:2000::/64
R2(config-router-af)#  network 2001:db8:acad:2001::/64
R2(config-router-af)#  neighbor 2001:db8:acad:1012::1 activate
R2(config-router-af)#  neighbor 2001:db8:acad:1023::3 activate
R2(config-router-af)# exit-address-family
R2(config-router)#end
R2#
*Oct 28 08:45:11.635: %BGP-5-ADJCHANGE: neighbor 10.1.2.1 Up
R2#
*Oct 28 08:45:12.747: %SYS-5-CONFIG_I: Configured from console by console
*Oct 28 08:45:13.627: %BGP-5-ADJCHANGE: neighbor 2001:DB8:ACAD:1012::1 Up
R2#
```

```
R3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#router bgp 300
R3(config-router)# bgp router-id 3.3.3.3
R3(config-router)# no bgp default ipv4-unicast
R3(config-router)# neighbor 10.1.3.1 remote-as 1000
R3(config-router)# neighbor 10.1.3.129 remote-as 1000
R3(config-router)# neighbor 10.2.3.2 remote-as 500
R3(config-router)# neighbor 2001:db8:acad:1013::1 remote-as 1000
R3(config-router)# neighbor 2001:db8:acad:1014::1 remote-as 1000
R3(config-router)# neighbor 2001:db8:acad:1023::2 remote-as 500
R3(config-router)# address-family ipv4 unicast
R3(config-router-af)#  network 192.168.3.0 mask 255.255.255.224
R3(config-router-af)#  network 192.168.3.64 mask 255.255.255.192
R3(config-router-af)#  neighbor 10.1.3.1 activate
R3(config-router-af)#  neighbor 10.1.3.129 activate
R3(config-router-af)#  neighbor 10.2.3.2 activate
R3(config-router-af)# exit-address-family
R3(config-router)# address-family ipv6 unicast
R3(config-router-af)#  network 2001:db8:acad:3000::/64
R3(config-router-af)#  network 2001:db8:acad:3001::/64
R3(config-router-af)#  neighbor 2001:db8:acad:1013::1 activate
R3(config-router-af)#  neighbor 2001:db8:acad:1014::1 activate
R3(config-router-af)#  neighbor 2001:db8:acad:1023::2 activate
R3(config-router-af)# exit-address-family
R3(config-router)#end
*Oct 28 08:46:17.271: %BGP-5-ADJCHANGE: neighbor 2001:DB8:ACAD:1023::2 Up
R3(config-router)#end
*Oct 28 08:46:18.987: %BGP-5-ADJCHANGE: neighbor 2001:DB8:ACAD:1013::1 Up
R3(config-router)#e
*Oct 28 08:46:21.095: %BGP-5-ADJCHANGE: neighbor 10.1.3.1 Up
R3(config-router)#
*Oct 28 08:46:22.351: %BGP-5-ADJCHANGE: neighbor 10.2.3.2 Up
R3(config-router)#
*Oct 28 08:46:24.175: %BGP-5-ADJCHANGE: neighbor 2001:DB8:ACAD:1014::1 Up
R3(config-router)#
```

Step 3: Verify the neighbors (R2)

```
R2#show bgp all neighbors
For address family: IPv4 Unicast
BGP neighbor is 10.1.2.1,  remote AS 1000, external link
  BGP version 4, remote router ID 1.1.1.1
  BGP state = Established, up for 00:03:02
  Last read 00:00:50, last write 00:00:26, hold time is 180, keepalive interval is 60 seconds
  Neighbor sessions:
    1 active, is not multisession capable (disabled)
  Neighbor capabilities:
    Route refresh: advertised and received(new)
    Four-octets ASN Capability: advertised and received
    Address family IPv4 Unicast: advertised and received
    Enhanced Refresh Capability: advertised and received
    Multisession Capability:
    Stateful switchover support enabled: NO for session 1
  Message statistics:
    InQ depth is 0
    OutQ depth is 0
```

```
R2#show bgp ipv4 unicast summary
BGP router identifier 2.2.2.2, local AS number 500
BGP table version is 7, main routing table version 7
6 network entries using 864 bytes of memory
10 path entries using 800 bytes of memory
5/3 BGP path/bestpath attribute entries using 680 bytes of memory
4 BGP AS-PATH entries using 96 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 2440 total bytes of memory
BGP activity 12/0 prefixes, 20/0 paths, scan interval 60 secs

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
10.1.2.1        4         1000      12      12        7    0    0 00:07:44        4
10.2.3.3        4          300      13      15        7    0    0 00:06:33        4
R2#
```

```
R2#show bgp ipv6 unicast summary
BGP router identifier 2.2.2.2, local AS number 500
BGP table version is 7, main routing table version 7
6 network entries using 1008 bytes of memory
10 path entries using 1040 bytes of memory
5/3 BGP path/bestpath attribute entries using 680 bytes of memory
4 BGP AS-PATH entries using 96 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 2824 total bytes of memory
BGP activity 12/0 prefixes, 20/0 paths, scan interval 60 secs

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
2001:DB8:ACAD:1012::1
                4         1000       9       8        7    0    0 00:03:08        4
2001:DB8:ACAD:1023::3
                4          300       9       8        7    0    0 00:02:04        4
```

Step 4: Verify the BGP tables (R2)

```
R2#show bgp ipv4 unicast
BGP table version is 7, local router ID is 2.2.2.2
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *    192.168.1.0/27   10.2.3.3                               0 300 1000 i
 *>                    10.1.2.1                 0             0 1000 i
 *    192.168.1.64/26  10.2.3.3                               0 300 1000 i
 *>                    10.1.2.1                 0             0 1000 i
 *>   192.168.2.0/27   0.0.0.0                   0         32768 i
 *>   192.168.2.64/26  0.0.0.0                   0         32768 i
 *    192.168.3.0/27   10.1.2.1                               0 1000 300 i
 *>                    10.2.3.3                 0             0 300 i
 *    192.168.3.64/26  10.1.2.1                               0 1000 300 i
 *>                    10.2.3.3                 0             0 300 i
R2#show bgp ipv6 unicast
BGP table version is 7, local router ID is 2.2.2.2
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *    2001:DB8:ACAD:1000::/64
                       2001:DB8:ACAD:1023::3
                                                              0 300 1000 i
 *>                    2001:DB8:ACAD:1012::1
                                              0             0 1000 i
 *    2001:DB8:ACAD:1001::/64
                       2001:DB8:ACAD:1023::3
                                                              0 300 1000 i
 *>                    2001:DB8:ACAD:1012::1
                                              0             0 1000 i
 *>   2001:DB8:ACAD:2000::/64
                       ::                     0         32768 i
 *>   2001:DB8:ACAD:2001::/64
                       ::                     0         32768 i
```

```
R2#show bgp ipv4 unicast 192.168.1.0/27
BGP routing table entry for 192.168.1.0/27, version 2
Paths: (2 available, best #2, table default)
  Advertised to update-groups:
     1
  Refresh Epoch 1
  300 1000
    10.2.3.3 from 10.2.3.3 (3.3.3.3)
      Origin IGP, localpref 100, valid, external
      rx pathid: 0, tx pathid: 0
  Refresh Epoch 1
  1000
    10.1.2.1 from 10.1.2.1 (1.1.1.1)
      Origin IGP, metric 0, localpref 100, valid, external, best
      rx pathid: 0, tx pathid: 0x0
R2#show bgp ipv6 unicast 2001:db8:acad:1000::/64
BGP routing table entry for 2001:DB8:ACAD:1000::/64, version 2
Paths: (2 available, best #2, table default)
  Advertised to update-groups:
     1
  Refresh Epoch 1
  300 1000
    2001:DB8:ACAD:1023::3 (FE80::C803:1AFF:FE34:38) from 2001:DB8:ACAD:1023::3 (3.3.3.3)
      Origin IGP, localpref 100, valid, external
      rx pathid: 0, tx pathid: 0
  Refresh Epoch 1
  1000
    2001:DB8:ACAD:1012::1 (FE80::C801:21FF:FE38:38) from 2001:DB8:ACAD:1012::1 (1.1.1.1)
      Origin IGP, metric 0, localpref 100, valid, external, best
      rx pathid: 0, tx pathid: 0x0
```

Step 5: Check what R2 advertises

```
R2#show bgp ipv4 unicast neighbors 10.1.2.1 advertised-routes
BGP table version is 7, local router ID is 2.2.2.2
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>  192.168.1.0/27   10.1.2.1                 0             0 1000 i
 *>  192.168.1.64/26  10.1.2.1                 0             0 1000 i
 *>  192.168.2.0/27   0.0.0.0                  0         32768 i
 *>  192.168.2.64/26  0.0.0.0                  0         32768 i
 *>  192.168.3.0/27   10.2.3.3                 0             0 300 i
 *>  192.168.3.64/26  10.2.3.3                 0             0 300 i

Total number of prefixes 6
R2#show bgp ipv6 unicast neighbors 2001:db8:acad:1012::1 advertised-routes
BGP table version is 7, local router ID is 2.2.2.2
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>  2001:DB8:ACAD:1000::/64
                      2001:DB8:ACAD:1012::1
                                             0             0 1000 i
 *>  2001:DB8:ACAD:1001::/64
                      2001:DB8:ACAD:1012::1
                                             0             0 1000 i
 *>  2001:DB8:ACAD:2000::/64
                      ::                     0         32768 i
 *>  2001:DB8:ACAD:2001::/64
                      ::                     0         32768 i
 *>  2001:DB8:ACAD:3000::/64
                      2001:DB8:ACAD:1023::3
                                             0             0 300 i
 *>  2001:DB8:ACAD:3001::/64
                      2001:DB8:ACAD:1023::3
 --More--
```

Step 6: Check the routing tables

```
R2#show ip route bgp
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
B        192.168.1.0/27 [20/0] via 10.1.2.1, 00:14:42
B        192.168.1.64/26 [20/0] via 10.1.2.1, 00:14:42
      192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
B        192.168.3.0/27 [20/0] via 10.2.3.3, 00:14:10
B        192.168.3.64/26 [20/0] via 10.2.3.3, 00:14:10
R2#show ipv6 route bgp
IPv6 Routing Table - default - 13 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, R - RIP, H - NHRP, I1 - ISIS L1
       I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary, D - EIGRP
       EX - EIGRP external, ND - ND Default, NDp - ND Prefix, DCE - Destination
       NDr - Redirect, O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1
       OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, l - LISP
B   2001:DB8:ACAD:1000::/64 [20/0]
     via FE80::C801:21FF:FE38:38, GigabitEthernet2/0
B   2001:DB8:ACAD:1001::/64 [20/0]
     via FE80::C801:21FF:FE38:38, GigabitEthernet2/0
B   2001:DB8:ACAD:3000::/64 [20/0]
     via FE80::C803:1AFF:FE34:38, GigabitEthernet3/0
B   2001:DB8:ACAD:3001::/64 [20/0]
     via FE80::C803:1AFF:FE34:38, GigabitEthernet3/0
```

Step 7: IPv6 summarization (R1)

```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#router bgp 1000
R1(config-router)# address-family ipv6 unicast
R1(config-router-af)#  aggregate-address 2001:db8:acad:1000::/52 summary-only
R1(config-router-af)#end
R1#
*Oct 28 09:03:02.907: %SYS-5-CONFIG_I: Configured from console by console
```

```
R2#show bgp ipv6 unicast
BGP table version is 12, local router ID is 2.2.2.2
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *    2001:DB8:ACAD:1000::/52
                       2001:DB8:ACAD:1023::3
                                                              0 300 1000 i
 *>                    2001:DB8:ACAD:1012::1
                                            0                 0 1000 i
 *>   2001:DB8:ACAD:2000::/64
                       ::                   0             32768 i
 *>   2001:DB8:ACAD:2001::/64
                       ::                   0             32768 i
 *    2001:DB8:ACAD:3000::/64
                       2001:DB8:ACAD:1012::1
                                                              0 1000 300 i
 *>                    2001:DB8:ACAD:1023::3
                                            0                 0 300 i
 *    2001:DB8:ACAD:3001::/64
                       2001:DB8:ACAD:1012::1
                                                              0 1000 300 i
 *>                    2001:DB8:ACAD:1023::3
                                            0                 0 300 i
R2#
R2#show ipv6 route bgp
IPv6 Routing Table - default - 12 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, R - RIP, H - NHRP, I1 - ISIS L1
       I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary, D - EIGRP
       EX - EIGRP external, ND - ND Default, NDp - ND Prefix, DCE - Destination
       NDr - Redirect, O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1
       OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, l - LISP
B    2001:DB8:ACAD:1000::/52 [20/0]
     via FE80::C801:21FF:FE38:38, GigabitEthernet2/0
B    2001:DB8:ACAD:3000::/64 [20/0]
     via FE80::C803:1AFF:FE34:38, GigabitEthernet3/0
B    2001:DB8:ACAD:3001::/64 [20/0]
     via FE80::C803:1AFF:FE34:38, GigabitEthernet3/0
```

```
R3#show bgp ipv6 unicast
BGP table version is 14, local router ID is 3.3.3.3
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *    2001:DB8:ACAD:1000::/52
                       2001:DB8:ACAD:1023::2
                                                              0 500 1000 i
 *                     2001:DB8:ACAD:1014::1
                                            0                 0 1000 i
 *>                    2001:DB8:ACAD:1013::1
                                            0                 0 1000 i
 *    2001:DB8:ACAD:2000::/64
                       2001:DB8:ACAD:1014::1
                                                              0 1000 500 i
 *                     2001:DB8:ACAD:1013::1
                                                              0 1000 500 i
 *>                    2001:DB8:ACAD:1023::2
                                            0                 0 500 i
 *    2001:DB8:ACAD:2001::/64
                       2001:DB8:ACAD:1014::1
                                                              0 1000 500 i
 *                     2001:DB8:ACAD:1013::1
                                                              0 1000 500 i
 *>                    2001:DB8:ACAD:1023::2
                                            0                 0 500 i
 *>   2001:DB8:ACAD:3000::/64
                       ::                   0             32768 i
 *>   2001:DB8:ACAD:3001::/64
                       ::                   0             32768 i
```

```
R3#show ipv6 route bgp
IPv6 Routing Table - default - 14 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, R - RIP, H - NHRP, I1 - ISIS L1
       I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary, D - EIGRP
       EX - EIGRP external, ND - ND Default, NDp - ND Prefix, DCE - Destination
       NDr - Redirect, O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1
       OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, l - LISP
B    2001:DB8:ACAD:1000::/52 [20/0]
     via FE80::C801:21FF:FE38:0, Serial1/0
B    2001:DB8:ACAD:2000::/64 [20/0]
     via FE80::C802:32FF:FE58:54, GigabitEthernet2/0
B    2001:DB8:ACAD:2001::/64 [20/0]
     via FE80::C802:32FF:FE58:54, GigabitEthernet2/0
```

Conclusion: The above practical Implementing MP-BGP has been Implemented Successfully.

c) Implement eBGP for IPv4

Step 1: Remove the old BGP configuration

```
R1#enable
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#no router bgp 1000
R1(config)#end
*Oct 28 09:28:47.791: %BGP-5-ADJCHANGE: neighbor 10.1.2.2 Down Neighbor deleted
*Oct 28 09:28:47.795: %BGP_SESSION-5-ADJCHANGE: neighbor 10.1.2.2 IPv4 Unicast topology base removed from session  Neighbor deleted
*Oct 28 09:28:47.799: %BGP-5-ADJCHANGE: neighbor 10.1.3.3 Down Neighbor deleted
*Oct 28 09:28:47.803: %BGP_SESSION-5-ADJCHANGE: neighbor 10.1.3.3 IPv4 Unicast topology base removed from session  Neighbor deleted
*Oct 28 09:28:47.807: %BGP-5-ADJCHANGE: neighbor 10.1.3.130 Down Neighbor deleted
*Oct 28 09:28:47.807: %BGP_SESSION-5-ADJCHANGE: neighbor 10.1.3.130 IPv4 Unicast topology base removed from session  Neighbor deleted
*Oct 28 09:28:47.811: %BGP-5-ADJCHANGE: neighbor 2001:DB8:ACAD:1012::2 Down Unknown path error
*Oct 28 09:28:47.815: %BGP_SESSION-5-ADJCHANGE: neighbor 2001:DB8:ACAD:1012::2 IPv6 Unicast topology base removed from session  Unknown path error
*Oct 28 09:28:47.819: %BGP-5-ADJCHANGE: neighbor 2001:DB8:ACAD:1013::3 Down Unknown path error
*Oct 28 09:
R1(config)#end28:47.819: %BGP_SESSION-5-ADJCHANGE: neighbor 2001:DB8:ACAD:1013::3 IPv6 Unicast topology base removed from session  Unknown path error
```

Activate Windo
Go to Settings to ac

```
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#no router bgp 500
R2(config)#
*Oct 28 09:30:32.711: %BGP-5-ADJCHANGE: neighbor 10.2.3.3 Down Neighbor deleted
*Oct 28 09:30:32.715: %BGP_SESSION-5-ADJCHANGE: neighbor 10.2.3.3 IPv4 Unicast topology base removed from session  Neighbor deleted
*Oct 28 09:30:32.719: %BGP-5-ADJCHANGE: neighbor 2001:DB8:ACAD:1023::3 Down Unknown path error
*Oct 28 09:30:32.723: %BGP_SESSION-5-ADJCHANGE: neighbor 2001:DB8:ACAD:1023::3 IPv6 Unicast topology base removed from session  Unknown path error
R2(config)#end
R2#
```

Activate Windo
Go to Settings to a

```
R3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#no router bgp 300
R3(config)#end
R3#
*Oct 28 09:32:02.471: %SYS-5-CONFIG_I: Configured from console by console
```

Step 2: Configure eBGP (R1 = AS 1000, R2 = AS 500, R3 = AS 300)

```
R1(config)#configure terminal
                  ^
% Invalid input detected at '^' marker.

R1(config)#router bgp 1000
R1(config-router)# bgp router-id 1.1.1.1
R1(config-router)# neighbor 10.1.2.2 remote-as 500
R1(config-router)# neighbor 10.1.3.3 remote-as 300
R1(config-router)# neighbor 10.1.3.130 remote-as 300
R1(config-router)# network 192.168.1.0 mask 255.255.255.224
R1(config-router)# network 192.168.1.64 mask 255.255.255.192
R1(config-router)#end
R1#
*Oct 28 09:33:18.803: %SYS-5-CONFIG_I: Configured from console by console
R1#
```

```
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#router bgp 500
R2(config-router)# bgp router-id 2.2.2.2
R2(config-router)# neighbor 10.1.2.1 remote-as 1000
R2(config-router)# neighbor 10.2.3.3 remote-as 300
R2(config-router)# network 192.168.2.0 mask 255.255.255.224
R2(config-router)# network 192.168.2.64 mask 255.255.255.192
R2(config-router)#end
R2#
*Oct 28 09:34:09.635: %SYS-5-CONFIG_I: Configured from console by console
R2#
*Oct 28 09:34:13.279: %BGP-5-ADJCHANGE: neighbor 10.1.2.1 Up
R2#
```

```
R3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#router bgp 300
R3(config-router)# bgp router-id 3.3.3.3
R3(config-router)# neighbor 10.1.3.1 remote-as 1000
R3(config-router)# neighbor 10.1.3.129 remote-as 1000
R3(config-router)# neighbor 10.2.3.2 remote-as 500
R3(config-router)# network 192.168.3.0 mask 255.255.255.224
R3(config-router)# network 192.168.3.64 mask 255.255.255.192
R3(config-router)#end
R3#
*Oct 28 09:35:03.951: %SYS-5-CONFIG_I: Configured from console by console
R3#
*Oct 28 09:35:09.671: %BGP-5-ADJCHANGE: neighbor 10.2.3.2 Up
*Oct 28 09:35:10.299: %BGP-5-ADJCHANGE: neighbor 10.1.3.1 Up
*Oct 28 09:35:10.303: %BGP-5-ADJCHANGE: neighbor 10.1.3.129 Up
```

```
R2#show ip bgp summary
BGP router identifier 2.2.2.2, local AS number 500
BGP table version is 7, main routing table version 7
6 network entries using 864 bytes of memory
10 path entries using 800 bytes of memory
5/3 BGP path/bestpath attribute entries using 680 bytes of memory
4 BGP AS-PATH entries using 96 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 2440 total bytes of memory
BGP activity 6/0 prefixes, 10/0 paths, scan interval 60 secs

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
10.1.2.1        4         1000       7       7        7    0    0 00:01:50        4
10.2.3.3        4          300       7       7        7    0    0 00:00:53        4
R2#show ip bgp
BGP table version is 7, local router ID is 2.2.2.2
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *   192.168.1.0/27   10.2.3.3                               0 300 1000 i
 *>                   10.1.2.1                 0             0 1000 i
 *   192.168.1.64/26  10.2.3.3                               0 300 1000 i
 *>                   10.1.2.1                 0             0 1000 i
 *>  192.168.2.0/27   0.0.0.0                   0         32768 i
 *>  192.168.2.64/26  0.0.0.0                   0         32768 i
 *   192.168.3.0/27   10.1.2.1                               0 1000 300 i
 *>                   10.2.3.3                 0             0 300 i
 *   192.168.3.64/26  10.1.2.1                               0 1000 300 i
 *>                   10.2.3.3                 0             0 300 i
```

```
R2#show ip route bgp
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
B        192.168.1.0/27 [20/0] via 10.1.2.1, 00:00:46
B        192.168.1.64/26 [20/0] via 10.1.2.1, 00:00:46
      192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
B        192.168.3.0/27 [20/0] via 10.2.3.3, 00:00:45
B        192.168.3.64/26 [20/0] via 10.2.3.3, 00:00:45
```

Step 3: Summarization (atomic aggregate)

```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#router bgp 1000
R1(config-router)# aggregate-address 192.168.1.0 255.255.255.0 summary-only
R1(config-router)#end
R1#
*Oct 28 09:38:21.579: %SYS-5-CONFIG_I: Configured from console by console
R1#
```

```
R3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#router bgp 300
R3(config-router)# aggregate-address 192.168.3.0 255.255.255.0 summary-only
R3(config-router)#end
R3#
*Oct 28 09:39:18.375: %SYS-5-CONFIG_I: Configured from console by console
R3#
```

```
R2#show ip route bgp
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

B     192.168.1.0/24 [20/0] via 10.1.2.1, 00:03:36
B     192.168.3.0/24 [20/0] via 10.2.3.3, 00:02:35
R2#show ip bgp
BGP table version is 17, local router ID is 2.2.2.2
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *   192.168.1.0      10.2.3.3                               0 300 1000 i
 *>                   10.1.2.1                 0             0 1000 i
 *>  192.168.2.0/27   0.0.0.0                   0         32768 i
 *>  192.168.2.64/26  0.0.0.0                   0         32768 i
 *   192.168.3.0      10.1.2.1                               0 1000 300 i
 *>                   10.2.3.3                 0             0 300 i
R2#
```

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 6 subnets, 3 masks
C        10.1.2.0/24 is directly connected, GigabitEthernet2/0
L        10.1.2.1/32 is directly connected, GigabitEthernet2/0
C        10.1.3.0/25 is directly connected, Serial1/0
L        10.1.3.1/32 is directly connected, Serial1/0
C        10.1.3.128/25 is directly connected, Serial1/1
L        10.1.3.129/32 is directly connected, Serial1/1
      192.168.1.0/24 is variably subnetted, 5 subnets, 4 masks
B        192.168.1.0/24 [200/0] via 0.0.0.0, 00:04:59, Null0
C        192.168.1.0/27 is directly connected, Loopback0
L        192.168.1.1/32 is directly connected, Loopback0
 --More--
```

Step 4: AS-SET

```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface s1/0
R1(config-if)# shutdown
R1(config-if)#interface s1/1
R1(config-if)# shutdown
R1(config-if)#exit
R1(config)#router bgp 1000
R1(config-router)#$te-address 192.168.1.0 255.255.255.0 summary-only
R1(config-router)#end
*Oct 28 09:45:09.551: %BGP-5-NBR_RESET: Neighbor 10.1.3.3 reset (Interface flap)
*Oct 28 09:45:09.579: %BGP-5-ADJCHANGE: neighbor 10.1.3.3 Down Interface flap
*Oct 28 09:45:09.579: %BGP_SESSION-5-ADJCHANGE: neighbor 10.1.3.3 IPv4 Unicast topology base removed from session  Interface flap
*Oct 28 09:45:09.939: %BGP-5-NBR_RESET: Neighbor 10.1.3.130 reset (Interface flap)
*Oct 28 09:45:09.963: %BGP-5-ADJCHANGE: neighbor 10.1.3.130 Down Interface flap
*Oct 28 09:45:09.963: %BGP_SESSION-5-ADJCHANGE: neighbor 10.1.3.130 IPv4 Unicast topology base removed from session  Interface flap
R1(config-router)#end
*Oct 28 09:45:11.507: %LINK-5-CHANGED: Interface Serial1/0, changed state to administratively down
*Oct 28 09:45:11.899: %LINK-5-CHANGED: Interface Serial1/1, changed state to administratively down
*Oct 28 09:45:12.507: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/0, changed state to down
R1(config-router)#end
*Oct 28 09:45:12.899: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/1, changed state to down
R1(config-router)#end
```

```
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#router bgp 500
R2(config-router)# aggregate-address 192.168.1.0 255.255.255.0 summary-only
R2(config-router)#end
R2#
*Oct 28 09:46:10.767: %SYS-5-CONFIG_I: Configured from console by console
R2#
```

```
R3#show ip bgp
BGP table version is 24, local router ID is 3.3.3.3
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>  192.168.1.0      10.2.3.2                 0             0 500 i
 *>  192.168.2.0/27   10.2.3.2                 0             0 500 i
 *>  192.168.2.64/26  10.2.3.2                 0             0 500 i
 s>  192.168.3.0/27   0.0.0.0                  0         32768 i
 *>  192.168.3.0      0.0.0.0                  0         32768 i
 s>  192.168.3.64/26  0.0.0.0                  0         32768 i
R3#
```

```
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#router bgp 500
R2(config-router)#$te-address 192.168.1.0 255.255.255.0 summary-only
R2(config-router)#$address 192.168.1.0 255.255.255.0 as-set summary-only
R2(config-router)#end
R2#
*Oct 28 09:48:29.791: %SYS-5-CONFIG_I: Configured from console by console
R2#
```

```
R3#show ip bgp
BGP table version is 30, local router ID is 3.3.3.3
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>  192.168.1.0      10.2.3.2                 0             0 500 1000 i
 *>  192.168.2.0/27   10.2.3.2                 0             0 500 i
 *>  192.168.2.64/26  10.2.3.2                 0             0 500 i
 s>  192.168.3.0/27   0.0.0.0                  0         32768 i
 *>  192.168.3.0      0.0.0.0                  0         32768 i
 s>  192.168.3.64/26  0.0.0.0                  0         32768 i
R3#
```

Step 5: Default route

```
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#router bgp 500
R2(config-router)# neighbor 10.1.2.1 default-originate
R2(config-router)#end
R2#
*Oct 28 09:51:08.959: %SYS-5-CONFIG_I: Configured from console by console
R2#
```

```
R1#show ip route bgp
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

      192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
B        192.168.2.0/27 [20/0] via 10.1.2.2, 00:16:42
B        192.168.2.64/26 [20/0] via 10.1.2.2, 00:16:42
B     192.168.3.0/24 [20/0] via 10.1.2.2, 00:06:56
```

Conclusion: The above practical Implementing eBGP for IPv4 has been Implemented Successfully.

d) Implement BGP Path Manipulation

Step 1: Reset and bring the serial links back (R1, R2, R3)

```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface s1/0
R1(config-if)# no shutdown
R1(config-if)#interface s1/1
R1(config-if)# no shutdown
R1(config-if)#end
R1#
*Oct 28 09:54:43.283: %LINK-3-UPDOWN: Interface Serial1/0, change
*Oct 28 09:54:43.399: %SYS-5-CONFIG_I: Configured from console by
R1#
```

```
R1#enable
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface s1/0
R1(config-if)# no shutdown
R1(config-if)#interface s1/1
R1(config-if)# no shutdown
R1(config-if)#exit
R1(config)#no router bgp 1000
R1(config)#end
*Oct 28 10:03:46.939: %BGP-5-ADJCHANGE: neighbor 10.1.2.2 Down Neighbor deleted
*Oct 28 10:03:46.943: %BGP_SESSION-5-ADJCHANGE: neighbor 10.1.2.2 IPv4 Unicast topology base removed from session  Neighbor deleted
*Oct 28 10:03:46.947: %BGP-5-ADJCHANGE: neighbor 10.1.3.3 Down Neighbor deleted
*Oct 28 10:03:46.947: %BGP_SESSION-5-ADJCHANGE: neighbor 10.1.3.3 IPv4 Unicast topology base removed from session  Neighbor deleted
*Oct 28 10:03:46.951: %BGP-5-ADJCHANGE: neighbor 10.1.3.130 Down Neighbor deleted
*Oct 28 10:03:46.955: %BGP_SESSION-5-ADJCHANGE: neighbor 10.1.3.130 IPv4 Unicast topology base removed from session  Neighbor deleted
R1(config)#end
```

```
R2#enable
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#no router bgp 500
R2(config)#end
*Oct 28 10:05:18.075: %BGP-5-ADJCHANGE: neighbor 10.2.3.3 Down Neighbor deleted
*Oct 28 10:05:18.075: %BGP_SESSION-5-ADJCHANGE: neighbor 10.2.3.3 IPv4 Unicast topology base removed from session  Neighbor deleted
R2(config)#
```

```
R3#enable
R3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#no router bgp 300
R3(config)#end
R3#
*Oct 28 10:06:52.775: %SYS-5-CONFIG_I: Configured from console by console
R3#
```

Step 2: eBGP base (R1 = AS 6500, R2 = AS 500, R3 = AS 300)

```
R1(config)#configure terminal
                  ^
% Invalid input detected at '^' marker.

R1(config)#router bgp 6500
R1(config-router)# bgp router-id 1.1.1.1
R1(config-router)# neighbor 10.1.2.2 remote-as 500
R1(config-router)# neighbor 10.1.3.3 remote-as 300
R1(config-router)# neighbor 10.1.3.130 remote-as 300
R1(config-router)# network 192.168.1.0 mask 255.255.255.224
R1(config-router)# network 192.168.1.64 mask 255.255.255.192
R1(config-router)#end
R1#
*Oct 28 10:08:07.211: %SYS-5-CONFIG_I: Configured from console by console
R1#
```

```
R2(config)#configure terminal
                  ^
% Invalid input detected at '^' marker.

R2(config)#router bgp 500
R2(config-router)# bgp router-id 2.2.2.2
R2(config-router)# neighbor 10.1.2.1 remote-as 6500
R2(config-router)# neighbor 10.2.3.3 remote-as 300
R2(config-router)# network 192.168.2.0 mask 255.255.255.224
R2(config-router)# network 192.168.2.64 mask 255.255.255.192
R2(config-router)#end
R2#
*Oct 28 10:08:57.563: %SYS-5-CONFIG_I: Configured from console by console
R2#
*Oct 28 10:09:02.219: %BGP-5-ADJCHANGE: neighbor 10.1.2.1 Up
R2#
```

```
R3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#router bgp 300
R3(config-router)# bgp router-id 3.3.3.3
R3(config-router)# neighbor 10.1.3.1 remote-as 6500
R3(config-router)# neighbor 10.1.3.129 remote-as 6500
R3(config-router)# neighbor 10.2.3.2 remote-as 500
R3(config-router)# network 192.168.3.0 mask 255.255.255.224
R3(config-router)# network 192.168.3.64 mask 255.255.255.192
R3(config-router)#end
*Oct 28 10:09:55.203: %BGP-5-ADJCHANGE: neighbor 10.1.3.1 Up
*Oct 28 10:09:55.243: %BGP-5-ADJCHANGE: neighbor 10.2.3.2 Up
R3(config-router)#
*Oct 28 10:09:58.159: %BGP-5-ADJCHANGE: neighbor 10.1.3.129 Up
R3(config-router)#
```

```
R1#show ip bgp summary
BGP router identifier 1.1.1.1, local AS number 6500
BGP table version is 7, main routing table version 7
6 network entries using 864 bytes of memory
14 path entries using 1120 bytes of memory
5/3 BGP path/bestpath attribute entries using 680 bytes of memory
4 BGP AS-PATH entries using 96 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 2760 total bytes of memory
BGP activity 6/0 prefixes, 14/0 paths, scan interval 60 secs

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
10.1.2.2        4          500       7       7        7    0    0 00:01:51        4
10.1.3.3        4          300       7       7        7    0    0 00:00:58        4
10.1.3.130      4          300       7       7        7    0    0 00:00:55        4
R1#show ip bgp
BGP table version is 7, local router ID is 1.1.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>  192.168.1.0/27   0.0.0.0                  0         32768 i
 *>  192.168.1.64/26  0.0.0.0                  0         32768 i
 *   192.168.2.0/27   10.1.3.130                           0 300 500 i
 *                    10.1.3.3                              0 300 500 i
 *>                   10.1.2.2                 0            0 500 i
 *   192.168.2.64/26  10.1.3.130                           0 300 500 i
 *                    10.1.3.3                              0 300 500 i
 *>                   10.1.2.2                 0            0 500 i
 *   192.168.3.0/27   10.1.2.2                              0 500 300 i
 *                    10.1.3.130               0            0 300 i
 *>                   10.1.3.3                 0            0 300 i
 *   192.168.3.64/26  10.1.2.2                              0 500 300 i
 *                    10.1.3.130               0            0 300 i
 *>                   10.1.3.3                 0            0 300 i
R1#
```

Step 3: ACL filter (R3 sends R1 only its own networks)

```
R1#show ip bgp | i 300
 *    192.168.2.0/27   10.1.3.130                           0 300 500 i
 *                     10.1.3.3                              0 300 500 i
 *    192.168.2.64/26  10.1.3.130                           0 300 500 i
 *                     10.1.3.3                              0 300 500 i
 *    192.168.3.0/27   10.1.2.2                              0 500 300 i
 *                     10.1.3.130               0            0 300 i
 *>                   10.1.3.3                 0            0 300 i
 *    192.168.3.64/26  10.1.2.2                              0 500 300 i
 *                     10.1.3.130               0            0 300 i
 *>                   10.1.3.3                 0            0 300 i
R1#
```

```
R3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#ip access-list extended ALLOWED_TO_R1
R3(config-ext-nacl)# permit ip host 192.168.3.0 host 255.255.255.224
R3(config-ext-nacl)# permit ip host 192.168.3.64 host 255.255.255.192
R3(config-ext-nacl)#exit
R3(config)#router bgp 300
R3(config-router)# neighbor 10.1.3.1 distribute-list ALLOWED_TO_R1 out
R3(config-router)# neighbor 10.1.3.129 distribute-list ALLOWED_TO_R1 out
R3(config-router)#end
R3#clear ip bgp * out
*Oct 28 10:13:38.591: %SYS-5-CONFIG_I: Configured from console by console
R3#clear ip bgp * out
*Oct 28 10:13:39.935: %SYS-5-CONFIG_I: Configured from console by console
R3#clear ip bgp * out
R3#
```

```
R1#show ip bgp | i 300
 *    192.168.3.0/27   10.1.2.2                              0 500 300 i
 *                     10.1.3.130               0             0 300 i
 *>                    10.1.3.3                 0             0 300 i
 *    192.168.3.64/26  10.1.2.2                              0 500 300 i
 *                     10.1.3.130               0             0 300 i
 *>                    10.1.3.3                 0             0 300 i
R1#
```

Step 4: Prefix-list filter (R1 accepts only R2's own networks)

```
R1#show ip bgp | i 500
 *>   192.168.2.0/27   10.1.2.2                  0             0 500 i
 *>   192.168.2.64/26  10.1.2.2                  0             0 500 i
 *    192.168.3.0/27   10.1.2.2                                0 500 300 i
 *    192.168.3.64/26  10.1.2.2                                0 500 300 i
```

```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ip prefix-list ALLOWED_FROM_R2 seq 5 permit 192.168.2.0/24 le 27
R1(config)#router bgp 6500
R1(config-router)# neighbor 10.1.2.2 prefix-list ALLOWED_FROM_R2 in
R1(config-router)#end
R1#clear ip bgp * in
*Oct 28 10:19:08.059: %SYS-5-CONFIG_I: Configured from console by console
R1#
```

```
R1#show ip bgp | i 500
 *>   192.168.2.0/27   10.1.2.2                  0             0 500 i
 *>   192.168.2.64/26  10.1.2.2                  0             0 500 i
 *    192.168.3.0/27   10.1.2.2                                0 500 300 i
 *    192.168.3.64/26  10.1.2.2                                0 500 300 i
R1#
```

Step 5: AS-PATH filter (R1 sends R2 only R1's own networks)

```
R2#show ip bgp | i 6500
 *    192.168.1.0/27   10.2.3.3                              0 300 6500 i
 *>                    10.1.2.1                 0             0 6500 i
 *    192.168.1.64/26  10.2.3.3                              0 300 6500 i
 *>                    10.1.2.1                 0             0 6500 i
 *    192.168.3.0/27   10.1.2.1                              0 6500 300 i
 *    192.168.3.64/26  10.1.2.1                              0 6500 300 i
R2#
```

```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ip as-path access-list 1 permit ^$
R1(config)#router bgp 6500
R1(config-router)# neighbor 10.1.2.2 filter-list 1 out
R1(config-router)#end
R1#clear ip bgp * out
*Oct 28 10:22:35.675: %SYS-5-CONFIG_I: Configured from console by console
R1#
```

```
R2#show ip bgp | i 6500
 *    192.168.1.0/27   10.2.3.3                              0 300 6500 i
 *>                    10.1.2.1                 0             0 6500 i
 *    192.168.1.64/26  10.2.3.3                              0 300 6500 i
 *>                    10.1.2.1                 0             0 6500 i
R2#
```

Step 6: Local preference (R1 prefers 10.1.3.130)

```
R1#show ip route bgp
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

      192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
B        192.168.2.0/27 [20/0] via 10.1.2.2, 00:19:04
B        192.168.2.64/26 [20/0] via 10.1.2.2, 00:19:04
      192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
B        192.168.3.0/27 [20/0] via 10.1.3.3, 00:19:03
B        192.168.3.64/26 [20/0] via 10.1.3.3, 00:19:03
R1#
```

```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#$ist PREFERRED_IPV4_PATH seq 5 permit 192.168.3.0/24 le 27
R1(config)#route-map USE_THIS_PATH_FOR_IPV4 permit 10
R1(config-route-map)# match ip address prefix-list PREFERRED_IPV4_PATH
R1(config-route-map)# set local-preference 250
R1(config-route-map)#exit
R1(config)#router bgp 6500
R1(config-router)# neighbor 10.1.3.130 route-map USE_THIS_PATH_FOR_IPV4 in
R1(config-router)#end
R1#clear ip bgp * in
*Oct 28 10:30:13.143: %SYS-5-CONFIG_I: Configured from console by console
R1#
```

```
R1#show ip route bgp
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

      192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
B        192.168.2.0/27 [20/0] via 10.1.2.2, 00:20:51
B        192.168.2.64/26 [20/0] via 10.1.2.2, 00:20:51
      192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
B        192.168.3.0/27 [20/0] via 10.1.3.3, 00:20:50
B        192.168.3.64/26 [20/0] via 10.1.3.3, 00:20:50
R1#show ip bgp
BGP table version is 7, local router ID is 1.1.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>  192.168.1.0/27   0.0.0.0                  0         32768 i
 *>  192.168.1.64/26  0.0.0.0                  0         32768 i
 *>  192.168.2.0/27   10.1.2.2                 0             0 500 i
 *>  192.168.2.64/26  10.1.2.2                 0             0 500 i
 *   192.168.3.0/27   10.1.2.2                               0 500 300 i
 *                    10.1.3.130               0             0 300 i
 *>                   10.1.3.3                 0             0 300 i
 *   192.168.3.64/26  10.1.2.2                               0 500 300 i
 *                    10.1.3.130               0             0 300 i
 *>                   10.1.3.3                 0             0 300 i
R1#
```

Conclusion: The above practical Implementing BGP Path Manipulation has been Implemented Successfully.

PRACTICAL-10

Aim:

a) Implement IPsec Site-to-Site VPNs

b) Implement GRE over IPsec Site-to-Site VPNs

c) Implement VRF Lite

Theory:

IPsec Site-to-Site VPN

IPsec is a set of protocols that protects IP traffic between two sites. IKE phase 1 builds a secure channel between the two routers using an ISAKMP policy and a pre-shared key. IKE phase 2 builds the IPsec tunnel that encrypts the data, using a transform set. An access list defines the interesting traffic, which is the traffic that is encrypted, and a crypto map ties the peer, the access list and the transform set together on the outside interface.

GRE over IPsec

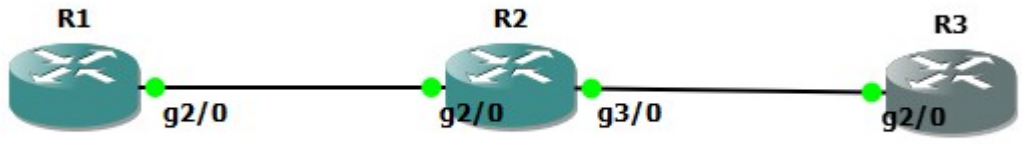
An IPsec site-to-site VPN carries only unicast traffic, so routing protocols such as OSPF cannot run over it. GRE can carry multicast and broadcast traffic but does not encrypt. GRE over IPsec combines both: the GRE tunnel carries the routing protocol and IPsec in transport mode encrypts the GRE packets. It can be configured with a crypto map or with an IPsec profile applied to the tunnel interface.

VRF-Lite

By default all interfaces of a router share one global routing table. Virtual Routing and Forwarding (VRF) creates separate virtual routing tables on the same router, so different customers can use the same IP addresses without mixing their traffic. VRF-Lite is VRF without MPLS.

a) Implement IPsec Site-to-Site VPNs

Topology



Addressing Table

Device	Interface	IPv4 Address
R1	G2/0	64.100.0.2/30
	Loopback1	10.10.1.1/24
	Loopback2	10.10.2.1/24
R2	G2/0	64.100.0.1/30
	G3/0	64.100.1.1/30
	Loopback0	209.165.200.225/27

Device	Interface	IPv4 Address
R3	G2/0	64.100.1.2/30
	Loopback5	10.10.5.1/24
	Loopback16	10.10.16.1/24
	Loopback17	10.10.17.1/24

Step 1: Configure the basic settings on R1, R2 and R3

```
R1#enable
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#hostname R1
R1(config)#no ip domain lookup
R1(config)#interface g2/0
R1(config-if)# ip address 64.100.0.2 255.255.255.252
R1(config-if)# no shutdown
R1(config-if)#interface loopback 1
R1(config-if)# ip address 10.10.1.1 255.255.255.0
R1(config-if)#interface loopback 2
R1(config-if)# ip address 10.10.2.1 255.255.255.0
R1(config-if)#exit
R1(config)#ip route 0.0.0.0 0.0.0.0 64.100.0.1
R1(config)#end
*Oct 29 14:50:30.999: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback1, changed state to up
*Oct 29 14:50:31.415: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback2, changed state to up
R1(config)#end
*Oct 29 14:50:32.771: %LINK-3-UPDOWN: Interface GigabitEthernet2/0, changed state to up
*Oct 29 14:50:33.771: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet2/0, changed state to up
R1(config)#
```

```
R2#enable
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#hostname R2
R2(config)#no ip domain lookup
R2(config)#interface g2/0
R2(config-if)# ip address 64.100.0.1 255.255.255.252
R2(config-if)# no shutdown
R2(config-if)#interface g3/0
R2(config-if)# ip address 64.100.1.1 255.255.255.252
R2(config-if)# no shutdown
R2(config-if)#interface loopback 0
R2(config-if)# ip address 209.165.200.225 255.255.255.224
R2(config-if)#exit
R2(config)#ip route 0.0.0.0 0.0.0.0 Loopback0
%Default route without gateway, if not a point-to-point interface, may impact performance
R2(config)#ip route 10.10.0.0 255.255.252.0 64.100.0.2
R2(config)#ip route 10.10.4.0 255.255.252.0 64.100.1.2
R2(config)#ip route 10.10.16.0 255.255.248.0 64.100.1.2
R2(config)#end
*Oct 29 14:53:36.783: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
R2(config)#end
*Oct 29 14:53:38.167: %LINK-3-UPDOWN: Interface GigabitEthernet2/0, changed state to up
*Oct 29 14:53:38.571: %LINK-3-UPDOWN: Interface GigabitEthernet3/0, changed state to up
*Oct 29 14:53:39.167: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet2/0, changed state to up
R2(config)#end
*Oct 29 14:53:39.571: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet3/0, changed state to up
R2(config)#
```

```
R3#enable
R3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#hostname R3
R3(config)#no ip domain lookup
R3(config)#interface g2/0
R3(config-if)# ip address 64.100.1.2 255.255.255.252
R3(config-if)# no shutdown
R3(config-if)#interface loopback 5
R3(config-if)# ip address 10.10.5.1 255.255.255.0
R3(config-if)#interface loopback 16
R3(config-if)# ip address 10.10.16.1 255.255.255.0
R3(config-if)#interface loopback 17
R3(config-if)# ip address 10.10.17.1 255.255.255.0
R3(config-if)#exit
R3(config)#ip route 0.0.0.0 0.0.0.0 64.100.1.1
R3(config)#end
*Oct 29 14:55:04.971: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback5, changed state to up
*Oct 29 14:55:05.383: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback16, changed state to up
*Oct 29 14:55:05.715: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback17, changed state to up
R3(config)#end
*Oct 29 14:55:06.747: %LINK-3-UPDOWN: Interface GigabitEthernet2/0, changed state to up
*Oct 29 14:55:07.747: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet2/0, changed state to up
R3(config)#
```

Step 2: Verify the basic configuration

```
R1(config)#do show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        unassigned      YES unset  administratively down down
Serial1/0              unassigned      YES unset  administratively down down
Serial1/1              unassigned      YES unset  administratively down down
Serial1/2              unassigned      YES unset  administratively down down
Serial1/3              unassigned      YES unset  administratively down down
GigabitEthernet2/0     64.100.0.2      YES manual up                    up
Loopback1              10.10.1.1       YES manual up                    up
Loopback2              10.10.2.1       YES manual up                    up
```

```
R1#ping 10.10.5.1 source 10.10.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.5.1, timeout is 2 seconds:
Packet sent with a source address of 10.10.1.1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 152/184/228 ms
R1#ping 10.10.16.1 source 10.10.2.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.16.1, timeout is 2 seconds:
Packet sent with a source address of 10.10.2.1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 116/169/216 ms
R1#
```

Step 3: Configure the VPN on R1 and R3

```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#crypto isakmp policy 10
R1(config-isakmp)# encryption aes 256
R1(config-isakmp)# hash sha256
R1(config-isakmp)# authentication pre-share
R1(config-isakmp)# group 14
R1(config-isakmp)# lifetime 3600
R1(config-isakmp)# exit
R1(config)#crypto isakmp key cisco123 address 64.100.1.2
R1(config)#crypto ipsec transform-set S2S-VPN esp-aes 256 esp-sha256-hmac
R1(cfg-crypto-trans)# exit
R1(config)#crypto ipsec security-association lifetime seconds 1800
R1(config)#ip access-list extended S2S-VPN-ACL
R1(config-ext-nacl)# permit ip 10.10.0.0 0.0.3.255 10.10.4.0 0.0.3.255
R1(config-ext-nacl)# permit ip 10.10.0.0 0.0.3.255 10.10.16.0 0.0.7.255
R1(config-ext-nacl)# exit
R1(config)#crypto map S2S-CMAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
        and a valid access list have been configured.
R1(config-crypto-map)# match address S2S-VPN-ACL
R1(config-crypto-map)# set peer 64.100.1.2
R1(config-crypto-map)# set pfs group14
R1(config-crypto-map)# set transform-set S2S-VPN
R1(config-crypto-map)# set security-association lifetime seconds 900
R1(config-crypto-map)# exit
R1(config)#interface g2/0
R1(config-if)# crypto map S2S-CMAP
R1(config-if)#end
*Oct 29 15:02:36.895: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
R1(config-if)#
```

```
R3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#crypto isakmp policy 10
R3(config-isakmp)# encryption aes 256
R3(config-isakmp)# hash sha256
R3(config-isakmp)# authentication pre-share
R3(config-isakmp)# group 14
R3(config-isakmp)# lifetime 3600
R3(config-isakmp)# exit
R3(config)#crypto isakmp key cisco123 address 64.100.0.2
R3(config)#crypto ipsec transform-set S2S-VPN esp-aes 256 esp-sha256-hmac
R3(cfg-crypto-trans)# exit
R3(config)#crypto ipsec security-association lifetime seconds 1800
R3(config)#ip access-list extended S2S-VPN-ACL
R3(config-ext-nacl)# permit ip 10.10.4.0 0.0.3.255 10.10.0.0 0.0.3.255
R3(config-ext-nacl)# permit ip 10.10.16.0 0.0.7.255 10.10.0.0 0.0.3.255
R3(config-ext-nacl)# exit
R3(config)#crypto map S2S-CMAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
        and a valid access list have been configured.
R3(config-crypto-map)# match address S2S-VPN-ACL
R3(config-crypto-map)# set peer 64.100.0.2
R3(config-crypto-map)# set pfs group14
R3(config-crypto-map)# set transform-set S2S-VPN
R3(config-crypto-map)# set security-association lifetime seconds 900
R3(config-crypto-map)# exit
R3(config)#interface g2/0
R3(config-if)# crypto map S2S-CMAP
R3(config-if)#end
*Oct 29 15:03:59.063: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
R3(config-if)#
```

Step 4: Verify the VPN

```
R1#show crypto map
Crypto Map IPv4 "S2S-CMAP" 10 ipsec-isakmp
        Peer = 64.100.1.2
        Extended IP access list S2S-VPN-ACL
            access-list S2S-VPN-ACL permit ip 10.10.0.0 0.0.3.255 10.10.4.0 0.0.3.255
            access-list S2S-VPN-ACL permit ip 10.10.0.0 0.0.3.255 10.10.16.0 0.0.7.255
        Current peer: 64.100.1.2
        Security association lifetime: 4608000 kilobytes/900 seconds
        Responder-Only (Y/N): N
        PFS (Y/N): Y
        DH group:  group14
        Transform sets={
                S2S-VPN:  { esp-256-aes esp-sha256-hmac  } ,
        }
        Interfaces using crypto map S2S-CMAP:
                GigabitEthernet2/0
```

```
R3#show crypto map
*Oct 29 15:06:42.555: %SYS-5-CONFIG_I: Configured from console by console
R3#show crypto map
Crypto Map IPv4 "S2S-CMAP" 10 ipsec-isakmp
        Peer = 64.100.0.2
        Extended IP access list S2S-VPN-ACL
            access-list S2S-VPN-ACL permit ip 10.10.4.0 0.0.3.255 10.10.0.0 0.0.3.255
            access-list S2S-VPN-ACL permit ip 10.10.16.0 0.0.7.255 10.10.0.0 0.0.3.255
        Current peer: 64.100.0.2
        Security association lifetime: 4608000 kilobytes/900 seconds
        Responder-Only (Y/N): N
        PFS (Y/N): Y
        DH group:  group14
        Transform sets={
                S2S-VPN:  { esp-256-aes esp-sha256-hmac  } ,
        }
        Interfaces using crypto map S2S-CMAP:
                GigabitEthernet2/0
```

```
R1#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst             src             state          conn-id status

IPv6 Crypto ISAKMP SA

R1#ping 64.100.1.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 64.100.1.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 152/189/296 ms
R1#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst             src             state          conn-id status

IPv6 Crypto ISAKMP SA
```

```
R1#ping 10.10.5.1 source 10.10.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.5.1, timeout is 2 seconds:
Packet sent with a source address of 10.10.1.1
..!!!
Success rate is 60 percent (3/5), round-trip min/avg/max = 160/166/180 ms
R1#ping 10.10.16.1 source 10.10.2.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.16.1, timeout is 2 seconds:
Packet sent with a source address of 10.10.2.1
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 144/168/188 ms
R1#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst             src             state          conn-id status
64.100.1.2      64.100.0.2      QM_IDLE           1001 ACTIVE

IPv6 Crypto ISAKMP SA

R1#show crypto ipsec sa

interface: GigabitEthernet2/0
    Crypto map tag: S2S-CMAP, local addr 64.100.0.2

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (10.10.0.0/255.255.252.0/0/0)
   remote ident (addr/mask/prot/port): (10.10.4.0/255.255.252.0/0/0)
   current_peer 64.100.1.2 port 500
     PERMIT, flags={origin_is_acl,}
    #pkts encaps: 3, #pkts encrypt: 3, #pkts digest: 3
    #pkts decaps: 3, #pkts decrypt: 3, #pkts verify: 3
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0

     local crypto endpt.: 64.100.0.2, remote crypto endpt.: 64.100.1.2
     path mtu 1500, ip mtu 1500, ip mtu idb GigabitEthernet2/0
     current outbound spi: 0xFDF31E7B(4260568699)
     PFS (Y/N): Y, DH group: group14

     inbound esp sas:
      spi: 0xED332D81(3979554177)
 --More--
```

Conclusion: The above practical Implementing IPsec Site-to-Site VPNs has been Implemented Successfully.

b) Implement GRE over IPsec Site-to-Site VPNs

Topology

The topology and the addressing are the same as in part (a). A GRE tunnel is added between R1 and R3.

Addressing Table

Device	Interface	IPv4 Address
R1	Tunnel1	172.16.1.1/30
R3	Tunnel1	172.16.1.2/30

Step 1: Remove the VPN settings of part (a) on R1 and R3

```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface g2/0
R1(config-if)# no crypto map
R1(config-if)# exit
R1(config)#no crypto map S2S-CMAP
R1(config)#no ip access-list extended S2S-VPN-ACL
R1(config)#no crypto ipsec transform-set S2S-VPN
R1(config)#no crypto isakmp key cisco123 address 64.100.1.2
R1(config)#no crypto isakmp policy 10
R1(config)#end
*Oct 29 15:17:15.779: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is OFF
R1(config)#
```

```
R3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#interface g2/0
R3(config-if)# no crypto map
R3(config-if)# exit
R3(config)#no crypto map S2S-CMAP
R3(config)#no ip access-list extended S2S-VPN-ACL
R3(config)#no crypto ipsec transform-set S2S-VPN
R3(config)#no crypto isakmp key cisco123 address 64.100.0.2
R3(config)#no crypto isakmp policy 10
R3(config)#end
*Oct 29 15:18:46.407: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is OFF
R3(config)#
```

Step 2: Check the traffic before the tunnel (R1)

```
R1#ping 10.10.16.1 source 10.10.2.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.16.1, timeout is 2 seconds:
Packet sent with a source address of 10.10.2.1
!
*Oct 29 15:20:44.867: %SYS-5-CONFIG_I: Configured from console by console!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 172/248/384 ms
R1#traceroute 10.10.16.1 source 10.10.2.1
Type escape sequence to abort.
Tracing the route to 10.10.16.1
VRF info: (vrf in name/id, vrf out name/id)
  1 64.100.0.1 120 msec 96 msec 64 msec
  2 64.100.1.2 196 msec 256 msec 264 msec
R1#
```

Step 3: Configure GRE over IPsec on R1 using a crypto map

```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#crypto isakmp policy 10
R1(config-isakmp)# encryption aes 256
R1(config-isakmp)# hash sha256
R1(config-isakmp)# authentication pre-share
R1(config-isakmp)# group 14
R1(config-isakmp)# lifetime 3600
R1(config-isakmp)# exit
R1(config)#crypto isakmp key ciscol23 address 64.100.1.2
R1(config)#crypto ipsec transform-set GRE-VPN esp-aes 256 esp-sha256-hmac
R1(cfg-crypto-trans)# mode transport
R1(cfg-crypto-trans)# exit
R1(config)#ip access-list extended GRE-VPN-ACL
R1(config-ext-nacl)# permit gre host 64.100.0.2 host 64.100.1.2
R1(config-ext-nacl)# exit
R1(config)#crypto map GRE-CMAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
        and a valid access list have been configured.
R1(config-crypto-map)# match address GRE-VPN-ACL
R1(config-crypto-map)# set transform-set GRE-VPN
R1(config-crypto-map)# set peer 64.100.1.2
R1(config-crypto-map)# exit
R1(config)#interface g2/0
R1(config-if)# crypto map GRE-CMAP
R1(config-if)# exit
R1(config)#interface tunnel 1
R1(config-if)# bandwidth 4000
R1(config-if)# ip address 172.16.1.1 255.255.255.252
R1(config-if)# ip mtu 1400
R1(config-if)# tunnel source 64.100.0.2
R1(config-if)# tunnel destination 64.100.1.2
R1(config-if)#end
*Oct 29 15:22:33.191: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
R1(config-if)#end
*Oct 29 15:22:34.295: %LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel1, changed state to up
R1(config-if)#
```

Step 4: Configure GRE over IPsec on R3 using an IPsec profile

```
R3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#crypto isakmp policy 10
R3(config-isakmp)# encryption aes 256
R3(config-isakmp)# hash sha256
R3(config-isakmp)# authentication pre-share
R3(config-isakmp)# group 14
R3(config-isakmp)# lifetime 3600
R3(config-isakmp)# exit
R3(config)#crypto isakmp key ciscol23 address 64.100.0.2
R3(config)#crypto ipsec transform-set GRE-VPN esp-aes 256 esp-sha256-hmac
R3(cfg-crypto-trans)# mode transport
R3(cfg-crypto-trans)# exit
R3(config)#crypto ipsec profile GRE-PROFILE
R3(ipsec-profile)# set transform-set GRE-VPN
R3(ipsec-profile)# exit
R3(config)#interface tunnel 1
R3(config-if)# bandwidth 4000
R3(config-if)# ip address 172.16.1.2 255.255.255.252
R3(config-if)# ip mtu 1400
R3(config-if)# tunnel source 64.100.1.2
R3(config-if)# tunnel destination
*Oct 29 15:23:46.075: %SYS-5-CONFIG_I: Configured from console by console64.100.0.2
R3(config-if)# tunnel protection ipsec profile GRE-PROFILE
R3(config-if)#end
*Oct 29 15:23:48.379: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
R3(config-if)#end
*Oct 29 15:23:48.415: %LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel1, changed state to up
R3(config-if)#
```

Step 5: Check that no traffic is encrypted yet (R1)

```
R1#show crypto ipsec sa | include encrypt|decrypt
    #pkts encaps: 0, #pkts encrypt: 0, #pkts digest: 0
    #pkts decaps: 0, #pkts decrypt: 0, #pkts verify: 0
R1#
```

Step 6: Run OSPF over the tunnel on R1 and R3

```
    #pkts decaps: 0, #pkts decrypt: 0, #pkts verify: 0
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface loopback 1
R1(config-if)# ip ospf network point-to-point
R1(config-if)#interface loopback 2
R1(config-if)# ip ospf network point-to-point
R1(config-if)#exit
R1(config)#router ospf 123
R1(config-router)# router-id 1.1.1.1
R1(config-router)# auto-cost reference-bandwidth 1000
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
R1(config-router)# network 10.10.1.0 0.0.0.255 area 0
R1(config-router)# network 10.10.2.0 0.0.0.255 area 0
R1(config-router)# network 172.16.1.0 0.0.0.3 area 0
R1(config-router)#end
R1#
*Oct 29 15:27:06.947: %SYS-5-CONFIG_I: Configured from console by console
R1#
```

```
R3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#interface loopback 5
R3(config-if)# ip ospf network point-to-point
R3(config-if)#interface loopback 16
R3(config-if)# ip ospf network point-to-point
R3(config-if)#interface loopback 17
R3(config-if)# ip ospf network point-to-point
R3(config-if)#exit
R3(config)#router ospf 123
R3(config-router)# router-id 3.3.3.1
R3(config-router)# auto-cost reference-bandwidth 1000
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
R3(config-router)# network 10.10.5.0 0.0.0.255 area 0
R3(config-router)# network 10.10.16.0 0.0.0.255 area 0
R3(config-router)# network 10.10.17.0 0.0.0.255 area 0
R3(config-router)# network 172.16.1.0 0.0.0.3 area
*Oct 29 15:28:22.375: %SYS-5-CONFIG_I: Configured from console by console0
R3(config-router)#end
*Oct 29 15:28:25.051: %OSPF-5-ADJCHG: Process 123, Nbr 1.1.1.1 on Tunnel1 from LOADING to FULL, Loading Done
R3(config-router)#end
```

Step 7: Verify the GRE over IPsec tunnel

On R1

```
R1#show interfaces tunnel 1
Tunnel1 is up, line protocol is up
  Hardware is Tunnel
  Internet address is 172.16.1.1/30
  MTU 17916 bytes, BW 4000 Kbit/sec, DLY 50000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation TUNNEL, loopback not set
  Keepalive not set
  Tunnel source 64.100.0.2, destination 64.100.1.2
  Tunnel protocol/transport GRE/IP
    Key disabled, sequencing disabled
    Checksumming of packets disabled
  Tunnel TTL 255, Fast tunneling enabled
  Tunnel transport MTU 1476 bytes
  Tunnel transmit bandwidth 8000 (kbps)
  Tunnel receive bandwidth 8000 (kbps)
  Last input 00:00:08, output 00:00:06, output hang never
  Last clearing of "show interface" counters 00:07:45
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue: 0/0 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
 --More--
```

```
R1#show crypto session
Crypto session current status

Interface: GigabitEthernet2/0
Session status: UP-ACTIVE
Peer: 64.100.1.2 port 500
  IKEv1 SA: local 64.100.0.2/500 remote 64.100.1.2/500 Active
  IPSEC FLOW: permit 47 host 64.100.0.2 host 64.100.1.2
        Active SAs: 2, origin: crypto map
```

```
R1#show ip ospf interface brief
Interface    PID   Area            IP Address/Mask    Cost  State Nbrs F/C
Lo1          123   0               10.10.1.1/24       1     P2P   0/0
Lo2          123   0               10.10.2.1/24       1     P2P   0/0
Tu1          123   0               172.16.1.1/30      250   P2P   1/1
R1#
```

```
R1#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
3.3.3.1           0   FULL/  -        00:00:38    172.16.1.2      Tunnel1
R1#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is 64.100.0.1 to network 0.0.0.0

      10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
O        10.10.5.0/24 [110/251] via 172.16.1.2, 00:46:12, Tunnel1
O        10.10.16.0/24 [110/251] via 172.16.1.2, 00:46:12, Tunnel1
O        10.10.17.0/24 [110/251] via 172.16.1.2, 00:46:12, Tunnel1
R1#show ip route 172.16.0.0
Routing entry for 172.16.0.0/16, 2 known subnets
  Attached (2 connections)
  Variably subnetted with 2 masks
C        172.16.1.0/30 is directly connected, Tunnel1
L        172.16.1.1/32 is directly connected, Tunnel1
```

On R3

```
R3#show interfaces tunnel 1
*Oct 29 16:20:07.071: %SYS-5-CONFIG_I: Configured from console by console
R3#show interfaces tunnel 1
Tunnel1 is up, line protocol is up
  Hardware is Tunnel
  Internet address is 172.16.1.2/30
  MTU 17874 bytes, BW 4000 Kbit/sec, DLY 50000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation TUNNEL, loopback not set
  Keepalive not set
  Tunnel source 64.100.1.2, destination 64.100.0.2
  Tunnel protocol/transport GRE/IP
    Key disabled, sequencing disabled
    Checksumming of packets disabled
  Tunnel TTL 255, Fast tunneling enabled
  Tunnel transport MTU 1434 bytes
  Tunnel transmit bandwidth 8000 (kbps)
  Tunnel receive bandwidth 8000 (kbps)
  Tunnel protection via IPSec (profile "GRE-PROFILE")
  Last input 00:00:06, output never, output hang never
  Last clearing of "show interface" counters 00:56:27
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue: 0/0 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
 --More--
```

```
R3#show crypto session
Crypto session current status

Interface: Tunnel1
Session status: UP-ACTIVE
Peer: 64.100.0.2 port 500
  IKEv1 SA: local 64.100.1.2/500 remote 64.100.0.2/500 Active
  IPSEC FLOW: permit 47 host 64.100.1.2 host 64.100.0.2
        Active SAs: 2, origin: crypto map

R3#show ip ospf interface brief
Interface    PID   Area            IP Address/Mask    Cost  State Nbrs F/C
Lo5          123   0               10.10.5.1/24       1     P2P   0/0
Lo16         123   0               10.10.16.1/24      1     P2P   0/0
Lo17         123   0               10.10.17.1/24      1     P2P   0/0
Tu1          123   0               172.16.1.2/30      250   P2P   1/1
R3#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           0   FULL/  -        00:00:30    172.16.1.1      Tunnel1
R3#
```

```
R3#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

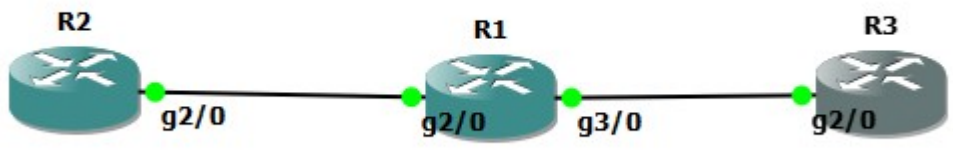
Gateway of last resort is 64.100.1.1 to network 0.0.0.0

      10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
O        10.10.1.0/24 [110/251] via 172.16.1.1, 00:54:43, Tunnel1
O        10.10.2.0/24 [110/251] via 172.16.1.1, 00:54:43, Tunnel1
R3#show ip route 172.16.0.0
Routing entry for 172.16.0.0/16, 2 known subnets
  Attached (2 connections)
  Variably subnetted with 2 masks
C        172.16.1.0/30 is directly connected, Tunnel1
L        172.16.1.2/32 is directly connected, Tunnel1
R3#
```

Conclusion: The above practical Implementing GRE over IPsec Site-to-Site VPNs has been Implemented Successfully.

c) Implement VRF Lite

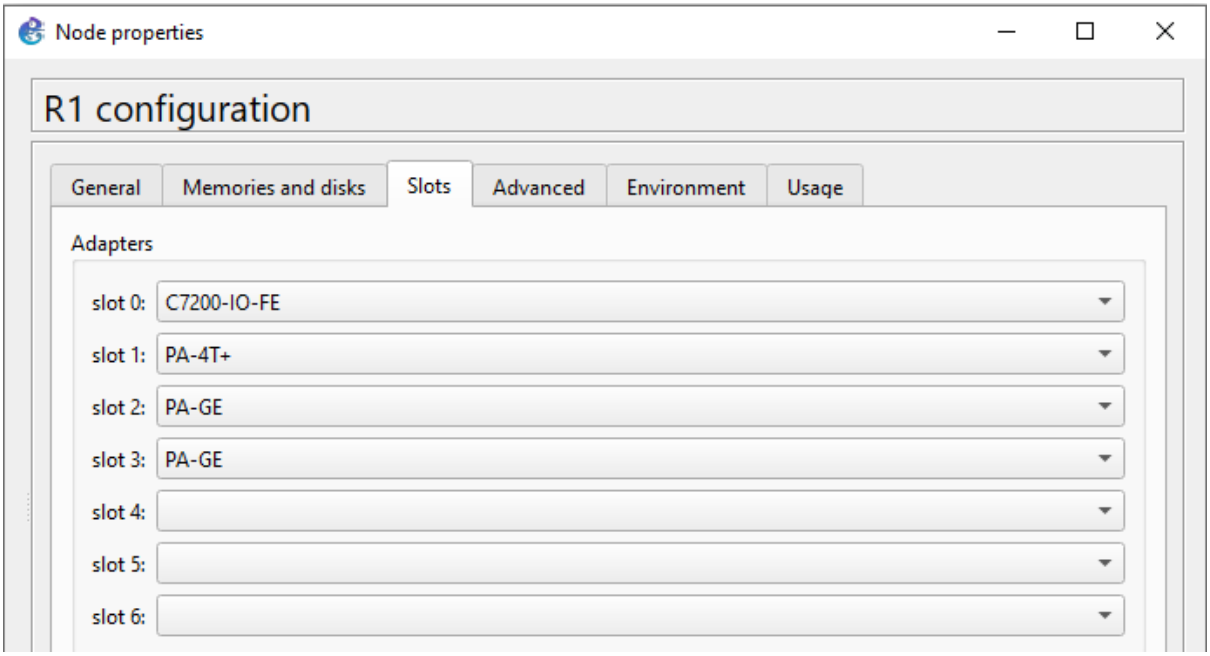
Topology

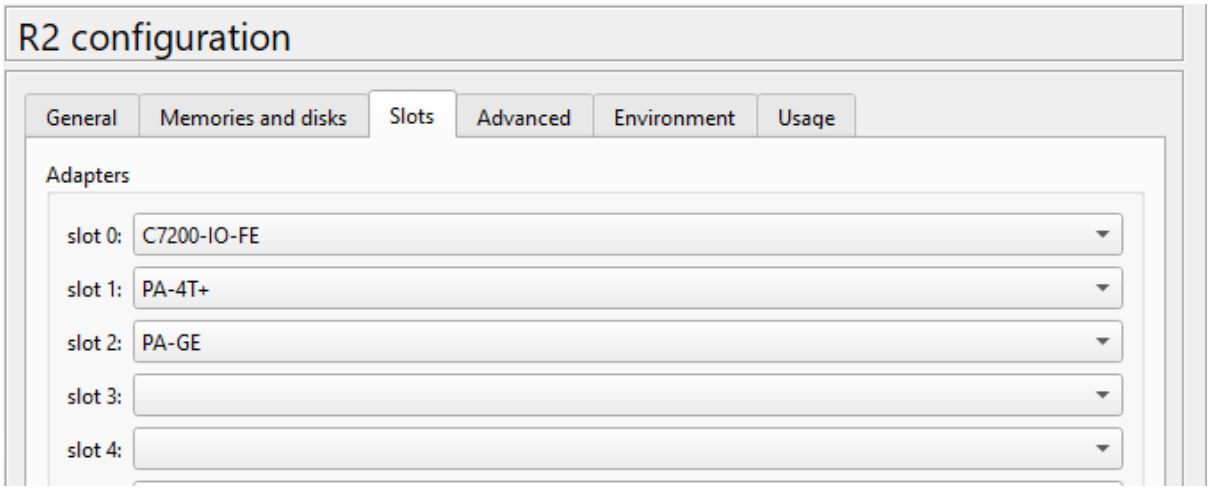


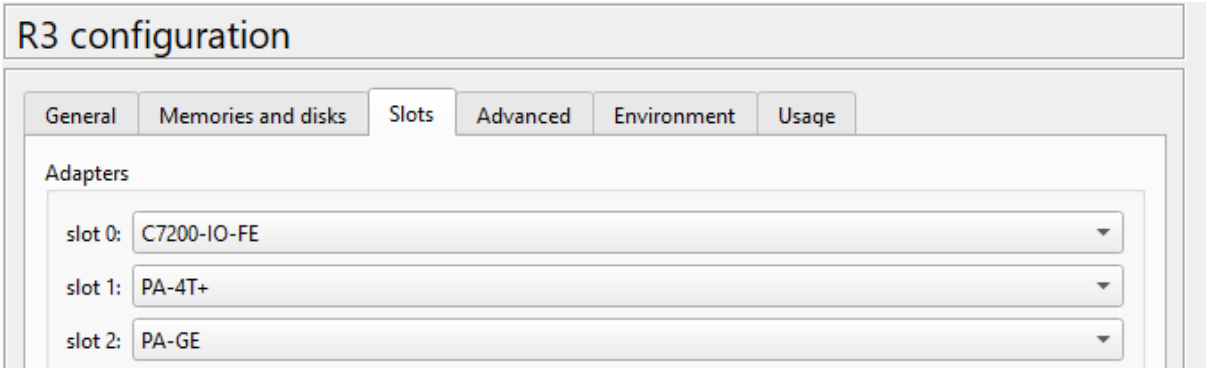
Addressing Table

Device	Interface	IPv4 Address	IPv6 Address
R1	G2/0 (VRF Customer_A)	10.1.2.1/24	2001:db8:acad:1012::1/64
	G3/0 (VRF Customer_B)	10.1.2.1/24	2001:db8:acad:1012::1/64
R2	G2/0	10.1.2.2/24	2001:db8:acad:1012::2/64
	Loopback0	192.168.2.1/24	2001:db8:acad:2000::1/64
R3	G2/0	10.1.2.2/24	2001:db8:acad:1012::2/64
	Loopback0	192.168.3.1/24	2001:db8:acad:3000::1/64

Step 1: Configure the router slots in GNS3







Step 2: Configure the basic settings on R1, R2 and R3

```
R1#enable
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#hostname R1
R1(config)#no ip domain lookup
R1(config)#ipv6 unicast-routing
R1(config)#end
R1#
*Oct 30 08:16:24.159: %SYS-5-CONFIG_I: Configured from console by console
R1#
```

```
R2#enable
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#hostname R2
R2(config)#no ip domain lookup
R2(config)#ipv6 unicast-routing
R2(config)#interface g2/0
R2(config-if)# ip address 10.1.2.2 255.255.255.0
R2(config-if)# ipv6 address fe80::2:1 link-local
R2(config-if)# ipv6 address 2001:db8:acad:1012::2/64
R2(config-if)# no shutdown
R2(config-if)# exit
R2(config)#interface loopback 0
R2(config-if)# ip address 192.168.2.1 255.255.255.0
R2(config-if)# ipv6 address fe80::2:2 link-local
R2(config-if)# ipv6 address 2001:db8:acad:2000::1/64
R2(config-if)# exit
R2(config)#ip route 0.0.0.0 0.0.0.0 g2/0 10.1.2.1
R2(config)#ipv6 route ::/0 g2/0 2001:db8:acad:1012::1
R2(config)#end
*Oct 30 08:17:33.339: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
R2(config)#end
*Oct 30 08:17:35.095: %LINK-3-UPDOWN: Interface GigabitEthernet2/0, changed state to up
*Oct 30 08:17:36.095: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet2/0, changed state to up
R2(config)#
```

```
R3#enable
R3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#hostname R3
R3(config)#no ip domain lookup
R3(config)#ipv6 unicast-routing
R3(config)#interface g2/0
R3(config-if)# ip address 10.1.2.2 255.255.255.0
R3(config-if)# ipv6 address fe80::3:1 link-local
R3(config-if)# ipv6 address 2001:db8:acad:1012::2/64
R3(config-if)# no shutdown
R3(config-if)# exit
R3(config)#interface loopback 0
R3(config-if)# ip address 192.168.3.1 255.255.255.0
R3(config-if)# ipv6 address fe80::3:2 link-local
R3(config-if)# ipv6 address 2001:db8:acad:3000::1/64
R3(config-if)# exit
R3(config)#ip route 0.0.0.0 0.0.0.0 g2/0 10.1.2.1
R3(config)#ipv6 route ::/0 g2/0 2001:db8:acad:1012::1
R3(config)#end
*Oct 30 08:18:55.311: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
R3(config)#end
*Oct 30 08:18:57.071: %LINK-3-UPDOWN: Interface GigabitEthernet2/0, changed state to up
*Oct 30 08:18:58.071: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet2/0, changed state to up
R3(config)#end
```

Step 3: Verify the basic configuration

```
R2#show ip interface brief
*Oct 30 08:22:05.739: %SYS-5-CONFIG_I: Configured from console by console
R2#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        unassigned      YES unset  administratively down down
Serial1/0              unassigned      YES unset  administratively down down
Serial1/1              unassigned      YES unset  administratively down down
Serial1/2              unassigned      YES unset  administratively down down
Serial1/3              unassigned      YES unset  administratively down down
GigabitEthernet2/0     10.1.2.2        YES manual up                    up
Loopback0              192.168.2.1     YES manual up                    up
R2#
```

```
R3#show ip interface brief
*Oct 30 08:24:33.923: %SYS-5-CONFIG_I: Configured from console by console
R3#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        unassigned      YES unset  administratively down down
Serial1/0              unassigned      YES unset  administratively down down
Serial1/1              unassigned      YES unset  administratively down down
Serial1/2              unassigned      YES unset  administratively down down
Serial1/3              unassigned      YES unset  administratively down down
GigabitEthernet2/0     10.1.2.2        YES manual up                    up
Loopback0              192.168.3.1     YES manual up                    up
R3#
```

Step 4: Create the VRFs and assign the interfaces on R1

```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#vrf definition Customer_A
R1(config-vrf)# address-family ipv4
R1(config-vrf-af)# exit-address-family
R1(config-vrf)# address-family ipv6
R1(config-vrf-af)# exit-address-family
R1(config-vrf)# exit
R1(config)#vrf definition Customer_B
R1(config-vrf)# address-family ipv4
R1(config-vrf-af)# exit-address-family
R1(config-vrf)# address-family ipv6
R1(config-vrf-af)# exit-address-family
R1(config-vrf)# exit
R1(config)#interface g2/0
R1(config-if)# vrf forwarding Customer_A
R1(config-if)# ip address 10.1.2.1 255.255.255.0
R1(config-if)# ipv6 address fe80::1:1 link-local
R1(config-if)# ipv6 address 2001:db8:acad:1012::1/64
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)#interface g3/0
R1(config-if)# vrf forwarding Customer_B
R1(config-if)# ip address 10.1.2.1 255.255.255.0
R1(config-if)# ipv6 address fe80::1:2 link-local
R1(config-if)# ipv6 address 2001:db8:acad:1012::1/64
R1(config-if)# no shutdown
R1(config-if)#end
*Oct 30 08:26:29.243: %LINK-3-UPDOWN: Interface GigabitEthernet2/0, changed state to up
*Oct 30 08:26:29.975: %LINK-3-UPDOWN: Interface GigabitEthernet3/0, changed state to up
*Oct 30 08:26:30.243: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet2/0, changed state to up
R1(config-if)#end
R1#
*Oct 30 08:26:30.975: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet3/0, changed state to up
R1#
*Oct 30 08:26:31.403: %SYS-5-CONFIG_I: Configured from console by console
R1#
```

Step 5: Verify the VRFs

```
R1#show ip vrf interfaces
Interface              IP-Address      VRF                              Protocol
Gi2/0                  10.1.2.1        Customer_A                       up
Gi3/0                  10.1.2.1        Customer_B                       up
R1#show ip route vrf Customer_A

Routing Table: Customer_A
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        10.1.2.0/24 is directly connected, GigabitEthernet2/0
L        10.1.2.1/32 is directly connected, GigabitEthernet2/0
R1#
```

```
R1#show ipv6 route vrf Customer_B
IPv6 Routing Table - Customer_B - 3 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, R - RIP, H - NHRP, I1 - ISIS L1
       I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary, D - EIGRP
       EX - EIGRP external, ND - ND Default, NDp - ND Prefix, DCE - Destination
       NDr - Redirect, O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1
       OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, l - LISP
C   2001:DB8:ACAD:1012::/64 [0/0]
     via GigabitEthernet3/0, directly connected
L   2001:DB8:ACAD:1012::1/128 [0/0]
     via GigabitEthernet3/0, receive
L   FF00::/8 [0/0]
     via Null0, receive
R1#ping vrf Customer_A 10.1.2.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.2.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 60/98/180 ms
R1#
```

```
R1#ping vrf Customer_B 10.1.2.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.2.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 52/102/176 ms
R1#ping vrf Customer_A 2001:db8:acad:1012::2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:1012::2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 48/90/208 ms
R1#ping vrf Customer_B 2001:db8:acad:1012::2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:1012::2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 56/110/252 ms
R1#
```

Step 6: Check that the distant networks are not reachable yet

```
R1#ping vrf Customer_A 192.168.2.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.2.1, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
R1#ping vrf Customer_B 192.168.3.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.1, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
R1#ping vrf Customer_A 2001:db8:acad:2000::1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:2000::1, timeout is 2 seconds:

% No valid route for destination
Success rate is 0 percent (0/1)
R1#ping vrf Customer_B 2001:db8:acad:3000::1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:3000::1, timeout is 2 seconds:

% No valid route for destination
Success rate is 0 percent (0/1)
R1#
```

Step 7: Configure the static routes inside each VRF on R1

```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ip route vrf Customer_A 192.168.2.0 255.255.255.0 g2/0 10.1.2.2
R1(config)#$vrf Customer_A 2001:db8:acad:2000::/64 g2/0 2001:db8:acad:1012::2
R1(config)#ip route vrf Customer_B 192.168.3.0 255.255.255.0 g3/0 10.1.2.2
R1(config)#$vrf Customer_B 2001:db8:acad:3000::/64 g3/0 2001:db8:acad:1012::2
R1(config)#end
R1#
*Oct 30 08:37:25.331: %SYS-5-CONFIG_I: Configured from console by console
R1#
```

Step 8: Verify reachability and isolation

On R1 (the networks of each customer are reachable)

```
R1#ping vrf Customer_A 192.168.2.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.2.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 68/96/180 ms
R1#ping vrf Customer_B 192.168.3.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 44/89/152 ms
R1#ping vrf Customer_A 2001:db8:acad:2000::1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:2000::1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 64/87/112 ms
R1#ping vrf Customer_B 2001:db8:acad:3000::1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:3000::1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 60/99/164 ms
R1#
```

On R1 (the networks of the other customer are not reachable)

```
R1#ping vrf Customer_A 192.168.3.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.1, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
R1#ping vrf Customer_B 192.168.2.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.2.1, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
R1#
```

On R2 and R3

```
R2#ping 10.1.2.1 source loopback0
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.2.1, timeout is 2 seconds:
Packet sent with a source address of 192.168.2.1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 76/107/136 ms
R2#ping 2001:db8:acad:1012::1 source loopback0
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:1012::1, timeout is 2 seconds:
Packet sent with a source address of 2001:DB8:ACAD:2000::1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 68/92/120 ms
R2#
```

```
R3#ping 10.1.2.1 source loopback0
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.2.1, timeout is 2 seconds:
Packet sent with a source address of 192.168.3.1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 40/88/176 ms
R3#ping 2001:db8:acad:1012::1 source loopback0
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:1012::1, timeout is 2 seconds:
Packet sent with a source address of 2001:DB8:ACAD:3000::1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 56/84/104 ms
R3#
```

Conclusion: The above practical Implementing VRF-Lite has been Implemented Successfully.